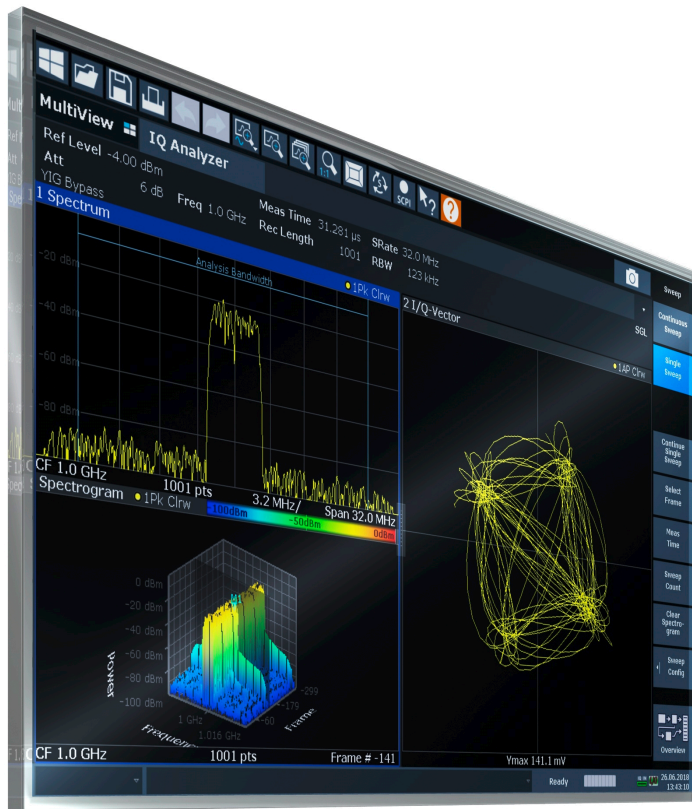


R&S®FSV/A3000 I/Q Analyzer and I/Q Input Interfaces User Manual



1178853602
Version 16

ROHDE & SCHWARZ
Make ideas real



This manual applies to the following R&S®FSV3000 and R&S®FSVA3000 models with firmware version 2.50 and higher:

- R&S®FSV3004 (1330.5000K04) / R&S®FSVA3004 (1330.5000K05)
- R&S®FSV3007 (1330.5000K07) / R&S®FSVA3007 (1330.5000K08)
- R&S®FSV3013 (1330.5000K13) / R&S®FSVA3013 (1330.5000K14)
- R&S®FSV3030 (1330.5000K30) / R&S®FSVA3030 (1330.5000K31)
- R&S®FSV3044 (1330.5000K43) / R&S®FSVA3044 (1330.5000K44)
- R&S®FSV3050 (1330.5000K50) / R&S®FSVA3050 (1330.5000K51)

In addition to the base unit, the following options are described:

- R&S®FSV3-B4, OCXO (1330.3794.02)
- R&S®FSV3-B5/B5E, Additional Interfaces (1330.3820.02/03)
- R&S®FSV3-B10, external generator control (1330.3859.02)
- R&S®FSV3-B13, high-pass filter (1313.0761.02)
- R&S®FSV3-B24, preamplifier (1330.4049.XX)
- R&S®FSV3-B25, electronic attenuator (1330.4078.02)
- R&S®FSV3-B40, 40 MHz analysis bandwidth extension (1330.4103.02)
- R&S®FSV3-B114, Enhanced computing power (1330.4910.02)
- R&S®FSV3-B200, 200 MHz analysis bandwidth extension (1330.4132.02)
- R&S®FSV3-B400, 400 MHz analysis bandwidth extension (1330.7154.02) / R&S®FSV3-U400 (1330.7183.02)
- R&S®FSV3-B600, 600 MHz analysis bandwidth extension (1346.5004.02)
- R&S®FSV3-B601, 600 MHz analysis bandwidth extension (for frequencies > 7.5 GHz, 1346.5762.02)
- R&S®FSV3-B1000, 1 GHz analysis bandwidth extension (for frequencies > 7.5 GHz, 1346.3699.02)
- R&S®FSV3-B1001, 1 GHz analysis bandwidth extension (1346.5779.02)
- R&S®FSV3-B271, analog baseband interface (1330.4190.02)
- R&S®FSV3-K575, I/Q noise cancellation (1346.6769.xx)

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Throughout this manual, products from Rohde & Schwarz are indicated without the ® symbol, e.g. R&S®FSV/A3000 is indicated as R&S FSV/A3000 and refers to both the R&S FSV3000 and the R&S FSVA3000. Products of the R&S®SMW family, e.g. R&S®SMW200A, are indicated as R&S SMW.

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1 Documentation overview

This section provides an overview of the R&S FSV/A user documentation. Unless specified otherwise, you find the documents at:

www.rohde-schwarz.com/manual/FSVA3000

www.rohde-schwarz.com/manual/FSV3000

Further documents are available at:

www.rohde-schwarz.com/product/FSVA3000

www.rohde-schwarz.com/product/FSV3000

1.1 Getting started manual

Introduces the R&S FSV/A and describes how to set up and start working with the product. Includes basic operations, typical measurement examples, and general information, e.g. safety instructions, etc.

A printed version is delivered with the instrument. A PDF version is available for download on the Internet.

1.2 User manuals and help

Separate user manuals are provided for the base unit and the firmware applications:

- **Base unit manual**
Contains the description of all instrument modes and functions. It also provides an introduction to remote control, a complete description of the remote control commands with programming examples, and information on maintenance, instrument interfaces and error messages. Includes the contents of the getting started manual.
- **Firmware application manual**
Contains the description of the specific functions of a firmware application, including remote control commands. Basic information on operating the R&S FSV/A is not included.

The contents of the user manuals are available as help in the R&S FSV/A. The help offers quick, context-sensitive access to the complete information for the base unit and the firmware applications.

All user manuals are also available for download or for immediate display on the Internet.

1.3 Printed safety instructions

Provides safety information in many languages. The printed document is delivered with the product.

1.4 Videos

Find various videos on Rohde & Schwarz products and test and measurement topics on YouTube: <https://www.youtube.com/@Rohde-Schwarz>

1.5 Service manual

Describes the performance test for checking compliance with rated specifications, firmware update, troubleshooting, adjustments, installing options and maintenance.

The service manual is available for registered users on the global Rohde & Schwarz information system (GLORIS):

[R&S®FSVA3000/FSV3000 Service manual](#)

1.6 Instrument health guide

Provides information for avoiding electrical and mechanical damage to the instrument. It is available for download from the product page on the internet.

See www.rohde-schwarz.com/brochure-datasheet/FSV3000.

1.7 Instrument security procedures

Deals with security issues when working with the R&S FSV/A in secure areas. It is available for download on the internet.

1.8 Specifications and brochures

The specifications document, also known as the data sheet, contains the technical specifications of the R&S FSV/A. It also lists the firmware applications and their order numbers, and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

See www.rohde-schwarz.com/brochure-datasheet/FSV3000 /

www.rohde-schwarz.com/brochure-datasheet/FSVA3000

1.9 Application notes, application cards, white papers, etc.

These documents deal with special applications or background information on particular topics.

See www.rohde-schwarz.com/application/FSV3000 /

www.rohde-schwarz.com/application/FSVA3000

1.10 Release notes and open-source acknowledgment (OSA)

The release notes list new features, improvements and known issues of the current software version, and describe the software installation.

The software uses several valuable open source software packages. An open source acknowledgment document provides verbatim license texts of the used open source software.

See www.rohde-schwarz.com/firmware/FSV3000 /

www.rohde-schwarz.com/firmware/FSVA3000

2 Welcome to the I/Q Analyzer application

The R&S FSV3 I/Q Analyzer is a firmware application that adds functionality to perform I/Q data acquisition and analysis to the R&S FSV/A.

The R&S FSV3 I/Q Analyzer features:

- Acquisition of analog I/Q data
- Import of stored I/Q data from other applications
- Spectrum, magnitude, I/Q vector and separate I and Q component analysis of any I/Q data on the instrument
- Export of I/Q data to other applications

This user manual contains a description of the functionality that the application provides, including remote control operation.

All functions not discussed in this manual are the same as in the base unit and are described in the R&S FSV/A User Manual. The latest version is available for download at the product homepage <http://www.rohde-schwarz.com/product/FSVA3000>.

Additional information

Several application notes discussing I/Q analysis are available from the Rohde & Schwarz website:

[1EF85: Converting R&S I/Q data files](#)

[1EF92: Wideband Signal Analysis](#)

[1MA257: Wideband mm-Wave Signal Generation and Analysis](#)

[1EF84: Differential measurements with Spectrum Analyzers and Probes](#)

Installation

The R&S FSV3 I/Q Analyzer application is part of the standard base unit and requires no further installation.

2.1 Starting the I/Q Analyzer application

The I/Q Analyzer is an application on the R&S FSV/A.

To activate the I/Q Analyzer application

1. Select the [MODE] key.
A dialog box opens that contains all applications currently available on your R&S FSV/A.
2. Select the "I/Q Analyzer" item.



The R&S FSV/A opens a new channel for the I/Q Analyzer application.

The measurement is started immediately with the default settings.

It can be configured in the I/Q Analyzer "Overview" dialog box, which is displayed when you select the "Overview" softkey from any menu (see [Section 5.1, "Configuration overview"](#), on page 41).

Multiple Channels and Sequencer Function

When you activate an application, a new channel is created which determines the measurement settings for that application (channel). The same application can be activated with different measurement settings by creating several channels for the same application.

The number of channels that can be configured at the same time depends on the available memory on the instrument.

Only one measurement can be performed at any time, namely the one in the currently active channel. However, in order to perform the configured measurements consecutively, a Sequencer function is provided.

If activated, the measurements configured in the currently defined channels are performed one after the other in the order of the tabs. The currently active measurement is indicated by a  symbol in the tab label.

The result displays of the individual channels are updated in the tabs (as well as the "MultiView") as the measurements are performed. Sequential operation itself is independent of the currently *displayed* tab.

For details on the Sequencer function see the R&S FSV/A User Manual.

2.2 Understanding the display information

The following figure shows a measurement diagram during I/Q Analyzer operation. All different information areas are labeled. They are explained in more detail in the following sections.



Figure 2-1: Screen elements in the I/Q Analyzer application

- 1+4 = Window title bar with diagram-specific (trace) information
- 2 = Channel bar for firmware and measurement settings
- 3 = Diagram area
- 5 = Diagram footer with diagram-specific information, depending on result display
- 6 = Instrument status bar with error messages and date/time display

Channel bar information

In the I/Q Analyzer application, the R&S FSV/A shows the following settings:

Table 2-1: Information displayed in the channel bar for the I/Q Analyzer application

"Ref Level"	Reference level
"m.+el.Att"	Mechanical and electronic RF attenuation
"SAN: Att"	Attenuation at the analyzer when using an optional external frontend
"ExtFE: Att"	External frontend attenuation when measuring external frontend input.
"Ref Offset"	Reference level offset
"Freq"	Center frequency
"Meas Time"	Measurement time
"Rec Length"	Defined record length (number of samples to capture)

"SRate"	Defined sample rate for data acquisition
"RBW"	(Spectrum evaluation only) Resolution bandwidth calculated from the sample rate and record length

In addition, the channel bar also displays information on instrument settings that affect the measurement results even though this is not immediately apparent from the display of the measured values (e.g. transducer or trigger settings). This information is displayed only when applicable for the current measurement.

For details see the R&S FSV/A Getting Started manual.

Window title bar information

For each diagram, the header provides the following information:



Figure 2-2: Window title bar information in the I/Q Analyzer application

- 1 = Window number
- 2 = Window type
- 3 = Trace color
- 4 = Trace number
- 5 = Detector
- 6 = Trace mode

Diagram footer information

The information in the diagram footer (beneath the diagram) depends on the evaluation:

- Center frequency
- Number of sweep points
- Range per division (x-axis)
- Span (Spectrum)

Status bar information

Global instrument settings, the instrument status and any irregularities are indicated in the status bar beneath the diagram.

Furthermore, the progress of the current operation is displayed in the status bar.

3 Measurement and result displays

Access: "Overview" > "Display Config"

Or: [MEAS] > "Display Config"

The I/Q Analyzer can capture I/Q data. The I/Q data that was captured by or imported to the R&S FSV/A can then be evaluated in various different result displays. Select the result displays using the SmartGrid functions.

Up to 6 evaluations can be displayed in the I/Q Analyzer at any time, including several graphical diagrams, marker tables or peak lists.

For details on working with the SmartGrid see the R&S FSV/A Getting Started manual.

Measurements in the time and frequency domain

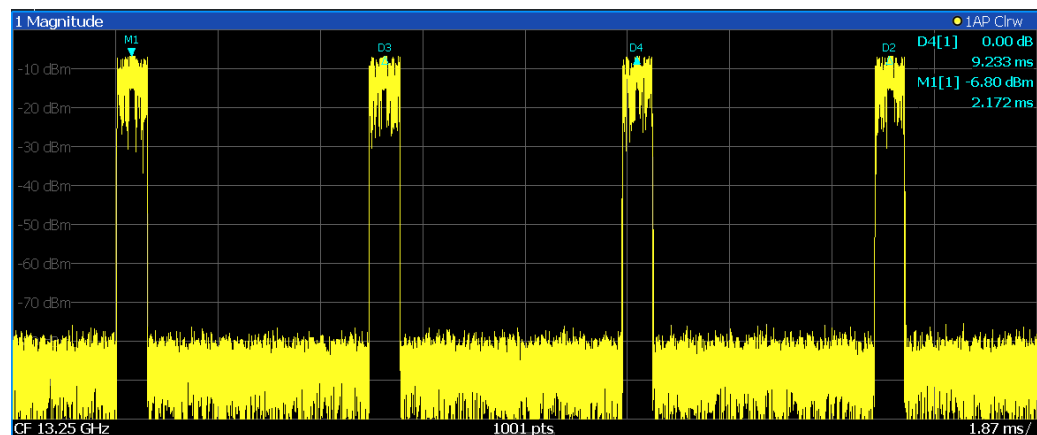
The time and frequency domain measurements and the available results are described in detail in the R&S FSV/A User Manual.

Result displays for I/Q data:

Magnitude.....	14
Spectrum.....	15
I/Q-Vector.....	15
Real/Imag (I/Q).....	16
Phase vs. Time.....	16
Marker Table.....	17
Marker Peak List.....	17

Magnitude

Shows the level values in time domain.



Remote command:

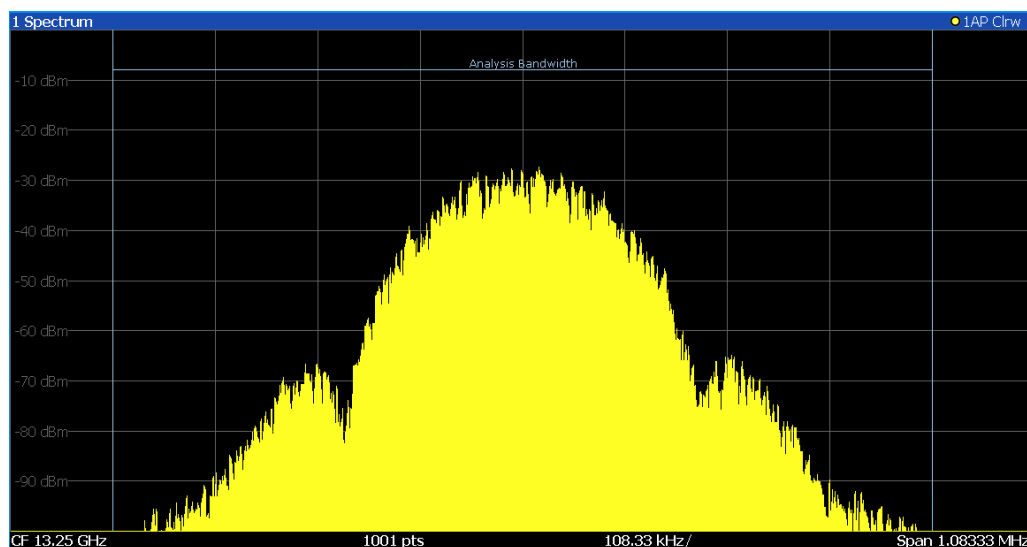
LAY:ADD:WIND? '1', RIGH, MAGN, see LAYout:ADD[:WINDow]? on page 259

Results:

TRACe<n>[:DATA]? on page 337

Spectrum

Displays the frequency spectrum of the captured I/Q samples.



Remote command:

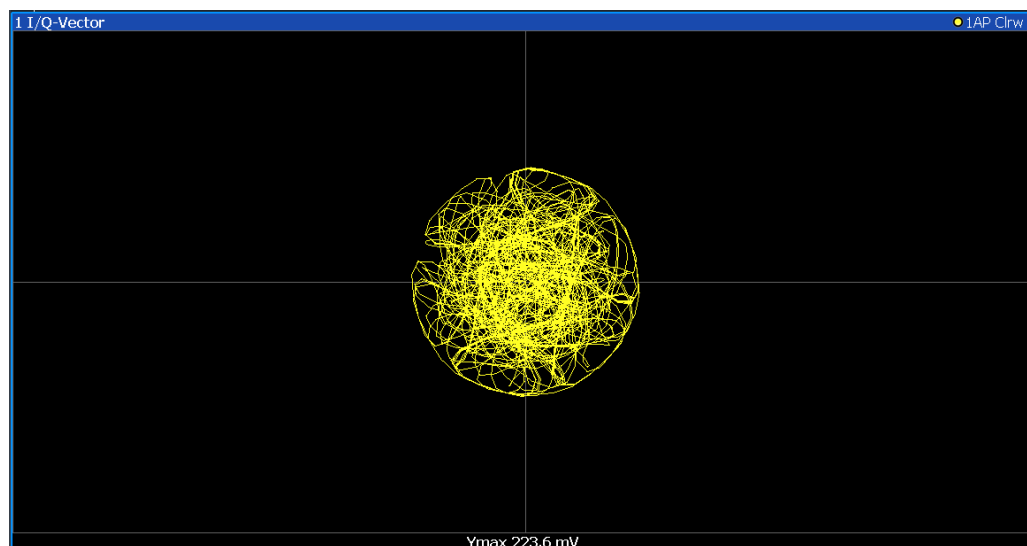
LAY:ADD:WIND? '1', RIGH, FREQ, see [LAYout:ADD\[:WINDow\]? on page 259](#)

Results:

[TRACe<n>\[:DATA\]? on page 337](#)

I/Q-Vector

Displays the captured samples in an I/Q-plot. The samples are connected by a line.



Note: For the I/Q vector result display, the number of I/Q samples to record ("Record Length") must be identical to the number of trace points to be displayed ("Sweep Points"; for I/Q Analyzer: 10001). For record lengths outside the valid range of sweep points the diagram does not show valid results.

Remote command:

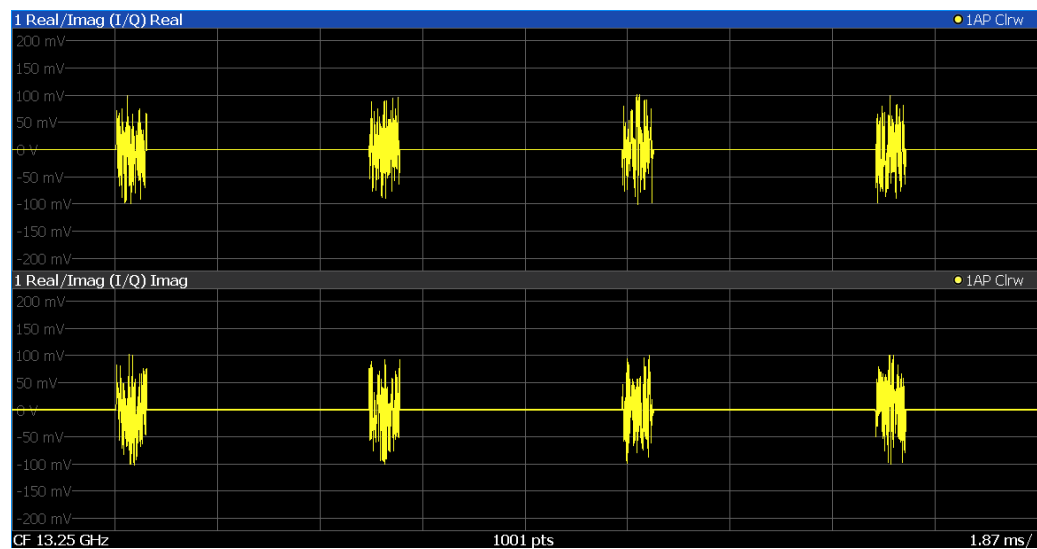
LAY:ADD:WIND? '1', RIGH, VECT, see [LAYout:ADD\[:WINDow\]?](#) on page 259

Results:

[TRACe<n>\[:DATA\]?](#) on page 337

Real/Imag (I/Q)

Displays the I and Q values in separate diagrams.



Remote command:

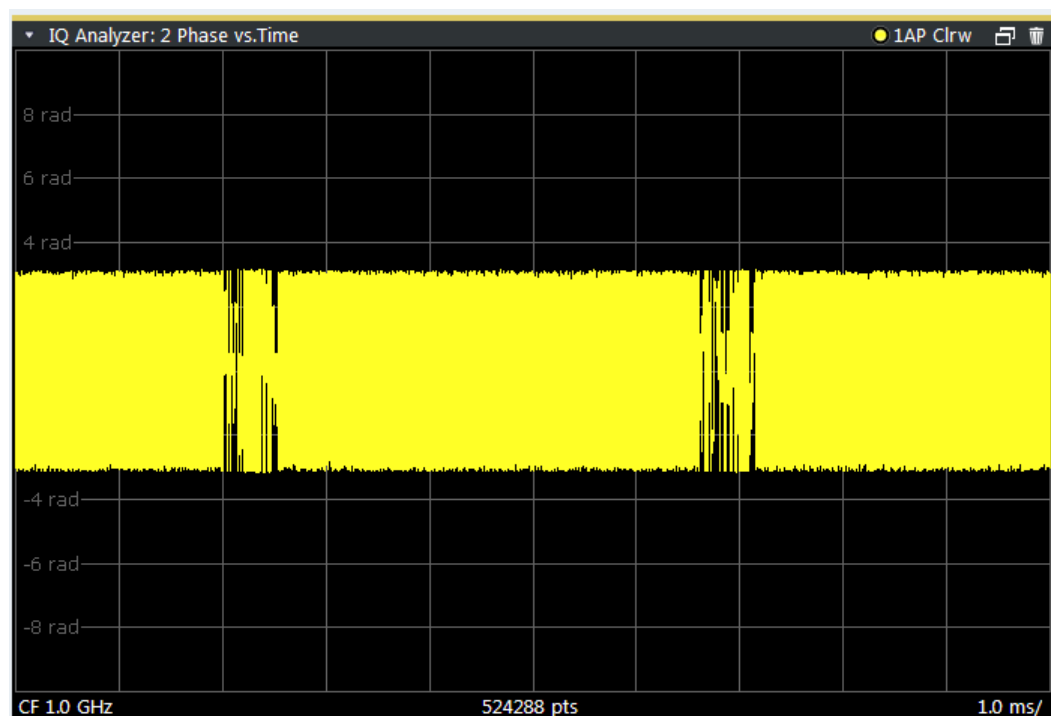
LAY:ADD:WIND? '1', RIGH, RIM, see [LAYout:ADD\[:WINDow\]?](#) on page 259

Results:

[TRACe<n>\[:DATA\]?](#) on page 337

Phase vs. Time

Shows the phase values in the time domain.



Remote command:

LAY:ADD? '1', RIGH, PHASe, see [LAYout:ADD\[:WINDow\]?](#) on page 259

Marker Table

Displays a table with the current marker values for the active markers.

This table is displayed automatically if configured accordingly.

2 Marker						
Type	Ref	Trc	Stimulus	Response	Function	Function Result
N1		1	13.197 GHz	-25.87 dBm	Count	13.197057
D1	N1	1	-7.942 GHz	-49.41 dB		
D2	N1	2	-3.918 GHz	-21.90 dB		
D3	N1	3	4.024 GHz	-21.99 dB		

Tip: To navigate within long marker tables, simply scroll through the entries with your finger on the touchscreen.

Remote command:

LAY:ADD? '1', RIGH, MTAB, see [LAYout:ADD\[:WINDow\]?](#) on page 259

Results:

[CALCulate<n>:MARKer<m>:X](#) on page 293

[CALCulate<n>:MARKer<m>:Y?](#) on page 344

Marker Peak List

The marker peak list determines the frequencies and levels of peaks in the spectrum or time domain. How many peaks are displayed can be defined, as well as the sort order. In addition, the detected peaks can be indicated in the diagram. The peak list can also be exported to a file for analysis in an external application.

3 Marker Peak List			
Wnd	No	X-Value	Y-Value
2	1	1.086245 ms	-75.810 dBm
2	2	2.172490 ms	-6.797 dBm
2	3	3.258736 ms	-76.448 dBm
2	4	4.831918 ms	-76.676 dBm
2	5	6.255274 ms	-76.482 dBm
2	6	6.798397 ms	-6.800 dBm
2	7	9.233084 ms	-76.519 dBm
2	8	10.075861 ms	-76.172 dBm
2	9	11.405574 ms	-6.801 dBm

Tip: To navigate within long marker peak lists, simply scroll through the entries with your finger on the touchscreen.

Remote command:

LAY:ADD? '1',RIGH, PEAK, see [LAYout:ADD\[:WINDow\]? on page 259](#)

Results:

[CALCulate<n>:MARKer<m>:X on page 293](#)

[CALCulate<n>:MARKer<m>:Y? on page 344](#)

4 Basics on I/Q data acquisition and processing

Some background knowledge on basic terms and principles used when describing I/Q data acquisition on the R&S FSV/A in general, and in the I/Q Analyzer application in particular, is provided here for a better understanding of the required configuration settings.

The I/Q Analyzer provides various possibilities to acquire the I/Q data to be analyzed:

- Capturing analog I/Q data from the "RF Input" connector
- Importing I/Q data from a file

Background information for all these scenarios and more is provided in the following sections.

- [Processing analog I/Q data from RF input](#)..... 19
- [Basics on input from I/Q data files](#)..... 25
- [Processing data from the Analog Baseband interface](#)..... 27
- [Basics on FFT](#)..... 32
- [Concept of I/Q noise cancellation](#)..... 38

4.1 Processing analog I/Q data from RF input

Complex baseband data

In the telephone systems of the past, baseband data was transmitted unchanged as an analog signal. In modern phone systems and in radio communication, however, the baseband data is modulated on a carrier frequency, which is then transmitted. The receiver must demodulate the data based on the carrier frequency. When using modern modulation methods (e.g. QPSK, QAM etc.), the baseband signal becomes complex. Complex data (or: I/Q data) consists of an imaginary (I) and a real (Q) component.

Sweep vs sampling

The standard Spectrum application on the R&S FSV/A performs frequency sweeps on the input signal and measurements in the frequency and time domain. Other applications on the R&S FSV/A, such as the I/Q Analyzer, sample and process the individual I and Q components of the complex signal.

I/Q Analyzer - processing complex data from RF input

The I/Q Analyzer is a standard application used to capture and analyze I/Q data on the R&S FSV/A. By default, it assumes the I/Q data is modulated on a carrier frequency and input via the "RF Input" connector on the R&S FSV/A.

The A/D converter samples the IF signal at a rate of 200 MHz. The digital signal is down-converted to the complex baseband, lowpass-filtered, and the sample rate is reduced. The analog filter stages in the analyzer cause a frequency response which adds to the modulation errors. An **equalizer filter** before the **resampler** compensates for this frequency response. The continuously adjustable sample rates are realized using an optimal decimation filter and subsequent resampling on the set sample rate.

A dedicated memory (**capture buffer**) is available in the R&S FSV/A for a maximum of 400 Msamples (400*1000*1000) of complex samples (pairs of I and Q data). The number of complex samples to be captured can be defined (for restrictions refer to [Section 4.1.1, "Sample rate and maximum usable I/Q bandwidth for RF input"](#), on page 20).

The block diagram in [Figure 4-1](#) shows the analyzer hardware from the IF section to the processor.

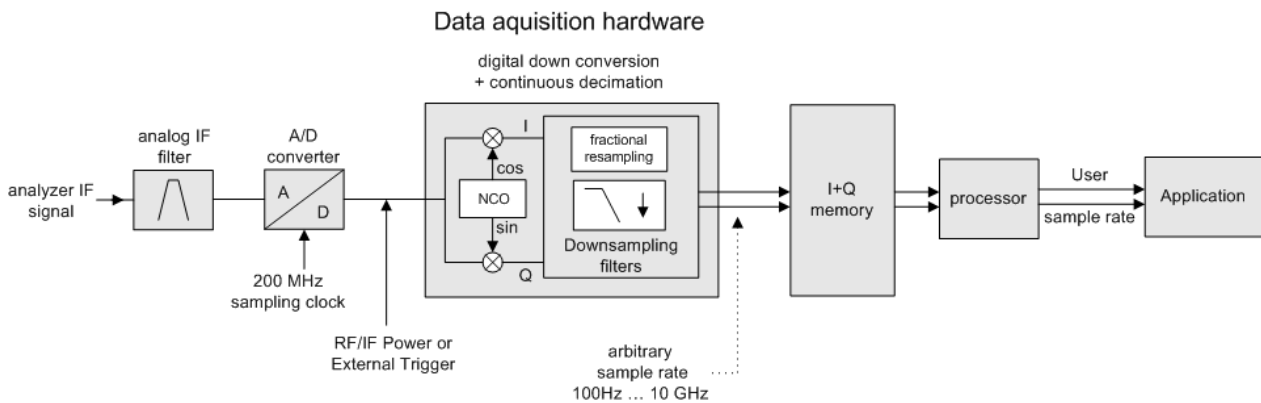


Figure 4-1: Block diagram illustrating the R&S FSV/A signal processing for analog I/Q data (without bandwidth extension options)

4.1.1 Sample rate and maximum usable I/Q bandwidth for RF input

Definitions

- **Input sample rate (ISR):** the sample rate of the useful data provided by the device connected to the input of the R&S FSV/A
- (User, Output) **Sample rate (SR):** the user-defined sample rate (e.g. in the "Data Acquisition" dialog box in the "I/Q Analyzer" application) which is used as the basis for analysis or output
- **Usable I/Q (Analysis) bandwidth:** the bandwidth range in which the signal remains undistorted in regard to amplitude characteristic and group delay; this range can be used for accurate analysis by the R&S FSV/A
- **Record length:** Number of I/Q samples to capture during the specified measurement time; calculated as the measurement time multiplied by the sample rate

For the I/Q data acquisition, digital decimation filters are used internally in the R&S FSV/A. The passband of these digital filters determines the *maximum usable I/Q bandwidth*. In consequence, signals within the usable I/Q bandwidth (passband)

remain unchanged, while signals outside the usable I/Q bandwidth (passband) are suppressed. Usually, the suppressed signals are noise, artifacts, and the second IF side band. If frequencies of interest to you are also suppressed, try to increase the output sample rate, which increases the maximum usable I/Q bandwidth.



Bandwidth extension options

You can extend the maximum usable I/Q bandwidth provided by the R&S FSV/A in the basic installation by adding options. These options can either be included in the initial installation (B-options) or updated later (U-options). The maximum bandwidth provided by the individual option is indicated by its number, for example, B40 extends the bandwidth to 40 MHz.

Note that the U-options as of U40 always require all lower-bandwidth options as a prerequisite, while the B-options already include them.

As a rule, the usable I/Q bandwidth is proportional to the output sample rate. Yet, when the I/Q bandwidth reaches the bandwidth of the analog IF filter (at very high output sample rates), the curve breaks.

• Bandwidth extension options	21
• Relationship between sample rate, record length and usable I/Q bandwidth	21
• R&S FSV/A without additional bandwidth extension options	22
• R&S FSV/A with I/Q bandwidth extension option B40 or U40	23
• R&S FSV/A with I/Q bandwidth extension option B200	24
• R&S FSV/A with I/Q bandwidth extension option B400	24
• R&S FSV/A with I/Q bandwidth extension option B600	24
• R&S FSV/A with I/Q bandwidth extension option B601	24
• R&S FSV/A with I/Q bandwidth extension option B1000	25
• R&S FSV/A with I/Q bandwidth extension option B1001	25

4.1.1.1 Bandwidth extension options

Max. usable I/Q BW	Required B-option	Required U-options
40 MHz	B40	
200 MHz	B200	
400 MHz	B400	B200+U400
600 MHz	B600	
600 MHz	B601	
1000 MHz	B1000	B600+U1006
1000 MHz	B1001	

4.1.1.2 Relationship between sample rate, record length and usable I/Q bandwidth

Up to the maximum bandwidth, the following rule applies:

$$\text{Usable I/Q bandwidth} = 0.8 * \text{Output sample rate}$$

Regarding the record length, the following rule applies:

$$\text{Record length} = \text{Measurement time} * \text{sample rate}$$

Maximum record length for RF input

The absolute maximum record length (AbsMaxRecordLength), that is, the maximum number of samples that can be captured, is 100 Msamples (with option B114: 800 Msamples).

When using bandwidth extension options R&S FSV3-B600/B601/B1000/B1001, the maximum record length depends on the analysis bandwidth.

Table 4-1: Maximum record length with I/Q bandwidth extension option B600/B601/B1000/B1001

Analysis band-width *)	Max. meas time	Maximum record length
80 Hz to 400 MHz	<Capture-Length> / <SampleRate>	AbsMaxRecordLength
400 MHz to 800 MHz (B600/B601: 400 MHz to 600 MHz)	<Capture-Length> / <SampleRate> with B114: 819.2 ms	AbsMaxRecordLength * <SampleRate> / (1024*10 ⁶) For sample rates ≥2048 MHz: AbsMaxRecordLength
>800 MHz to 1000 MHz	<Capture-Length> / <SampleRate> with B114: 409.6 ms	AbsMaxRecordLength * <SampleRate> / (2048*10 ⁶) For sample rates ≥2048 MHz: AbsMaxRecordLength
*) If you restrict the maximum bandwidth to 40 MHz, 200 MHz, or 400 MHz manually (" Maximum Bandwidth " on page 78), the maximum record length is AbsMaxRecordLength.		

4.1.1.3 R&S FSV/A without additional bandwidth extension options

Sample rate: 100 Hz - 10 GHz

Maximum I/Q bandwidth: 28 MHz

Table 4-2: Maximum I/Q bandwidth

Sample rate	Maximum I/Q bandwidth
100 Hz to 35 MHz	Proportional up to maximum 28 MHz
35 MHz to 10 GHz	28 MHz

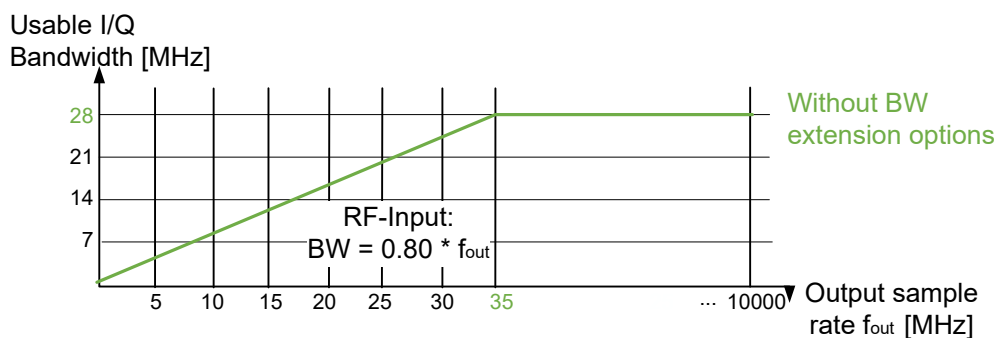


Figure 4-2: Relationship between maximum usable I/Q bandwidth and output sample rate without bandwidth extensions

I/Q bandwidths for RF input

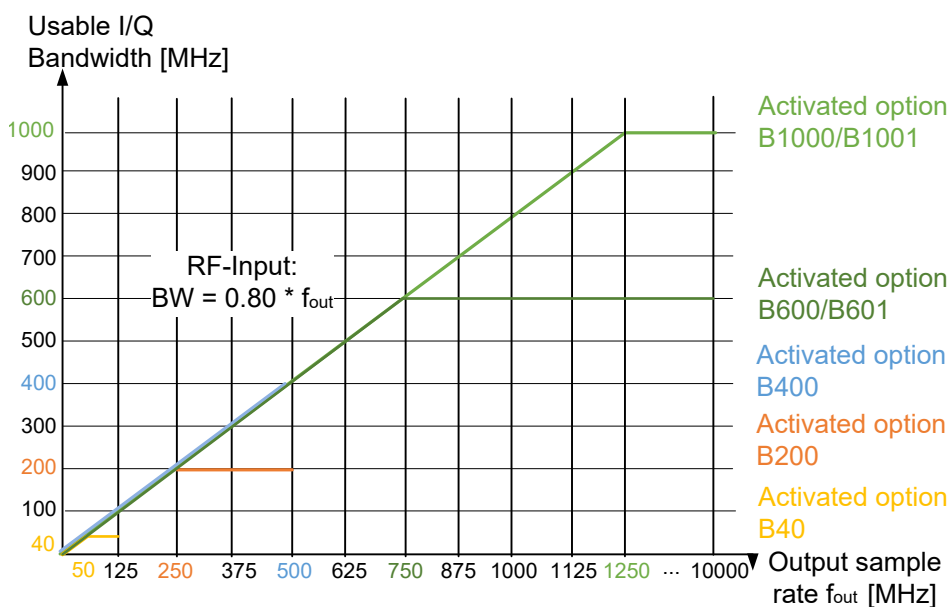


Figure 4-3: Relationship between maximum usable I/Q bandwidth and output sample rate with optional bandwidth extensions

4.1.1.4 R&S FSV/A with I/Q bandwidth extension option B40 or U40

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 40 MHz

Sample rate	Maximum I/Q bandwidth
100 Hz to 50 MHz	Proportional up to maximum 40 MHz
50 MHz to 10 GHz	40 MHz

4.1.1.5 R&S FSV/A with I/Q bandwidth extension option B200

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 200 MHz

Sample rate	Maximum I/Q bandwidth
100 Hz to 250 MHz	Proportional up to maximum 200 MHz
250 MHz to 10 GHz	200 MHz

4.1.1.6 R&S FSVA with I/Q bandwidth extension option B400

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 400 MHz

Sample rate	Maximum I/Q bandwidth
100 Hz to 500 MHz	Proportional up to maximum 400 MHz
500 MHz to 10 GHz	400 MHz

4.1.1.7 R&S FSVA with I/Q bandwidth extension option B600

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 600 MHz

Note that using the bandwidth extension option R&S FSV3-B600, an I/Q bandwidth larger than 400 MHz is only available for frequency ranges above 7.5 GHz.

Center frequency	Sample rate	Maximum I/Q bandwidth
≤7.5 GHz	100 Hz to 500 MHz	Proportional up to maximum 400 MHz
	500 MHz to 10 GHz	400 MHz
>7.5 GHz	100 Hz to 750 MHz	Proportional up to maximum 600 MHz
	750 MHz to 10 GHz	600 MHz

4.1.1.8 R&S FSVA with I/Q bandwidth extension option B601

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 600 MHz

Sample rate	Maximum I/Q bandwidth
100 Hz to 750 MHz	Proportional up to maximum 600 MHz
750 MHz to 10 GHz	600 MHz

4.1.1.9 R&S FSVA with I/Q bandwidth extension option B1000

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 1000 MHz

Note that using the bandwidth extension option R&S FSV3-B1000, an I/Q bandwidth larger than 400 MHz is only available for frequency ranges above 7.5 GHz.

Center frequency	Sample rate	Maximum I/Q bandwidth
≤7.5 GHz	100 Hz to 500 MHz	Proportional up to maximum 400 MHz
	500 MHz to 10 GHz	400 MHz
>7.5 GHz	100 Hz to 1250 MHz	Proportional up to maximum 1000 MHz
	1250 MHz to 10 GHz	1000 MHz

4.1.1.10 R&S FSVA with I/Q bandwidth extension option B1001

Sample rate: 100 Hz - 10 GHz

Maximum bandwidth: 1000 MHz

Using the bandwidth extension option R&S FSV3-B1001, an I/Q bandwidth up to 1 GHz is available for all supported input frequencies.

Sample rate	Maximum I/Q bandwidth
100 Hz to 1250 MHz	Proportional up to maximum 1000 MHz
1250 MHz to 10 GHz	1000 MHz

4.2 Basics on input from I/Q data files

The I/Q data to be evaluated in a particular R&S FSV/A application can not only be captured by the application itself, it can also be loaded from a file, provided it has the correct format. The file is then used as the input source for the application.

For example, you can capture I/Q data using the I/Q Analyzer application, store it to a file, and then analyze the signal parameters for that data later using the Pulse application (if available).

The I/Q data file must be in one of the following supported formats:

- .iq.tar
- .iqw
- .csv
- .mat
- .wv
- .aid

(For details, see [Section C, "Reference: supported I/Q file formats"](#), on page 358)



Only a single data stream can be used as input, even if multiple streams are stored in the file.



An application note on converting Rohde & Schwarz I/Q data files is available from the Rohde & Schwarz website:

[1EF85: Converting R&S I/Q data files](#)

For I/Q file input, the stored I/Q data remains available as input for any number of subsequent measurements. When the data is used as an input source, the data acquisition settings in the current application (attenuation, center frequency, measurement bandwidth, sample rate) can be ignored. As a result, these settings cannot be changed in the current application. Only the measurement time can be decreased, in order to perform measurements on an extract of the available data (from the beginning of the file) only.



For some file formats that do not provide the sample rate and measurement time or record length, you must define these parameters manually. Otherwise the traces are not visible in the result displays.

For details, see [Table C-1](#).

When using input from an I/Q data file, the [RUN SINGLE] function starts a single measurement (i.e. analysis) of the stored I/Q data, while the [RUN CONT] function repeatedly analyzes the same data from the file.



Sample iq.tar files

If you have the optional R&S FSV/A VSA application (R&S FSV3-K70), some sample `iq.tar` files are provided in the `C:/R_S/Instr/user/vsa/DemoSignals` directory on the R&S FSV/A.

Furthermore, you can create your own `iq.tar` files in the I/Q Analyzer, see [Section 8.3, "How to export and import I/Q data"](#), on page 139.

Pre-trigger and post-trigger samples

In applications that use pre-triggers or post-triggers, if no pre-trigger or post-trigger samples are specified in the I/Q data file, or too few trigger samples are provided to satisfy the requirements of the application, the missing pre- or post-trigger values are filled up with zeros. Superfluous samples in the file are dropped, if necessary. For pre-trigger samples, values are filled up or omitted at the beginning of the capture buffer, for post-trigger samples, values are filled up or omitted at the end of the capture buffer.

4.3 Processing data from the Analog Baseband interface

By default, (analog) I/Q data is captured from the standard "RF Input" connector on the front panel of the R&S FSV/A. Alternatively, analog baseband signals can also be captured via the optional **"Analog Baseband" interface**, if installed.

- [\(Analog\) baseband input 50 ohm connectors \(optional\)](#)..... 27
- [Analog baseband input](#)..... 28
- [I/Q processing modes](#)..... 29
- [Sample rates and bandwidths for analog baseband signals](#).....31
- [Average power consumption](#).....32

4.3.1 (Analog) baseband input 50 ohm connectors (optional)

The "Analog Baseband" interface option provides four "Baseband input" BNC connectors on the front panel of the R&S FSV/A for analog I and Q signals.



The upper BNC connectors BASEBAND INPUT I and BASEBAND INPUT Q are used to input:

- Single-ended signals
- The positive signal input for differential signals
- Input from active Rohde & Schwarz probes (see specifications document)

The lower BNC connectors $\bar{I}$ and $\bar{Q}$ are used to input the negative signal for differential signals.



Complex signal input ($I+jQ$)

For complex signal input ($I+jQ$), always use two identical cables for the I and Q connectors (same length, same type, same manufacturer). Otherwise, time delay or gain imbalance can occur between the different cables, which cannot be calibrated.

All connectors have a fixed impedance of 50 Ω . Do not overload the input. For maximum allowed values, see the specifications document.

Input via the "Analog Baseband" interface can be enabled in the I/Q Analyzer or in one of the optional applications that process I/Q data (where available).

4.3.2 Analog baseband input

The "Analog Baseband" interface can be used as an alternative data input source for measurements with the R&S FSV/A.

An analog baseband signal is input at the "Baseband Input I" and "Baseband Input Q" connectors and processed from there.

Complex spectrum analysis

If the input is already available as a complex baseband signal (I and Q signals), the "Analog Baseband" interface allows you to analyze the complex spectrum of the baseband signal. This is useful for measurements in the early stages of signal processing or radio transmission, when the analog baseband signal has not yet been modulated.

Low IF signals

I/Q input that has already been modulated ("Low IF signal") is down-converted digitally.

Data acquisition

The "Analog Baseband" interface of the R&S FSV/A can process both single-ended (unbalanced) and differential (balanced) input. The signal is input to the R&S FSV/A via the connectors of the "Analog Baseband" interface. If necessary, the I and Q values in the input can be swapped. This is useful, for instance, if the connections are mixed up or the data is inverted by the device under test.

Voltage levels - full scale level

For RF input, the maximum expected voltage level is defined by the reference level. For analog baseband input, the maximum expected voltage level *for each component* (I or Q) is defined by the **full scale level**. The full scale level defines the maximum power you can input at the "Baseband Input" connectors without clipping the signal.

The full scale level can be defined manually or automatically, such that the power of I and Q does not exceed the reference level.

If probes are connected, the possible full scale values are adapted according to the probe's attenuation and maximum allowed voltage.

For details on probes, see the R&S FSV/A User Manual.

When converting the measured voltage into dBm, an impedance of 50 Ω is assumed.

Triggering

The following trigger sources are supported for analog baseband input (see ["Trigger Source"](#) on page 69):

- External
- Baseband power
- Time
- I/Q Power
- Power sensor



Gating

Gating is not supported for analog baseband input.

Calibration

A special calibration signal is available for analog baseband input and can be activated in the general instrument settings. If activated, an internal DC or AC calibration signal is input to the "Analog Baseband" interface.

For details, see the R&S FSV/A User Manual.

4.3.3 I/Q processing modes

The "Analog Baseband" interface provides different methods of processing the baseband input (I/Q modes), depending on the measurement requirements.

Complex baseband mode (I+jQ)

In the (default) *complex baseband mode*, the analog input signal is assumed to be a complex baseband signal. There is no need to equalize any IF filter or mix the signal into the complex baseband. The analog hardware just has to ensure that the final I/Q data stored in the capture buffer has the correct sample rate for the application.

The analog baseband input signal is brought to the desired sample rate using a down-sampling filter and fractional resampling. No level compensation is necessary. The resulting data can be processed by the selected application.

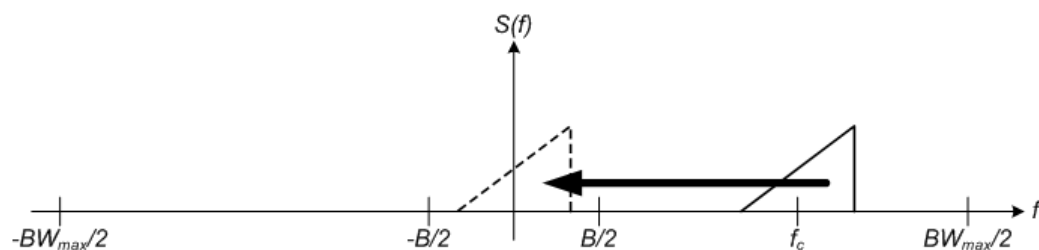


Figure 4-4: Spectrum in complex baseband (I+jQ) mode

The complex spectrum of the input signal is displayed. The center frequency does not need to be moved. However, it can be, as long as the selected spectrum remains within the maximum analysis bandwidth (see [Section 4.3.4, "Sample rates and bandwidths for analog baseband signals"](#), on page 31).

Low IF mode (I or Q)

In *low IF mode*, the real signal from the selected input component (I or Q) is assumed to be a modulated carrier with a specific center frequency. The signal is down-converted to a selected center frequency (= low IF frequency) using an NCO. The center frequency must be higher than 0 Hz so that no part of the negative mirrored spectrum lies within the analysis bandwidth. (The center frequency must be different to 0 Hz, as in this case real baseband mode is assumed, see ["Real baseband mode \(I or Q only\)"](#) on page 30.) Furthermore, select the center frequency such that the displayed

spectrum remains within the maximum analysis bandwidth (see [Section 4.3.4, "Sample rates and bandwidths for analog baseband signals"](#), on page 31).

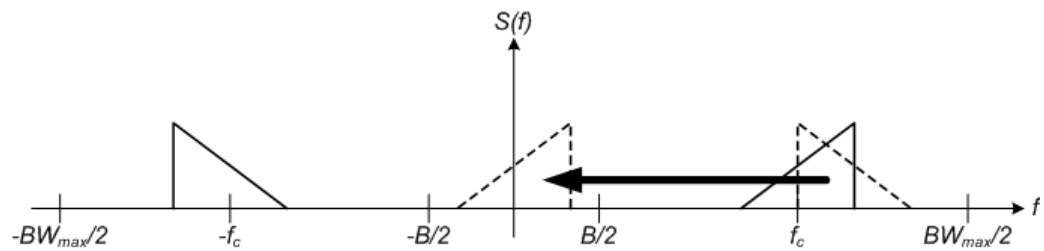


Figure 4-5: Spectrum in low IF mode

Compared to the initial complex baseband signal that was input, the down-converted I or Q component contains only half the spectrum (i.e. one sideband less) after passing the filter. The power is thus reduced by one half (or: -3 dB). This power loss is compensated for by increasing the power of the resulting spectrum by +3 dB.

The digitized data is brought to the desired sample rate using a downsampling filter and fractional resampling.

This processing mode corresponds to the common RF spectrum analysis, applied to the analog baseband input.

Real baseband mode (I or Q only)

As mentioned above, a center frequency of 0 Hz is not allowed for low IF mode. In this case, the input signal is assumed to be a real baseband signal, so no down-conversion is performed. Thus, this mode resembles an oscilloscope. The spectrum result display always starts at 0 and has a maximum span of half the sample rate. (Half of the captured samples are from the other component, which is not displayed in this mode.) The Real/Imag result display shows only one diagram (namely the one for the selected component).

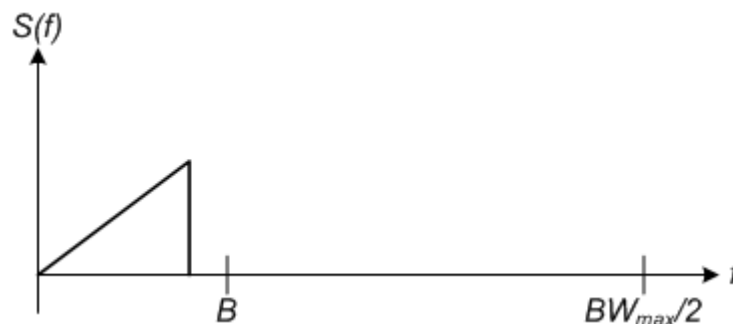


Figure 4-6: Spectrum in real baseband mode

This mode is useful for pulse measurements, for example.

4.3.4 Sample rates and bandwidths for analog baseband signals

The analog baseband input is sampled internally by the R&S FSV/A at a rate of 512 MHz. As a result, 512 megasamples of I values and 512 megasamples of Q values can be obtained per second. The actual sample rate required by the application, however, can be lower, in which case the data is downsampled. Depending on the application used to process the data, the required sample rate is defined by the application itself or by the user.

The sample rate also determines the analysis bandwidth. The analysis bandwidth is the range in which the signal remains undistorted in regard to amplitude characteristic and group delay which can be used for accurate analysis by the R&S FSV/A. The sample rate and the analysis bandwidth are interdependent and are adapted according to the following formula in the I/Q Analyzer:

$$\text{Analysis bandwidth} = 0.8 * \text{sample rate}$$

(For I or Q only: $\text{Analysis bandwidth} = 0.4 * \text{sample rate}$)

See also [Section 4.1.1, "Sample rate and maximum usable I/Q bandwidth for RF input"](#), on page 20.

The standard R&S FSV/A equipped with the optional "Analog Baseband" interface can analyze a maximum bandwidth of 200 MHz input *per connector*. That makes a 400 MHz analysis bandwidth for a complex baseband signal.



The bandwidth extension options for RF input have no effect on analog baseband input.

Spectrum limits

The analog baseband spectrum to be analyzed depends both on the analysis bandwidth and on the center frequency, which defines the middle of the spectrum. The spectrum should always remain within an ideal span (see [Table 4-3](#)) to avoid effects from unwanted signal components (e.g. mirrored sidebands). Thus, always select the maximum analysis bandwidth and the position of the center frequency such that the spectrum remains within the specified limits. You are not forced by the R&S FSV/A to do so, but a warning message is displayed if the limits are exceeded.

Table 4-3: Spectrum limits depending on I/Q mode

I/Q Mode	Complex baseband (I+jQ)	Low-IF (I / Q)	Real Baseband (I / Q)
Analysis bandwidth BW	$BW_{\max} = +400 \text{ MHz}$	$BW_{\max} = +200 \text{ MHz}$	$BW_{\max} = +200 \text{ MHz}$
Center frequency f_c	$-BW_{\max}/2 < f_c < BW_{\max}/2$ i.e.: -200 MHz to +200 MHz	$0 < f_c < BW_{\max}$ i.e.: +1 Hz to +200 MHz	$f_c = 0 \text{ Hz}$
Span	= Sample rate	= Sample rate	= Sample rate / 2

4.3.5 Average power consumption

The "Analog Baseband" interface can be used to capture two different signals: one proportional to the voltage and one proportional to the current of a DUT. The *average power consumption* can then be calculated from the captured I/Q signal. To avoid processing large amounts of I/Q data, the R&S FSV/A provides an internal calculation of the average power consumption for remote operation according to the following equation:

$$P_{avg} = \frac{1}{NofSamples} \sum_{n=0}^{NofSamples-1} P(n)$$

$$P(n) = A * V(n) * I(n) - B * V(n) * V(n)$$

with:

- V(n): I data of the instrument
- I(n): Q data of the instrument
- A: conversion factor A
- B: conversion factor B

Remote commands:

[TRACe:IQ:APCon\[:STATe\]](#) on page 182

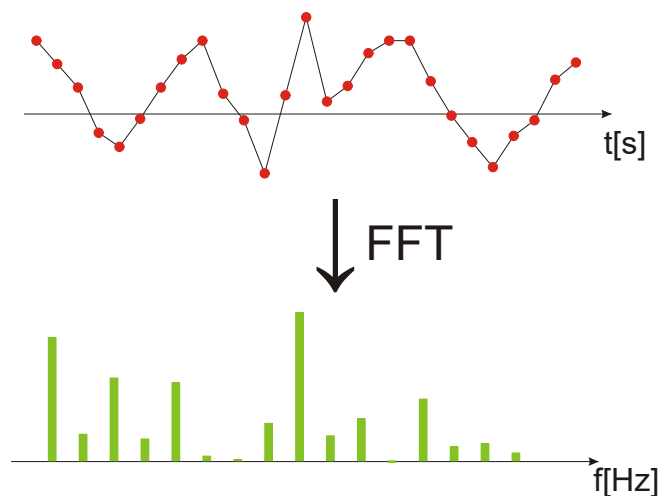
[TRACe:IQ:APCon:A](#) on page 182

[TRACe:IQ:APCon:B](#) on page 182

[TRACe:IQ:APCon:RESult?](#) on page 183

4.4 Basics on FFT

The I/Q Analyzer measures the power of the signal input over time. To convert the time domain signal to a frequency spectrum, an FFT (Fast Fourier Transformation) is performed which converts a vector of input values into a discrete spectrum of frequencies.



4.4.1 Window functions

The Fourier transformation is not performed on the entire captured data in one step. Only a limited number of samples is used to calculate an individual result. This process is called windowing.

After sampling in the time domain, each window is multiplied with a specific window function. Windowing helps minimize the discontinuities at the end of the measured signal interval and thus reduces the effect of spectral leakage, increasing the frequency resolution.

Various different window functions are provided in the R&S FSV/A to suit different input signals. Each of the window functions has specific characteristics, including some advantages and some trade-offs. Consider these characteristics to find the optimum solution for the measurement task.



Ignoring the window function - rectangular window

The rectangular window function is in effect not a function at all, it maintains the original sampled data. This may be useful to minimize the required bandwidth. However, be aware that if the window does not contain exactly one period of your signal, heavy sidelobes may occur, which do not exist in the original signal.

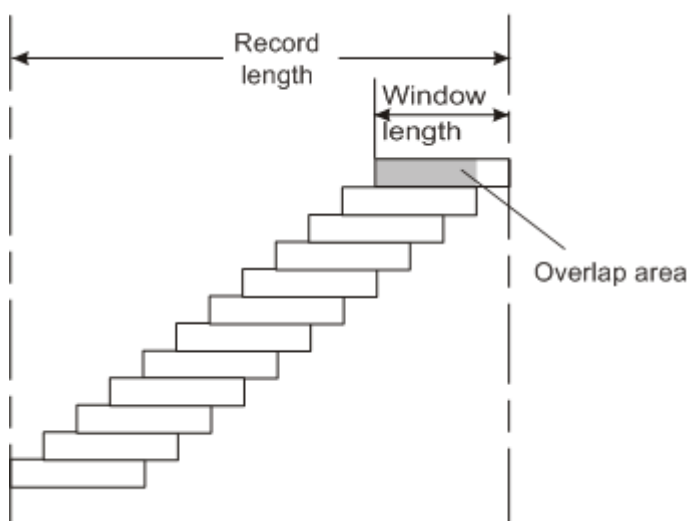
Table 4-4: Characteristics of typical FFT window functions

Window type	Frequency resolution	Magnitude resolution	Sidelobe suppression	Measurement recommendation
Rectangular	Best	Worst	Worst	No function applied. Separation of two tones with almost equal amplitudes and a small frequency distance
Blackman-Harris (default)	Good	Good	Good	Harmonic detection and spurious emission detection

Window type	Frequency resolution	Magnitude resolution	Sidelobe suppression	Measurement recommendation
Gauss (Alpha = 0.4)	Good	Good	Good	Weak signals and short duration
Flattop	Worst	Best	Good	Accurate single tone measurements
5-Term	Good	Good	Best	Measurements with very high dynamic range

4.4.2 Overlapping

The I/Q Analyzer calculates multiple FFTs per measurement by dividing one captured record into several windows. Furthermore, the I/Q Analyzer allows consecutive windows to overlap. Overlapping "reuses" samples that were already used to calculate the preceding FFT result.



In advanced FFT mode with averaging, the overlapping factor can be set freely. The higher the overlap factor, the more windows are used. This leads to more individual results and improves detection of transient signal effects. However, it also extends the duration of the calculation. The size of the window can be defined manually according to the record length, the overlap factor, and the FFT length.

An FFT overlap of 67%, for example, means the second FFT calculation uses the last 67% of the data of the first FFT. It uses only 33% new data. The third FFT still covers 33% of the first FFT and 67% of the second FFT, and so on.

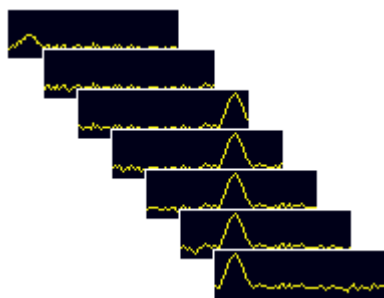


Figure 4-7: Overlapping FFTs

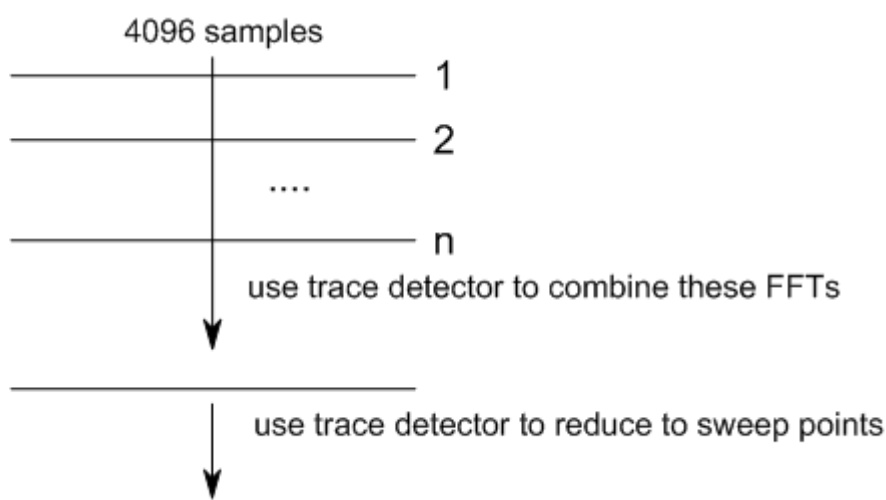
In "Manual" or "Auto" FFT mode, an FFT length of 4096 and a window length of 4096 (or the record length, if shorter) is used to calculate the spectrum.

Combining results - trace detector

If the record length permits, multiple overlapping windows are calculated and combined to create the final spectrum using the selected trace detector. If necessary, the trace detector is also used to reduce the number of calculated frequency points (defined by the FFT length) to the defined number of sweep points. By default, the Autopeak trace detector is used.



Since the frequency points are reduced to the number of sweep points, using a detector other than "Auto Peak" and fewer than 4096 sweep points can lead to false level results.



4.4.3 Dependencies between FFT parameters

FFT analysis in the R&S FSV/A is highly configurable. Several parameters, including the resolution bandwidth, record length, and FFT length, are user-definable. Note, however, that several parameters are correlated and not all can be configured independently of the others.

Record Length

Defines the number of I/Q samples to capture. By default, the number of sweep points is used. The record length is calculated as the measurement time multiplied by the sample rate.

If you change the record length, the [Meas Time](#) is automatically changed, as well.

For FFTs using only a single window ("Single" mode), the record length (which is then identical to the FFT length) must not exceed 512k.

FFT Length

Defines the number of frequency points determined by each FFT calculation. The more points are used, the higher the resolution in the spectrum becomes, but the longer the calculation takes.

In "Auto" or "Manual" mode, an FFT length of 4096 is used.

In advanced FFT mode, the FFT length is user-definable. If you use the arrow keys or the rotary knob to change the FFT length, the value is incremented or decremented by powers of 2. If you enter the value manually, any integer value from 3 to 524288 is available.

If the FFT length is longer than the [Window Length](#) the sample data is filled up with zeros up to the FFT length. The FFT is then performed using interpolated frequency points.

For an FFT length that is not a power of 2, a DFT (discrete Fourier transform) is performed, which requires more time for calculation, but avoids the effects of interpolation.

To display all calculated frequency points (defined by the FFT length), the number of sweep points is set to the FFT length automatically in advanced FFT mode.

Window Length

Defines the number of samples to be included in a single window in averaging mode. (In single mode, the window length corresponds to the ["Record Length"](#) on page 79.)

Values from 3 to 4096 are available in "Manual" mode; in "Advanced" FFT mode, values from 3 to 524288 are available. However, the window length must not be longer than the [FFT Length](#).

If the window length is shorter than the [FFT Length](#), the sample data is filled up with zeros up to the FFT length.

If the window length is longer than the [Record Length](#) (that is, not enough samples are available), a window length the size of the [Record Length](#) is used for calculation.

The window length and the [Window Overlap](#) determine how many FFT calculations must be performed for each record in averaging mode (see ["Transformation Algorithm"](#) on page 80).

4.4.4 Frequency resolution of FFT results - RBW

The **resolution bandwidth** defines the minimum frequency separation at which the individual components of a spectrum can be distinguished. Small values result in high precision, as the distance between two distinguishable frequencies is small. Higher values decrease the precision, but increase measurement speed.

The RBW is determined by the following equation:

$$RBW = \text{Normalized Bandwidth} * \frac{\text{Sample Rate}}{\text{Window Length}}$$

Equation 4-1: Definition of RBW

(Note: The normalized bandwidth is a fixed value that takes the noise bandwidth of the window function into consideration.)

If a higher spectral resolution is required, the number of samples must be increased by using a higher sample rate or longer record length.

The minimum achievable RBW depends on the sample rate and record length, according to the following equation:

$$RBW_{min} = \frac{\text{Normalized Bandwidth} * \text{Sample Rate}}{\min(4096, \text{Record Length})}$$

To simplify operation, some parameters are coupled and automatically calculated, such as record length and RBW.

RBW mode

Depending on the selected RBW mode, the resolution bandwidth is either determined automatically or can be defined manually.

Auto mode:

This is the default mode in the I/Q Analyzer. The RBW is determined automatically depending on the [Sample Rate](#) and [Window Length](#), where the window length corresponds to the [Record Length](#), or a maximum of 4096.

If the record length is larger than the window length, multiple windows are combined; the FFT length is 4096.

A Flatop window function is used.

Manual mode:

The RBW is user-definable.

The [Window Length](#) is adapted to comply with [Equation 4-1](#). Since only window lengths with integer values can be employed, the [Sample Rate](#) is adapted, if necessary, to obtain an integer window length value.

If the record length is larger than the window length, multiple windows are combined; the FFT length is 4096.

A Flatop window function is used.

Advanced FFT mode

The RBW is determined by the [advanced FFT parameters](#), depending on the selected [FFT calculation methods](#) method.

4.4.5 FFT calculation methods

FFT calculation can be performed using different methods.

Single

In single mode, one FFT is calculated for the entire record length, that means the window length is identical to the record length.

If the defined [FFT Length](#) is larger than the record length, zeros are appended to the captured data to reach the FFT length.



Figure 4-8: FFT parameters for single FFT calculation

Averaging

In averaging mode, several overlapping FFTs are calculated for each record; the results are combined to determine the final FFT result for the record.

The number of FFTs to be combined is determined by the [Window Overlap](#) and the [Window Length](#).

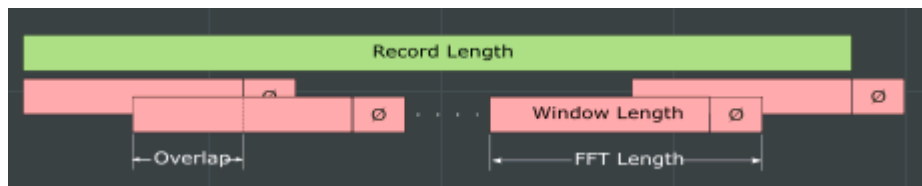


Figure 4-9: FFT parameters for averaged FFT calculation

4.5 Concept of I/Q noise cancellation

This function requires the R&S FSV3-K575 option and a repetitive input signal. It is only available in selected applications.

Noise components

Figure 4-10 illustrates the components of a signal measured by an R&S FSV/A. In this sample setup, a reference signal is input to a DUT, and the signal output is measured by the R&S FSV/A. The measured signal on the R&S FSV/A contains not only noise contributions from the signal path up to the R&S FSV/A input (i.e., the external contributions), but also instrument-inherent noise. The wideband receiver noise n_{RX} increases with increasing bandwidth and can have a major impact on the instrument-residual EVM.

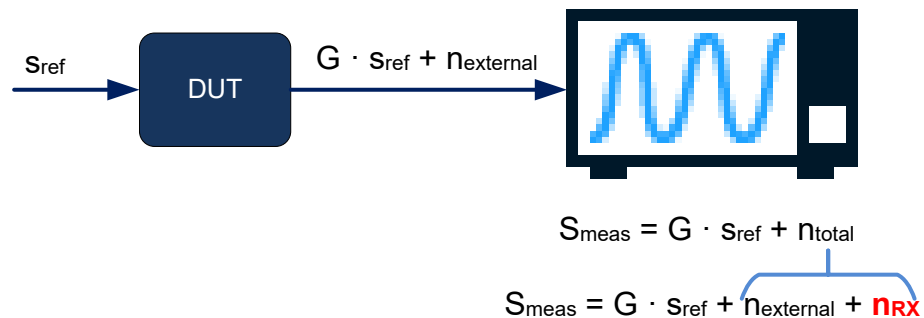


Figure 4-10: Components of measured signal at the R&S FSV/A

S_{ref} = reference signal
 G = Gain due to the DUT
 n_{external} = external noise components (noise at the RF port of the receiver)
 S_{meas} = measured signal
 n_{total} = total noise
 n_{RX} = noise caused by receiver

The R&S FSV3-K575 I/Q noise cancellation option aims at reducing this wideband receiver noise, while leaving external noise contributions (not caused by the instrument) untouched. The result is a corrected signal that contains only external noise contributions.

$$S_{\text{corrected}} = G \cdot S_{\text{ref}} + n_{\text{external}}$$

Using this function, the residual EVM of the signal analyzer can be improved, even for very low signal levels.

Correction process

The I/Q noise cancellation process requires a repetitive input signal.

1. A single data capture is performed on the input signal to obtain a signal that contains all noise contributions (S_{meas}).
2. The application attempts to estimate the reference signal.
If synchronization fails, the I/Q noise cancellation process is not started. An error message is displayed in the status bar.
For tips on possible causes for synchronization failure, see [Section 9.1, "Troubleshooting I/Q noise cancellation"](#), on page 142.

3. A defined number of data captures is performed and the values are averaged. Since the signal is repetitive, averaging removes the total noise almost completely. The result is an ideal (noise-free) signal.

$$s_{avg} = G \cdot s_{ref}$$

4. The total noise is determined as

$$n_{total} = s_{meas} - s_{avg}$$

5. The total noise power (N_{total}) is calculated from n_{total} .
6. Using an internal, terminated (calibration) input signal, the receiver noise power (N_{RX}) is measured.
7. The ratio of the external noise to the total noise is determined as

$$w^2 = \frac{N_{external}}{N_{total}} = \frac{(N_{total} - N_{RX})}{N_{total}}$$

Where w is a weighting factor

8. For the most recently captured I/Q data, the measured noise components are added to the ideal, noise-free signal to obtain a corrected signal without internal noise components.

The "Magnitude Capture" display shows the most recently captured I/Q data without I/Q noise cancellation.

All other measurement results are based on the corrected signal. For statistical evaluation, the entire I/Q noise cancellation process is counted as one measurement.

An additional function is available to export the corrected signal data.

I/Q noise cancellation via remote control

If you have an input file (.wv or .iq.tar) with the reference signal, you can remove the internal noise from the measured raw I/Q data in the I/Q Analyzer application remotely. The resulting data is stored in a specified iq.tar file, but it is not displayed in the result displays. You can load the file in any other I/Q application for analysis.



Set the same sample rate and level settings in the I/Q Analyzer as in the application you require for analysis.

5 Configuration

Access: [MODE] > "I/Q Analyzer"

The I/Q Analyzer is a special application on the R&S FSV/A.

When you switch to an I/Q Analyzer channel the first time, a set of parameters is passed on from the currently active application. After initial setup, the parameters for the channel are stored upon exiting and restored upon re-entering the channel. Thus, you can switch between applications quickly and easily.

When you activate a channel for the I/Q Analyzer application, data acquisition from the input signal is started automatically with the default configuration. The "I/Q Analyzer" menu is displayed and provides access to the most important configuration functions.

The remote commands required to perform these tasks are described in [Section 10](#), "Remote commands to perform measurements with I/Q data", on page 144.



Importing and Exporting I/Q Data

The I/Q data to be evaluated in the I/Q Analyzer application can not only be captured by the I/Q Analyzer itself, it can also be imported to the R&S FSV/A, provided it has the correct format. Furthermore, the captured I/Q data from the I/Q Analyzer can be exported for further analysis in external applications.

For details see [Section 7](#), "I/Q data import and export", on page 132.

• Configuration overview	41
• Import/export functions	43
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• Amplitude	56
• Frequency settings	67
• Trigger settings	68
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5.1 Configuration overview



Access: all menus

Throughout the channel configuration, an overview of the most important currently defined settings is provided in the "Overview".



Multiple access paths to functionality

The easiest way to configure a channel is via the "Overview" dialog box, which is available from all menus.

Alternatively, you can access the individual dialog boxes from the corresponding menu items, or via tools in the toolbars, if available.

In this documentation, only the most convenient method of accessing the dialog boxes is indicated - usually via the "Overview".

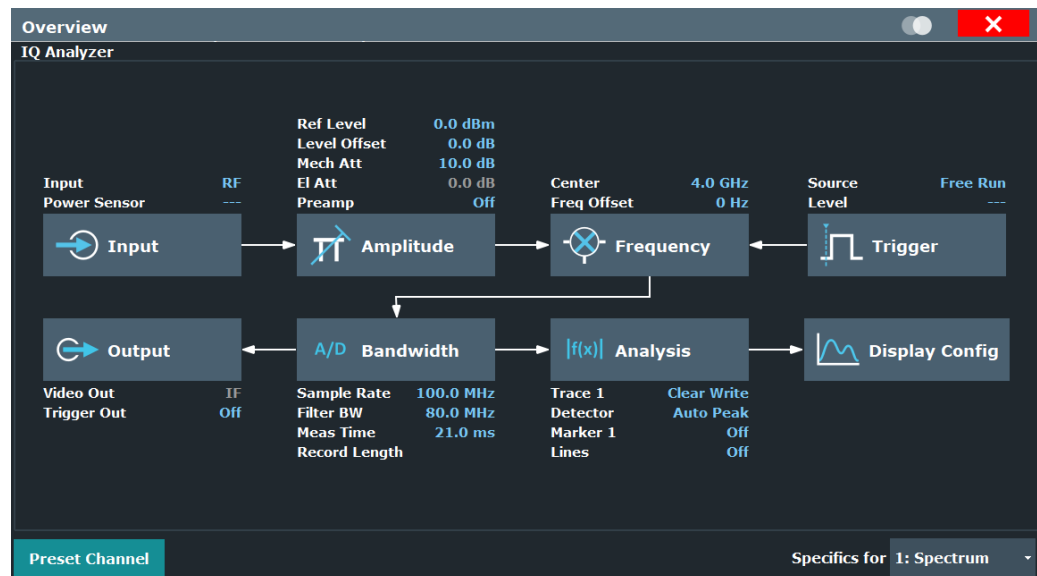


Figure 5-1: Configuration Overview for I/Q Analyzer primary

In addition to the main measurement settings, the "Overview" provides quick access to the main settings dialog boxes. The individual configuration steps are displayed in the order of the data flow. Thus, you can easily configure an entire channel from input over processing to output and analysis by stepping through the dialog boxes as indicated in the "Overview".

The "Overview" for the I/Q Analyzer provides quick access to the following configuration dialog boxes (listed in the recommended order of processing):

1. Input settings
See [Section 5.3.1, "Input source settings"](#), on page 47
2. Amplitude settings
See [Section 5.4, "Amplitude"](#), on page 56
3. Frequency settings
See [Section 5.5, "Frequency settings"](#), on page 67
4. Optionally, Trigger/Gate settings
See [Section 5.6, "Trigger settings"](#), on page 68
5. Bandwidth settings

See [Section 5.7, "Data acquisition and bandwidth settings"](#), on page 76

6. Optionally, output settings
See [Section 5.3.2, "Output settings"](#), on page 55
7. Analysis settings and functions
See [Section 6, "Analysis"](#), on page 89
8. Display configuration
See [Section 5.8, "Display configuration"](#), on page 84

To configure settings

- Select any button in the "Overview" to open the corresponding dialog box.
Select a setting in the channel bar (at the top of the channel tab) to change a specific setting.

For step-by-step instructions on configuring I/Q Analyzer measurements, see [Section 8.1, "How to perform measurements in the I/Q Analyzer application"](#), on page 137.

Preset Channel

Select "Preset Channel" in the lower left-hand corner of the "Overview" to restore all measurement settings *in the current channel* to their default values.

Note: Do not confuse "Preset Channel" with the [Preset] key, which restores the entire instrument to its default values and thus closes *all channels* on the R&S FSV/A (except for the default channel)!

Remote command:

[SYSTem:PRESet:CHANnel \[:EXEC\]](#) on page 154

Specific Settings for

The channel can contain several windows for different results. Thus, the settings indicated in the "Overview" and configured in the dialog boxes vary depending on the selected window.

Select an active window from the "Specific Settings for" selection list that is displayed in the "Overview" and in all window-specific configuration dialog boxes.

The "Overview" and dialog boxes are updated to indicate the settings for the selected window.

5.2 Import/export functions



Access: "Save"/ "Open" icon in the toolbar > "Import" / "Export"



Access (Import I/Q data): [Input Output] > "Input Source Config" > "I/Q File"

The R&S FSV/A provides various evaluation methods for the results of the performed measurements. However, you may want to evaluate the data with further, external applications. In this case, you can export the measurement data to a standard format file (ASCII or XML). Some of the data stored in these formats can also be re-imported to the R&S FSV/A for further evaluation later, for example in other applications.

The following data types can be exported (depending on the application):

- Trace data
- Table results, such as result summaries, marker peak lists etc.
- I/Q data (in applications that process I/Q data)



I/Q data can only be imported and exported in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

See the corresponding user manuals for those applications for details.



Exporting I/Q data is only possible in single sweep mode ([Continuous Sweep / Run Cont](#)).

Importing I/Q data is also possible in continuous sweep mode.

For more information about importing I/Q data files, see [Section 5.3.1.2, "Settings for input from I/Q data files"](#), on page 50.

Import.....	44
Export.....	44
L Export Trace to ASCII File.....	44
L File Type.....	45
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L Column Separator.....	46
L File Explorer.....	46
L Export Configuration.....	46
L I/Q Export.....	46
L File Type.....	47
L File Explorer.....	47



Import

Access: "Save/Recall" > Import



Provides functions to import data.

For more information about importing I/Q data files, see [Section 5.3.1.2, "Settings for input from I/Q data files"](#), on page 50.



Export

Access: "Save/Recall" > Export

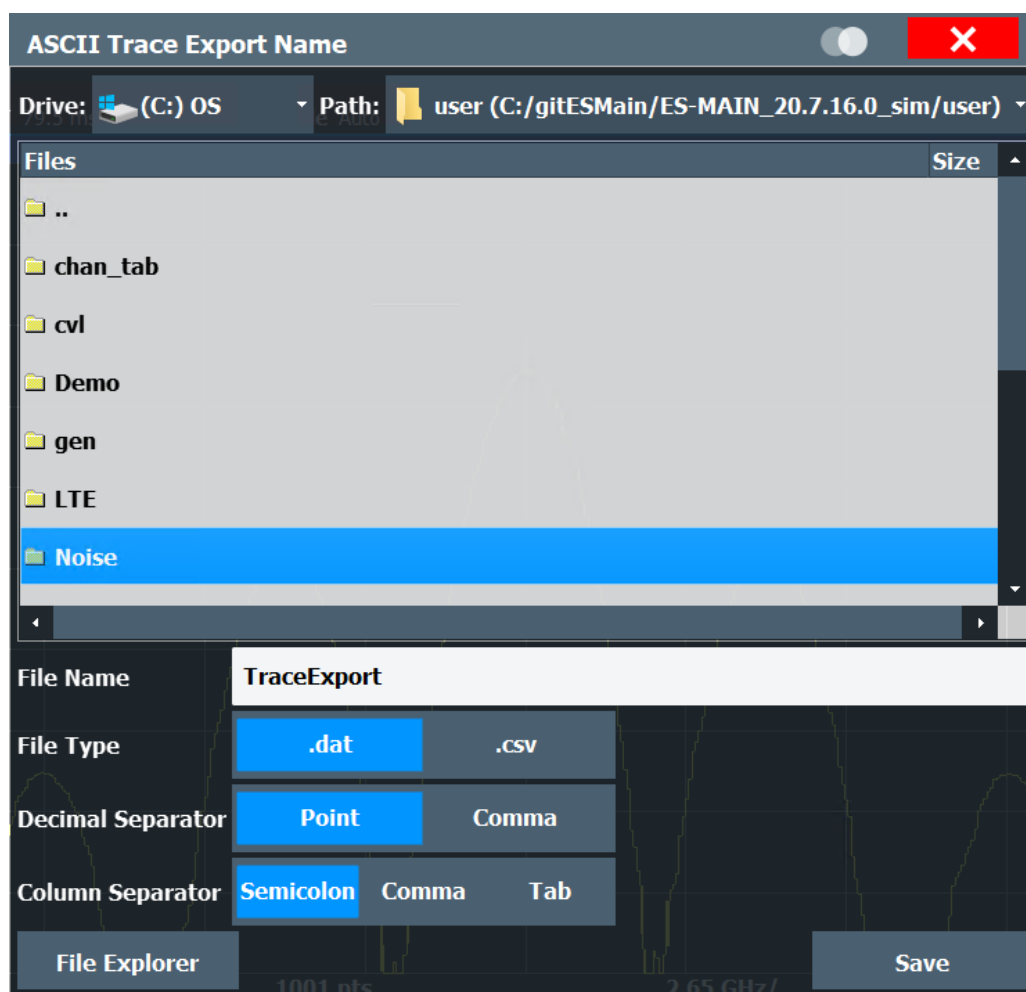


Opens a submenu to configure data export.

Export Trace to ASCII File ← Export

Saves the selected trace or all traces in the currently active result display to the specified file and directory in the selected ASCII format.

"File Explorer": Instead of using the file manager of the R&S FSV/A firmware, you can also use the Microsoft Windows File Explorer to manage files.



Note: Secure user mode.

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

[MMEMory:STORe<n>:TRACe](#) on page 342

File Type ← Export Trace to ASCII File ← Export

Determines the format of the ASCII file to be imported or exported.

Depending on the external program in which the data file was created or is evaluated, a comma-separated list (CSV) or a plain data format (DAT) file is required.

Remote command:

[FORMat:DEXPort:FORMat](#) on page 341

Decimal Separator ← Export Trace to ASCII File ← Export

Defines the decimal separator for floating-point numerals for the data export/import files. Evaluation programs require different separators in different languages.

Remote command:

`FORMat:DEXPort:DSEParator` on page 340

Column Separator ← Export Trace to ASCII File ← Export

Selects the character that separates columns in the exported ASCII file. The character can be either a semicolon, a comma or a tabulator (tab).

Example for semicolon:

```
Type;FSV3007;Version;1.00;Date;01.Jan 3000;
```

Example for comma:

```
Type,FSV3007,
Version,1.00,
Date,01.Jan 3000,
```

Example for tabulator (tab after the last column is not visible):

```
Type      FSV3007
Version    1.00
Date      01.Jan 3000
```

The selected column separator setting remains the same, even after a preset.

Remote command:

`FORMat:DEXPort:CSEParator` on page 340

File Explorer ← Export Trace to ASCII File ← Export

Opens the Microsoft Windows File Explorer.

Remote command:

not supported

Export Configuration ← Export

Opens the "Traces" dialog box to configure the trace and data export settings.

I/Q Export ← Export

Opens a file selection dialog box to define an export filename to which the I/Q data is stored. This function is only available in single sweep mode.

It is not available in the Spectrum application, only in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

For details, see the description in the R&S FSV/A I/Q Analyzer User Manual ("Importing and Exporting I/Q Data").

Note: Storing large amounts of I/Q data (several Gigabytes) can exceed the available (internal) storage space on the R&S FSV/A. In this case, it can be necessary to use an external storage medium.

Note: Secure user mode.

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

[MMEMory:STORe<n>:IQ:STATe](#) on page 345

[MMEMory:STORe<n>:IQ:COMMeNt](#) on page 345

File Type ← I/Q Export ← Export

The I/Q data file can be stored in one of the following supported formats:

- .iq.tar
- .iqw
- .csv
- .mat
- .aid

Note: Not all applications support all formats.

For details on formats, see [Section C, "Reference: supported I/Q file formats"](#), on page 358.

Remote command:

The file type is determined by the file extension of the file name parameter. If no file extension is provided, the file type is assumed to be .iq.tar. For .mat files, Matlab® v4 is assumed.

File Explorer ← I/Q Export ← Export

Opens the Microsoft Windows File Explorer.

Remote command:

not supported

5.3 Data input and output settings

Access: "Overview" > "Input"/ "Output"

The R&S FSV/A can analyze signals from different input sources and provide various types of output (such as noise source control or trigger signals).

- [Input source settings](#).....47
- [Output settings](#).....55

5.3.1 Input source settings

Access: "Overview" > "Input/Frontend" > "Input Source"

The input source determines which data the R&S FSV/A analyzes.

The default input source for the R&S FSV/A is "Radio Frequency", i.e. the signal at the "RF Input" connector of the R&S FSV/A. If no additional options are installed, this is the only available input source.



Further input sources

The I/Q Analyzer application application can also process input from the following optional sources:

- Probes
- External generator
- Power sensors
- External Mixer
- External frontend

For details, see the R&S FSV/A User Manual.

Radio frequency input.....	48
Settings for input from I/Q data files.....	50
Analog baseband input settings.....	52

5.3.1.1 Radio frequency input

Access: "Overview" > "Input/Frontend" > "Input Source" > "Radio Frequency"

Input Source

Power Sensor

External Generator

Probes

Radio Frequency

On

Off

Input Coupling

AC

DC

Direct Path

Auto

Off

YIG-Preselector

On

Off

Impedance Matching

Impedance

50Ω

75Ω

User

Value

100.0 Ohm

Pad Type

Series-R

MLP

Radio Frequency State.....	48
Input Coupling.....	48
Impedance.....	49
Direct Path.....	49
YIG-Preselector.....	50
Input Connector.....	50

Radio Frequency State

Activates input from the "RF Input" connector.

Remote command:

INPut<ip>:SElect on page 158

Input Coupling

The RF input of the R&S FSV/A can be coupled by alternating current (AC) or direct current (DC).

For an active external frontend, input coupling is always AC.

Not available for input from the optional "Analog Baseband" interface.

AC coupling blocks any DC voltage from the input signal. AC coupling is activated by default to prevent damage to the instrument. Very low frequencies in the input signal can be distorted.

However, some specifications require DC coupling. In this case, you must protect the instrument from damaging DC input voltages manually. For details, refer to the specifications document.

Remote command:

[INPut:COUPling](#) on page 157

Impedance

The R&S FSV/A has an internal impedance of 50 Ω . However, some applications use other impedance values. To match the impedance of an external application to the impedance of the R&S FSV/A, an *impedance matching pad* can be inserted at the input. If the type and impedance value of the used matching pad is known to the R&S FSV/A, it can convert the measured units accordingly so that the results are calculated correctly.

For an active external frontend, impedance is always 50 Ω .

The impedance conversion does not affect the level of the output signals (such as IF, video, demod).

"50 Ω "	(Default:) no conversion takes place
"75 Ω "	The 50 Ω input impedance is transformed to a higher impedance using a 75 Ω adapter of the selected "Pad Type": "Series-R" (default) or "MLP" (Minimum Loss Pad)
"User"	The 50 Ω input impedance is transformed to a user-defined impedance value according to the selected "Pad Type": "Series-R" (default) or "MLP" (Minimum Loss Pad)

Remote command:

[INPut:IMPedance](#) on page 157

[INPut:IMPedance:PTYPe](#) on page 158

For Analog Baseband input:

[INPut:IQ:IMPedance](#) on page 180

[INPut:IQ:IMPedance:PTYPe](#) on page 180

Direct Path

Enables or disables the use of the direct path for small frequencies.

In spectrum analyzers, passive analog mixers are used for the first conversion of the input signal. In such mixers, the LO signal is coupled into the IF path due to its limited isolation. The coupled LO signal becomes visible at the RF frequency 0 Hz. This effect is referred to as LO feedthrough.

To avoid the LO feedthrough the spectrum analyzer provides an alternative signal path to the A/D converter, referred to as the *direct path*. By default, the direct path is selected automatically for RF frequencies close to zero. However, this behavior can be disabled. If "Direct Path" is set to "Off", the spectrum analyzer always uses the analog mixer path.

For an active external frontend, the direct path is always used automatically for frequencies close to zero.

"Auto" (Default) The direct path is used automatically for frequencies close to zero.

"Off" The analog mixer path is always used.

Remote command:

[INPut:DPATH](#) on page 157

YIG-Preselector

Enables or disables the YIG-preselector.

This setting requires an additional option R&S FSV3-B11 on the R&S FSV/A.

An internal YIG-preselector at the input of the R&S FSV/A ensures that image frequencies are rejected. However, image rejection is only possible for a restricted bandwidth. To use the maximum bandwidth for signal analysis, you can disable the YIG-preselector at the input of the R&S FSV/A, which can lead to image-frequency display.

By default, the the YIG-"Preselector" is off.

Note: Note that the YIG-preselector is active only on frequencies greater than 7.5 GHz. Therefore, switching the YIG-preselector on or off has no effect if the frequency is below that value.

For frequencies above 50 GHz (requires option R&S FSV3-B54G, for R&S FSVA3050 only), the YIG-preselector is automatically switched off (internally, not indicated in the display). In this case, image frequencies can occur, as specified in the specifications document.

Remote command:

[INPut:FILTer:YIG\[:STATe\]](#) on page 157

Input Connector

Determines which connector the input data for the measurement is taken from.

For more information on the optional "Analog Baseband" interface, see the R&S FSV/A I/Q Analyzer and I/Q Input user manual.

"RF" (Default:) The "RF Input" connector

"RF Probe" The "RF Input" connector with an adapter for a modular probe
This setting is only available if a probe is connected to the "RF Input" connector.
It is not available for an active external frontend.

Remote command:

[INPut:CONNector](#) on page 156

5.3.1.2 Settings for input from I/Q data files

Access: "Overview" > "Input/Frontend" > "Input Source" > "I/Q File"

Or: [INPUT/OUTPUT] > "Input Source Config" > "Input Source" > "I/Q File"

Input Source

Radio Frequency

On

Off

TAP C

I/Q File

Input File

S:\GR-14DS\GR-14DS4\F\FSW_Sirius\input\IQ\vse_all_cmd_iq.iq.tar

Select File

Saved by:

IqTarWriteComplexFloat32 (MATLAB 8.4.0.150421 (R2014b))

Comment:

Test iq.tar file with 2 streams. Each stream has exactly two points. Stream1 at 500-

Date & Time:

2015-08-13 16:15:54

Sample Rate:

1 MHz

Number of Samples:

25304

Duration of Signal:

25.304 ms

Number of Channels:

2

Selected Channel:

Auto

File Repetitions:

1

For details, see [Section 4.2, "Basics on input from I/Q data files"](#), on page 25.

I/Q Input File State	51
Select I/Q data file	51
Selected Channel	52
File Repetitions	52

I/Q Input File State

Enables input from the selected I/Q input file.

If enabled, the application performs measurements on the data from this file. Thus, most measurement settings related to data acquisition (attenuation, center frequency, measurement bandwidth, sample rate) cannot be changed. The measurement time can only be decreased to perform measurements on an extract of the available data only.

Note: Even when the file input is disabled, the input file remains selected and can be enabled again quickly by changing the state.

Remote command:
[INPut<ip>:SElect](#) on page 158

Select I/Q data file

Opens a file selection dialog box to select an input file that contains I/Q data.

The I/Q data file must be in one of the following supported formats:

- .iq.tar
- .iqw
- .csv
- .mat
- .wv
- .aid

For details on formats, see [Section C, "Reference: supported I/Q file formats"](#), on page 358.

The file type is determined by the file extension. If no file extension is provided, the file type is assumed to be .iq.tar. For .mat files, Matlab® v4 is assumed.

Note: Only a single data stream or channel can be used as input, even if multiple streams or channels are stored in the file.

Note: For some file formats that do not provide the sample rate and measurement time or record length, you must define these parameters manually. Otherwise the traces are not visible in the result displays.

The default storage location for I/Q data files is `C:\R_S\INSTR\USER`.

Remote command:

`INPut:FILE:PATH` on page 159

Selected Channel

Only available for files that contain more than one data stream from multiple channels: selects the data stream to be used as input for the currently selected channel.

In "Auto" mode (default), the first data stream in the file is used as input for the channel. Applications that support multiple data streams use the first data stream in the file for the first input stream, the second for the second stream etc.

Remote command:

`MMEMory:LOAD:IQ:STReam` on page 160

`MMEMory:LOAD:IQ:STReam:AUTO` on page 160

`MMEMory:LOAD:IQ:STReam:LIST?` on page 161

File Repetitions

Determines how often the data stream is repeatedly copied in the I/Q data memory to create a longer record. If the available memory is not sufficient for the specified number of repetitions, the largest possible number of complete data streams is used.

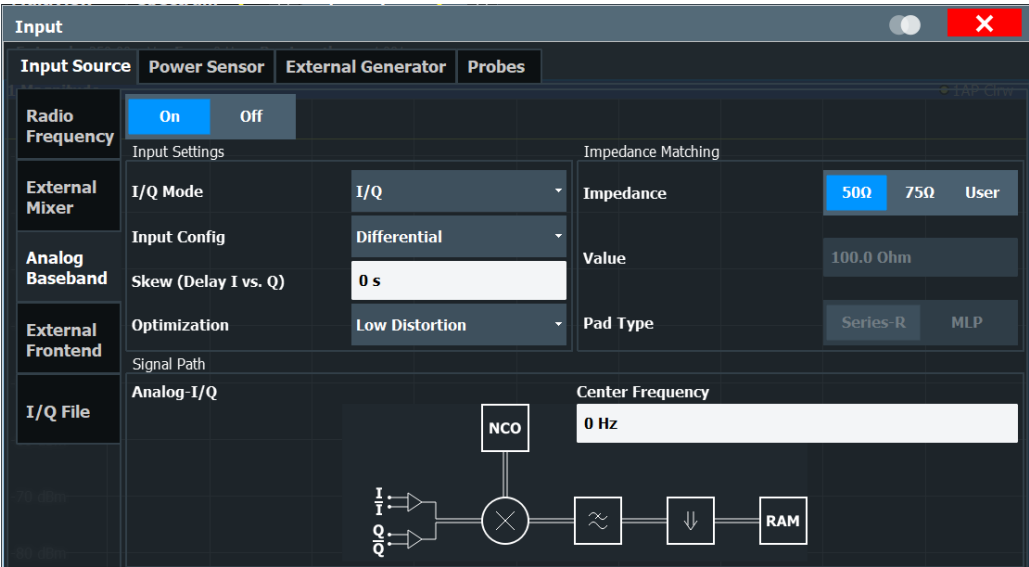
Remote command:

`TRACe:IQ:FILE:REPetition:COUNt` on page 161

5.3.1.3 Analog baseband input settings

Access: [INPUT/OUTPUT] > "Input Source Config" > "Analog Baseband" tab

The following settings and functions are available to provide input via the optional "Analog Baseband" interface in the applications that support it.



Analog Baseband Input State.....	53
I/Q Mode.....	53
Input Configuration.....	54
Skew (Delay I vs. Q).....	54
Optimization.....	54
Impedance.....	54
Center Frequency.....	55

Analog Baseband Input State
 Enables or disable the use of the "Analog Baseband" input source for measurements. "Analog Baseband" is only available if the optional "Analog Baseband" is installed.
 Remote command:
[INPut<ip>:SELEct](#) on page 158

I/Q Mode
 Defines the format of the input signal.
 For more information, see [Section 4.3.3, "I/Q processing modes"](#), on page 29.

"I + jQ" The input signal is filtered and resampled to the sample rate of the application.
 Two inputs are required for a complex signal, one for the in-phase component, and one for the quadrature component.

"I Only / Low IF I" The input signal at the "Baseband Input I" connector is filtered and resampled to the sample rate of the application.
 If the center frequency is set to 0 Hz, the real baseband signal is displayed without down-conversion (**Real Baseband I**).
 If a center frequency greater than 0 Hz is set, the input signal is down-converted with the center frequency (**Low IF I**).

"Q Only / Low IF Q"

The input signal at the "Baseband Input Q" connector is filtered and resampled to the sample rate of the application.

If the center frequency is set to 0 Hz, the real baseband signal is displayed without down-conversion (**Real Baseband Q**).

If a center frequency greater than 0 Hz is set, the input signal is down-converted with the center frequency (**Low IF Q**).

Remote command:

`INPut:IQ:TYPE` on page 181

Input Configuration

Defines whether the input is provided as a differential signal via all four Analog Baseband connectors or as a plain I/Q signal via two single-ended lines.

Note: Both single-ended and differential probes are supported as input; however, since only one connector is occupied by a probe, the "Single-ended" setting must be used for all probes.

"Single-ended" I, Q data only

"Differential" I, Q and inverse I,Q data

Remote command:

`INPut:IQ:BALanced[:STATe]` on page 179

Skew (Delay I vs. Q)

Defines a delay of the I vector in respect to the Q vector. A skew can be used to compensate a physical impairment of the two analog signals I and Q.

Remote command:

`INPut:IQ:SKEW` on page 181

Optimization

Selects the priority for signal processing when using input from the optional "Analog Baseband" interface.

"Low distortion"

(Default:) Optimized for low distortion by avoiding intermodulation

"Low noise"

Optimized for high sensitivity and low noise levels

If this setting is selected, "Low noise" is indicated in the channel information bar.

Remote command:

`INPut:IQ:OPTimize` on page 180

Impedance

The R&S FSV/A has an internal impedance of 50 Ω . However, some applications use other impedance values. To match the impedance of an external application to the impedance of the R&S FSV/A, an *impedance matching pad* can be inserted at the input. If the type and impedance value of the used matching pad is known to the R&S FSV/A, it can convert the measured units accordingly so that the results are calculated correctly.

For an active external frontend, impedance is always 50 Ω .

The impedance conversion does not affect the level of the output signals (such as IF, video, demod).

"50Ω"	(Default:) no conversion takes place
"75Ω"	The 50 Ω input impedance is transformed to a higher impedance using a 75 Ω adapter of the selected "Pad Type": "Series-R" (default) or "MLP" (Minimum Loss Pad)
"User"	The 50 Ω input impedance is transformed to a user-defined impedance value according to the selected "Pad Type": "Series-R" (default) or "MLP" (Minimum Loss Pad)

Remote command:

[INPut:IMPedance](#) on page 157

[INPut:IMPedance:PTYPe](#) on page 158

For Analog Baseband input:

[INPut:IQ:IMPedance](#) on page 180

[INPut:IQ:IMPedance:PTYPe](#) on page 180

Center Frequency

Defines the center frequency for analog baseband input.

For real-type baseband input (I or Q only), the center frequency is always 0 Hz.

Note: If the analysis bandwidth to either side of the defined center frequency exceeds the minimum frequency or the maximum frequency, an error is displayed. In this case, adjust the center frequency or the analysis bandwidth.

For details on frequency ranges and the analysis bandwidth, see [Section 4.3, "Processing data from the Analog Baseband interface"](#), on page 27.

Remote command:

[\[SENSe:\]FREQuency:CENTer](#) on page 235

5.3.2 Output settings

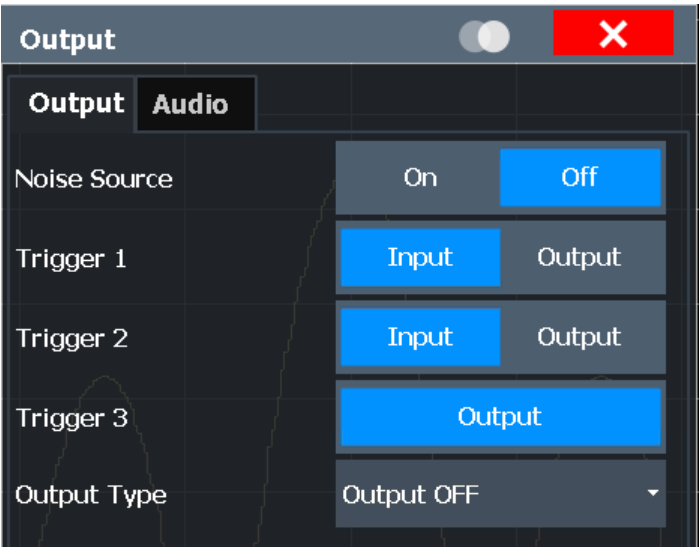
Access: [Input/Output] > "Output"

The R&S FSV/A can provide output to special connectors for other devices.

For details on connectors, refer to the R&S FSV/A Getting Started manual, "Front / Rear Panel View" sections.



How to provide trigger signals as output is described in detail in the R&S FSV3000/FSVA3000 base unit user manual.



Noise Source Control.....

56

Noise Source Control

Enables or disables the 28 V voltage supply for an external noise source connected to the "Noise source control / Power sensor") connector. By switching the supply voltage for an external noise source on or off in the firmware, you can enable or disable the device as required.

External noise sources are useful when you are measuring power levels that fall below the noise floor of the R&S FSV/A itself, for example when measuring the noise level of an amplifier.

In this case, you can first connect an external noise source (whose noise power level is known in advance) to the R&S FSV/A and measure the total noise power. From this value, you can determine the noise power of the R&S FSV/A. Then when you measure the power level of the actual DUT, you can deduct the known noise level from the total power to obtain the power level of the DUT.

Remote command:

[DIAGnostic:SERVice:NSource](#) on page 222

5.4 Amplitude

Access: "Overview" > "Amplitude"

Amplitude settings are identical to the Spectrum application, except for a new scaling function for I/Q Vector and Real/Imag results (see "[Y-Axis Max](#)" on page 66).

For background information on amplitude settings see the R&S FSV/A User Manual.

5.4.1 Amplitude settings

Access: "Overview" > "Amplitude"

Amplitude settings determine how the R&S FSV/A must process or display the expected input power levels.

Configuring amplitude settings allows you to:

- Adapt the instrument hardware to the expected maximum signal level by setting the [Reference Level](#) to this maximum
- Consider an external attenuator or preamplifier (using the "Offset").
- Optimize the SNR of the measurement for low signal levels by configuring the [Reference Level](#) as high as possible without introducing compression, clipping or overload. Use early amplification by the preamplifier and a low attenuation.
- Optimize the SNR for high signal levels and ensure that the instrument hardware is not damaged, using high attenuation and AC coupling (for DC input voltage).
- Adapt the reference impedance for power results when measuring in a 75-Ohm system by connecting an external matching pad to the RF input.



Using external frontends

For an active external frontend, the amplitude settings refer to the RF input at the external frontend, not the levels at the RF input of the R&S FSV/A. You can configure the attenuation at the external frontend and the analyzer separately. Electronic attenuation, preamplifier and noise cancellation are not available. Input coupling is always AC. Impedance is always 50 Ω . The "Auto level" function is not available for R&S FE44S and R&S FE50DTR frontends using the "Frequency Band Configuration": "IF High".

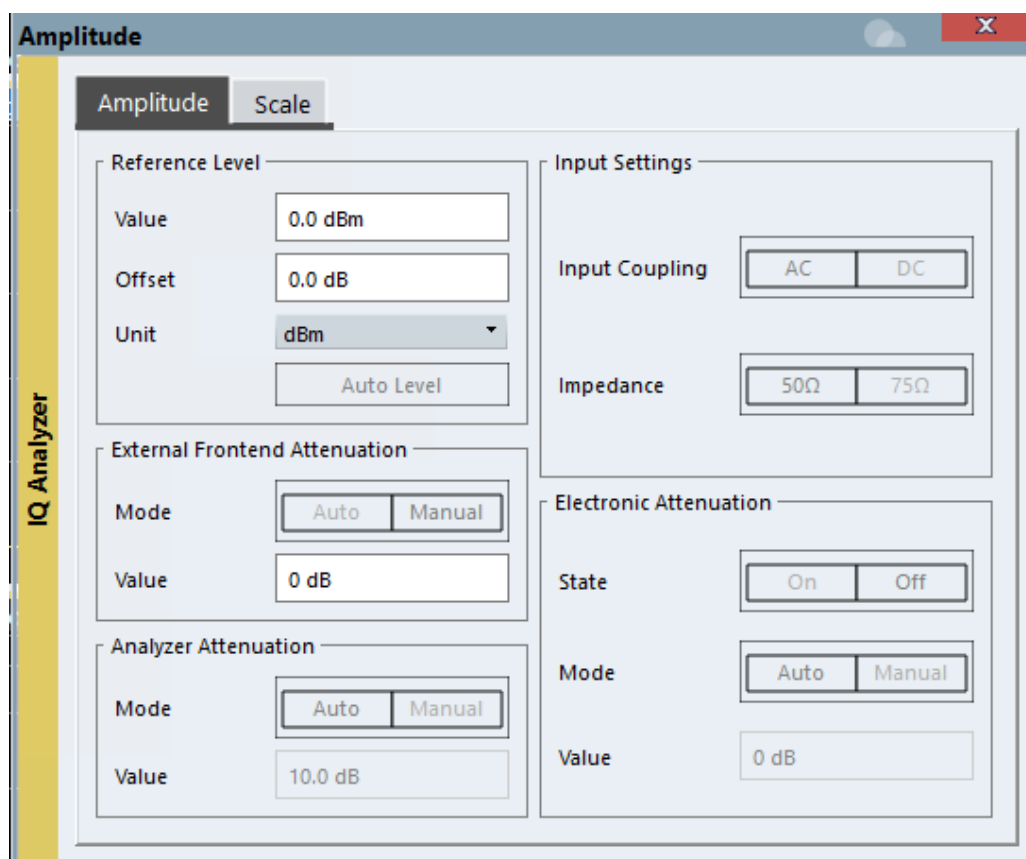


Figure 5-2: Amplitude settings for active external frontend

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Reference Level

Defines the expected maximum reference level. Signal levels above this value are possibly not measured correctly. Signals above the reference level are indicated by an "IF Overload" or "OVLD" status display.

The reference level can also be used to scale power diagrams; the reference level is then used for the calculation of the maximum on the y-axis.

Since the hardware of the R&S FSV/A is adapted according to this value, it is recommended that you set the reference level close above the expected maximum signal level. Thus you ensure an optimal measurement (no compression, good signal-to-noise ratio).

Note that for input from the External Mixer (R&S FSV3-B21) the maximum reference level also depends on the conversion loss; see the R&S FSV3000/ FSV/A3000 base unit user manual for details.

For an active external frontend, the reference level refers to the RF input at the external frontend, not the levels at the RF input of the R&S FSV/A. The hardware is adjusted to the defined reference level optimally for input signals with a crest factor of 10 dB. Thus, the required reference level for an optimal measurement can differ depending on the crest factor of the input signal.

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALE]:RLEVel`
on page 226

Shifting the Display (Offset) ← Reference Level

Defines an arithmetic level offset. This offset is added to the measured level. In some result displays, the scaling of the y-axis is changed accordingly.

Define an offset if the signal is attenuated or amplified before it is fed into the R&S FSV/A so the application shows correct power results. All displayed power level results are shifted by this value.

The setting range is ± 200 dB in 0.01 dB steps.

Note, however, that the *internal* reference level (used to adjust the hardware settings to the expected signal) ignores any "Reference Level Offset". Thus, it is important to keep in mind the actual power level the R&S FSV/A must handle. Do not rely on the displayed reference level (internal reference level = displayed reference level - offset).

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALE]:RLEVel:OFFSet` on page 226

Unit ← Reference Level

The R&S FSV/A measures the signal voltage at the RF input.

In the default state, the level is displayed at a power level of 1 mW (= dBm). Via the known input impedance (50 Ω or 75 Ω , see "[Impedance](#)" on page 49), conversion to other units is possible.

Remote command:

`INPut:IMPedance` on page 157
`CALCulate<n>:UNIT:POWer` on page 226

Setting the Reference Level Automatically (Auto Level) ← Reference Level

Automatically determines a reference level which ensures that no overload occurs at the R&S FSV/A for the current input data. At the same time, the internal attenuators and the preamplifier (for analog baseband input: the full-scale level) are adjusted. As a result, the signal-to-noise ratio is optimized, while signal compression and clipping are minimized.

To determine the required reference level, a level measurement is performed on the R&S FSV/A.

If necessary, you can optimize the reference level further. Decrease the attenuation level manually to the lowest possible value before an overload occurs, then decrease the reference level in the same way.

You can change the measurement time for the level measurement if necessary (see "(Meas Time Manual)" on page 87).

Note: Using external frontends. The "Auto level" function is not available for R&S FE44S and R&S FE50DTR frontends using the "Frequency Band Configuration": "IF High".

Remote command:

[SENSe:]ADJust:LEVel on page 257

External Frontend Attenuation

If an external frontend is active, you can configure the attenuation for the external frontend and the analyzer separately.

For more information, see the R&S FSV/A User Manual.

When using an external frontend, only mechanical attenuation is available.

The RF attenuation can be set automatically as a function of the selected reference level (Auto mode). Automatic attenuation ensures that no overload occurs at the RF Input connector for the current reference level. It is the default setting.

In "Manual" mode, you can set the RF attenuation in 1 dB steps (down to 0 dB).

Other entries are rounded to the next integer value. The range is specified in the specifications document. If the defined reference level cannot be set for the defined RF attenuation, the reference level is adjusted accordingly and the warning "limit reached" is displayed.

NOTICE! Risk of hardware damage due to high power levels. When decreasing the attenuation manually, ensure that the power level does not exceed the maximum level allowed at the RF input, as an overload can lead to hardware damage.

Remote command:

INPut:ATTenuation on page 227

INPut:ATTenuation:AUTO on page 228

Attenuation Mode / Value

Defines the attenuation applied to the RF input of the R&S FSV/A.

The RF attenuation can be set automatically as a function of the selected reference level (Auto mode). Automatic attenuation ensures that no overload occurs at the RF Input connector for the current reference level. It is the default setting.

Note: Using an external frontend. If an external frontend is active, you can configure the attenuation for the external frontend and the analyzer separately. When using an external frontend, only mechanical attenuation is available.

For more information, see the R&S FSV3000/ FSVA3000 base unit user manual.

By default and when no (optional) [electronic attenuation](#) is available, mechanical attenuation is applied.

In "Manual" mode, you can set the RF attenuation in 1 dB steps (down to 0 dB). Other entries are rounded to the next integer value. The range is specified in the specifications document. If the defined reference level cannot be set for the defined RF attenuation, the reference level is adjusted accordingly and the warning "limit reached" is displayed.

NOTICE! Risk of hardware damage due to high power levels. When decreasing the attenuation manually, ensure that the power level does not exceed the maximum level allowed at the RF input, as an overload can lead to hardware damage.

Remote command:

`INPut:ATTenuation` on page 227

`INPut:ATTenuation:AUTO` on page 228

Defining attenuation for the analyzer when using an external frontend:

`INPut:SANalyzer:ATTenuation:AUTO` on page 230

`INPut:SANalyzer:ATTenuation` on page 229

Optimization

Selects the priority for signal processing *after* the RF attenuation has been applied.

This function is only available under the following conditions:

- One of the following options that provide a separate wideband processing path in the R&S FSV/A is installed:
Bandwidth extension R&S FSV3-B200/-B400
- An I/Q bandwidth that requires the wideband path is used.

"Low distortion"

(Default:) Optimized for low distortion by avoiding intermodulation

"Low noise"

Optimized for high sensitivity and low noise levels

If this setting is selected, "Low noise" is indicated in the channel information bar.

Remote command:

`INPut:ATTenuation:AUTO:MODE` on page 228

Using Electronic Attenuation

If the (optional) Electronic Attenuation hardware is installed on the R&S FSV/A, you can also activate an electronic attenuator.

In "Auto" mode, the settings are defined automatically; in "Manual" mode, you can define the mechanical and electronic attenuation separately.

For an active external frontend, electronic attenuation is not available.

Note: Electronic attenuation is not available for stop frequencies (or center frequencies in zero span) above 7 GHz.

In "Auto" mode, RF attenuation is provided by the electronic attenuator as much as possible to reduce the amount of mechanical switching required. Mechanical attenuation can provide a better signal-to-noise ratio, however.

When you switch off electronic attenuation, the RF attenuation is automatically set to the same mode (auto/manual) as the electronic attenuation was set to. Thus, the RF attenuation can be set to automatic mode, and the full attenuation is provided by the mechanical attenuator, if possible.

The electronic attenuation can be varied in 1 dB steps. If the electronic attenuation is on, the mechanical attenuation can be varied in 5 dB steps. Other entries are rounded to the next lower integer value.

If the defined reference level cannot be set for the given attenuation, the reference level is adjusted accordingly and the warning "limit reached" is displayed in the status bar.

Remote command:

`INPut:EATT:STATe` on page 229

`INPut:EATT:AUTO` on page 229

`INPut:EATT` on page 228

Input Settings

Some input settings affect the measured amplitude of the signal, as well.

The parameters "Input Coupling" and "Impedance" are identical to those in the "Input" settings.

See [Section 5.3.1, "Input source settings"](#), on page 47.

Preamplifier ← Input Settings

If the (optional) internal preamplifier hardware is installed on the R&S FSV/A, a preamplifier can be activated for the RF input signal.

You can use a preamplifier to analyze signals from DUTs with low output power.

For an active external frontend, a preamplifier is not available.

For R&S FSV/A, the following settings are available:

"Off"	Deactivates the preamplifier.
"15 dB"	The RF input signal is amplified by about 15 dB.
"30 dB"	The RF input signal is amplified by about 30 dB.
"On"	Using the preamplifier with the option number 1330.3465.02: the input signal is amplified by 30 dB if the preamplifier is activated.

For R&S FSV/A3044 models, the preamplifier is only available under the following conditions:

- In zero span, the maximum center frequency is 43.5 GHz
- For frequency spans, the maximum stop frequency is 43.5 GHz
- For I/Q measurements, the maximum center frequency depends on the analysis bandwidth:

$$f_{center} \leq 43.5 \text{ GHz} - (<Analysis_bw> / 2)$$

If any of the conditions no longer apply after you change a setting, the preamplifier is automatically deactivated.

Remote command:

`INPut:GAIN:STATe` on page 231

`INPut:GAIN[:VALue]` on page 231

5.4.2 Amplitude settings for analog baseband input

Access: "Overview" > "Amplitude"

The following settings and functions are available to define amplitude settings for input via the optional "Analog Baseband" interface in the applications that support it.



The input settings provided here are identical to those in the "Input Source" > "Analog Baseband" tab, see [Section 5.3.1.3, "Analog baseband input settings"](#), on page 52.

Reference Level.....	63
L Shifting the Display (Offset).....	64
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Full Scale Level Mode / Value.....	65

Reference Level

Defines the expected maximum reference level. Signal levels above this value are possibly not measured correctly. Signals above the reference level are indicated by an "IF Overload" or "OVLD" status display.

The reference level can also be used to scale power diagrams; the reference level is then used for the calculation of the maximum on the y-axis.

Since the hardware of the R&S FSV/A is adapted according to this value, it is recommended that you set the reference level close above the expected maximum signal level. Thus you ensure an optimal measurement (no compression, good signal-to-noise ratio).

Note that for input from the External Mixer (R&S FSV3-B21) the maximum reference level also depends on the conversion loss; see the R&S FSV3000/ FSVA3000 base unit user manual for details.

For an active external frontend, the reference level refers to the RF input at the external frontend, not the levels at the RF input of the R&S FSV/A. The hardware is adjusted to the defined reference level optimally for input signals with a crest factor of 10 dB. Thus, the required reference level for an optimal measurement can differ depending on the crest factor of the input signal.

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RLEVel`
on page 226

Shifting the Display (Offset) ← Reference Level

Defines an arithmetic level offset. This offset is added to the measured level. In some result displays, the scaling of the y-axis is changed accordingly.

Define an offset if the signal is attenuated or amplified before it is fed into the R&S FSV/A so the application shows correct power results. All displayed power level results are shifted by this value.

The setting range is ± 200 dB in 0.01 dB steps.

Note, however, that the *internal* reference level (used to adjust the hardware settings to the expected signal) ignores any "Reference Level Offset". Thus, it is important to keep in mind the actual power level the R&S FSV/A must handle. Do not rely on the displayed reference level (internal reference level = displayed reference level - offset).

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RLEVel:OFFSet` on page 226

Unit ← Reference Level

The R&S FSV/A measures the signal voltage at the RF input.

In the default state, the level is displayed at a power level of 1 mW (= dBm). Via the known input impedance (50 Ω or 75 Ω , see "[Impedance](#)" on page 49), conversion to other units is possible.

Remote command:

`INPut:IMPedance` on page 157
`CALCulate<n>:UNIT:POWer` on page 226

Setting the Reference Level Automatically (Auto Level) ← Reference Level

Automatically determines a reference level which ensures that no overload occurs at the R&S FSV/A for the current input data. At the same time, the internal attenuators and the preamplifier (for analog baseband input: the full-scale level) are adjusted. As a result, the signal-to-noise ratio is optimized, while signal compression and clipping are minimized.

To determine the required reference level, a level measurement is performed on the R&S FSV/A.

If necessary, you can optimize the reference level further. Decrease the attenuation level manually to the lowest possible value before an overload occurs, then decrease the reference level in the same way.

You can change the measurement time for the level measurement if necessary (see "[\(Meas Time Manual\)](#)" on page 87).

Note: Using external frontends. The "Auto level" function is not available for R&S FE44S and R&S FE50DTR frontends using the "Frequency Band Configuration": "IF High".

Remote command:

`[SENSe:]ADJust:LEVel` on page 257

Full Scale Level Mode / Value

The full scale level defines the maximum power you can input at the Baseband Input connector without clipping the signal.

The full scale level can be defined automatically according to the reference level, or manually.

For manual input, the following values can be selected:

- 0.25 V
- 0.5 V
- 1 V
- 2 V

If probes are connected, the possible full scale values are adapted according to the probe's attenuation and maximum allowed power.

For details on probes, see the R&S FSV/A User Manual.

Remote command:

For Analog Baseband input:

`INPut:IQ:FULLscale:AUTO` on page 179

`INPut:IQ:FULLscale[:LEVel]` on page 179

5.4.3 Scaling the y-axis

The individual scaling settings that affect the vertical axis are described here.

Access: "Overview" > "Amplitude" > "Scale" tab

Or: [AMPT] > "Scale Config"

The screenshot shows the 'Amplitude' configuration window with the 'Scale' tab selected. The 'Range' section contains two input fields: 'Range' with a value of '100.0 dB' and a dropdown arrow, and 'Ref Level Position' with a value of '100.0 %'. The 'Scaling' section features three radio buttons: 'Logarithmic' (which is selected and highlighted in blue), 'Linear Percent', and 'Linear with Unit'. At the bottom of the scaling section are two buttons: 'Absolute' (selected and highlighted in blue) and 'Relative'.

Range.....	66
Ref Level Position.....	66
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Y-Axis Max.....	66

Range

Defines the displayed y-axis range in dB.

The default value is 100 dB.

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]` on page 232

Ref Level Position

Defines the reference level position, i.e. the position of the maximum AD converter value on the level axis in %.

0 % corresponds to the lower and 100 % to the upper limit of the diagram.

Values from -120 % to +280 % are available.

Larger values are useful for small scales, such as a power range of 10 dB or 20 dB, and low signal levels, for example 60 dB below the reference level. In this case, large reference level position values allow you to see the trace again.

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RPOSition` on page 234

Scaling

Defines the scaling method for the y-axis.

"Logarithmic"	Logarithmic scaling (only available for logarithmic units - dB..., and A, V, Watt)
"Linear with Unit"	Linear scaling in the unit of the measured signal
"Linear Percent"	Linear scaling in percentages from 0 to 100
"Absolute"	The labeling of the level lines refers to the absolute value of the reference level (not available for "Linear Percent")
"Relative"	The scaling is in dB, relative to the reference level (only available for logarithmic units - dB...). The upper line of the grid (reference level) is always at 0 dB.

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y:SPACing` on page 234

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:MODE` on page 233

Y-Axis Max

Defines the maximum value of the y-axis in the currently selected diagram in either direction (in Volts). Thus, the y-axis scale starts at -<Y-Axis Max> and ends at +<Y-Axis Max>.

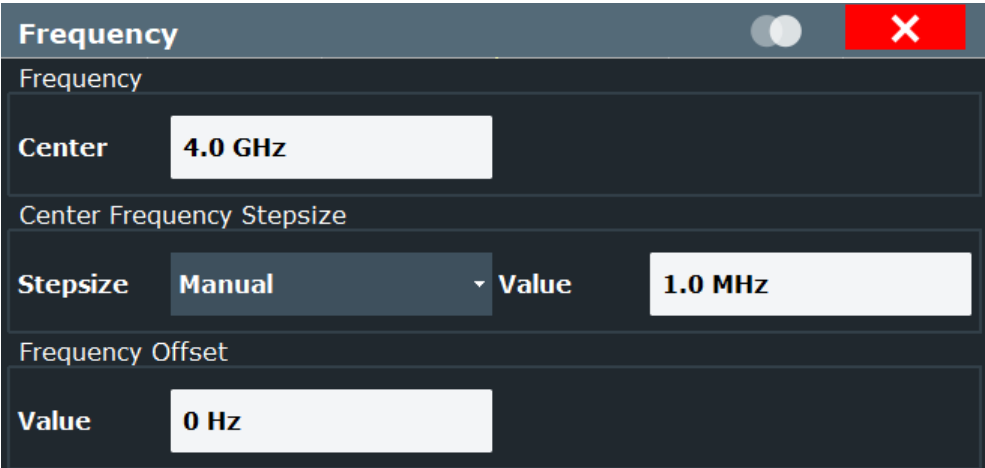
This command is only available if the evaluation mode for the I/Q Analyzer is set to "I/Q-Vector" or "Real/Imag (I/Q)".

Remote command:

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]` on page 232

5.5 Frequency settings

Access: "Overview" > "Frequency"



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Center Frequency

Defines the center frequency of the signal in Hertz.

The allowed range of values for the center frequency depends on the frequency span.

span > 0: $\text{span}_{\text{min}}/2 \leq f_{\text{center}} \leq f_{\text{max}} - \text{span}_{\text{min}}/2$

f_{max} and span_{min} depend on the instrument and are specified in the specifications document.

Remote command:

`[SENSe:]FREQuency:CENTer` on page 235

Center Frequency Stepsize

Defines the step size by which the center frequency is increased or decreased using the arrow keys.

The step size can be coupled to another value or it can be manually set to a fixed value.

"= Center"	Sets the step size to the value of the center frequency. The used value is indicated in the "Value" field.
"Manual"	Defines a fixed step size for the center frequency. Enter the step size in the "Value" field.

Remote command:

[SENSe:] FREQuency:CENTer:STEP on page 235

Frequency Offset

Shifts the displayed frequency range along the x-axis by the defined offset.

This parameter has no effect on the instrument's hardware, on the captured data, or on data processing. It is simply a manipulation of the final results in which absolute frequency values are displayed. Thus, the x-axis of a spectrum display is shifted by a constant offset if it shows absolute frequencies. However, if it shows frequencies relative to the signal's center frequency, it is not shifted.

A frequency offset can be used to correct the display of a signal that is slightly distorted by the measurement setup, for example.

The allowed values range from -1 THz to 1 THz. The default setting is 0 Hz.

Remote command:

[SENSe:] FREQuency:OFFSet on page 236

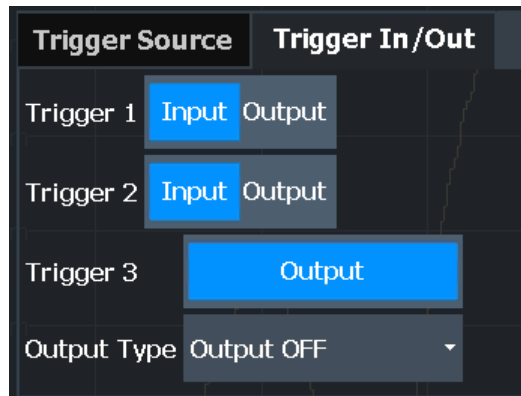
5.6 Trigger settings

Access: "Overview" > "Trigger"(> "Trigger In/Out")

Trigger settings determine when the input signal is measured.

Trigger	
Trigger Source	Trigger In/Out
Source	IF Power
Level	-20.0 dBm
Offset	0.0 s
Hysteresis	3.0 dB
Drop-Out Time	0.0 s
Slope	Rising Falling
Holdoff	0.0 s

External triggers from one of the "TRIGGER INPUT/OUTPUT" connectors on the R&S FSV/A are configured in a separate tab of the dialog box.



Conventional gating as in the Spectrum application is not available for the I/Q Analyzer; however, a special gating mode is available in remote control, see [Section 10.4.4.3, "Configuring I/Q gating"](#), on page 245.

For step-by-step instructions on configuring triggered measurements, see the R&S FSV/A User Manual.

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Trigger Source

The trigger settings define the beginning of a measurement.

Trigger Source ← Trigger Source

Selects the trigger source. If a trigger source other than "Free Run" is set, "TRG" is displayed in the channel bar and the trigger source is indicated.

For gated measurements in sweep mode, only external triggers are allowed as the gating trigger source.

For an active external frontend, only external triggers, I/Q power triggers and (periodic) time triggers are supported as trigger sources.

Note: If the 600 MHz or 1 GHz bandwidth extension options (B600/B601/B1000/B1001) are active, only an external trigger, IF power trigger or no trigger is available for bandwidths ≥ 400 MHz. If any other trigger is active and the analysis bandwidth is increased above 400 MHz (thus activating the B600/B601/B1000/B1001 option), the trigger is automatically deactivated.

Remote command:

TRIGger[:SEquence]:SOURce on page 241

Free Run ← Trigger Source ← Trigger Source

No trigger source is considered. Data acquisition is started manually or automatically and continues until stopped explicitly.

Remote command:

TRIG:SOUR IMM, see TRIGger[:SEquence]:SOURce on page 241

External Trigger 1/2 ← Trigger Source ← Trigger Source

Data acquisition starts when the TTL signal fed into the specified input connector meets or exceeds the specified trigger level.

(See "Trigger Level" on page 72).

Note: "External Trigger 1" automatically selects the trigger signal from the "Trigger 1 Input / Output" connector on the front panel.

In the I/Q Analyzer application, only "External Trigger 1" is supported.

For details, see the "Instrument Tour" section in the R&S FSV/A Getting Started manual.

"External Trigger 1"

Trigger signal from the "Trigger 1 Input / Output" connector.

"External Trigger 2"

Trigger signal from the "Trigger 2 Input / Output" connector.

Note: Connector must be configured for "Input" in the "Output" configuration

(See the R&S FSV3000/ FSVA3000 base unit user manual).

Remote command:

TRIG:SOUR EXT, TRIG:SOUR EXT2

See TRIGger[:SEquence]:SOURce on page 241

IF Power ← Trigger Source ← Trigger Source

The R&S FSV/A starts capturing data as soon as the trigger level is exceeded around the third intermediate frequency.

For frequency sweeps, the third IF represents the start frequency. The trigger threshold depends on the defined trigger level, as well as on the RF attenuation and preamplification. A reference level offset, if defined, is also considered. The trigger bandwidth at the intermediate frequency depends on the RBW and sweep type. For details on available trigger levels and trigger bandwidths, see the instrument specifications document.

For measurements on a fixed frequency (e.g. zero span or I/Q measurements), the third IF represents the center frequency.

This trigger source is only available for RF input.

The available trigger levels depend on the RF attenuation and preamplification. A reference level offset, if defined, is also considered.

For details on available trigger levels and trigger bandwidths, see the specifications document.

Remote command:

TRIG:SOUR IFP, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

Baseband Power ← Trigger Source ← Trigger Source

Defines triggering on the baseband power for baseband input.

Available for input from the optional "Analog Baseband" interface.

Remote command:

TRIG:SOUR BBP, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

I/Q Power ← Trigger Source ← Trigger Source

This trigger source is only available in the I/Q Analyzer application and in applications that process I/Q data.

If the R&S FSV3-B600/B601/B1000/B1001 bandwidth extension option is active, this trigger is not available for bandwidths ≥ 400 MHz.

Triggers the measurement when the magnitude of the sampled I/Q data exceeds the trigger threshold.

The trigger bandwidth corresponds to the bandwidth setting for I/Q data acquisition.

(See ["Analysis Bandwidth"](#) on page 78).

Remote command:

TRIG:SOUR IQP, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

RF Power ← Trigger Source ← Trigger Source

Defines triggering of the measurement via signals which are outside the displayed measurement range.

For this purpose, the instrument uses a level detector at the first intermediate frequency.

The "RF Power" trigger is available for input signals above 500 MHz. The trigger (6 dB-) bandwidth is ± 250 MHz. For the R&S FSV3000 and input signals between 500 MHz and 7.5 GHz, the trigger bandwidth depends on the instrument's serial number, see specifications document.

When using the [Direct Path](#), the RF power trigger is not available.

The resulting trigger level at the RF input depends on the RF attenuation and preamplification. For details on available trigger levels, see the instrument's specifications document.

Note: If the input signal contains frequencies outside of this range (e.g. for fullspan measurements), the sweep can be aborted. A message indicating the allowed input frequencies is displayed in the status bar.

A "Trigger Offset", "Trigger Polarity" and "Trigger Holdoff" (to improve the trigger stability) can be defined for the RF trigger, but no "Hysteresis".

Not available for input from the optional "Analog Baseband" interface.

If the trigger source "RF Power" is selected and you enable baseband input, the trigger source is automatically switched to "Free Run".

If the bandwidth extension option R&S FSV3-B600/B601/B1000/B1001 is active, this trigger is not available for bandwidths ≥ 400 MHz.

Remote command:

TRIG:SOUR RFP, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

Power Sensor ← Trigger Source ← Trigger Source

Uses an external power sensor as a trigger source. This option is only available if a power sensor is connected and configured.

Note: For Rohde & Schwarz power sensors, the "Gate Mode" *Lvl* is not supported. The signal sent by these sensors merely reflects the instant the level is first exceeded, rather than a time period. However, only time periods can be used for gating in level mode. Thus, the trigger impulse from the sensors is not long enough for a fully gated measurement; the measurement cannot be completed.

Remote command:

TRIG:SOUR PSE, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

Time ← Trigger Source ← Trigger Source

Triggers in a specified repetition interval.

See ["Repetition Interval"](#) on page 72.

Remote command:

TRIG:SOUR TIME, see [TRIGger\[:SEquence\]:SOURce](#) on page 241

Trigger Level ← Trigger Source

Defines the trigger level for the specified trigger source.

For details on supported trigger levels, see the instrument specifications document.

For time triggers, the repetition interval is defined. See ["Repetition Interval"](#) on page 72.

Remote command:

[TRIGger\[:SEquence\]:LEVel:IFPower](#) on page 240

[TRIGger<undef>\[:SEquence\]:LEVel:IQPower](#) on page 240

[TRIGger\[:SEquence\]:LEVel\[:EXternal<port>\]](#) on page 240

For baseband input only:

[TRIGger\[:SEquence\]:LEVel:BBPower](#) on page 239

Repetition Interval ← Trigger Source

Defines the repetition interval for a time trigger.

The shortest interval is 1 μ s. The granularity of the repetition interval is 1/256 MHz.

Set the repetition interval to the exact pulse period, burst length, frame length or other repetitive signal characteristic. If the required interval cannot be set with the available granularity, configure a multiple of the interval that can be set. Thus, the trigger remains synchronized to the signal.

Remote command:

[TRIGger\[:SEquence\]:TIME:RINterval](#) on page 242

Drop-Out Time ← Trigger Source

Defines the time that the input signal must stay below the trigger level before triggering again.

Note: For input from the optional "Analog Baseband" interface using the baseband power trigger (BBP), the default drop out time is set to 100 ns. This avoids unintentional trigger events (as no hysteresis can be configured in this case).

Remote command:

[TRIGger\[:SEquence\]:DTIME](#) on page 238

Trigger Offset ← Trigger Source

Defines the time offset between the trigger event and the start of the sweep.

Offset > 0:	Start of the sweep is delayed
Offset < 0:	Sweep starts earlier (pretrigger) Only possible for zero span (e.g. I/Q Analyzer application) and gated trigger switched off Maximum allowed range limited by the sweep time: $\text{Pretrigger}_{\text{max}} = \text{sweep time}_{\text{max}}$

Tip: To determine the trigger point in the sample (for "External" or "IF Power" trigger source), use the [TRACe:IQ:TPISample?](#) command.

Remote command:

[TRIGger\[:SEquence\]:HOLDoff\[:TIME\]](#) on page 238

Slope ← Trigger Source

For all trigger sources except time, you can define whether triggering occurs when the signal rises to the trigger level or falls down to it.

For gated measurements in "Edge" mode, the slope also defines whether the gate starts on a falling or rising edge.

Remote command:

[TRIGger\[:SEquence\]:SLOPe](#) on page 241

Hysteresis ← Trigger Source

Defines the distance in dB to the trigger level that the trigger source must exceed before a trigger event occurs. Setting a hysteresis avoids unwanted trigger events caused by noise oscillation around the trigger level.

This setting is only available for "IF Power" trigger sources. The range of the value is between 3 dB and 50 dB with a step width of 1 dB.

Remote command:

[TRIGger\[:SEquence\]:IFPower:HYSTeresis](#) on page 239

Trigger Holdoff ← Trigger Source

Defines the minimum time (in seconds) that must pass between two trigger events. Trigger events that occur during the holdoff time are ignored.

Remote command:

[TRIGger\[:SEquence\]:IFPower:HOLDoFF](#) on page 239

Correct TPIS ← Trigger Source

If enabled, the trigger event is set to the actual trigger point within the sample (TPIS). See also [TRACe:IQ:TPISample?](#) on page 253.

This setting is available for all trigger sources except "Free Run".

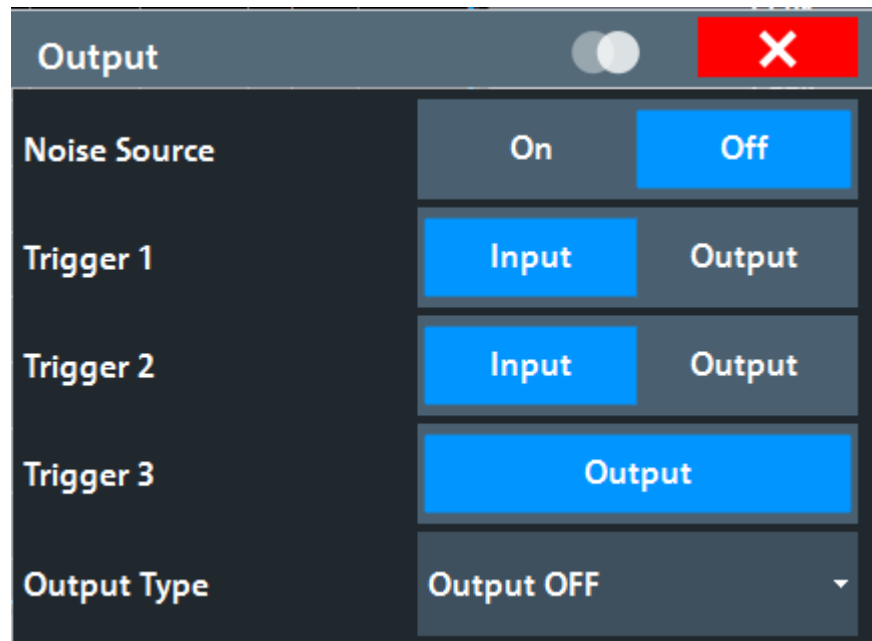
TPIS correction is not supported if user frequency response correction (R&S FSV3-K544) is active. If you activate frequency response correction, TPIS correction is automatically deactivated in all channels.

For details see frequency response correction in the R&S FSV/A user manual.

Note that when you enable this setting, the displayed trace no longer matches the instrument settings. Thus, the  yellow star icon is displayed on the tab label. Start a new measurement for the correction to take effect and to update the display.

Remote command:

[TRIGger\[:SEquence\]:CTPIs](#) on page 237

Trigger 1/2


Output	
Noise Source	On Off
Trigger 1	Input Output
Trigger 2	Input Output
Trigger 3	Output
Output Type	Output OFF

The trigger input and output functionality depends on how the variable "Trigger Input/Output" connectors are used.

Note: Providing trigger signals as output is described in detail in the R&S FSV3000/FSVA3000 base unit user manual.

"Trigger 1" "Trigger 1" is input only.

"Trigger 2"	Defines the usage of the variable "Trigger Input/Output" connector on the front panel
"Trigger 3"	Defines the usage of the variable "Trigger 3 Input/Output" connector on the rear panel
"Input"	The signal at the connector is used as an external trigger source by the R&S FSV/A. Trigger input parameters are available in the "Trigger" dialog box.
"Output"	The R&S FSV/A sends a trigger signal to the output connector to be used by connected devices. Further trigger parameters are available for the connector.

Remote command:

[OUTPut:TRIGger<tp>:DIRection](#) on page 243

Output Type ← Trigger 1/2

Type of signal to be sent to the output

Note: Trigger output and B1000 bandwidth extension.

If the B1000 bandwidth extension is installed and an analysis bandwidth larger than 40 MHz or analog baseband input is used, trigger output is only available using manual triggering ("Output Type": "User Defined").

"Output Off"	Deactivates the output. (Only for "Trigger 3", for which only output is supported.)
"Device Triggered"	(Default) Sends a trigger when the R&S FSV/A triggers.
"Trigger Armed"	Sends a (high level) trigger when the R&S FSV/A is in "Ready for trigger" state. This state is indicated by a status bit in the <code>STATUS:OPERation</code> register (bit 5), as well as by a low-level signal at the "AUX" port (pin 9). For details, see the description of the <code>STATUS:OPERation</code> register in the R&S FSV3000/ FSVA3000 base unit user manual and the description of the "AUX" port in the R&S FSV/A Getting Started manual.
"User Defined"	Sends a trigger when you select "Send Trigger". In this case, further parameters are available for the output signal.

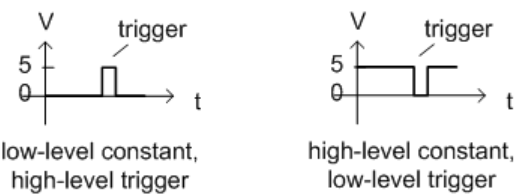
Remote command:

[OUTPut:TRIGger<tp>:OTYPe](#) on page 244

Level ← Output Type ← Trigger 1/2

Defines whether a high (1) or low (0) constant signal is sent to the trigger output connector (for "Output Type": "User Defined").

The trigger pulse level is always opposite to the constant signal level defined here. For example, for "Level" = "High", a constant high signal is output to the connector until you select the [Send Trigger](#) function. Then, a low pulse is provided.



Remote command:
[OUTPut:TRIGger<tp>:LEVel](#) on page 243

Pulse Length ← Output Type ← Trigger 1/2
 Defines the duration of the pulse (pulse width) sent as a trigger to the output connector.
 Remote command:
[OUTPut:TRIGger<tp>:PULSe:LENGth](#) on page 244

Send Trigger ← Output Type ← Trigger 1/2
 Sends a user-defined trigger to the output connector immediately.
 Note that the trigger pulse level is always opposite to the constant signal level defined by the output [Level](#) setting. For example, for "Level" = "High", a constant high signal is output to the connector until you select the "Send Trigger" function. Then, a low pulse is sent.
 Which pulse level is sent is indicated by a graphic on the button.
 Remote command:
[OUTPut:TRIGger<tp>:PULSe:IMMediate](#) on page 244

5.7 Data acquisition and bandwidth settings

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5.7.1 Data acquisition

Access: "Overview" > "Bandwidth" > "Data Acquisition" tab
 The data acquisition settings define which parts of the input signal are captured for further evaluation in the applications.

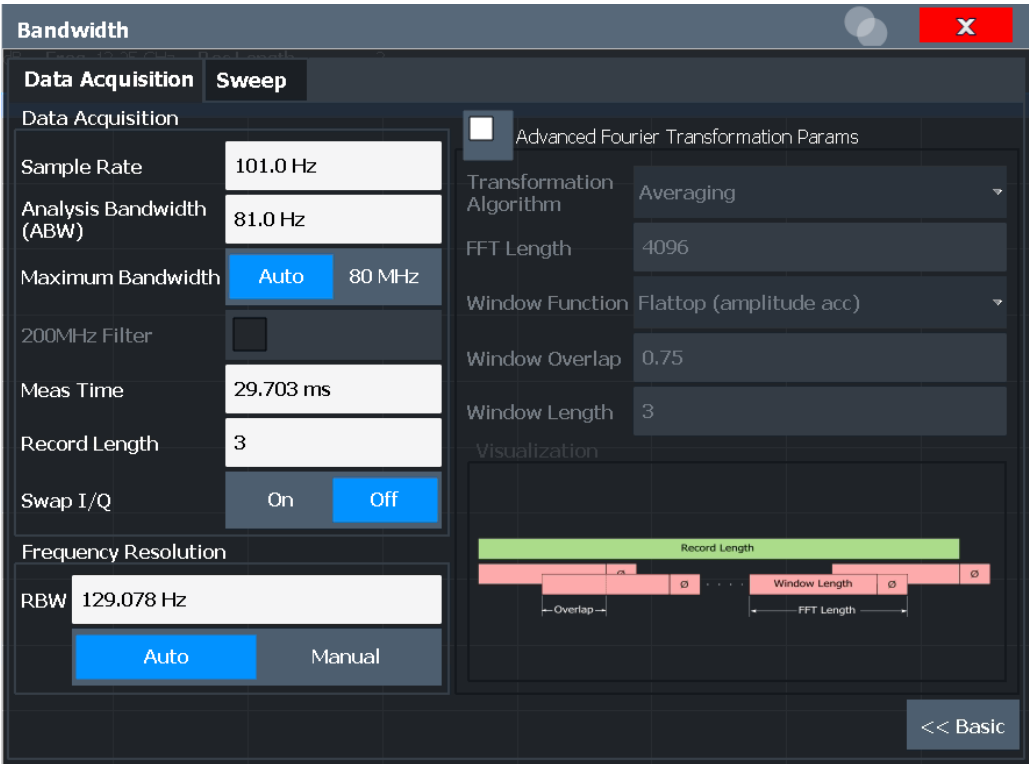


Figure 5-3: Data acquisition settings with advanced FFT parameters

The remote commands required to perform these tasks are described in [Section 10.4.5, "Configuring data acquisition"](#), on page 247.

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Sample Rate
 Defines the I/Q data sample rate of the R&S FSV/A. This value depends on the defined [Analysis Bandwidth](#).

Up to the [Maximum Bandwidth](#), the following rule applies:

sample rate = analysis bandwidth / 0.8

For details on the dependencies see [Section 4.1.1, "Sample rate and maximum usable I/Q bandwidth for RF input"](#), on page 20

Remote command:

[TRACe: IQ:SRATe](#) on page 252

Analysis Bandwidth

Defines the flat, usable bandwidth of the final I/Q data. This value depends on the defined [Sample Rate](#).

Up to the [Maximum Bandwidth](#), the following rule applies:

*analysis bandwidth = 0.8 * sample rate*

Note: For input from the optional **"Analog Baseband" interface**: If the frequency range defined by the analysis bandwidth and the center frequency exceeds the minimum frequency (0 Hz for low IF evaluation) or the maximum frequency (for I+jQ evaluation), an error is displayed. In this case, adjust the center frequency or the analysis bandwidth to exclude possible unwanted signal components.

For details on frequency ranges and the analysis bandwidth see [Section 4.3, "Processing data from the Analog Baseband interface"](#), on page 27.

Note: R&S FSV/A with B1000 and using an external frontend. To make use of the extended analysis bandwidth using B1000 (not B1001), the external frontend must downconvert all RF input signals to an IF frequency above 7.5 GHz. In this case, you must set the "Frequency Band Configuration" to "IF High".

For details, see the "External Frontend Control" section in the R&S FSV/A user manual.

Remote command:

[TRACe: IQ:BWIDth](#) on page 251

Maximum Bandwidth

Defines the maximum bandwidth to be used by the R&S FSV/A for I/Q data acquisition. Which options are available depends on which bandwidth extension options are installed.

This setting is only available if a bandwidth extension option larger than 40 MHz is installed on the R&S FSV/A. Otherwise, the maximum bandwidth is determined automatically.

"Auto"	(Default:) All installed bandwidth extension options are enabled. The currently available maximum bandwidth is allowed. (See Section 4.1.1, "Sample rate and maximum usable I/Q bandwidth for RF input" , on page 20).
"40 MHz"	Restricts the analysis bandwidth to a maximum of 40 MHz. Larger bandwidth extension options are disabled.
"200 MHz"	Restricts the analysis bandwidth to a maximum of 200 MHz. Larger bandwidth extension options are disabled.
"400 MHz"	Restricts the analysis bandwidth to a maximum of 400 MHz. Larger bandwidth extension options are disabled.
"600 MHz"	Restricts the analysis bandwidth to a maximum of 600 MHz. Larger bandwidth extension options are disabled.

Remote command:

[TRACe: IQ:WBAND\[:STATe\]](#) on page 253

[TRACe: IQ:WBAND:MBWidth](#) on page 254

Meas Time

Defines the I/Q acquisition time. By default, the measurement time is calculated as the number of I/Q samples ("Record Length") divided by the sample rate. If you change the measurement time, the [Record Length](#) is automatically changed, as well.

Remote command:

[\[SENSe:\] SWEep:TIME](#) on page 271

Record Length

Defines the number of I/Q samples to record. By default, the number of sweep points is used. The record length is calculated as the measurement time multiplied by the sample rate. If you change the record length, the [Meas Time](#) is automatically changed, as well.

Note: For the I/Q vector result display, the number of I/Q samples to record ("Record Length") must be identical to the number of trace points to be displayed ("Sweep Points"). Thus, the sweep points are not editable for this result display. If the "Record Length" is edited, the sweep points are adapted automatically.

Remote command:

[TRACe:IQ:RLENgth](#) on page 251

[TRACe:IQ:SET](#) on page 251

Swap I/Q

Activates or deactivates the inverted I/Q modulation. If the I and Q parts of the signal from the DUT are interchanged, the R&S FSV/A can do the same to compensate for it.

On	I and Q signals are interchanged Inverted sideband, $Q+j*I$
Off	I and Q signals are not interchanged Normal sideband, $I+j*Q$

Remote command:

[\[SENSe:\] SWAPiQ](#) on page 250

RBW

Defines the resolution bandwidth for Spectrum results. The available RBW values depend on the sample rate and record length.

(See [Section 4.4.4, "Frequency resolution of FFT results - RBW"](#), on page 37).

Depending on the selected RBW mode, the value is either determined automatically or can be defined manually. As soon as you enter a value in the input field, the RBW mode is changed to "Manual".

If the "Advanced Fourier Transformation Params" option is enabled, advanced FFT mode is selected and the RBW cannot be defined directly.

Note that the RBW is correlated with the [Sample Rate](#) and [Record Length](#) (and possibly the [Window Function](#) and [Window Length](#)). Changing any one of these parameters may cause a change to one or more of the other parameters. For more information see [Section 4.4, "Basics on FFT"](#), on page 32.

"Auto mode"	(Default) The RBW is determined automatically depending on the Sample Rate and Record Length .
"Manual mode"	The RBW can be defined by the user. The user-defined RBW is used and the Window Length (and possibly Sample Rate) are adapted accordingly.
"Advanced FFT mode"	This mode is used if the "Advanced Fourier Transformation Params" option is enabled. The RBW is determined by the advanced FFT parameters .

Remote command:

[\[SENSe:\] IQ:BWIDth:MODE](#) on page 248

[\[SENSe:\] IQ:BWIDth:RESolution](#) on page 248

Advanced FFT mode / Basic Settings

Shows or hides the "Advanced Fourier Transformation" parameters in the "Data Acquisition" dialog box.

Note that if the advanced FFT mode is used, the [RBW](#) settings are not available.

Transformation Algorithm ← Advanced FFT mode / Basic Settings

Defines the FFT calculation method.

"Single"	One FFT is calculated for the entire record length; if the FFT Length is larger than the record length, zeros are appended to the captured data.
"Averaging"	Several overlapping FFTs are calculated for each record; the results are combined to determine the final FFT result for the record. The number of FFTs to be averaged is determined by the Window Overlap and the Window Length .

Remote command:

[\[SENSe:\] IQ:FFT:ALGorithm](#) on page 249

FFT Length ← Advanced FFT mode / Basic Settings

Defines the number of frequency points determined by each FFT calculation. The more points are used, the higher the resolution in the spectrum becomes, but the longer the calculation takes.

Note: If you enter the value manually, any integer value from 3 to 524288 is available.

Remote command:

[\[SENSe:\] IQ:FFT:LENGth](#) on page 249

Window Function ← Advanced FFT mode / Basic Settings

In the I/Q analyzer you can select one of several FFT window types.

The following window types are available:

- Blackman-Harris
- Flattop
- Gauss
- Rectangular
- 5-Term

Remote command:
[\[SENSe:\] IQ:FFT:WINDow:TYPE](#) on page 250

Window Overlap ← **Advanced FFT mode / Basic Settings**
Defines the part of a single FFT window that is re-calculated by the next FFT calculation when using multiple FFT windows.

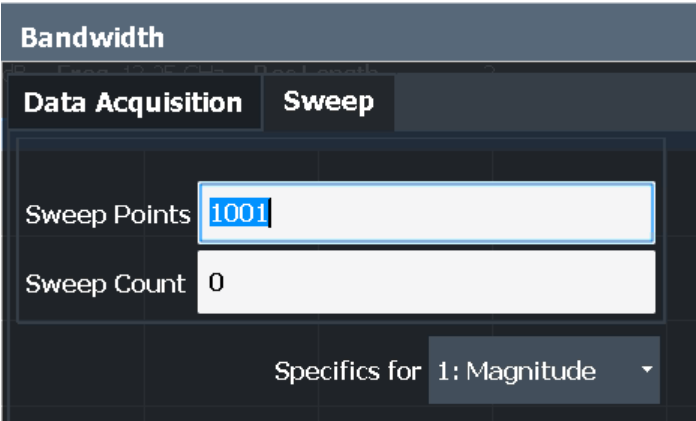
Remote command:
[\[SENSe:\] IQ:FFT:WINDow:OVERlap](#) on page 250

Window Length ← **Advanced FFT mode / Basic Settings**
Defines the number of samples to be included in a single FFT window in averaging mode. (In single mode, the window length corresponds to the ["Record Length"](#) on page 79.)
However, the window length may not be longer than the [FFT Length](#).

Remote command:
[\[SENSe:\] IQ:FFT:WINDow:LENGth](#) on page 249

5.7.2 Sweep settings

Access: "Overview" > "Bandwidth" > "Sweep" tab



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Sweep Points

In the I/Q Analyzer application, a specific frequency bandwidth is swept for a specified measurement time. During this time, a defined number of samples (= "Record Length") are captured. These samples are then evaluated by the applications. Therefore, in this case the number of sweep points does not define the amount of data to be acquired, but rather the number of trace points that are evaluated and displayed in the result diagrams.

Note: As opposed to previous versions of the I/Q Analyzer, the sweep settings are now window-specific.

For some result displays, the sweep points may not be editable as they are determined automatically, or restrictions may apply. For the I/Q vector result display, the number of I/Q samples to record ("Record Length") must be identical to the number of trace points to be displayed ("Sweep Points"). Thus, the sweep points are not editable for this result display. If the "Record Length" is edited, the sweep points are adapted automatically. For record lengths outside the valid range of sweep points, i.e. less than 101 points or more than 100001 points, the diagram does not show valid results.

Using fewer than 4096 sweep points with a detector other than "Auto Peak" may lead to wrong level results. For details see ["Combining results - trace detector"](#) on page 35.

Remote command:

[\[SENSe:\] SWEEp\[:WINDow<n>\]:POINTs](#) on page 270

Sweep/Average Count

Defines the number of sweeps to be performed in the single sweep mode. Values from 0 to 200000 are allowed. If the values 0 or 1 are set, one sweep is performed.

The sweep count is applied to all the traces in all diagrams.

If the trace modes "Average", "Max Hold" or "Min Hold" are set, this value also determines the number of averaging or maximum search procedures.

In continuous sweep mode, if "Sweep Count" = 0 (default), averaging is performed over 10 sweeps. For "Sweep Count" = 1, no averaging, maxhold or minhold operations are performed.

Remote command:

[\[SENSe:\] SWEEp:COUNT](#) on page 270

Continuous Sweep / Run Cont

After triggering, starts the sweep and repeats it continuously until stopped. This is the default setting.

While the measurement is running, "Continuous Sweep" and [RUN CONT] are highlighted. The running measurement can be aborted by selecting the highlighted softkey or key again. The results are not deleted until a new measurement is started.

Note: Sequencer. If the Sequencer is active, "Continuous Sweep" only controls the sweep mode for the currently selected channel. However, the sweep mode only takes effect the next time the Sequencer activates that channel, and only for a channel-defined sequence. In this case, a channel in continuous sweep mode is swept repeatedly.

If the Sequencer is active in MSRT mode, the "Continuous Sweep" function does not start data capturing. It merely affects trace averaging over multiple sequences. In this case, trace averaging is performed.

Furthermore, [RUN CONT] controls the Sequencer, not individual sweeps. [RUN CONT] starts the Sequencer in continuous mode.

For details on the Sequencer, see the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`INITiate<n>:CONTinuous` on page 268

Single Sweep / Run Single

After triggering, starts the number of sweeps set in "Sweep Count". The measurement stops after the defined number of sweeps has been performed.

While the measurement is running, "Single Sweep" and [RUN SINGLE] are highlighted. The running measurement can be aborted by selecting the highlighted softkey or key again.

Note: Sequencer. If the Sequencer is active, "Single Sweep" only controls the sweep mode for the currently selected channel. However, the sweep mode only takes effect the next time the Sequencer activates that channel, and only for a channel-defined sequence. In this case, the Sequencer sweeps a channel in single sweep mode only once.

Furthermore, [RUN SINGLE] controls the Sequencer, not individual sweeps. [RUN SINGLE] starts the Sequencer in single mode.

If the Sequencer is off, only the evaluation for the currently displayed channel is updated.

For details on the Sequencer, see the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`INITiate<n>[:IMMediate]` on page 268

Continue Single Sweep

After triggering, repeats the number of sweeps set in "Sweep Count", without deleting the trace of the last measurement.

While the measurement is running, "Continue Single Sweep" and [RUN SINGLE] are highlighted. The running measurement can be aborted by selecting the highlighted softkey or key again.

Remote command:

`INITiate<n>:CONMeas` on page 267

Select Frame

Selects a specific frame, loads the corresponding trace from the memory, and displays it in the Spectrum window.

Note that activating a marker or changing the position of the active marker automatically selects the frame that belongs to that marker.

This function is only available in single sweep mode or if the sweep is stopped, and only if a spectrogram is selected.

The most recent frame is number 0, all previous frames have a negative number.

Remote command:

`CALCulate<n>:SPECTrogram:FRAMe:SElect` on page 281

Continue Frame

Determines whether the results of the previous sweeps are included in the analysis of the next sweeps for trace modes "Max Hold", "Min Hold", and "Average".

This function is available in single sweep mode only.

- **On**
When the average or peak values are determined for the new sweep, the results of the previous sweeps in the spectrogram are also considered.
- **Off**
The average or peak values are determined from the results of the newly swept frames only.

Remote command:

`CALCulate<n>:SPECTrogram:CONTInuous` on page 280

Frame Count

Determines how many frames are plotted during a single sweep (as opposed to a continuous sweep). The maximum number of possible frames depends on the history depth (see "History Depth" on page 96).

Remote command:

`CALCulate<n>:SPECTrogram:FRAMe:COUnT` on page 281

Clear Spectrogram

Resets the spectrogram result display and clears the history buffer.

This function is only available if a spectrogram is selected.

Remote command:

`CALCulate<n>:SPECTrogram:CLear[:IMMediate]` on page 280

5.8 Display configuration



Access: "Overview" > "Display Config"

The captured signal can be displayed using various evaluation methods. All evaluation methods available for the current application are displayed in the evaluation bar in SmartGrid mode.

For a description of the available evaluation methods see [Section 3, "Measurement and result displays"](#), on page 14.



Up to 6 evaluations can be displayed in the I/Q Analyzer at any time, including several graphical diagrams, marker tables or peak lists.

The selected evaluation method not only affects the result display in a window, but also the results of the trace data query in remote control (see `TRACe<n>[:DATA]?` on page 337).

5.9 Adjusting settings automatically

Access: [AUTO SET]

The screenshot shows the 'Auto Set Config' dialog box. It has a title bar with a toggle switch and a red close button. The main content area is divided into several sections:

- Meas Time:** A section with a 'Mode' dropdown set to 'Auto' and a 'Value' field set to '1.0 ms'.
- Hysteresis:** A section with 'Upper Value' and 'Lower Value' fields, both set to '1.0 dB'.
- Auto All:** A section with a 'Show Search Protokoll' button.
- Search Mode:** A section with two buttons: 'Fast' (selected) and 'Pulse Optimized'.

Some settings can be adjusted by the R&S FSV/A automatically according to the current measurement settings. To do so, a measurement is performed. You can configure this measurement.



Adjusting settings automatically during triggered measurements

When you select an auto adjust function, a measurement is performed to determine the optimal settings. If you select an auto adjust function for a triggered measurement, you are asked how the R&S FSV/A should behave:

- (Default:) The measurement for adjustment waits for the next trigger
- The measurement for adjustment is performed without waiting for a trigger. The trigger source is temporarily set to "Free Run". After the measurement is completed, the original trigger source is restored. The trigger level is adjusted as follows for "IF Power" and "RF Power" triggers:
Trigger level = Reference level - 15 dB

Remote command:

[SENSe:]ADJust:CONFigure:TRIGger on page 256

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Setting the Reference Level Automatically (Auto Level).....	86
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Upper Level Hysteresis.....	87

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Adjusting all Determinable Settings Automatically (Auto All)

Activates all automatic adjustment functions for the current measurement settings, including:

- [Auto Frequency](#)
- [Auto Level](#)

Note: Auto measurement. For some measurements, the "Auto All" function determines the required measurement parameters automatically. In this case, the progress of the auto measurement is indicated in a message box.

See the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`[SENSe:]ADJust:ALL` on page 255

Adjusting the Center Frequency Automatically (Auto Frequency)

The R&S FSV/A adjusts the center frequency automatically.

The optimum center frequency is the frequency with the highest S/N ratio in the frequency span. As this function uses the signal counter, it is intended for use with sinusoidal signals.

At the same time, the optimal reference level is also set (see "[Setting the Reference Level Automatically \(Auto Level\)](#)" on page 59).

Remote command:

`[SENSe:]ADJust:FREQuency` on page 257

Setting the Reference Level Automatically (Auto Level)

Automatically determines a reference level which ensures that no overload occurs at the R&S FSV/A for the current input data. At the same time, the internal attenuators and the preamplifier (for analog baseband input: the full-scale level) are adjusted. As a result, the signal-to-noise ratio is optimized, while signal compression and clipping are minimized.

To determine the required reference level, a level measurement is performed on the R&S FSV/A.

If necessary, you can optimize the reference level further. Decrease the attenuation level manually to the lowest possible value before an overload occurs, then decrease the reference level in the same way.

You can change the measurement time for the level measurement if necessary (see "[\(Meas Time Manual\)](#)" on page 87).

Note: Using external frontends. The "Auto level" function is not available for R&S FE44S and R&S FE50DTR frontends using the "Frequency Band Configuration": "IF High".

Remote command:

`[SENSe:]ADJust:LEVel` on page 257

Resetting the Automatic Measurement Time (Meas Time Auto)

Resets the measurement duration for automatic settings to the default value.

Remote command:

[\[SENSe:\]ADJust:CONFigure:LEVel:DURation:MODE](#) on page 255

(Meas Time Manual)

This function allows you to change the measurement duration for automatic setting adjustments. Enter the value in seconds.

Note: The maximum measurement duration depends on the currently selected measurement and the installed (optional) hardware. Thus, the measurement duration actually used to determine the automatic settings can be shorter than the value you define here.

Remote command:

[\[SENSe:\]ADJust:CONFigure:LEVel:DURation:MODE](#) on page 255

[\[SENSe:\]ADJust:CONFigure:LEVel:DURation](#) on page 255

Upper Level Hysteresis

When the reference level is adjusted automatically using the [Auto Level](#) function, the internal attenuators and the preamplifier are also adjusted. To avoid frequent adaptation due to small changes in the input signal, you can define a hysteresis. This setting defines an upper threshold that the signal must exceed (compared to the last measurement) before the reference level is adapted automatically.

Remote command:

[\[SENSe:\]ADJust:CONFigure:HYSTeresis:UPPer](#) on page 256

Lower Level Hysteresis

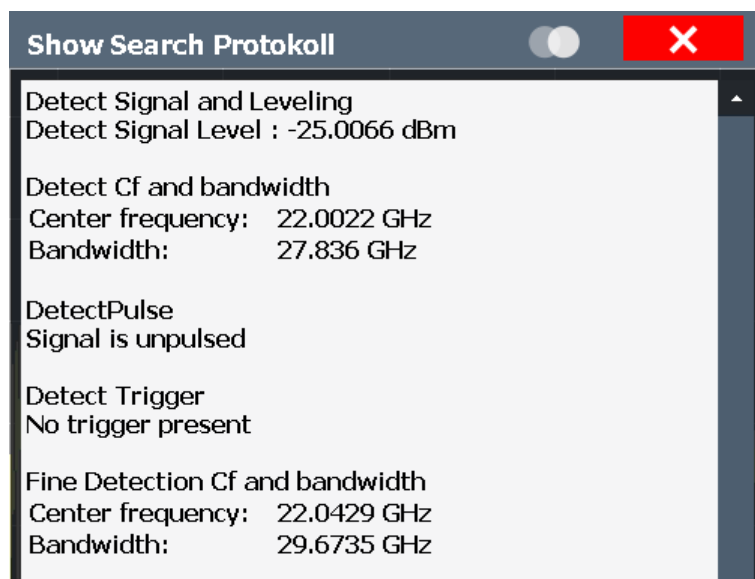
When the reference level is adjusted automatically using the [Auto Level](#) function, the internal attenuators and the preamplifier are also adjusted. To avoid frequent adaptation due to small changes in the input signal, you can define a hysteresis. This setting defines a lower threshold that the signal must fall below (compared to the last measurement) before the reference level is adapted automatically.

Remote command:

[\[SENSe:\]ADJust:CONFigure:HYSTeresis:LOWer](#) on page 256

Show Search Protocol

Displays the results of the automatic measurement performed to determine the optimal measurement configuration.



Search Mode

Determines the search mode for the automatic measurement performed to determine the optimal measurement configuration.

"Fast" The measurement is optimized for speed.

"Pulse optimized" The measurement is optimized to analyze pulse signals adequately.

Remote command:

[\[SENSe:\]ADJust:CONFigure:SMODE](#) on page 256

6 Analysis

Access: "Overview" > "Analysis"

General result analysis settings concerning the trace, markers etc. are identical to the analysis functions in the Spectrum application, except for the lines and special marker functions, which are not available for I/Q data.

The remote commands required to perform these tasks are described in [Section 6, "Analysis"](#), on page 89.

- [Trace settings](#).....89
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- [Trace / data export configuration](#).....99
- [Marker usage](#).....103

6.1 Trace settings

Access: "Overview" > "Analysis" > "Traces"

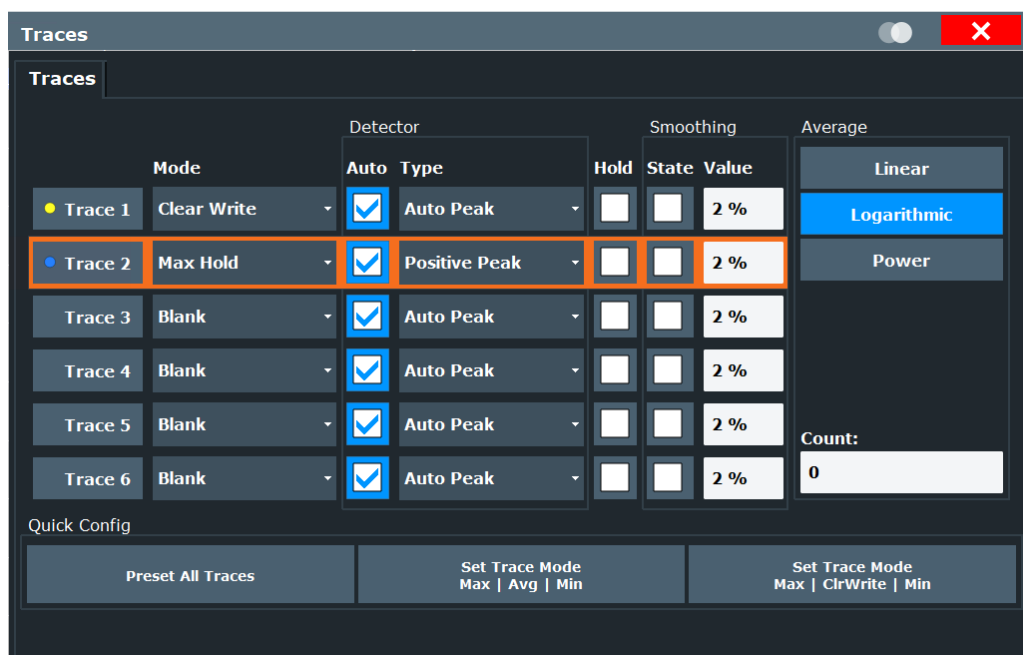
Or: [TRACE] > "Trace Config"

You can configure the settings for up to 6 individual traces in the same result display. Each trace is displayed in a different color, indicated in the window title bar and the trace settings.



Trace data can also be exported to an ASCII file for further analysis. For details see [Section 6.3, "Trace / data export configuration"](#), on page 99.

For I/Q Vector evaluation mode, only 1 trace is available and the detector is not editable.



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Trace 1/ Trace 2/ Trace 3/ Trace 4 (Softkeys).....	93
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Trace 1/Trace 2/Trace 3/Trace 4/Trace 5/Trace 6

Selects the corresponding trace for configuration. The currently selected trace is highlighted.

Remote command:

Selected via numeric suffix of: `TRACe<1...6>` commands

`DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>[:STATe]` on page 275

Trace Mode

Defines the update mode for subsequent traces.

- "Clear/ Write" Overwrite mode (default): the trace is overwritten by each sweep. All available detectors can be selected.
- "Max Hold" The maximum value is determined over several measurements and displayed. The R&S FSV/A saves the sweep result in the trace memory only if the new value is greater than the previous one. This mode is especially useful with modulated or pulsed signals. The signal spectrum is filled up upon each sweep until all signal components are detected in a kind of envelope. This mode is not available for statistics measurements.

"Min Hold"	The minimum value is determined from several measurements and displayed. The R&S FSV/A saves the sweep result in the trace memory only if the new value is lower than the previous one. This mode is useful for example for making an unmodulated carrier in a composite signal visible. Noise, interference signals or modulated signals are suppressed, whereas a CW signal is recognized by its constant level. This mode is not available for statistics measurements.
"Average"	The average is formed over several sweeps. The Sweep/Average Count determines the number of averaging procedures. This mode is not available for statistics measurements.
"View"	The current contents of the trace memory are frozen and displayed. Note: If a trace is frozen, you can change the measurement settings, apart from scaling settings, without impact on the displayed trace. The fact that the displayed trace no longer matches the current measurement settings is indicated by a yellow asterisk  on the tab label. If you change any parameters that affect the scaling of the diagram axes, the R&S FSV/A automatically adapts the trace data to the changed display range. Thus, you can zoom into the diagram after the measurement to show details of the trace.
"Blank"	Removes the selected trace from the display.

Remote command:

[DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:MODE](#) on page 273

Detector

Defines the trace detector to be used for trace analysis.

The trace detector is used to combine multiple FFT window results to create the final spectrum. (Note: in previous versions of the R&S FSV/A, the I/Q Analyzer always used the linear average detector.) If necessary, the trace detector is also used to reduce the number of calculated frequency points (defined by the FFT length) to the defined number of sweep points. By default, the Autopeak trace detector is used.

"Auto"	(default:) Selects the optimum detector for the selected trace and filter mode
"Type"	Defines the selected detector type. Note: If the EMI (R&S FSV3-K54) measurement option is installed, additional detectors are available, even if EMI measurement is not active. If you select a CISPR trace detector, the RBW filter type is automatically also set to CISPR. CISPR detectors are only available under the following conditions: • Time domain measurements and frequency measurements in sweep mode (not FFT mode, not power measurements, emission measurements, or statistics measurements) • Trigger mode "Free Run" or "External" (trigger offset ≥ 0 only for "External") • Gate mode: "Off"

Remote command:

[SENSe:] [WINDow<n>:] DETector<t>[:FUNCTION] on page 277

[SENSe:] [WINDow<n>:] DETector<t>[:FUNCTION]:AUTO on page 278

Hold

If activated, traces in "Min Hold", "Max Hold" and "Average" mode are not reset after specific parameter changes have been made.

Normally, the measurement is started again after parameter changes, before the measurement results are analyzed (e.g. using a marker). In all cases that require a new measurement after parameter changes, the trace is reset automatically to avoid false results (e.g. with span changes). For applications that require no reset after parameter changes, the automatic reset can be switched off.

The default setting is off.

Remote command:

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:MODE:HCONTinuous
on page 274

Smoothing

If enabled, the trace is smoothed by the specified value (between 1 % and 50 %).

The smoothing value is defined as a percentage of the display width. The larger the smoothing value, the greater the smoothing effect.

For more information, see the R&S FSV/A User Manual.

Remote command:

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing[:STATe]
on page 276

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing:APERture
on page 276

Average Mode

Defines the mode with which the trace is averaged over several sweeps.

This setting is generally applicable if trace mode "Average" is selected.

For FFT sweeps, the setting also affects the VBW (regardless of whether the trace is averaged).

(See the section on ACLR power measurements in the R&S FSV/A User Manual.)

How many sweeps are averaged is defined by the "Sweep/Average Count"
on page 82.

"Linear"	The power level values are converted into a linear unit (Volt) before averaging. After the averaging, the data is converted back into its original unit.
"Logarithmic"	For logarithmic scaling, the values are averaged in dBm. For linear scaling, the behavior is the same as with linear averaging.

"Power" Activates linear power averaging.
The power level values are converted into unit Watt before averaging.
After the averaging, the data is converted back into its original unit.
Use this mode to average power values in Volts or Amperes correctly.
In particular, for small VBW values (smaller than the RBW), use power averaging mode for correct power measurements in FFT sweep mode.

Remote command:

[SENSe:] AVERage<n>:TYPE on page 277

Predefined Trace Settings - Quick Config

Commonly required trace settings have been predefined and can be applied very quickly by selecting the appropriate button.

Function	Trace Settings	
Preset All Traces	Trace 1:	Clear Write
	Traces 2-6:	Blank
Set Trace Mode Max Avg Min	Trace 1:	Max Hold
	Trace 2:	Average
	Trace 3:	Min Hold
	Traces 4-6:	Blank
Set Trace Mode Max ClrWrite Min	Trace 1:	Max Hold
	Trace 2:	Clear Write
	Trace 3:	Min Hold
	Traces 4-6:	Blank

Trace 1/ Trace 2/ Trace 3/ Trace 4 (Softkeys)

Displays the "Traces" settings and focuses the "Mode" list for the selected trace.

Remote command:

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>[:STATe] on page 275

Copy Trace

Access: "Overview" > "Analysis" > "Traces" > "Copy Trace"

Copies trace data to another trace.

The first group of buttons (labeled "Trace 1" to "Trace 6") selects the source trace. The second group of buttons (labeled "Copy to Trace 1" to "Copy to Trace 6") selects the destination.

Remote command:

TRACe<n>:COPY on page 278

Trace Labels

Access: "Overview" > "Analysis" > "Traces" > "Trace Label" tab

You can define a descriptive label to active traces instead of the default "Trace <x>" label.

Note: This function is not available in all applications.

The labels are displayed in the diagram area. The font color corresponds to the color of the particular trace (for example, yellow trace: yellow font).

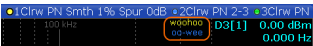


Figure 6-1: Example: the yellow and blue traces have a label

You can move the trace label to any position on the display by dragging it to the new position.

You can only configure labels for active traces and for traces whose "State" is enabled.

Traces	Export / Import	Copy Trace	Trace Math	Trace Label
	State Text			
Trace 1	☒			Trace 1
Trace 2	☒			Trace 2
Trace 3	☐			Trace 3
Trace 4	☐			Trace 4
Trace 5	☐			Trace 5
Trace 6	☐			Trace 6

Enable the state and enter a trace label.

Remote command:

`DISPlay[:WINDow<n>]:TRACe<t>:LABel[:STATe]` on page 272

`DISPlay[:WINDow<n>]:TRACe<t>:LABel:TEXT` on page 273

6.2 Spectrogram settings

Access: [TRACE] > "Spectrogram Config"

The individual settings available for spectrogram display are described here. For settings on color mapping, see [Section 6.2.2, "Color map settings"](#), on page 98.

Settings concerning the frames and how they are handled during a sweep are provided as additional sweep settings for spectrogram display.

See [Section 5.7.2, "Sweep settings"](#), on page 81.

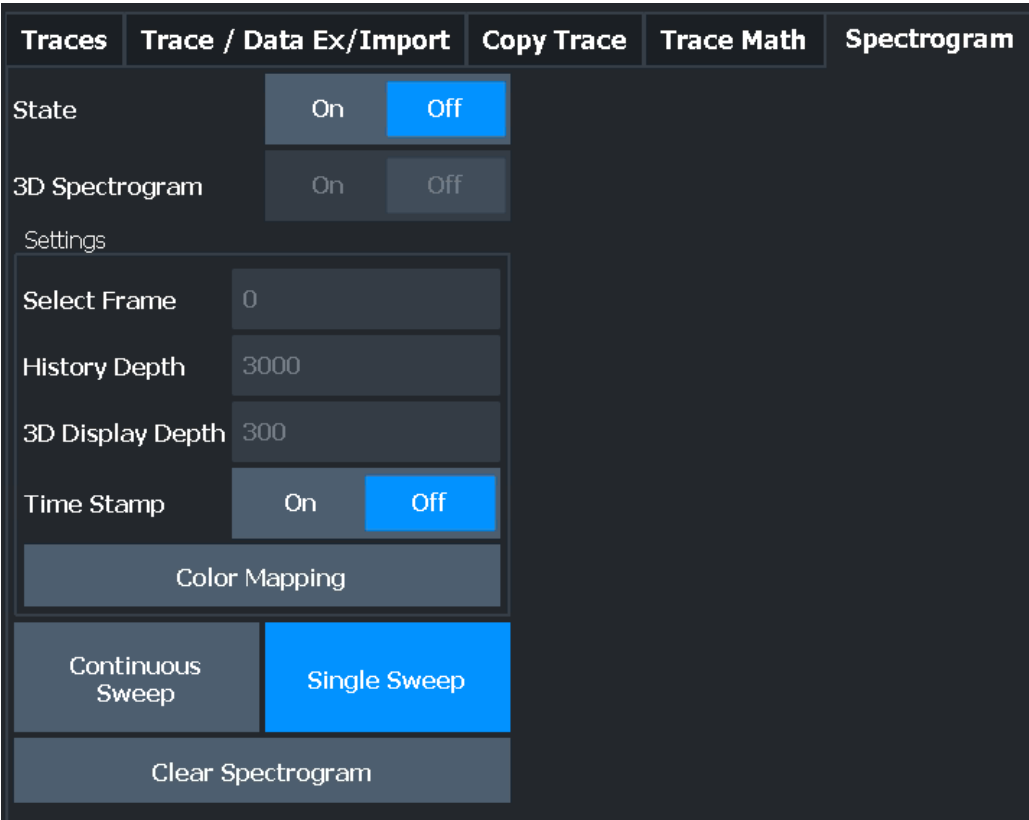
Search functions for spectrogram markers are described in [Section 6.4.2.2, "Marker search settings for spectrograms"](#), on page 111.

- [General spectrogram settings](#).....95
- [Color map settings](#).....98

6.2.1 General spectrogram settings

Access: [TRACE] > "Spectrogram Config"

This section describes general settings for spectrogram display.



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3D Spectrogram State.....	96
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Continuous Sweep / Run Cont.....	97
Single Sweep / Run Single.....	97
Clear Spectrogram.....	98

State

Activates and deactivates a Spectrogram subwindow.

"Off" Closes the Spectrogram subwindow.

Remote command:

[CALCulate<n>:SPECTrogram:LAYout](#) on page 282

3D Spectrogram State

Activates and deactivates a 3-dimensional spectrogram. As opposed to the common 2-dimensional spectrogram, the power is not only indicated by a color mapping, but also in a third dimension, the z-axis.

For details see the R&S FSV/A User Manual.

Remote command:

[CALCulate<n>:SPECTrogram:THReedim\[:STATe\]](#) on page 283

Select Frame

Selects a specific frame, loads the corresponding trace from the memory, and displays it in the Spectrum window.

Note that activating a marker or changing the position of the active marker automatically selects the frame that belongs to that marker.

This function is only available in single sweep mode or if the sweep is stopped, and only if a spectrogram is selected.

The most recent frame is number 0, all previous frames have a negative number.

Remote command:

[CALCulate<n>:SPECTrogram:FRAMe:SElect](#) on page 281

History Depth

Sets the number of frames that the R&S FSV/A stores in its memory.

The maximum number of frames depends on the ["Sweep Points"](#) on page 82.

For an overview of the maximum number of frames depending on the number of sweep points, see the R&S FSV/A User Manual.

If the memory is full, the R&S FSV/A deletes the oldest frames stored in the memory and replaces them with the new data.

Remote command:

[CALCulate<n>:SPECTrogram:HDEPth](#) on page 282

3-D Display Depth

Defines the number of frames displayed in a 3-dimensional spectrogram.

For details see the R&S FSV/A User Manual.

Time Stamp

Activates and deactivates the timestamp. The timestamp shows the system time while the measurement is running. In single sweep mode or if the sweep is stopped, the timestamp shows the time and date of the end of the sweep.

When active, the timestamp replaces the display of the frame number.

Remote command:

[CALCulate<n>:SPECTrogram:TSTamp\[:STATe\]](#) on page 284

[CALCulate<n>:SPECTrogram:TSTamp:DATA?](#) on page 283

Color Mapping

Opens the "Color Mapping" dialog.

For details see the R&S FSV/A User Manual.

Continuous Sweep / Run Cont

After triggering, starts the sweep and repeats it continuously until stopped. This is the default setting.

While the measurement is running, "Continuous Sweep" and [RUN CONT] are highlighted. The running measurement can be aborted by selecting the highlighted softkey or key again. The results are not deleted until a new measurement is started.

Note: Sequencer. If the Sequencer is active, "Continuous Sweep" only controls the sweep mode for the currently selected channel. However, the sweep mode only takes effect the next time the Sequencer activates that channel, and only for a channel-defined sequence. In this case, a channel in continuous sweep mode is swept repeatedly.

If the Sequencer is active in MSRT mode, the "Continuous Sweep" function does not start data capturing. It merely affects trace averaging over multiple sequences. In this case, trace averaging is performed.

Furthermore, [RUN CONT] controls the Sequencer, not individual sweeps. [RUN CONT] starts the Sequencer in continuous mode.

For details on the Sequencer, see the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`INITiate<n>:CONTinuous` on page 268

Single Sweep / Run Single

After triggering, starts the number of sweeps set in "Sweep Count". The measurement stops after the defined number of sweeps has been performed.

While the measurement is running, "Single Sweep" and [RUN SINGLE] are highlighted. The running measurement can be aborted by selecting the highlighted softkey or key again.

Note: Sequencer. If the Sequencer is active, "Single Sweep" only controls the sweep mode for the currently selected channel. However, the sweep mode only takes effect the next time the Sequencer activates that channel, and only for a channel-defined sequence. In this case, the Sequencer sweeps a channel in single sweep mode only once.

Furthermore, [RUN SINGLE] controls the Sequencer, not individual sweeps. [RUN SINGLE] starts the Sequencer in single mode.

If the Sequencer is off, only the evaluation for the currently displayed channel is updated.

For details on the Sequencer, see the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`INITiate<n>[:IMMediate]` on page 268

Clear Spectrogram

Resets the spectrogram result display and clears the history buffer.

This function is only available if a spectrogram is selected.

Remote command:

CALCulate<n>:SPECTrogram:CLEar[:IMMediate] on page 280

6.2.2 Color map settings

Access: "Overview" > "Analysis" > "Traces" > "Spectrogram" > "Color Mapping"

or: [TRACE] > "Spectrogram Config" > "Color Mapping"

In addition to the available color settings, the dialog box displays the current color map and provides a preview of the display with the current settings.

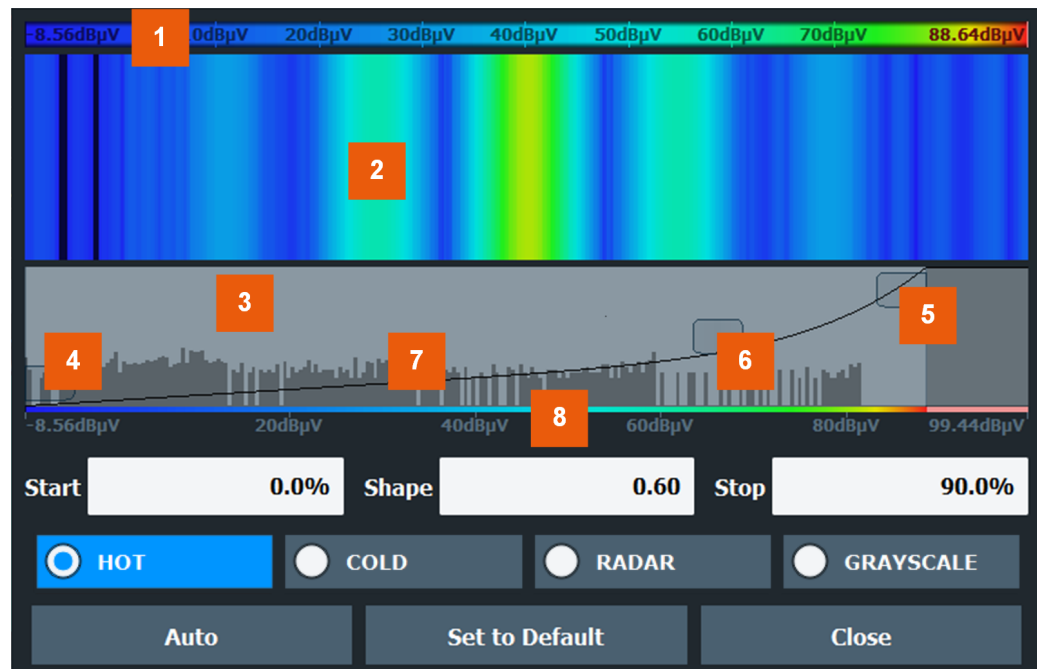


Figure 6-2: Color Mapping dialog box

- 1 = Color map: shows the current color distribution
- 2 = Preview pane: shows a preview of the spectrogram with any changes that you make to the color scheme
- 3 = Color curve pane: graphical representation of all settings available to customize the color scheme
- 4/5 = Color range start and stop sliders: define the range of the color map or amplitudes for the spectrogram
- 6 = Color curve slider: adjusts the focus of the color curve
- 7 = Histogram: shows the distribution of measured values
- 8 = Scale of the horizontal axis (value range)

Start / Stop.....	99
Shape.....	99
Hot/Cold/Radar/Grayscale.....	99

Auto.....	99
Set to Default.....	99
Close.....	99

Start / Stop

Defines the lower and upper boundaries of the value range of the spectrogram.

Remote command:

`DISPlay[:WINDow<n>]:SPECTrogram:COLor:LOWer` on page 285

`DISPlay[:WINDow<n>]:SPECTrogram:COLor:UPPer` on page 286

Shape

Defines the shape and focus of the color curve for the spectrogram result display.

"-1 to <0" More colors are distributed among the lower values

"0" Colors are distributed linearly among the values

">0 to 1" More colors are distributed among the higher values

Remote command:

`DISPlay[:WINDow<n>]:SPECTrogram:COLor:SHApe` on page 285

Hot/Cold/Radar/Grayscale

Sets the color scheme for the spectrogram.

Remote command:

`DISPlay[:WINDow<n>]:SPECTrogram:COLor[:STYLE]` on page 286

Auto

Defines the color range automatically according to the existing measured values for optimized display.

Set to Default

Sets the color mapping to the default settings.

Remote command:

`DISPlay[:WINDow<n>]:SPECTrogram:COLor:DEFault` on page 285

Close

Saves the changes and closes the dialog box.

6.3 Trace / data export configuration



Access: "Save" > "Export" > "Export Configuration"

Or: [TRACE] > "Trace Config" > "Trace / Data Export"

The R&S FSV/A provides various evaluation methods for the results of the performed measurements. However, if you want to evaluate the data with other, external applications, you can export the measurement data to an ASCII file.



The standard data management functions (e.g. saving or loading instrument settings) that are available for all R&S FSV/A applications are not described here.

See the R&S FSV3000/ FSVA3000 base unit user manual for a description of the standard functions.

Export all Traces and all Table Results.....	100
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Export Trace to ASCII File.....	101
L File Type.....	102
L Decimal Separator.....	102
L Column Separator.....	102
L File Explorer.....	102
Export Spectrogram to ASCII File.....	102

Export all Traces and all Table Results

Selects all displayed traces and result tables (e.g. "Result Summary", marker table etc.) in the current application for export to an ASCII file.

Alternatively, you can select one specific trace only for export (see [Trace to Export](#)).

The results are output in the same order as they are displayed on the screen: window by window, trace by trace, and table row by table row.

Remote command:

[FORMat:DEXPort:TRACes](#) on page 341

Include Instrument & Measurement Settings

Includes additional instrument and measurement settings in the header of the export file for result data.

Remote command:

[FORMat:DEXPort:HEADer](#) on page 341

Trace to Export

Defines an individual trace to be exported to a file.

This setting is not available if [Export all Traces](#) and [all Table Results](#) is selected.

Decimal Separator

Defines the decimal separator for floating-point numerals for the data export/import files. Evaluation programs require different separators in different languages.

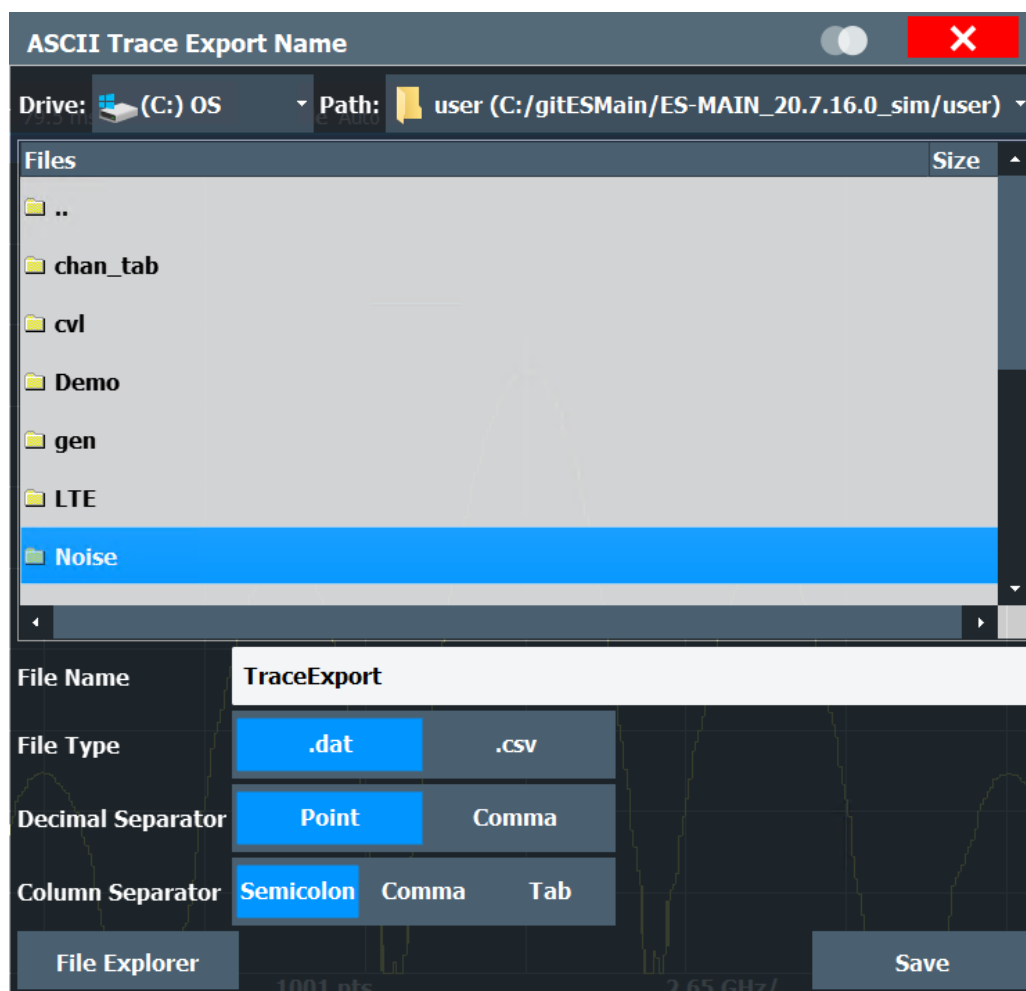
Remote command:

[FORMat:DEXPort:DSEPARATOR](#) on page 340

Export Trace to ASCII File

Saves the selected trace or all traces in the currently active result display to the specified file and directory in the selected ASCII format.

"File Explorer": Instead of using the file manager of the R&S FSV/A firmware, you can also use the Microsoft Windows File Explorer to manage files.



Note: Secure user mode.

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

`MMEMory:STORe<n>:TRACe` on page 342

File Type ← Export Trace to ASCII File

Determines the format of the ASCII file to be imported or exported.

Depending on the external program in which the data file was created or is evaluated, a comma-separated list (CSV) or a plain data format (DAT) file is required.

Remote command:

`FORMat:DEXPort:FORMat` on page 341

Decimal Separator ← Export Trace to ASCII File

Defines the decimal separator for floating-point numerals for the data export/import files. Evaluation programs require different separators in different languages.

Remote command:

`FORMat:DEXPort:DSEParator` on page 340

Column Separator ← Export Trace to ASCII File

Selects the character that separates columns in the exported ASCII file. The character can be either a semicolon, a comma or a tabulator (tab).

Example for semicolon:

```
Type;FSV3007;Version;1.00;Date;01.Jan 3000;
```

Example for comma:

```
Type,FSV3007,
Version,1.00,
Date,01.Jan 3000,
```

Example for tabulator (tab after the last column is not visible):

```
Type      FSV3007
Version    1.00
Date      01.Jan 3000
```

The selected column separator setting remains the same, even after a preset.

Remote command:

`FORMat:DEXPort:CSEParator` on page 340

File Explorer ← Export Trace to ASCII File

Opens the Microsoft Windows File Explorer.

Remote command:

not supported

Export Spectrogram to ASCII File

Opens a file selection dialog box and saves the selected spectrogram in ASCII format (.dat) to the specified file and directory.

If the spectrogram display is selected when you perform this function, the entire histogram buffer with all frames is exported to a file. The data corresponding to a particular frame begins with information about the frame number and the time that frame was recorded. For large history buffers the export operation can take some time.

For details on the file format, see the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

[MMEMory:STORe<n>:SPECTrogram](#) on page 341

6.4 Marker usage

Access: "Overview" > "Analysis"

The following marker settings and functions are available in the I/Q Analyzer application.



For "I/Q-Vector" displays markers are not available.



In the I/Q Analyzer application, the resolution with which the frequency can be measured with a marker is always the filter bandwidth, which is derived from the defined sample rate.

(See [Section 4.1.1, "Sample rate and maximum usable I/Q bandwidth for RF input"](#), on page 20).



Marker settings are now window-specific.

- [Marker settings](#).....103
- [Marker search settings and positioning functions](#).....108
- [Marker search settings for spectrograms](#).....116
- [Marker functions](#).....119

6.4.1 Marker settings

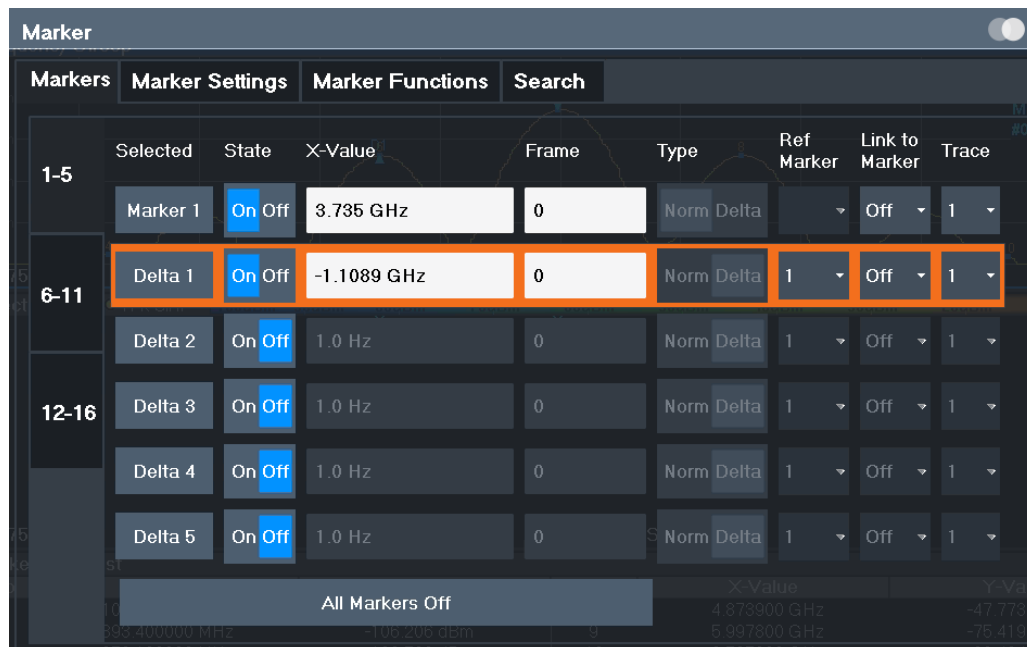
Or: [MKR] > "Marker Config"

The remote commands required to define these settings are described in [Section 10.7.3.1, "Setting up individual markers"](#), on page 287.

- [Individual marker setup](#).....104
- [General marker settings](#).....107

6.4.1.1 Individual marker setup

Up to 17 markers or delta markers can be activated for each window simultaneously. Initial marker setup is performed using the "Marker" dialog box.



The markers are distributed among 3 tabs for a better overview. By default, the first marker is defined as a normal marker, whereas all others are defined as delta markers with reference to the first marker. All markers are assigned to trace 1, but only the first marker is active.

Selected Marker.....	104
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Marker Position X-value.....	105
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Reference Marker.....	105
Linking to Another Marker.....	105
All Markers Off.....	106
Assigning the Marker to a Trace.....	106
Select Marker.....	106

Selected Marker

Marker name. The marker which is currently selected for editing is highlighted orange.

Remote command:

Marker selected via suffix <m> in remote commands.

Marker State

Activates or deactivates the marker in the diagram.

Remote command:

`CALCulate<n>:MARKer<m>[:STATe]` on page 292

`CALCulate<n>:DELTamarker<m>[:STATe]` on page 290

Marker Position X-value

Defines the position (x-value) of the marker in the diagram. For normal markers, the absolute position is indicated. For delta markers, the position relative to the reference marker is provided.

Remote command:

`CALCulate<n>:MARKer<m>:X` on page 293

`CALCulate<n>:DELTamarker<m>:X` on page 291

Frame (Spectrogram only)

Spectrogram frame the marker is assigned to.

Remote command:

`CALCulate<n>:MARKer<m>:SPECTrogram:FRAMe` on page 300

`CALCulate<n>:DELTamarker<m>:SPECTrogram:FRAMe` on page 305

Marker Type

Toggles the marker type.

The type for marker 1 is always "Normal", the type for delta marker 1 is always "Delta". These types cannot be changed.

Note: If normal marker 1 is the active marker, switching the "Mkr Type" activates an additional delta marker 1. For any other marker, switching the marker type does not activate an additional marker, it only switches the type of the selected marker.

"Normal" A normal marker indicates the absolute value at the defined position in the diagram.

"Delta" A delta marker defines the value of the marker relative to the specified reference marker (marker 1 by default).

Remote command:

`CALCulate<n>:MARKer<m>[:STATe]` on page 292

`CALCulate<n>:DELTamarker<m>[:STATe]` on page 290

Reference Marker

Defines a marker as the reference marker which is used to determine relative analysis results (delta marker values).

Remote command:

`CALCulate<n>:DELTamarker<m>:MREFerence` on page 290

Linking to Another Marker

Links the current marker to the marker selected from the list of active markers. If the x-axis value of the initial marker is changed, the linked marker follows to the same position on the x-axis. Linking is off by default.

Using this function you can set two markers on different traces to measure the difference (e.g. between a max hold trace and a min hold trace or between a measurement and a reference trace).

Remote command:

[CALCulate<n>:MARKer<ms>:LINK:TO:MARKer<md>](#) on page 292

[CALCulate<n>:DELTamarker<ms>:LINK:TO:MARKer<md>](#) on page 289

[CALCulate<n>:DELTamarker<m>:LINK](#) on page 288

All Markers Off

Deactivates all markers in one step.

Remote command:

[CALCulate<n>:MARKer<m>:AOFF](#) on page 291

Assigning the Marker to a Trace

The "Trace" setting assigns the selected marker to an active trace. The trace determines which value the marker shows at the marker position. If the marker was previously assigned to a different trace, the marker remains on the previous frequency or time, but indicates the value of the new trace.

If a trace is turned off, the assigned markers and marker functions are also deactivated.

Remote command:

[CALCulate<n>:MARKer<m>:TRACe](#) on page 293

Select Marker

The "Select Marker" function opens a dialog box to select and activate or deactivate one or more markers quickly.



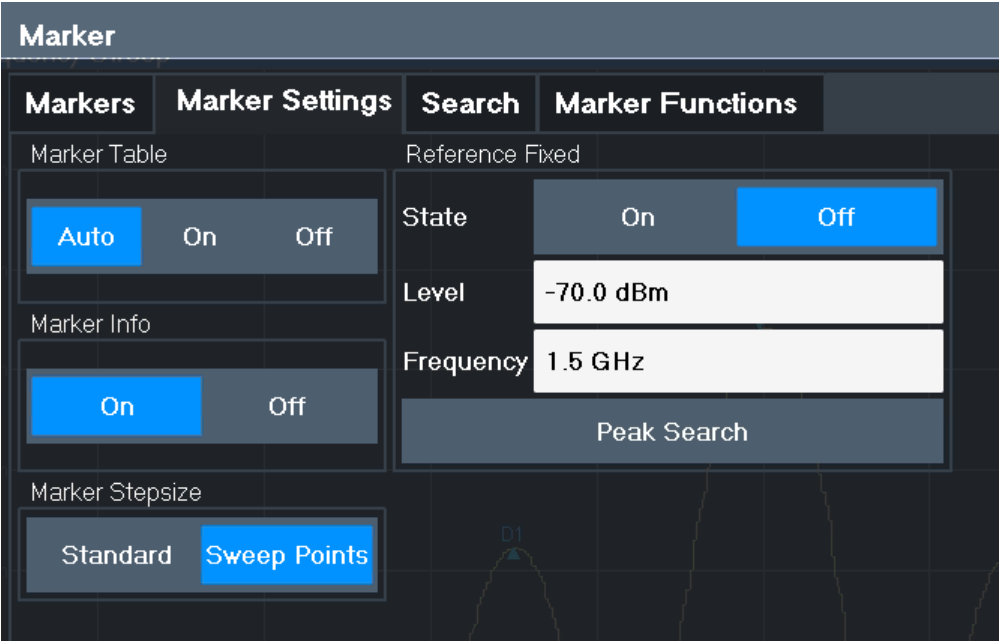
Remote command:

[CALCulate<n>:MARKer<m>\[:STATe\]](#) on page 292

[CALCulate<n>:DELTamarker<m>\[:STATe\]](#) on page 290

6.4.1.2 General marker settings

Some general marker settings allow you to influence the marker behavior for all markers.



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Marker Stepsize.....	108

Marker Table Display

Defines how the marker information is displayed.

- "On" Displays the marker information in a table in a separate area beneath the diagram.
- "Off" No separate marker table is displayed.
If [Marker Info](#) is active, the marker information is displayed within the diagram area.
- "Auto" (Default) If more than two markers are active, the marker table is displayed automatically.
If [Marker Info](#) is active, the marker information for up to two markers is displayed in the diagram area.

Remote command:
`DISPlay[:WINDow<n>]:MTABLE` on page 294

Marker Info

Turns the marker information displayed in the diagram on and off.

1AP Clrw	
M1[1]	81.13 dBμV 177.610 MHz
D2[1]	-22.18 dB -28.980 MHz

Remote command:

`DISPlay[:WINDow<n>]:MINFo[:STATe]` on page 294

Marker Stepsize

Defines the size of the steps that the marker position is moved using the rotary knob.

"Standard"	The marker position is moved in steps of (Span/1000), which corresponds approximately to the number of pixels for the default display of 1001 sweep points. This setting is most suitable to move the marker over a larger distance.
"Sweep Points"	The marker position is moved from one sweep point to the next. This setting is required for a very precise positioning if more sweep points are collected than the number of pixels that can be displayed on the screen. It is the default mode.

Remote command:

`CALCulate<n>:MARKer<m>:X:SSize` on page 295

6.4.2 Marker search settings and positioning functions

Access: "Overview" > "Analysis" > "Marker" > "Search"

or: [MKR TO]

Several functions are available to set the marker to a specific position very quickly and easily, or to use the current marker position to define another characteristic value. In order to determine the required marker position, searches may be performed. The search results can be influenced by special settings.

For more information on searching for signal peaks see [Section 6.4.4.3, "Marker peak list"](#), on page 126.



In I/Q Analyzer mode, the search settings for "Real/Imag (I/Q)" evaluation include an additional parameter, see ["Branch for Peaksearch"](#) on page 111.

The remote commands required to define these settings are described in [Section 10.7.3.5, "Positioning the marker"](#), on page 308.

- [Marker search settings](#)..... 109
- [Marker search settings for spectrograms](#)..... 111
- [Positioning functions](#)..... 114

6.4.2.1 Marker search settings

Access: [MKR TO] > "Search Config"

Markers are commonly used to determine peak values, i.e. maximum or minimum values, in the measured signal. Configuration settings allow you to influence the peak search results.



For Spectrograms, special marker settings are available, see [Section 6.4.2.2, "Marker search settings for spectrograms"](#), on page 111.

Marker	
Markers	Marker Settings
Search	
Peak Search	
Next Peak Mode	Left Absolute Right
Exclude LO	On Off
Peak Excursion	6.0 dB
Auto Max Peak	On Off
Auto Min Peak	On Off
Search Limits	
Left Limit	☐ 40.0 GHz
Right Limit	☐ 40.0 GHz
Threshold	☐ -120.0 dBm
Use Zoom Limits	On Off
Search Limits Off	

Search Mode for Next Peak.....	109
Exclude LO.....	110
Peak Excursion.....	110
Auto Max Peak Search / Auto Min Peak Search.....	110
Search Limits.....	110
L Search Limits (Left / Right).....	110
L Search Threshold.....	110
L Use Zoom Limits.....	111
L Search Limits Off.....	111
Branch for Peaksearch.....	111

Search Mode for Next Peak

Selects the search mode for the next peak search.

"Left"	Determines the next maximum/minimum to the left of the current peak.
"Absolute"	Determines the next maximum/minimum to either side of the current peak.
"Right"	Determines the next maximum/minimum to the right of the current peak.

Remote command:

[Section 10.7.3.5, "Positioning the marker"](#), on page 308

Exclude LO

If activated, restricts the frequency range for the marker search functions.

"On"	The minimum frequency included in the peak search range is $\geq 5 \times$ resolution bandwidth (RBW). Due to the interference by the first local oscillator to the first intermediate frequency at the input mixer, the LO is represented as a signal at 0 Hz. To avoid the peak marker jumping to the LO signal at 0 Hz, this frequency is excluded from the peak search.
"Off"	No restriction to the search range. The frequency 0 Hz is included in the marker search functions.

Remote command:

`CALCulate<n>:MARKer<m>:LOEXclude` on page 295

Peak Excursion

Defines the minimum level value by which a signal must rise or fall so that it is identified as a maximum or a minimum by the search functions.

Entries from 0 dB to 80 dB are allowed; the resolution is 0.1 dB. The default setting for the peak excursion is 6 dB.

For Analog Modulation Analysis, the unit and value range depend on the selected result display type.

Remote command:

`CALCulate<n>:MARKer<m>:PEXCursion` on page 296

Auto Max Peak Search / Auto Min Peak Search

If activated, a maximum or minimum peak search is performed automatically for marker 1 after each sweep.

Remote command:

`CALCulate<n>:MARKer<m>:MAXimum:AUTO` on page 309

`CALCulate<n>:MARKer<m>:MINimum:AUTO` on page 310

Search Limits

The search results can be restricted by limiting the search area or adding search conditions.

Search Limits (Left / Right) ← Search Limits

If activated, limit lines are defined and displayed for the search. Only results within the limited search range are considered.

Remote command:

`CALCulate<n>:MARKer<m>:X:SLIMits[:STATe]` on page 297

`CALCulate<n>:MARKer<m>:X:SLIMits:LEFT` on page 297

`CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT` on page 298

Search Threshold ← Search Limits

Defines an absolute threshold as an additional condition for the peak search. If enabled, only peaks that exceed the threshold are detected.

Remote command:

[CALCulate<n>:THReshold:STATe](#) on page 299

[CALCulate<n>:THReshold](#) on page 299

Use Zoom Limits ← Search Limits

If activated, the peak search is restricted to the active zoom area defined for a single zoom.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMits:ZOOM\[:STATe\]](#) on page 298

Search Limits Off ← Search Limits

Deactivates the search range limits.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMits\[:STATe\]](#) on page 297

[CALCulate<n>:THReshold:STATe](#) on page 299

Branch for Peaksearch

Defines which data is used for marker search functions in I/Q data.

This function is only available for the display configuration "Real/Imag (I/Q)" (see "[Real/Imag \(I/Q\)](#)" on page 16).

Note: The search settings apply to all markers, not only the currently selected one.

"Real"

Marker search functions are performed on the real trace of the I/Q measurement.

"Imag"

Marker search functions are performed on the imaginary trace of the I/Q measurement.

"Magnitude"

Marker search functions are performed on the magnitude of the I and Q data.

Remote command:

[CALCulate<n>:MARKer<m>:SEARch](#) on page 296

6.4.2.2 Marker search settings for spectrograms

Access: "Overview" > "Analysis" > "Markers" > "Search"

or: [MKR TO] > "Search Config"

Spectrograms show not only the current sweep results, but also the sweep history. Thus, when searching for peaks, you must define the search settings within a single time frame (x-direction) and within several time frames (y-direction).

These settings are only available for spectrogram displays.

Marker				
Markers	Marker Settings		Marker Functions	Search
Next X Search	Left	Absolute	Right	Search Limits Left Limit 0.0 Hz Right Limit 7.5 GHz Threshold -120.0 dBm Use Zoom Limits On Off Search Limits Off
Next Y Search	Up	Absolute	Down	
Search Type	X Search	Y Search	XY Search	
Search Area	Visible		Memory	
Exclude LO	On		Off	
Peak Excursion	6.0 dB			
Auto Max Peak	On	Off		
Auto Min Peak	On	Off		

Search Mode for Next Peak in X-Direction.....	112
Search Mode for Next Peak in Y-Direction.....	112
Marker Search Type.....	113
Marker Search Area.....	113
Peak Excursion.....	113
Search Limits.....	114
L Search Limits (Left / Right).....	114
L Search Threshold.....	114
L Use Zoom Limits.....	114
L Search Limits Off.....	114

Search Mode for Next Peak in X-Direction

Selects the search mode for the next peak search within the currently selected frame.

"Left"	Determines the next maximum/minimum to the left of the current peak.
"Absolute"	Determines the next maximum/minimum to either side of the current peak.
"Right"	Determines the next maximum/minimum to the right of the current peak.

Remote command:

[Section 10.7.3.5, "Positioning the marker"](#), on page 308

Search Mode for Next Peak in Y-Direction

Selects the search mode for the next peak search within all frames at the current marker position.

"Up"	Determines the next maximum/minimum above the current peak (in more recent frames).
------	---

"Absolute"	Determines the next maximum/minimum above or below the current peak (in all frames).
"Down"	Determines the next maximum/minimum below the current peak (in older frames).

Remote command:

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:ABOVE](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:ABOVE](#)
on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:BELOW](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:BELOW](#)
on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:NEXT](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:NEXT](#) on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:ABOVE](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:ABOVE](#)
on page 307

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:BELOW](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:BELOW](#)
on page 307

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:NEXT](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:NEXT](#) on page 308

Marker Search Type

Defines the type of search to be performed in the spectrogram.

"X-Search" Searches only within the currently selected frame.

"Y-Search" Searches within all frames but only at the current frequency position.

"XY-Search" Searches in all frames at all positions.

Remote command:

Defined by the search function, see [Section 10.7.3.4, "Marker search \(spectrograms\)"](#),
on page 299

Marker Search Area

Defines which frames the search is performed in.

"Visible" Only the visible frames are searched.

"Memory" All frames stored in the memory are searched.

Remote command:

[CALCulate<n>:MARKer<m>:SPECTrogram:SARea](#) on page 301

[CALCulate<n>:DELTamarker<m>:SPECTrogram:SARea](#) on page 305

Peak Excursion

Defines the minimum level value by which a signal must rise or fall so that it is identified as a maximum or a minimum by the search functions.

Entries from 0 dB to 80 dB are allowed; the resolution is 0.1 dB. The default setting for the peak excursion is 6 dB.

For Analog Modulation Analysis, the unit and value range depend on the selected result display type.

Remote command:

`CALCulate<n>:MARKer<m>:PEXCursion` on page 296

Search Limits

The search results can be restricted by limiting the search area or adding search conditions.

Search Limits (Left / Right) ← Search Limits

If activated, limit lines are defined and displayed for the search. Only results within the limited search range are considered.

Remote command:

`CALCulate<n>:MARKer<m>:X:SLIMits[:STATe]` on page 297

`CALCulate<n>:MARKer<m>:X:SLIMits:LEFT` on page 297

`CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT` on page 298

Search Threshold ← Search Limits

Defines an absolute threshold as an additional condition for the peak search. If enabled, only peaks that exceed the threshold are detected.

Remote command:

`CALCulate<n>:THReshold:STATe` on page 299

`CALCulate<n>:THReshold` on page 299

Use Zoom Limits ← Search Limits

If activated, the peak search is restricted to the active zoom area defined for a single zoom.

Remote command:

`CALCulate<n>:MARKer<m>:X:SLIMits:ZOOM[:STATe]` on page 298

Search Limits Off ← Search Limits

Deactivates the search range limits.

Remote command:

`CALCulate<n>:MARKer<m>:X:SLIMits[:STATe]` on page 297

`CALCulate<n>:THReshold:STATe` on page 299

6.4.2.3 Positioning functions

Access: [MKR ->]

The following functions set the currently selected marker to the result of a peak search or set other characteristic values to the current marker value.

Peak Search	115
Search Next Peak	115
Search Minimum	115
Search Next Minimum	115
Center Frequency = Marker Frequency	115
Reference Level = Marker Level	115

Peak Search

Sets the selected marker/delta marker to the maximum of the trace. If no marker is active, marker 1 is activated.

Remote command:

`CALCulate<n>:MARKer<m>:MAXimum[:PEAK]` on page 310

`CALCulate<n>:DELTamarker<m>:MAXimum[:PEAK]` on page 313

Search Next Peak

Sets the selected marker/delta marker to the next (lower) maximum of the assigned trace. If no marker is active, marker 1 is activated.

Remote command:

`CALCulate<n>:MARKer<m>:MAXimum:NEXT` on page 309

`CALCulate<n>:MARKer<m>:MAXimum:RIGHT` on page 310

`CALCulate<n>:MARKer<m>:MAXimum:LEFT` on page 309

`CALCulate<n>:DELTamarker<m>:MAXimum:NEXT` on page 312

`CALCulate<n>:DELTamarker<m>:MAXimum:RIGHT` on page 313

`CALCulate<n>:DELTamarker<m>:MAXimum:LEFT` on page 312

Search Minimum

Sets the selected marker/delta marker to the minimum of the trace. If no marker is active, marker 1 is activated.

Remote command:

`CALCulate<n>:MARKer<m>:MINimum[:PEAK]` on page 311

`CALCulate<n>:DELTamarker<m>:MINimum[:PEAK]` on page 314

Search Next Minimum

Sets the selected marker/delta marker to the next (higher) minimum of the selected trace. If no marker is active, marker 1 is activated.

For spectrogram displays, define which frame the next minimum is to be searched in.

Remote command:

`CALCulate<n>:MARKer<m>:MINimum:NEXT` on page 311

`CALCulate<n>:MARKer<m>:MINimum:LEFT` on page 311

`CALCulate<n>:MARKer<m>:MINimum:RIGHT` on page 311

`CALCulate<n>:DELTamarker<m>:MINimum:NEXT` on page 313

`CALCulate<n>:DELTamarker<m>:MINimum:LEFT` on page 313

`CALCulate<n>:DELTamarker<m>:MINimum:RIGHT` on page 314

Center Frequency = Marker Frequency

Sets the center frequency to the selected marker or delta marker frequency. A peak can thus be set as center frequency, for example to analyze it in detail with a smaller span.

Remote command:

`CALCulate<n>:MARKer<m>:FUNCTION:CENTer` on page 235

Reference Level = Marker Level

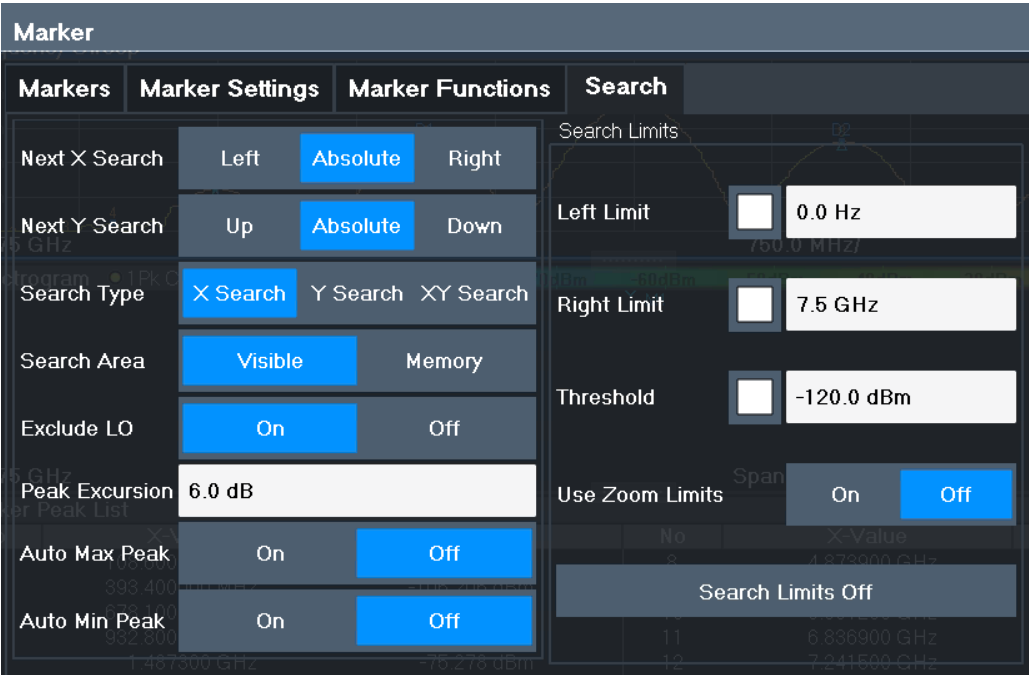
Sets the reference level to the selected marker level.

Remote command:
CALCulate<n>:MARKer<m>:FUNCTION:REFERENCE on page 225

6.4.3 Marker search settings for spectrograms

Access: "Overview" > "Analysis" > "Markers" > "Search"
or: [MKR TO] > "Search Config"

Spectrograms show not only the current sweep results, but also the sweep history. Thus, when searching for peaks, you must define the search settings within a single time frame (x-direction) and within several time frames (y-direction).
These settings are only available for spectrogram displays.



Search Mode for Next Peak in X-Direction.....	116
Search Mode for Next Peak in Y-Direction.....	117
Marker Search Type.....	117
Marker Search Area.....	117
Peak Excursion.....	118
Search Limits.....	118
L Search Limits (Left / Right).....	118
L Search Threshold.....	118
L Use Zoom Limits.....	118
L Search Limits Off.....	118

Search Mode for Next Peak in X-Direction
Selects the search mode for the next peak search within the currently selected frame.
"Left" Determines the next maximum/minimum to the left of the current peak.

"Absolute" Determines the next maximum/minimum to either side of the current peak.

"Right" Determines the next maximum/minimum to the right of the current peak.

Remote command:

[Section 10.7.3.5, "Positioning the marker"](#), on page 308

Search Mode for Next Peak in Y-Direction

Selects the search mode for the next peak search within all frames at the current marker position.

"Up" Determines the next maximum/minimum above the current peak (in more recent frames).

"Absolute" Determines the next maximum/minimum above or below the current peak (in all frames).

"Down" Determines the next maximum/minimum below the current peak (in older frames).

Remote command:

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:ABOVE](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:ABOVE](#)

on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:BELOW](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:BELOW](#)

on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:NEXT](#) on page 302

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:NEXT](#) on page 306

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:ABOVE](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:ABOVE](#)

on page 307

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:BELOW](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:BELOW](#)

on page 307

[CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:NEXT](#) on page 303

[CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:NEXT](#) on page 308

Marker Search Type

Defines the type of search to be performed in the spectrogram.

"X-Search" Searches only within the currently selected frame.

"Y-Search" Searches within all frames but only at the current frequency position.

"XY-Search" Searches in all frames at all positions.

Remote command:

Defined by the search function, see [Section 10.7.3.4, "Marker search \(spectrograms\)"](#), on page 299

Marker Search Area

Defines which frames the search is performed in.

"Visible" Only the visible frames are searched.

"Memory" All frames stored in the memory are searched.

Remote command:

[CALCulate<n>:MARKer<m>:SPECTrogram:SARea](#) on page 301

[CALCulate<n>:DELTamarker<m>:SPECTrogram:SARea](#) on page 305

Peak Excursion

Defines the minimum level value by which a signal must rise or fall so that it is identified as a maximum or a minimum by the search functions.

Entries from 0 dB to 80 dB are allowed; the resolution is 0.1 dB. The default setting for the peak excursion is 6 dB.

For Analog Modulation Analysis, the unit and value range depend on the selected result display type.

Remote command:

[CALCulate<n>:MARKer<m>:PEXCursion](#) on page 296

Search Limits

The search results can be restricted by limiting the search area or adding search conditions.

Search Limits (Left / Right) ← Search Limits

If activated, limit lines are defined and displayed for the search. Only results within the limited search range are considered.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMits\[:STATe\]](#) on page 297

[CALCulate<n>:MARKer<m>:X:SLIMits:LEFT](#) on page 297

[CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT](#) on page 298

Search Threshold ← Search Limits

Defines an absolute threshold as an additional condition for the peak search. If enabled, only peaks that exceed the threshold are detected.

Remote command:

[CALCulate<n>:THReshold:STATe](#) on page 299

[CALCulate<n>:THReshold](#) on page 299

Use Zoom Limits ← Search Limits

If activated, the peak search is restricted to the active zoom area defined for a single zoom.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMits:ZOOM\[:STATe\]](#) on page 298

Search Limits Off ← Search Limits

Deactivates the search range limits.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMits\[:STATe\]](#) on page 297

[CALCulate<n>:THReshold:STATe](#) on page 299

6.4.4 Marker functions

Some special marker functions are available in the I/Q Analyzer application.

- [Measuring the power in a channel \(band power marker\)](#)..... 119
- [Time domain power measurement](#)..... 122
- [Marker peak list](#)..... 126
- [Deactivating all marker functions](#)..... 130

6.4.4.1 Measuring the power in a channel (band power marker)

Access: "Overview" > "Analysis" > "Marker Functions" > "Band Power" > "Band Power Config"

or: [MKR FUNC] > "Select Marker Function" > "Band Power"

To determine the noise power in a transmission channel, you can use a noise marker and multiply the result with the channel bandwidth. However, the results are only accurate for flat noise.

Band power markers allow you to measure the integrated power for a defined span (band) around a marker (similar to ACP measurements). By default, 5 % of the current span is used. The span is indicated by limit lines in the diagram. You can easily change the span by moving the limit lines in the diagram. They are automatically aligned symmetrically to the marker frequency. They are also moved automatically if you move the marker on the screen.

The results can be displayed either as a power (dBm) or density (dBm/Hz) value and are indicated in the "marker table" for each band power marker.



Relative band power markers

The results for band power markers which are defined as *delta* markers and thus have a reference value can also be calculated as reference power values (in dB).

For Analog Modulation Analysis, relative band power markers are not available.

In this case, the result of the band power deltamarker is the difference between the absolute power in the band around the delta marker and the absolute power for the reference marker. The powers are subtracted logarithmically, so the result is a dB value.

[Relative band power (Delta2) in dB] = [absolute band power (Delta2) in dBm] - [absolute (band) power of reference marker in dBm]

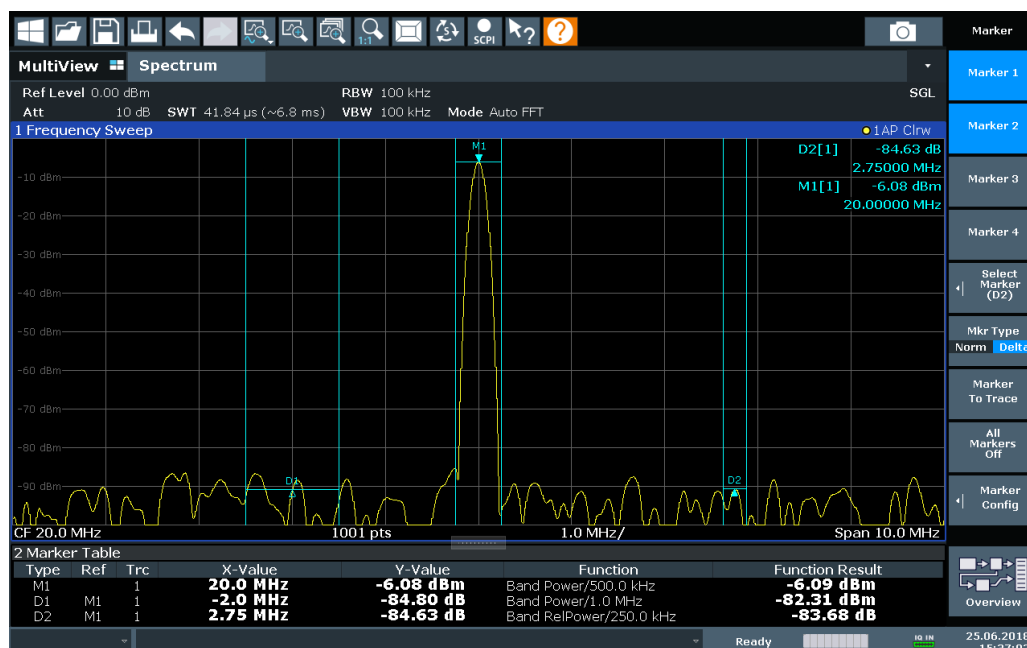
The measured power for the reference marker may be an absolute power at a single point (if the reference marker is not a band power marker), or the power in a band (if the reference marker is a band power marker itself).

If the reference marker for the band power marker is also a delta marker, the absolute power level for the reference marker is used for calculation.



For the I/Q Analyzer application, band power markers are only available for Spectrum displays.

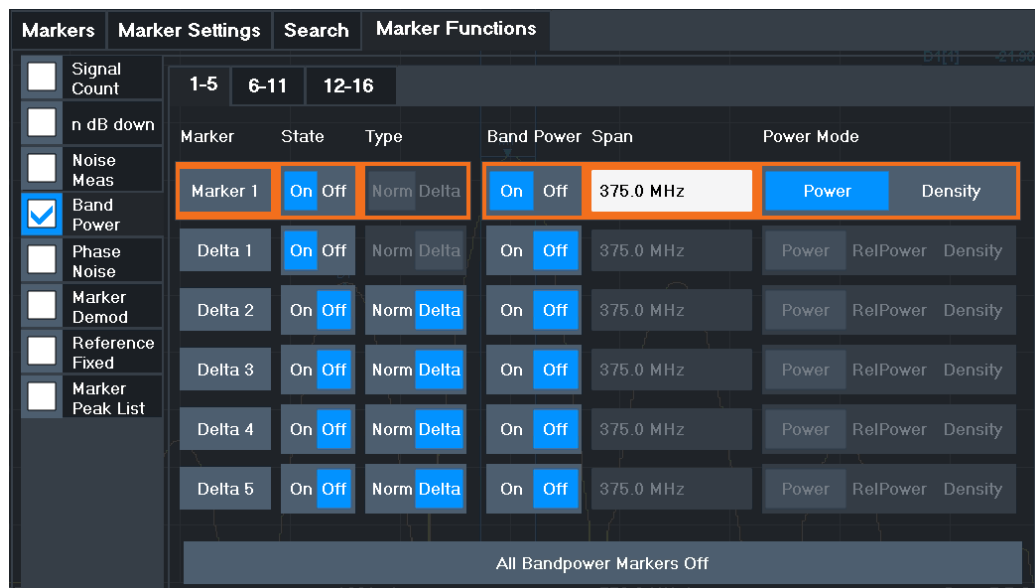
The entire band must lie within the display. If it is moved out of the display, the result cannot be calculated (indicated by "- - -" as the "Function Result"). However, the width of the band is maintained so that the band power can be calculated again when it returns to the display.



All markers can be defined as band power markers, each with a different span. When a band power marker is activated, if no marker is active yet, marker 1 is activated. Otherwise, the currently active marker is used as a band power marker (all other marker functions for this marker are deactivated).

If the detector mode for the marker trace is set to "Auto", the RMS detector is used.

The individual marker settings correspond to those defined in the "Marker" dialog box (see [Section 6.4.1.1, "Individual marker setup"](#), on page 104). Any settings to the marker state or type changed in the "Marker Function" dialog box are also changed in the "Marker" dialog box and vice versa.



Remote commands:

[CALCulate<n>:MARKer<m>:FUNCTION:BPOWer\[:STATe\]](#) on page 316

[CALCulate<n>:MARKer<m>:FUNCTION:BPOWer:RESult?](#) on page 315

Band Power Measurement State	121
Span	121
Power Mode	122
Switching All Band Power Measurements Off	122

Band Power Measurement State

Activates or deactivates band power measurement for the marker in the diagram.

Band power markers are only available for standard frequency measurements (not zero span) in the Spectrum application.

If activated, the markers display the power or density measured in the band around the current marker position.

For details see [Section 6.4.4.1, "Measuring the power in a channel \(band power marker\)"](#), on page 119.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:BPOWer\[:STATe\]](#) on page 316

[CALCulate<n>:DELTaMarker<m>:FUNCTION:BPOWer\[:STATe\]](#) on page 318

Span

Defines the span (band) around the marker for which the power is measured.

The span is indicated by lines in the diagram. You can easily change the span by moving the limit lines in the diagram. They are automatically aligned symmetrically to the marker frequency. They are also moved automatically if you move the marker on the screen.

Remote command:

`CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:SPAN` on page 316

`CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:SPAN` on page 317

Power Mode

Defines the mode of the power measurement result.

For Analog Modulation Analysis, the power mode is not editable for AM, FM, or PM spectrum results. In this case, the marker function does not determine a power value, but rather the deviation within the specified span.

"Power"	The result is an absolute power level. The power unit depends on the Unit setting.
"Relative Power"	This setting is only available for a delta band power marker. The result is the difference between the absolute power in the band around the delta marker and the absolute power for the reference marker (see "Reference Marker" on page 105). The powers are subtracted logarithmically, so the result is a dB value. <i>[Relative band power (Delta2) in dB] = [absolute band power (Delta2) in dBm] - [absolute (band) power of reference marker in dBm]</i> For details see "Relative band power markers" on page 119
"Density"	The result is a power level in relation to the bandwidth, displayed in dBm/Hz. Note: For the following amplitude units the band power marker result in the power density mode is not dBm/Hz. • Amplitude units W, dBpW: band power density in amplitude unit/Hz. • Amplitude units dBmV, dBμV, V, dBμA, A: band power density in amplitude unit/sqrt(Hz).

Remote command:

`CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:MODE` on page 315

`CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:MODE` on page 316

Switching All Band Power Measurements Off

Deactivates band power measurement for all markers.

Remote command:

`CALCulate<n>:MARKer<m>:FUNCTION:BPOWER[:STATE]` on page 316

`CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER[:STATE]` on page 318

6.4.4.2 Time domain power measurement

The Time Domain Power measurement determines the power of a signal in the time domain.

A time domain power measurement is only possible for zero span.

- [About the measurement](#)..... 123
- [Time domain power results](#)..... 123
- [Time domain power basics - range definition using limit lines](#)..... 124
- [Time domain power configuration](#)..... 124
- [How to measure powers in the time domain](#)..... 126

About the measurement

Using the Time Domain Power measurement function, the R&S FSV/A determines the power of the signal in zero span by summing up the power at the individual measurement points and dividing the result by the number of measurement points. Thus it is possible to measure the power of TDMA signals during transmission, for example, or during the muting phase. Both the mean power and the RMS power can be measured.

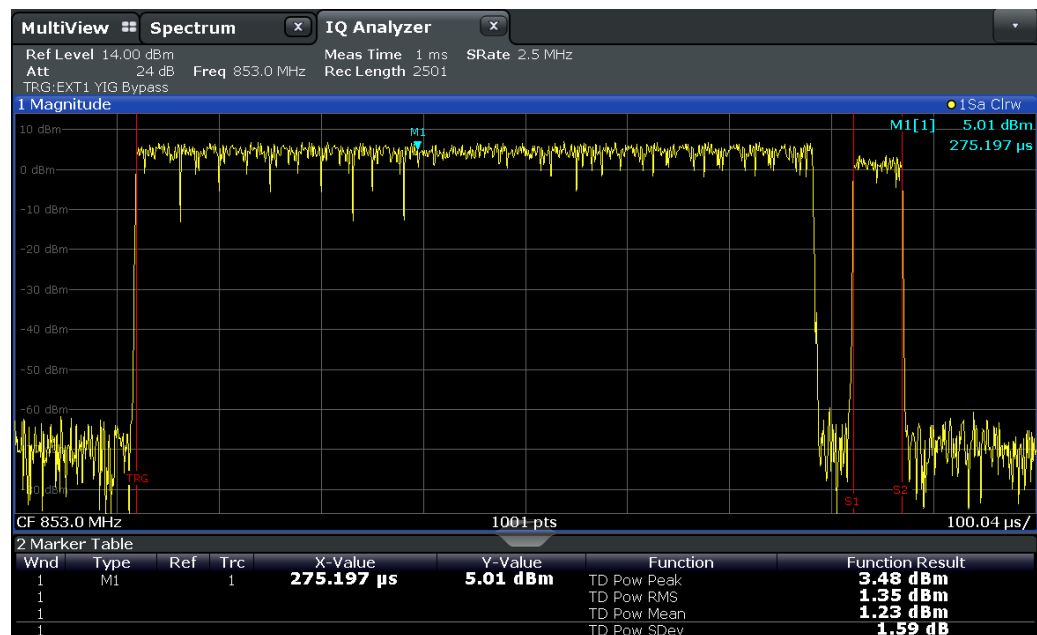
For this measurement, the sample detector is recommended. The sample detector is activated automatically if the detector is in auto mode.

Time domain power results

Several different power results can be determined simultaneously:

Mode	Description
Peak	Peak value from the points of the displayed trace or a trace segment.
RMS	RMS value from the points of the displayed trace or a trace segment.
Mean	Mean value from the points of the displayed trace or a trace segment. The linear mean value of the equivalent voltages is calculated. For example, to measure the mean power during a GSM burst.
Std Dev	The standard deviation of the measurement points from the mean value.

The result is displayed in the marker results, indicated by "Power" and the selected power mode, e.g. "RMS". The measured values are updated after each sweep or averaged over a user-defined number of sweeps (trace averaging).



The results can also be queried using the remote commands described in [Section 10.7.3.8, "Measuring the time domain power"](#), on page 322.

Time domain power basics - range definition using limit lines

The range of the measured signal to be evaluated for the power measurement can be restricted using limit lines. The left and right limit lines (S1, S2) define the evaluation range and are indicated by vertical red lines in the diagram. If activated, the power results are only calculated from the levels within the limit lines.

For example, if both the on and off phase of a burst signal are displayed, the measurement range can be limited to the transmission or to the muting phase. The ratio between signal and noise power of a TDMA signal for instance can be measured by using a measurement as a reference value and then varying the measurement range.



In order to get stable measurement results for a limited evaluation range, usually a trigger is required.

Time domain power configuration

Access: [MARK FUNC] > "Select Marker Function" > "Time Domain Power" > "Time Dom Power Config"

Results		Power		Limits	
Peak	On Off	Relative	On Off	State	On Off
RMS	On Off	Reference	Meas -> Reference	Left	-----
Mean	On Off	Peak Hold	On Off	Right	-----
Std Dev	On Off	Average	On Off		

The remote commands required to perform these tasks are described in [Section 10.7.3.8, "Measuring the time domain power"](#), on page 322.

Results	125
Power	125
Limit State	126
Left Limit / Right Limit	126

Results

Activates the power results to be evaluated from the displayed trace or a limited area of the trace.

"Peak"	Peak power over several measurements (uses trace averaging, Max Hold)
"RMS"	RMS value from the points of the displayed trace or a segment thereof.
"Mean"	Mean value from the points of the displayed trace or a segment thereof. The linear mean value of the equivalent voltages is calculated.
"Std Dev"	The standard deviation of the measurement points from the mean value. The measurement of the mean power is automatically switched on at the same time.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEak\[:STATe\]](#) on page 325
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEak:RESult?](#) on page 328
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS\[:STATe\]](#) on page 325
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS:RESult?](#) on page 329
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN\[:STATe\]](#) on page 325
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:RESult?](#) on page 327

Power

Enables or disables relative power measurement in the time domain power measurement.

"Relative"	Switches the time domain power results between absolute units and relative display. • "On": Values are shown relative to the stored reference. • "Off": Values are shown as absolute power.
"Reference" ("Meas → Reference")	Stores the current measurement as the reference for relative display.
"Peak Hold"	If enabled, the display retains the maximum value seen for each enabled result for successive sweeps.
"Average"	Averages the result values over successive sweeps for smoother, more stable readings.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MODE](#) on page 323
[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:REFERENCE:AUTO](#) on page 324

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLD](#) on page 324

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERAGE](#) on page 323

Limit State

Switches the limitation of the evaluation range on or off. Default setting is off.

If deactivated, the entire sweep time is evaluated. If switched on, the evaluation range is defined by the left and right limit. If only one limit is set, it corresponds to the left limit and the right limit is defined by the stop frequency. If the second limit is also set, it defines the right limit.

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMITS\[:STATe\]](#) on page 297

Left Limit / Right Limit

Defines a power level limit for line S1 (left) or S2 (right).

Remote command:

[CALCulate<n>:MARKer<m>:X:SLIMITS:LEFT](#) on page 297

[CALCulate<n>:MARKer<m>:X:SLIMITS:RIGHT](#) on page 298

How to measure powers in the time domain

The step-by-step procedure to measure powers in the time domain is described here in detail.

To measure the power in the time domain

1. Select a result window in the time domain, for example a "Magnitude" window.
2. Select [MARK FUNC].
3. From the "Select Marker Function" dialog box, select the "Time Domain Power" function.
4. Select the type of power measurement results to be determined by selecting the corresponding softkeys.
5. To restrict the power evaluation range, define limits:
 - a) Select "Time Dom Power Config" to display the "Time Domain Power" configuration dialog box.
 - b) Switch on the limits by setting the "Limit State" to "On".
The limit lines S1 and S2 are displayed.
 - c) Define the left limit (limit line S1), the right limit (S2), or both.
6. Start a sweep.

The measured powers are displayed in the marker results.

6.4.4.3 Marker peak list

Access: "Overview" > "Analysis" > "Marker Functions" > "Marker Peak List"

Or: [MKR FUNC] > "Marker Peak List"

A common measurement task is to determine peak values, i.e. maximum or minimum signal levels. The R&S FSV/A provides various peak search functions and applications:

- Setting a marker to a peak value once (Peak Search)
- Searching for a peak value within a restricted search area (Search Limits)
- Creating a "marker table" with all or a defined number of peak values for one sweep ("Marker Peak List")
- Updating the marker position to the current peak value automatically after each sweep (Auto Peak Search)
- Creating a fixed reference marker at the current peak value of a trace (Fixed Reference)

Peak search limits

The peak search can be restricted to a search area. The search area is defined by limit lines which are also indicated in the diagram. In addition, a minimum value (threshold) can be defined as a further search condition.

When is a peak a peak? - Peak excursion

During a peak search, noise values are detected as a peak if the signal is very flat or does not contain many peaks. Therefore, you can define a relative threshold ("Peak Excursion"). The signal level must increase by the threshold value before falling again before a peak is detected. To avoid identifying noise peaks as maxima or minima, enter a peak excursion value that is higher than the difference between the highest and the lowest value measured for the displayed inherent noise.

Effect of peak excursion settings (example)

The following figure shows a trace to be analyzed.

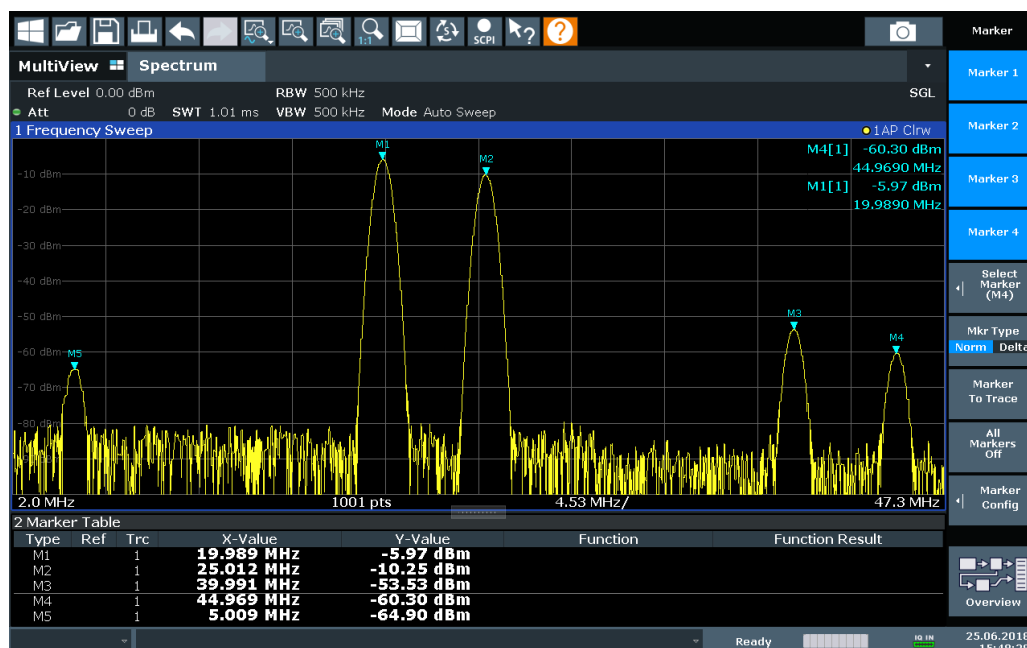


Figure 6-3: Trace example

The following table lists the peaks as indicated by the marker numbers in the diagram above, as well as the minimum decrease in amplitude to either side of the peak:

Marker #	Min. amplitude decrease to either side of the signal
1	80 dB
2	80 dB
3	55 dB
4	39 dB
5	32 dB

To eliminate the smaller peaks M3, M4 and M5 in the example above, a peak excursion of at least 60 dB is required. In this case, the amplitude must rise at least 60 dB before falling again before a peak is detected.

Marker peak list

The marker peak list determines the frequencies and levels of peaks in the spectrum. It is updated automatically after each sweep. How many peaks are displayed can be defined, as well as the sort order. In addition, the detected peaks can be indicated in the diagram. The peak list can also be exported to a file for analysis in an external application.

Automatic peak search

A peak search can be repeated automatically after each sweep to keep the maximum value as the reference point for a phase noise measurement. Automatic peak search is useful to track a drifting source. The delta marker 2, which shows the phase noise

measurement result, keeps the delta frequency value. Therefore, the phase noise measurement leads to reliable results in a certain offset although the source is drifting.

Using a peak as a fixed reference marker

Some results are analyzed in relation to a peak value, for example a carrier frequency level. In this case, the maximum level can be determined by an initial peak search and then be used as a reference point for further measurement results.

Marker

Markers **Marker Settings** **Marker Functions** **Search**

☐ n dB down State **On** **Off**

☐ Phase Noise Settings

☐ Reference Fixed Sort Mode **X Value** **Y Value**

☒ **Marker Peak List** Maximum Number of Peaks 50

Peak Excursion 6.0 dB

Display Marker Numbers **On** **Off**

Search Limits

Left Limit 0.0 Hz

Right Limit 20.0 MHz

Threshold -120.0 dB

Use Zoom Limits **On** **Off**

Search Limits Off

Export

Export Peak List Decimal Separator **Point** **Comma**

All Functions Off

Remote commands:

CALCulate<n>:MARKer<m>:FUNctIon:FPEaks:STATe on page 320

TRAC? LIST,

See TRACe<n>[:DATA]? on page 337

Peak List State.....	129
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Maximum Number of Peaks.....	130
Peak Excursion.....	130
Display Marker Numbers.....	130
Export Peak List.....	130

Peak List State

Activates/deactivates the marker peak list. If activated, the peak list is displayed and the peaks are indicated in the trace display.

For each listed peak, the frequency/time ("X-value") and level ("Y-Value") values are given.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:STATe](#) on page 320

Sort Mode

Defines whether the peak list is sorted according to the x-values or y-values. In either case, the values are sorted in ascending order.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:SORT](#) on page 320

Maximum Number of Peaks

Defines the maximum number of peaks to be determined and displayed.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:LIST:SIZE](#) on page 320

Peak Excursion

Defines the minimum level value by which a signal must rise or fall so that it is identified as a maximum or a minimum by the search functions.

Entries from 0 dB to 80 dB are allowed; the resolution is 0.1 dB. The default setting for the peak excursion is 6 dB.

For Analog Modulation Analysis, the unit and value range depend on the selected result display type.

Remote command:

[CALCulate<n>:MARKer<m>:PEXCursion](#) on page 296

Display Marker Numbers

By default, the marker numbers are indicated in the diagram so you can find the peaks from the list. However, for large numbers of peaks, the marker numbers can decrease readability; in this case, deactivate the marker number display.

Remote command:

[CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:ANNotation:LABel\[:STATe\]](#) on page 318

Export Peak List

The peak list can be exported to an ASCII file (.DAT) for analysis in an external application.

Remote command:

[MMEMory:STORe<n>:PEAK](#) on page 321

[FORMat:DEXPort:DSEParator](#) on page 340

6.4.4.4 Deactivating all marker functions

Access: "Overview" > "Analysis" > "Marker Functions" > "All Functions Off"

All special marker functions can be deactivated in one step.

Remote command:

7 I/Q data import and export

Baseband signals mostly occur as so-called complex baseband signals, i.e. a signal representation that consists of two channels; the inphase (I) and the quadrature (Q) channel. Such signals are referred to as I/Q signals. The complete modulation information and even distortion that originates from the RF, IF or baseband domains can be analyzed in the I/Q baseband.

Importing and exporting I/Q signals is useful for various applications:

- Generating and saving I/Q signals in an RF or baseband signal generator or in external software tools to analyze them with the R&S FSV/A later
The I/Q analyzer supports various data formats for import and export, see [Section C, "Reference: supported I/Q file formats"](#), on page 358.
- Capturing and saving I/Q signals with the R&S FSV/A to analyze them with the R&S FSV/A or an external software tool later
As opposed to storing trace data, which can be averaged or restricted to peak values, I/Q data is stored as it was captured, without further processing. Multi-channel data is not supported.
The data is stored as complex values in 32-bit floating-point format. By default, the I/Q data is stored in a format with the file extension `.iq.tar`.

For example, you can capture I/Q data using the I/Q Analyzer application and then perform analog demodulation on that data using the R&S FSV/A Analog Demodulation application, if available.



An application note on converting Rohde & Schwarz I/Q data files is available from the Rohde & Schwarz website:

[1EF85: Converting R&S I/Q data files](#)

The export functions are available in the "Save/Recall" menu which is displayed when you select the "Save" or "Open" icon in the toolbar (see [Section 5.2, "Import/export functions"](#), on page 43). The import functions are available in the "I/Q File" dialog box (see [Section 5.3.1.2, "Settings for input from I/Q data files"](#), on page 50).

- [Export functions](#)..... 132

7.1 Export functions



Access: "Save" icon in the toolbar > "Export"

The R&S FSV/A provides various evaluation methods for the results of the performed measurements. However, possibly you want to evaluate the data with further, external applications. In this case, you can export the measurement data to a standard format file (ASCII or XML). Some of the data stored in these formats can also be re-imported to the R&S FSV/A for further evaluation later, for example in other applications.

The following data types can be exported (depending on the application):

- Trace data

- Table results, such as result summaries or marker peak lists
- I/Q data

The following data types can be imported (depending on the application):

- I/Q data



I/Q data can only be imported and exported in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

See the corresponding user manuals for those applications for details.



Exporting I/Q data is only possible in single sweep mode ([Continuous Sweep / Run Cont](#)).

Importing I/Q data is also possible in continuous sweep mode.

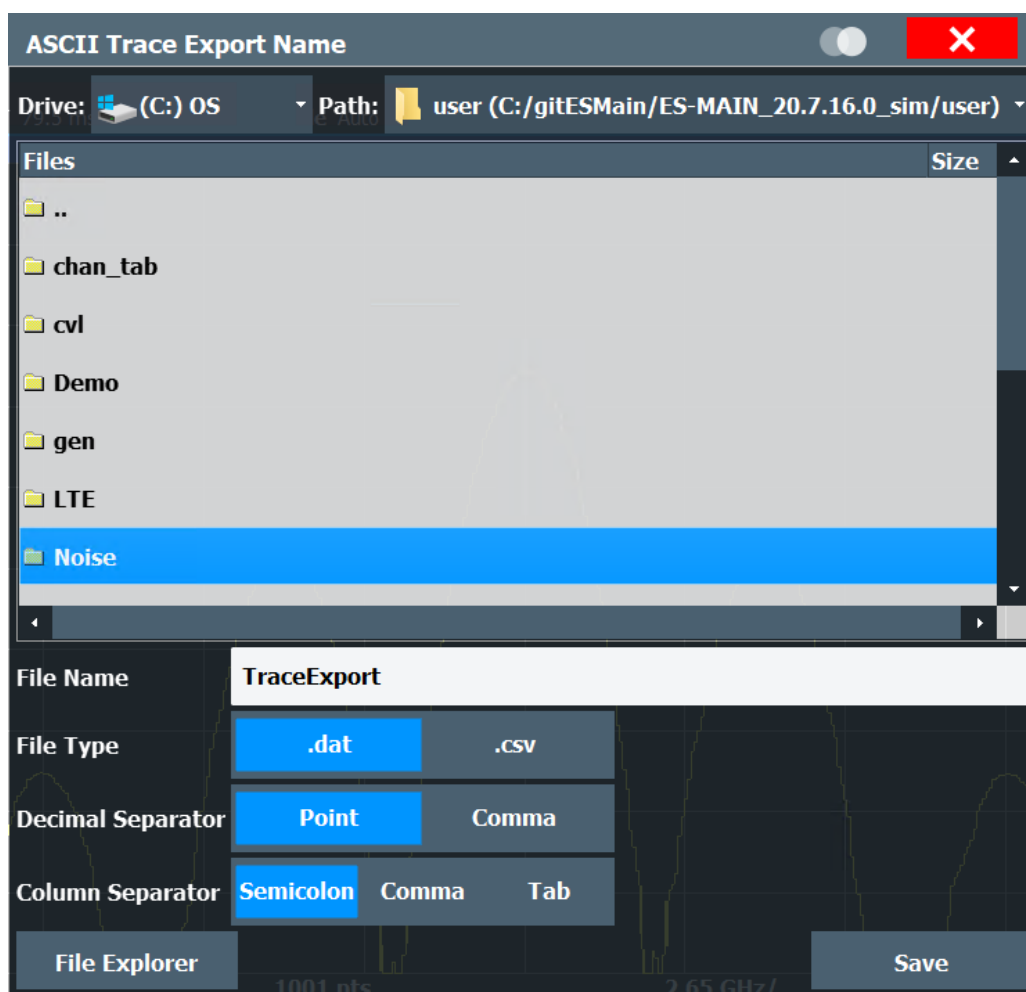
For more information about importing I/Q data files, see [Section 5.3.1.2, "Settings for input from I/Q data files"](#), on page 50.

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Export Trace to ASCII File

Saves the selected trace or all traces in the currently active result display to the specified file and directory in the selected ASCII format.

"File Explorer": Instead of using the file manager of the R&S FSV/A firmware, you can also use the Microsoft Windows File Explorer to manage files.



Note: Secure user mode.

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

[MMEMory:STORe<n>:TRACe](#) on page 342

File Type ← Export Trace to ASCII File

Determines the format of the ASCII file to be imported or exported.

Depending on the external program in which the data file was created or is evaluated, a comma-separated list (CSV) or a plain data format (DAT) file is required.

Remote command:

[FORMat:DEXPort:FORMat](#) on page 341

Decimal Separator ← Export Trace to ASCII File

Defines the decimal separator for floating-point numerals for the data export/import files. Evaluation programs require different separators in different languages.

Remote command:

[FORMat:DEXPort:DSEParator](#) on page 340

Column Separator ← Export Trace to ASCII File

Selects the character that separates columns in the exported ASCII file. The character can be either a semicolon, a comma or a tabulator (tab).

Example for semicolon:

```
Type;FSV3007;Version;1.00;Date;01.Jan 3000;
```

Example for comma:

```
Type,FSV3007,
Version,1.00,
Date,01.Jan 3000,
```

Example for tabulator (tab after the last column is not visible):

```
Type      FSV3007
Version    1.00
Date      01.Jan 3000
```

The selected column separator setting remains the same, even after a preset.

Remote command:

[FORMat:DEXPort:CSEParator](#) on page 340

File Explorer ← Export Trace to ASCII File

Opens the Microsoft Windows File Explorer.

Remote command:

not supported

Export Configuration

Opens the "Traces" dialog box to configure the trace and data export settings.

See [Section 6.3, "Trace / data export configuration"](#), on page 99.

I/Q Export

Opens a file selection dialog box to define an export filename to which the I/Q data is stored. This function is only available in single sweep mode.

It is not available in the Spectrum application, only in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

For details, see the description in the R&S FSV/A I/Q Analyzer User Manual ("Importing and Exporting I/Q Data").

Note: Storing large amounts of I/Q data (several Gigabytes) can exceed the available (internal) storage space on the R&S FSV/A. In this case, it can be necessary to use an external storage medium.

Note: Secure user mode.

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Remote command:

[MMEMory:STORe<n>:IQ:STATe](#) on page 345

[MMEMory:STORe<n>:IQ:COMMeNt](#) on page 345

File Type ← I/Q Export

The I/Q data file can be stored in one of the following supported formats:

- .iq.tar
- .iqw
- .csv
- .mat
- .aid

Note: Not all applications support all formats.

For details on formats, see [Section C, "Reference: supported I/Q file formats"](#), on page 358.

Remote command:

The file type is determined by the file extension of the file name parameter. If no file extension is provided, the file type is assumed to be .iq.tar. For .mat files, Matlab® v4 is assumed.

File Explorer ← I/Q Export

Opens the Microsoft Windows File Explorer.

Remote command:

not supported

8 How to work with I/Q data

The following step-by-step procedures demonstrate in detail how to perform various tasks when working with I/Q data.

- [How to perform measurements in the I/Q Analyzer application](#)..... 137
- [How to capture analog baseband input via the optional Analog Baseband interface](#)..... 137
- [How to export and import I/Q data](#)..... 139

8.1 How to perform measurements in the I/Q Analyzer application

The following step-by-step instructions demonstrate how to capture I/Q data on the R&S FSV/A and how to analyze data in the I/Q Analyzer application.

- [How to analyze data in the I/Q Analyzer](#)..... 137

8.1.1 How to analyze data in the I/Q Analyzer

1. Select [MODE] and select the "I/Q Analyzer" application.
2. Select "Overview" to display the "Overview" for an I/Q Analyzer measurement.
3. Select "Display Config" and select up to six displays that are of interest to you. Arrange them on the display to suit your preferences.
4. Exit the SmartGrid mode and select "Overview" to display the "Overview" again.
5. Select "Analysis" in the "Overview" to make use of the advanced analysis functions in the displays.
 - Configure a trace to display the average over a series of sweeps (on the "Trace" tab; if necessary, increase the "Average Count").
 - Configure markers and delta markers to determine deviations and offsets within the signal (on the "Marker" tab).

8.2 How to capture analog baseband input via the optional Analog Baseband interface

Analog baseband signals can also be captured via the optional "Analog Baseband" interface, if installed.

1. Connect the device that provides analog baseband input to the "Baseband Input" connectors at the front of the R&S FSV/A.

How to capture analog baseband input via the optional Analog Baseband interface

For single-ended input signals, use the I or Q connector, or both.

For differential input signals, connect the positive input to the I and Q connectors, and the negative input to the $\bar{I}$ and $\bar{Q}$ connectors.

2. Press [INPUT/OUTPUT] of the R&S FSV/A.
3. Select "Input Source Config".
4. Switch to the "Analog Baseband" tab to configure the "Analog Baseband" interface.
5. Set the state of the "Analog Baseband" signal source to "On".
6. Select the "I/Q Mode" depending on the signal at the input connectors, or how you want to interpret it.
7. If necessary, change the input configuration setting depending on whether a single-ended or differential signal is being input.
Note that both differential and single-ended active probes are supported. However, since a probe only uses a single connector (either "Baseband Input I" or Q), the input configuration must be set to single-ended. The type of probe is indicated in the "Probes" subtab of the Input dialog box.
8. If necessary, for example due to a physical impairment of the two analog signals, define a skew of the I vector in respect to the Q vector for correct analysis.
9. If only one component of the input signal is of interest (I/Q mode: "I only/ Low IF I" or "Q only/ Low IF Q"), define how to interpret the signal: as modulated or real data. For modulated data, change the "Center Frequency" to use for down-conversion.
10. Press [Ampt].
11. Select "Amplitude Config".
12. Define the reference level for the input. If a probe is connected, consider the probe's attenuation when defining the reference level.
13. Select the maximum power level you expect to input at the "Baseband Input" connector as the "Full scale level", or select "Auto" mode to have it set automatically according to the selected reference level.
14. Optionally, select "Trigger" and define a trigger for data acquisition, for example a "Baseband Power" trigger to start capturing data only when a specific input power is exceeded.
15. Press [MEAS CONFIG].
16. Select "Data Acquisition" to configure the signal capture.
 - "Sample rate" selected for analysis data or "Analysis Bandwidth" (the bandwidth range in which the signal remains unchanged by the digital decimation filter and thus remains undistorted; this range can be used for accurate analysis by the R&S FSV/A); both values are correlated
 - "Measurement Time:" how long the signal is to be captured

- "Record Length": the number of samples to be captured (also defined by sample rate and measurement time)
 - If necessary, for example due to a mixed up connection or inverse data from the connected device, swap the I and Q values for correct analysis ("Swap I/Q").
17. Select "Display Config" and select up to six displays that are of interest to you. To analyze the complex spectrum of the analog baseband signal, for instance, select the Spectrum result display (and the I/Q mode "I+jQ" in the input settings). The displayed span corresponds to the selected sample rate. Arrange the windows on the display to suit your preferences.
 18. Exit the SmartGrid mode.
 19. Start a new sweep with the defined settings.

8.3 How to export and import I/Q data



I/Q data can only be exported in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

Capturing and exporting I/Q data

1. Press [PRESET].
2. Press [MODE] and select the I/Q Analyzer application or any other application that supports I/Q data.
3. Configure the data acquisition.
4. Press [RUN SINGLE] to perform a single sweep measurement.
5. Select the  "Save" icon in the toolbar.
6. Select "I/Q Export".
7. In the file selection dialog box, select a storage location and enter a file name.
8. Select "Save".

The captured data is stored to a file.

The file type is determined by the file extension. If no file extension is provided, the file type is assumed to be `.iq.tar`. For `.mat` files, Matlab® v4 is assumed.

Using exported I/Q data as an input source

1. Press [MODE] and select the I/Q Analyzer application.
2. If necessary, switch to single sweep mode by pressing [RUN SINGLE].
3. Select "Input/Frontend" and switch to the "Input Source" > "I/Q File" tab.

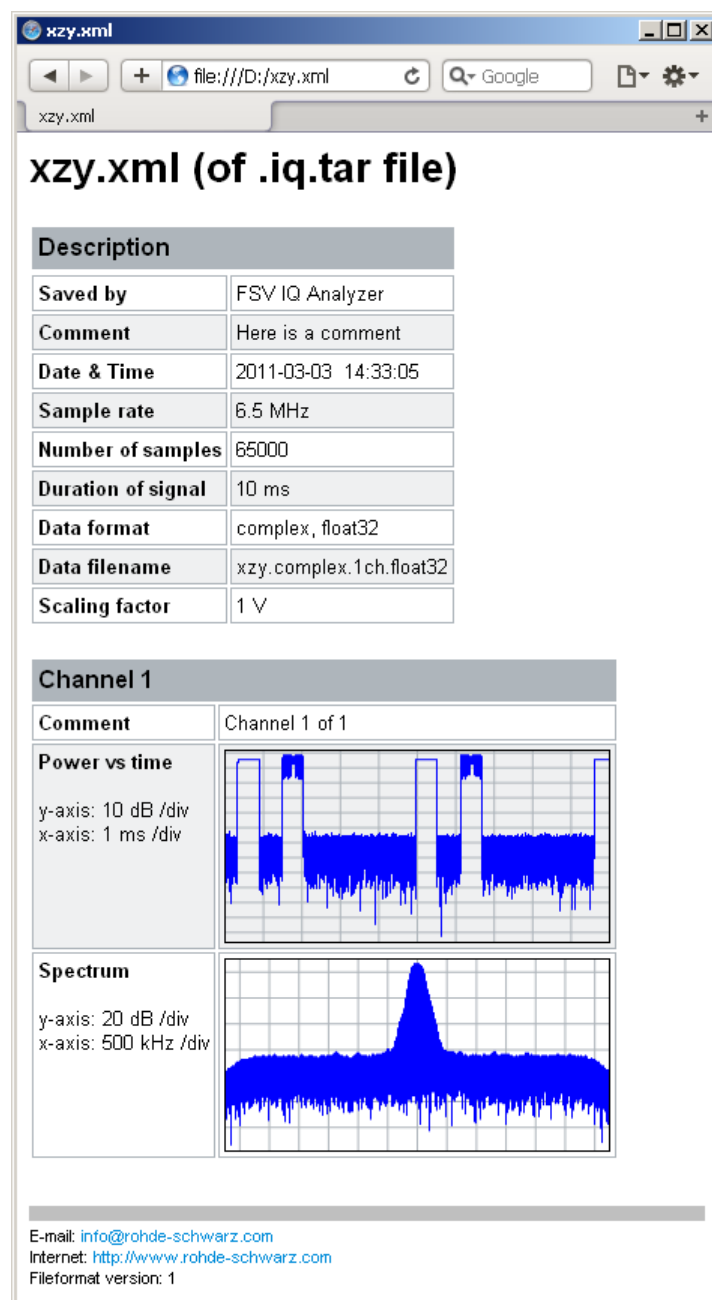
4. Select "Select File".
5. In the file selection dialog box, select the file that contains the exported I/Q data (`.iq.tar` extension).
6. Set the I/Q file state to "On".
7. Select the "Frequency" tab to define the input signal's center frequency.
8. Start a new measurement with the data from the file.
 - To perform a single sweep measurement, press [RUN SINGLE].
 - To perform a continuous sweep measurement, press [RUN CONT].

Previewing the I/Q data in a web browser

The `iq-tar` file format allows you to preview the I/Q data in a web browser.

1. Use an archive tool (e.g. WinZip® or PowerArchiver®) to unpack the `iq-tar` file into a folder.
2. Locate the folder using Windows Explorer.
3. Open your web browser.

4. Drag the I/Q parameter XML file, e.g. `example.xml`, into your web browser.



9 Troubleshooting the measurement

If errors or problems occur, try the following methods to correct the measurement.

9.1 Troubleshooting I/Q noise cancellation

If I/Q noise cancellation fails, the main cause is failed synchronization. Check the following possible issues.



Error status bit

If an error occurs during I/Q noise cancellation, bit 7 in the `STATUS:QUESTIONABLE:SYNC:CONDITION` status register is set (`BIT_K575_FAILED`).

See `STATUS:QUESTIONABLE:SYNC:CONDITION?` on page 347.

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Capture length is too short

Increase the capture length. If no trigger is used, try a capture length that is longer than twice the signal length.

The EVM is too high

Check the signal description to ensure that the reference signal is estimated correctly. If the EVM is close to or exceeds the defined limits, I/Q noise cancellation may not be possible.

You can detect exceeded limits as red entries in the "Result Summary".

PPDUs:	Min	Mean	Limit
EVM All Carriers	11.40	11.42*	7.94
	-18.86	-18.85*	-22.00

Frequency offset too large

If the frequency offset is too large, I/Q noise cancellation may not be possible.

You can detect exceeded center frequency error limits as red entries in the "Result Summary".

Center Freq Error	100094....	100095.... ±25000....	100095.... ±25000....
Symbol Clock Error	46.04	46.04* ±25.00	46.04* ±25.00

Sample rate error too large

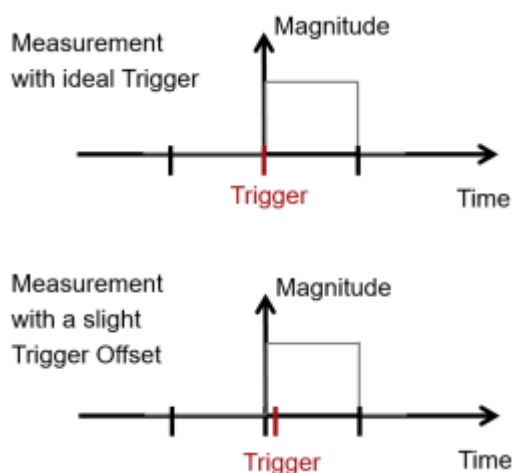
Check the signal capture settings to ensure that the input sample rate is defined correctly. If possible, synchronize the R&S FSV/A and the DUT using an external reference.

If the sample rate error is too large, I/Q noise cancellation may not be possible.

You can detect exceeded sample rate (symbol clock) error limits as red entries in the "Result Summary".

Trigger uncertainty

Due to trigger uncertainty, a trigger can occur after the first relevant sample. You can define a negative trigger offset so the measurement includes pretrigger samples and thus, the full signal.



Previous application warnings

If the application shows messages in the status bar before the I/Q noise cancellation has even started, it may not be possible.

For example, if the payload is too short (fewer than 5 symbols), the EVM results are not reliable and I/Q noise cancellation may not be possible.

Short payload (<5 symbols). EVM results not reliable

Check the status bar for errors or messages.

10 Remote commands to perform measurements with I/Q data

The following commands are specific to performing measurements in the I/Q Analyzer application in a remote environment. The R&S FSV/A must already be set up for remote operation in a network as described in the base unit manual.



Note that basic tasks that are also performed in the base unit in the same way are not described here. For a description of such tasks, see the R&S FSV/A User Manual.

In particular, this includes:

- Basic instrument configuration, e.g. checking the system configuration, customizing the screen layout, or configuring networks and remote operation
- Using the common status registers



SCPI Recorder - automating tasks with remote command scripts

The I/Q Analyzer application also supports the SCPI Recorder functionality.

Using the SCPI Recorder functions, you can create a SCPI script directly on the instrument and then export the script for use on the controller. You can also edit or write a script manually, using a suitable editor on the controller. For manual creation, the instrument supports you by showing the corresponding command syntax for the current setting value.

For details see the "Network and Remote Operation" section in the R&S FSV/A User Manual.

The following tasks specific to the I/Q Analyzer application are described here:

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10.1 Introduction

Commands are program messages that a controller (e.g. a PC) sends to the instrument or software. They operate its functions ('setting commands' or 'events') and request information ('query commands'). Some commands can only be used in one way, others work in two ways (setting and query). If not indicated otherwise, the commands can be used for settings and queries.

The syntax of a SCPI command consists of a header and, usually, one or more parameters. To use a command as a query, you have to append a question mark after the last header element, even if the command contains a parameter.

A header contains one or more keywords, separated by a colon. Header and parameters are separated by a "white space" (ASCII code 0 to 9, 11 to 32 decimal, e.g. blank). If there is more than one parameter for a command, they are separated by a comma from one another.

Only the most important characteristics that you need to know when working with SCPI commands are described here. For a more complete description, refer to the user manual of the R&S FSV/A.



Remote command examples

Note that some remote command examples mentioned in this general introduction are possibly not supported by this particular application.

10.1.1 Conventions used in descriptions

The following conventions are used in the remote command descriptions:

- *Command usage*
If not specified otherwise, commands can be used both for setting and for querying parameters.
If a command can be used for setting or querying only, or if it initiates an event, the usage is stated explicitly.
- *Parameter usage*
If not specified otherwise, a parameter can be used to set a value, and it is the result of a query.
Parameters required only for setting are indicated as "Setting parameters".
Parameters required only to refine a query are indicated as "Query parameters".
Parameters that are only returned as the result of a query are indicated as "Return values".
- *Conformity*
Commands that are taken from the SCPI standard are indicated as "SCPI confirmed". All commands used by the R&S FSV/A follow the SCPI syntax rules.
- *Asynchronous commands*
A command which does not automatically finish executing before the next command starts executing (overlapping command) is indicated as an "Asynchronous command".
- *Reset values (*RST)*

Default parameter values that are used directly after resetting the instrument (*RST command) are indicated as "**RST" values, if available.

- *Default unit*
The default unit is used for numeric values if no other unit is provided with the parameter.
- *Manual operation*
If the result of a remote command can also be achieved in manual operation, a link to the description is inserted.

10.1.2 Long and short form

The keywords have a long and a short form. You can use either the long or the short form, but no other abbreviations of the keywords.

The short form is emphasized in uppercase letters. Note however, that this emphasis only serves the purpose to distinguish the short from the long form in the manual. For the instrument, the case does not matter.

Example:

SENSe:FREQuency:CENTer is the same as SENS:FREQ:CENT.

10.1.3 Numeric suffixes

Some keywords have a numeric suffix if the command can be applied to multiple instances of an object. In that case, the suffix selects a particular instance (e.g. a measurement window).

Numeric suffixes are indicated by angular brackets (<n>) next to the keyword.

If you do not quote a suffix for keywords that support one, a 1 is assumed.

Example:

DISPlay[:WINDow<1...4>]:ZOOM:STATe enables the zoom in a particular measurement window, selected by the suffix at WINDow.

DISPlay:WINDow4:ZOOM:STATe ON refers to window 4.

10.1.4 Optional keywords

Some keywords are optional and are only part of the syntax because of SCPI compliance. You can include them in the header or not.



If an optional keyword has a numeric suffix and you need to use the suffix, you have to include the optional keyword. Otherwise, the suffix of the missing keyword is assumed to be the value 1.

Optional keywords are emphasized with square brackets.

Example:

Without a numeric suffix in the optional keyword:

```
[SENSe:]FREQuency:CENTer is the same as FREQuency:CENTer
```

With a numeric suffix in the optional keyword:

```
DISPlay[:WINDow<1...4>]:ZOOM:STATe
```

DISPlay:ZOOM:STATe ON enables the zoom in window 1 (no suffix).

DISPlay:WINDow4:ZOOM:STATe ON enables the zoom in window 4.

10.1.5 Alternative keywords

A vertical stroke indicates alternatives for a specific keyword. You can use both keywords to the same effect.

Example:

```
[SENSe:]BANDwidth|BWIDth[:RESolution]
```

In the short form without optional keywords, BAND 1MHZ would have the same effect as BWID 1MHZ.

10.1.6 SCPI parameters

Many commands feature one or more parameters.

If a command supports more than one parameter, they are separated by a comma.

Example:

```
LAYout:ADD:WINDow Spectrum,LEFT,MTABLE
```

Parameters can have different forms of values.

- [Numeric values](#)..... 147
- [Boolean](#)..... 148
- [Character data](#)..... 149
- [Character strings](#)..... 149
- [Block data](#)..... 149

10.1.6.1 Numeric values

Numeric values can be entered in any form, i.e. with sign, decimal point or exponent. For physical quantities, you can also add the unit. If the unit is missing, the command uses the basic unit.

Example:

With unit: SENSe:FREQuency:CENTer 1GHZ

Without unit: SENSe:FREQuency:CENTer 1E9 would also set a frequency of 1 GHz.

Values exceeding the resolution of the instrument are rounded up or down.

If the number you have entered is not supported (e.g. for discrete steps), the command returns an error.

Instead of a number, you can also set numeric values with a text parameter in special cases.

- **MIN/MAX**
Defines the minimum or maximum numeric value that is supported.
- **DEF**
Defines the default value.
- **UP/DOWN**
Increases or decreases the numeric value by one step. The step size depends on the setting. Sometimes, you can customize the step size with a corresponding command.

Querying numeric values

When you query numeric values, the system returns a number. For physical quantities, it applies the basic unit (e.g. Hz for frequencies). The number of digits after the decimal point depends on the type of numeric value.

Example:

Setting: `SENSe:FREQuency:CENTer 1GHZ`

Query: `SENSe:FREQuency:CENTer?` would return `1E9`

Sometimes, numeric values are returned as text.

- **INF/NINF**
Infinity or negative infinity. Represents the numeric values `9.9E37` or `-9.9E37`.
- **NAN**
Not a number. Represents the numeric value `9.91E37`. NAN is returned if errors occur.

10.1.6.2 Boolean

Boolean parameters represent two states. The "on" state (logically true) is represented by "ON" or the numeric value 1. The "off" state (logically untrue) is represented by "OFF" or the numeric value 0.

Querying Boolean parameters

When you query Boolean parameters, the system returns either the value 1 ("ON") or the value 0 ("OFF").

Example:

Setting: `DISPlay:WINDow:ZOOM:STATe ON`

Query: `DISPlay:WINDow:ZOOM:STATe?` would return `1`

10.1.6.3 Character data

Character data follows the syntactic rules of keywords. You can enter text using a short or a long form. For more information, see [Section 10.1.2, "Long and short form"](#), on page 146.

Querying text parameters

When you query text parameters, the system returns its short form.

Example:

Setting: `SENSe:BANDwidth:RESolution:TYPE NORMal`

Query: `SENSe:BANDwidth:RESolution:TYPE?` would return `NORM`

10.1.6.4 Character strings

Strings are alphanumeric characters. They have to be in straight quotation marks. You can use a single quotation mark (') or a double quotation mark (").

Example:

`INSTRument:DELeTe 'Spectrum'`

10.1.6.5 Block data

Block data is a format which is suitable for the transmission of large amounts of data.

The ASCII character # introduces the data block. The next number indicates how many of the following digits describe the length of the data block. The data bytes follow. During the transmission of these data bytes, all end or other control signs are ignored until all bytes are transmitted. #0 specifies a data block of indefinite length. The use of the indefinite format requires an `NL^END` message to terminate the data block. This format is useful when the length of the transmission is not known or if speed or other considerations prevent segmentation of the data into blocks of definite length.

10.2 Common suffixes

In the I/Q Analyzer application, the following common suffixes are used in remote commands:

Table 10-1: Common suffixes used in remote commands in the I/Q Analyzer application

Suffix	Value range	Description
<m>	1 to 16	Marker
<n>	1 to 6	Window (in the currently selected channel)

Suffix	Value range	Description
<t>	1 to 6	Trace
	1 to 8	Limit line By default, the suffixes 1 to 8 correspond to the lines REM1 to REM8. If they are used but do not yet exist, they are created automatically. If a name has been set via <code>CALCulate:LIMit:NAME</code> , you can access the corresponding line using its index.

10.3 Activating I/Q Analyzer measurements

I/Q Analyzer measurements require a special channel on the R&S FSV/A. It can be activated using the common `INSTrument:CREate[:NEW]` or `INSTrument:CREate:REPLace` commands. In this case, some - but not all - parameters from the previously selected application are passed on to the I/Q Analyzer channel. In order to retain *all* relevant parameters from the current application for the I/Q measurement, use the `TRACe:IQ[:STATe]` command to change the application of the current channel.

A measurement is started immediately with the default settings when the channel is activated.



Different remote modes available

In remote control, two different modes for the I/Q Analyzer measurements are available:

- A quick mode for pure data acquisition
This mode is activated by default with the `TRACe:IQ[:STATe]` command. The evaluation functions are not available; however, performance is slightly improved.
- A more sophisticated mode for acquisition and analysis.
This mode is activated when a new channel is opened for the I/Q Analyzer application (`INST:CRE:NEW/ INST:CRE:REPL`) or by an additional command (see `TRACe:IQ:EVAL` on page 154).

Switching the data basis for measurement

For a description of remote commands required to perform measurements in the time and frequency domain, see the R&S FSV/A User Manual.

<code>INSTrument:CREate:DUPLicate</code>	151
<code>INSTrument:CREate[:NEW]</code>	151
<code>INSTrument:CREate:REPLace</code>	151
<code>INSTrument:DELeTe</code>	152
<code>INSTrument:LIST?</code>	152
<code>INSTrument:REName</code>	153
<code>INSTrument[:SELeCt]</code>	154

SYSTem:PRESet:CHANnel[:EXEC].....	154
TRACe:IQ:EVAL.....	154
TRACe:IQ[:STATe].....	155

INSTrument:CREate:DUPLicate

Duplicates the currently selected channel, i.e creates a new channel of the same type and with the identical measurement settings. The name of the new channel is the same as the copied channel, extended by a consecutive number (e.g. "IQAnalyzer" -> "IQAnalyzer 2").

The channel to be duplicated must be selected first using the `INST:SEL` command.

Example:

```
INST:SEL 'IQAnalyzer'
```

```
INST:CRE:DUPL
```

Duplicates the channel named 'IQAnalyzer' and creates a new channel named 'IQAnalyzer2'.

Usage: Event

INSTrument:CREate[:NEW] <ChannelType>, <ChannelName>

Adds a measurement channel. You can configure up to 10 measurement channels at the same time (depending on available memory).

Parameters:

<ChannelType> Channel type of the new channel.
For a list of available channel types, see [INSTrument:LIST?](#) on page 152.

<ChannelName> String containing the name of the channel.
Note that you cannot assign an existing channel name to a new channel. If you do, an error occurs.
Channel names can have a maximum of 31 characters, and must be compatible with the Windows conventions for file names. In particular, they must not contain special characters such as ":", "*", "?".

Example:

```
INST:CRE SAN, 'Spectrum 2'
```

Adds a spectrum display named "Spectrum 2".

INSTrument:CREate:REPLace <ChannelName1>, <ChannelType>, <ChannelName2>

Replaces a channel with another one.

Setting parameters:

<ChannelName1> String containing the name of the channel you want to replace.

<ChannelType> Channel type of the new channel.
For a list of available channel types, see [INSTrument:LIST?](#) on page 152.

<ChannelName2> String containing the name of the new channel.

Note: If the specified name for a new channel already exists, the default name, extended by a sequential number, is used for the new channel (see [INSTrument:LIST?](#) on page 152).

Channel names can have a maximum of 31 characters, and must be compatible with the Windows conventions for file names. In particular, they must not contain special characters such as ":", "*", "?".

Example:

```
INST:CRE:REPL 'IQAnalyzer2', IQ, 'IQAnalyzer'
```

Replaces the channel named "IQAnalyzer2" by a new channel of type "IQ Analyzer" named "IQAnalyzer".

Usage:

Setting only

INSTrument:DELeTe <ChannelName>

Deletes a channel.

Setting parameters:

<ChannelName> String containing the name of the channel you want to delete.
A channel must exist to delete it.

Usage:

Setting only

INSTrument:LIST?

Queries all active channels. The query is useful to obtain the names of the existing channels, which are required to replace or delete the channels.

Return values:

<ChannelType>, For each channel, the command returns the channel type and
<ChannelName> channel name (see tables below).
Tip: to change the channel name, use the [INSTrument:REName](#) command.

Example:

```
INST:LIST?
```

Result for 3 channels:

```
'ADEM', 'Analog Demod', 'IQ', 'IQ  
Analyzer', 'IQ', 'IQ Analyzer2'
```

Usage:

Query only

Table 10-2: Available channel types and default channel names

Application	<ChannelType> Parameter	Default Channel Name*)
Spectrum	SANALYZER	Spectrum
5G NR (R&S FSV3-K144)	NR5G	5G NR
3GPP FDD BTS (R&S FSV3-K72)	BWCD	3G FDD BTS
Note: the default channel name is also listed in the table. If the specified name for a new channel already exists, the default name, extended by a sequential number, is used for the new channel.		

Application	<ChannelType> Parameter	Default Channel Name*)
3GPP FDD UE (R&S FSV3-K73)	MWCD	3G FDD UE
Amplifier Measurements (R&S FSV3-K18)	AMPLifier	Amplifier
AM/FM/PM Modulation Analysis	ADEM	Analog Demod
Bluetooth (R&S FSV3-K8)	BTO	Bluetooth
GSM (R&S FSV3-K10)	GSM	GSM
I/Q Analyzer	IQ	IQ Analyzer
LTE (R&S FSV3-K10x)	LTE	LTE
NB-IoT (R&S FSV3-K106)	NIOT	NB-IoT
Noise Figure Measurements	NOISE	Noise
OFDM VSA (R&S FSV3-K96)	OFDMVSA	OFDM VSA
Phase Noise (R&S FSV3-K40)	PNOISE	Phase Noise
Pulse (R&S FSV3-K6)	PULSE	Pulse
Vector Signal Analysis (VSA, R&S FSV3-K70)	DDEM	VSA
WLAN (R&S FSV3-K91)	WLAN	WLAN
Note: the default channel name is also listed in the table. If the specified name for a new channel already exists, the default name, extended by a sequential number, is used for the new channel.		

INSTrument:REName <ChannelName1>, <ChannelName2>

Renames a channel.

Setting parameters:

<ChannelName1> String containing the name of the channel you want to rename.

<ChannelName2> String containing the new channel name.
 Note that you cannot assign an existing channel name to a new channel. If you do, an error occurs.
 Channel names can have a maximum of 31 characters, and must be compatible with the Windows conventions for file names. In particular, they must not contain special characters such as ":", "*", "?".

Example:

```
INST:REN 'IQAnalyzer2','IQAnalyzer3'
```

Renames the channel with the name 'IQAnalyzer2' to 'IQAnalyzer3'.

Usage:

Setting only

INSTrument[:SElect] <ChannelType> | <ChannelName>

Activates a new channel with the defined channel type, or selects an existing channel with the specified name.

Also see

- [INSTrument:CREate\[:NEW\]](#) on page 151

Parameters:

<ChannelType> Channel type of the new channel.
For a list of available channel types see [INSTrument:LIST?](#) on page 152.

<ChannelName> String containing the name of the channel.

Example:
 INST IQ
 INST 'MyIQSpectrum'
 Selects the channel named 'MyIQSpectrum' (for example before executing further commands for that channel).

SYSTem:PRESet:CHANnel[:EXEC]

Restores the default instrument settings in the current channel.

Use `INST:SEL` to select the channel.

Example:
 INST:SEL 'Spectrum2'
 Selects the channel for "Spectrum2".
 SYST:PRESet:CHAN:EXEC
 Restores the factory default settings to the "Spectrum2" channel.

Usage: Event

Manual operation: See ["Preset Channel"](#) on page 43

TRACe:IQ:EVAL <State>

Turns I/Q data analysis on and off.

Before you can use this command, you have to turn on the I/Q data acquisition using `INST:CRE:NEW IQ` or `INST:CRE:REPL`, or using the [TRACe:IQ\[:STATe\]](#) command to replace the current channel while retaining the settings.

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
 Switches the function off
ON | 1
 Switches the function on

Example: `TRAC:IQ ON`
 Enables I/Q data acquisition
 `TRAC:IQ:EVAL ON`
 Enables the I/Q data analysis mode.

TRACe:IQ[:STATe] <State>

Activates the simple I/Q data acquisition mode (see ["Different remote modes available"](#) on page 150).

Executing this command also has the following effects:

- The sweep, amplitude, input and trigger settings from the measurement are retained.
- All measurements are turned off.
- All traces are set to "Blank" mode.
- The I/Q data analysis mode is turned off (`TRAC:IQ:EVAL OFF`).

Note: To turn trace display back on or to enable the evaluation functions of the I/Q Analyzer, execute the `TRAC:IQ:EVAL ON` command (see [TRACe:IQ:EVAL](#) on page 154).

Parameters:

<State> `ON | OFF | 0 | 1`
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on

Example: `TRAC:IQ ON`
 Switches on I/Q data acquisition

10.4 Configuring I/Q Analyzer measurements

The following commands configure the I/Q Analyzer measurements.

- [Configuring the data input and output](#)..... 155
- [Configuring the vertical axis \(amplitude, scaling\)](#)..... 225
- [Frequency](#)..... 235
- [Triggering](#)..... 237
- [Configuring data acquisition](#)..... 247
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10.4.1 Configuring the data input and output

The following commands are required to configure data input and output.

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• Input from I/Q data files.....	159
• Remote commands for external frontend control.....	161
• Configuring input via the optional Analog Baseband interface.....	178
• Using external mixers.....	183
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10.4.1.1 RF input

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INPut:IMPedance.....	157
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INPut<ip>:SELEct.....	158

INPut:ATTenuation:PROTection:RESet

Resets the attenuator and reconnects the RF input with the input mixer for the R&S FSV/A after an overload condition occurred and the protection mechanism intervened. The error status bit (bit 3 in the `STAT:QUES:POW` status register) and the `INPUT OVLD` message in the status bar are cleared.

The command works only if the overload condition has been eliminated first.

Example: `INP:ATT:PROT:RES`

INPut:CONNector <ConnType>

Determines which connector the input for the measurement is taken from.

If an external frontend is active, the connector is automatically set to `RF`.

Parameters:

<ConnType>	RF
	RF input connector
	RFProbe
	Active RF probe
	*RST: RF

Example: `INP:CONN RF`
Selects input from the RF input connector.

Manual operation: See "[Input Connector](#)" on page 50

INPut:COUPling <CouplingType>

Selects the coupling type of the RF input.

If an external frontend is active, the coupling is automatically set to AC.

Parameters:

<CouplingType> AC | DC

AC
AC coupling

DC
DC coupling

*RST: AC

Example: INP:COUP DC

Manual operation: See ["Input Coupling"](#) on page 48

INPut:DPATH <DirectPath>

Enables or disables the use of the direct path for frequencies close to 0 Hz.

If an external frontend is active, the direct path is always used.

Parameters:

<DirectPath> AUTO | OFF

AUTO | 1
(Default) the direct path is used automatically for frequencies close to 0 Hz.

OFF | 0
The analog mixer path is always used.

Example: INP:DPAT OFF

Manual operation: See ["Direct Path"](#) on page 49

INPut:FILTer:YIG[:STATE] <State>

Enables or disables the YIG filter.

Parameters:

<State> ON | OFF | 0 | 1

Example: INP:FILT:YIG OFF
Deactivates the YIG-preselector.

Manual operation: See ["YIG-Preselector"](#) on page 50

INPut:IMPedance <Impedance>

Selects the nominal input impedance of the RF input. In some applications, only 50 Ω are supported.

Parameters:

<Impedance>

50 | 75

numeric value

User-defined impedance from 50 Ohm to 100000000 Ohm
(=100 MOhm)

User-defined values are only available for the Spectrum application, the I/Q Analyzer, and some optional applications.

*RST: 50 Ω

Default unit: OHM

Example:

INP:IMP 75

Manual operation:See "[Impedance](#)" on page 49See "[Unit](#)" on page 59**INPut:IMPedance:PTYPe <PadType>**

Defines the type of matching pad used for impedance conversion for RF input.

Parameters:

<PadType>

SRESistor | MLPad

SRESistor

Series-R

MLPad

Minimum Loss Pad

*RST: SRESistor

Example:

INP:IMP 100

INP:IMP:PTYP MLP

Manual operation:See "[Impedance](#)" on page 49**INPut<ip>:SElect <Source>**

Selects the signal source for measurements, i.e. it defines which connector is used to input data to the R&S FSV/A.

If no additional input options are installed, only RF input or file input is supported.

Suffix:

<ip>

1..n

Parameters:

<Source>

RF

Radio Frequency ("RF INPUT" connector)

FIQ

I/Q data file

(selected by [INPut:FILE:PATH](#) on page 159)

AIQ

Analog Baseband signal (only available with optional "Analog Baseband" interface)

For details on Analog Baseband input, see [Section 4.3, "Processing data from the Analog Baseband interface"](#), on page 27.

*RST: RF

Manual operation: See ["Radio Frequency State"](#) on page 48
 See ["I/Q Input File State"](#) on page 51
 See ["Analog Baseband Input State"](#) on page 53

10.4.1.2 Input from I/Q data files

The input for measurements can be provided from I/Q data files. The commands required to configure the use of such files are described here.

For details see [Section 4.2, "Basics on input from I/Q data files"](#), on page 25.

Useful commands for retrieving results described elsewhere:

- [INPut<ip>:SElect](#) on page 158

Remote commands exclusive to input from I/Q data files:

INPut:FILE:PATH	159
MMEMory:LOAD:IQ:STReam	160
MMEMory:LOAD:IQ:STReam:AUTO	160
MMEMory:LOAD:IQ:STReam:LIST?	161
TRACe:IQ:FILE:REPetition:COUNt	161

INPut:FILE:PATH <FileName>[, <AnalysisBW>]

Selects the I/Q data file to be used as input for further measurements.

The I/Q data file must be in one of the following supported formats:

- .iq.tar
- .iqw
- .csv
- .mat
- .wv
- .aid

Only a single data stream or channel can be used as input, even if multiple streams or channels are stored in the file.

For some file formats that do not provide the sample rate and measurement time or record length, you must define these parameters manually. Otherwise the traces are not visible in the result displays.

For details, see [Table C-1](#).

Parameters:

- <FileName>** String containing the path and name of the source file.
The file type is determined by the file extension. If no file extension is provided, the file type is assumed to be `.iq.tar`. For `.mat` files, Matlab® v4 is assumed.
- <AnalysisBW>** Optionally: The analysis bandwidth to be used by the measurement. The bandwidth must be smaller than or equal to the bandwidth of the data that was stored in the file.
Default unit: HZ

Example:

```
INP:FILE:PATH 'C:\R_S\Instr\user\data.iq.tar'
```

Uses I/Q data from the specified file as input.

Example:

```
//Load an IQW file
INP:FILE:PATH 'C:\R_S\Instr\user\data.iqw'
INP:SEL FIQ

//Define the sample rate
TRAC:IQ:SRAT 10MHz
//Define the measurement time
SENSe:SWEp:TIME 0.001001
//Start the measurement
INIT:IMM
```

Manual operation: See ["Select I/Q data file"](#) on page 51

MMEMory:LOAD:IQ:STReam <Channel>

Only available for files that contain more than one data stream from multiple channels: selects the data stream to be used as input for the currently selected channel.

Automatic mode (`MMEMory:LOAD:IQ:STReam:AUTO`) is set to OFF.

Parameters:

- <Channel>** String containing the channel name.

Example:

```
MMEM:LOAD:IQ:STR?
//Result: 'Channel1','Channel2'
MMEM:LOAD:IQ:STR 'Channel2'
```

Manual operation: See ["Selected Channel"](#) on page 52

MMEMory:LOAD:IQ:STReam:AUTO <State>

Only available for files that contain more than one data stream from multiple channels: automatically defines which data stream in the file is used as input for the channel.

Parameters:

- <State>** ON | OFF | 0 | 1
OFF | 0
The data stream specified by `MMEMory:LOAD:IQ:STReam` is used as input for the channel.

ON | 1

The first data stream in the file is used as input for the channel. Applications that support multiple data streams use the first data stream in the file for the first input stream, the second for the second stream etc.

*RST: 1

Manual operation: See ["Selected Channel"](#) on page 52

MMEMory:LOAD:IQ:STReam:LIST?

Returns the available channels in the currently loaded input file.

Example: `MMEM:LOAD:IQ:STR?`
`//Result: 'Channel1','Channel2'`

Usage: Query only

Manual operation: See ["Selected Channel"](#) on page 52

TRACe:IQ:FILE:REPetition:COUNt <RepetitionCount>

Determines how often the data stream is repeatedly copied in the I/Q data memory. If the available memory is not sufficient for the specified number of repetitions, the largest possible number of complete data streams is used.

Parameters:

<RepetitionCount> integer

Example: `TRAC:IQ:FILE:REP:COUN 3`

Manual operation: See ["File Repetitions"](#) on page 52

10.4.1.3 Remote commands for external frontend control

The following commands are available and required only if the external frontend control option (R&S FSV3-K553) is installed.

Further commands for external frontend control described elsewhere:

- `INPut:SElect RF`; see [INPut<ip>:SElect](#) on page 158
- [\[SENSe:\]FREQuency:CENter](#) on page 235
- [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:Y\[:SCALe\]:RLEVel](#) on page 226
- [INPut:ATTenuation:AUTO](#) on page 228
- [INPut:ATTenuation](#) on page 227
- [INPut:SANalyzer:ATTenuation:AUTO](#) on page 230
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Commands for initial configuration

The following commands are required when you initially set up an external frontend.

[SENSe:]EFRontend:CONNection[:STATe].....	162
[SENSe:]EFRontend:CONNection:CONFig.....	163
[SENSe:]EFRontend:CONNection:CSTate?.....	163
[SENSe:]EFRontend:FREQuency:BAND:COUNT?.....	164
[SENSe:]EFRontend:FREQuency:BAND:LOWer?.....	164
[SENSe:]EFRontend:FREQuency:BAND:UPPer?.....	164
[SENSe:]EFRontend:FREQuency:BCONfig:AUTO.....	165
[SENSe:]EFRontend:FREQuency:BCONfig:LIST?.....	165
[SENSe:]EFRontend:FREQuency:BCONfig:SElect.....	165
[SENSe:]EFRontend:FREQuency:IFRequency:MAXimum?.....	166
[SENSe:]EFRontend:FREQuency:IFRequency:MINimum?.....	166
[SENSe:]EFRontend:FREQuency:IFRequency:SIDeband?.....	166
[SENSe:]EFRontend:FREQuency:IFRequency[:VALue]?.....	166
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[SENSe:]EFRontend:FREQuency:LOSCillator:MODE.....	167
[SENSe:]EFRontend:FREQuency:LOSCillator:OUTPut:FREQuency?.....	168
[SENSe:]EFRontend:FREQuency:LOSCillator:OUTPut:STATe.....	168
[SENSe:]EFRontend:FREQuency:REFerence.....	168
[SENSe:]EFRontend:FREQuency:REFerence:LIST?.....	169
[SENSe:]EFRontend:IDN?.....	169
[SENSe:]EFRontend:NETWork.....	169
[SENSe:]EFRontend[:STATe].....	170

[SENSe:]EFRontend:CONNection[:STATe] <State>

Queries the external frontend connection state in the firmware.

Note: to query the physical connection state of the external frontend, use [SENSe:]EFRontend:CONNection:CSTate? on page 163.

Parameters:

<State>

ON | OFF | 0 | 1

OFF | 0

The connection to the frontend is deactivated temporarily. The frontend is thus available for use elsewhere, for example by a signal generator. The measurement settings on the R&S FSV/A remain untouched.

ON | 1

Frontend connection enabled.

The frontend is reserved for exclusive use by the R&S FSV/A.

*RST: 0

Example:

```
//Global activation of external frontend
EFR ON
//Configure frontend
EFR:CONN:CONF "FE44S","123.456.789"
//Activate exclusive use of frontend by
R&S FSV/A.
EFR:CONN ON
```

[SENSe:]EFRontend:CONNection:CONFig <Type>, <IPAddress>[, <DeviceID>[, <SymbolicName>]]

Configures the connection to the external frontend.

Parameters:

<Type>	"FE44S" "FE50DTR"
	String in double quotes containing the type of frontend to be connected.
<IPAddress>	string in double quotes
	The IP address or computer name of the frontend connected to the R&S FSV/A via LAN. The IP address and computer name are indicated on the electronic ink display on the side panel of the frontend.
<DeviceID>	string in double quotes
	Unique device ID consisting of <type>-<serialnumber> Not required or relevant for the R&S FSV/A.
<SymbolicName>	string in double quotes
	Symbolic name of the external frontend. Not required or relevant for the R&S FSV/A.

Example:

```
//Global activation of external frontend
EFR ON
//Configure frontend
EFR:CONN:CONF "FE44S","123.456.789"
//Activate exclusive use of frontend by
R&S FSV/A.
EFR:CONN ON
```

[SENSe:]EFRontend:CONNection:CState?

Queries the status of the physical connection to the external frontend.

Return values:

<State>	ON OFF 0 1
	OFF 0
	Frontend not connected; connection error
	ON 1
	Frontend connected

Usage: Query only

[SENSe:]EFRontend:FREQuency:BAND:COUNT?

Queries the number of frequency bands provided by the selected frontend.

Return values:

<NoBands> integer
Number of frequency bands

Example: `//Query number of frequency bands
EFR:FREQ:BAND:COUN?
//Result: 2`

Usage: Query only

[SENSe:]EFRontend:FREQuency:BAND:LOWer?

Queries the start of the frequency range supported by the selected frontend frequency band.

Suffix:

 1..n
Band for multi-band frontends
Use [SENSe:]EFRontend:FREQuency:BAND:COUNT?
on page 164 to determine the number of available bands.

Return values:

<StartFreq> Start frequency of the specified band

Example: `//Query start frequency of second band
EFR:FREQ:BAND2:LOW?
//Result: 240000000000`

Usage: Query only

[SENSe:]EFRontend:FREQuency:BAND:UPPer?

Queries the end of the frequency range supported by the selected frontend frequency band.

Suffix:

 1..n
Band for multi-band frontends
Use [SENSe:]EFRontend:FREQuency:BAND:COUNT?
on page 164 to determine the number of available bands.

Return values:

<StopFreq> End frequency of the specified band

Example: `//Query end frequency of second band
EFR:FREQ:BAND2:UPP?
//Result: 440000000000`

Usage: Query only

[SENSe:]EFRontend:FREQUENCY:BCONfig:AUTO <State>

Determines whether the frequency band of the external frontend is configured automatically or manually.

Parameters:

<State>

ON | OFF | 0 | 1

OFF | 0

Uses the frequency band configured by [SENSe:]EFRontend:FREQUENCY:BCONfig:SElect on page 165.

ON | 1

Configures the frequency band automatically

For bandwidths ≤ 400 MHz, "IF Low" is used. For bandwidths larger than 400 MHz, "IF High" is used.

*RST: 1

Example:

//Configures the use of the IF high band manually.

EFR:FREQ:BCON:AUTO 0

EFR:FREQ:BCON:SEL "IF HIGH"

[SENSe:]EFRontend:FREQUENCY:BCONfig:LIST?

Returns the intermediate frequency (output) range of the external frontend.

Return values:

<BandConfigs>

string

"IF LOW"

A higher intermediate frequency is used on the external frontend, resulting in a higher input frequency at the R&S FSV/A.

"IF HIGH"

A lower intermediate frequency is used on the external frontend, resulting in a lower input frequency at the R&S FSV/A.

Example:

EFR:FREQ:BCON:LIST?

//Result: "IF HIGH", "IF LOW"

EFR:FREQ:BCON:SEL "IF HIGH"

Usage:

Query only

[SENSe:]EFRontend:FREQUENCY:BCONfig:SElect <BandConfig>

Defines the intermediate frequency (output) range of the external frontend.

Parameters:

<BandConfig>

"IF HIGH"

(R&S FE44S/ R&S FE50DTR)

A higher intermediate frequency is used on the external frontend, resulting in a higher input frequency at the R&S FSV/A.

"IF LOW"

(R&S FE44S/ R&S FE50DTR)

A lower intermediate frequency is used on the external frontend, resulting in a lower input frequency at the R&S FSV/A.

Example:

```
EFR:FREQ:BCON:LIST?
//Result: "IF HIGH", "IF LOW"
EFR:FREQ:BCON:SEL "IF HIGH"
```

[SENSe:]EFRontend:FREQuency:IFRequency:MAXimum?

Queries the maximum used intermediate frequency (IF) for frequency conversion for a frequency span.

Return values:

<IFFrequency> numeric

Example:

```
EFR:FREQ:IFR:MAX?
```

Usage:

Query only

[SENSe:]EFRontend:FREQuency:IFRequency:MINimum?

Queries the minimum used intermediate frequency (IF) for frequency conversion for a frequency span.

Return values:

<IFFrequency> numeric

Example:

```
EFR:FREQ:IFR:MIN?
```

Usage:

Query only

[SENSe:]EFRontend:FREQuency:IFRequency:SIDeband?

Queries the currently used sideband for frequency conversion.

Return values:

<Sideband> "USB" | "LSB"
"USB"
 Upper sideband
"LSB"
 Lower sideband

Example:

```
EFR:FREQ:IFR?
EFR:FREQ:IFR:SID?
```

Usage:

Query only

[SENSe:]EFRontend:FREQuency:IFRequency[:VALue]?

Queries the currently used intermediate frequency (IF) for frequency conversion.

To query the maximum and minimum IF for the selected frequency range, use `[SENSe:]EFRontend:FREQUENCY:IFrequency:MAXimum?` on page 166 and `[SENSe:]EFRontend:FREQUENCY:IFrequency:MINimum?` on page 166.

Return values:

<IFFrequency> numeric

Example: EFR:FREQ:IFR?

Usage: Query only

[SENSe:]EFRontend:FREQUENCY:LOSCillator:INPut:FREQUENCY?

Queries the frequency of the LO input for `[SENSe:]EFRontend:FREQUENCY:LOSCillator:MODE EXT`.

Return values:

<LOInFreq> Default unit: Hz

Example: The external frontend uses the external LO provided at the "LO IN" connector.

EFR:FREQ:LOSC:MODE EXT

Query the frequency that the external LO must be provided at.

EFR:FREQ:LOSC:INP:FREQ?

//Result: 10615000000

Usage: Query only

[SENSe:]EFRontend:FREQUENCY:LOSCillator:MODE <Type>

Determines whether the external frontend uses its internal LO or an external LO.

Parameters:

<Type> EXternal | Internal

EXternal

Uses the external LO provided at the LO input connector of the external frontend. Query the frequency at which the LO must be input to the external frontend using `[SENSe:]EFRontend:FREQUENCY:LOSCillator:INPut:FREQUENCY?` on page 167.

Internal

Uses the internal LO.

*RST: EXternal

Example: EFR:FREQ:LOSC:MODE EXT
EFR:FREQ:LOSC:INP:FREQ?
//Result: 10615000000

[SENSe:]EFRontend:FREQUENCY:LOSCillator:OUTPut:FREQUENCY?

Queries the frequency of the LO output for `[SENSe:]EFRontend:FREQUENCY:LOSCillator:OUTPut:STATe ON`.

Return values:

<LOOutFreq> Default unit: Hz

Example:

The external frontend uses the internal LO and provides it as output to the "LO OUT" connector.

```
EFR:FREQ:LOSC:MODE INT
```

```
EFR:FREQ:LOSC:OUTP:STAT ON
```

Query the frequency of the LO output.

```
EFR:FREQ:LOSC:OUTP:FREQ?
```

```
//Result: 10615000000
```

Usage:

Query only

[SENSe:]EFRontend:FREQUENCY:LOSCillator:OUTPut:STATe <State>

Enables or disables output of the LO by the external frontend. The output frequency is returned by `[SENSe:]EFRontend:FREQUENCY:LOSCillator:OUTPut:FREQUENCY?` on page 168.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

```
*RST: 0
```

Example:

The external frontend provides the LO as output at the "LO OUT" connector.

```
EFR:FREQ:LOSC:OUTP:STAT ON
```

Query the frequency of the LO output.

```
EFR:FREQ:LOSC:OUTP:FREQ?
```

```
//Result: 10615000000
```

[SENSe:]EFRontend:FREQUENCY:REFerence <Frequency>

Sets the reference frequency that is used for frequency conversion on the frontend. Depending on the connected type of frontend, different values are available. To determine which reference levels are available, use `[SENSe:]EFRontend:FREQUENCY:REFerence:LIST?` on page 169.

Parameters:

<Frequency> Default unit: HZ

Example:

```
//Query the available reference levels
EFR:FREQ:REF:LIST?
//Result: 100000000,640000000,1000000000
//Use 640 MHz reference
EFR:FREQ:REF 640000000
```

[SENSe:]EFRontend:FREQuency:REference:LIST?

Queries the available reference signals for the connected frontend type.

Return values:

<References> 100000000 | 640000000 | 1000000000

Example:

```
//Query the available reference levels
EFR:FREQ:REF:LIST?
//Result: 100000000,640000000,1000000000
//Use 640 MHz reference
EFR:FREQ:REF 640000000
```

Usage: Query only

[SENSe:]EFRontend:IDN?

Queries the device identification information (*IDN?) of the frontend.

Return values:

<DevInfo> string without quotes
 Rohde&Schwarz,<device type>,<part number>/<serial number>,<firmware version>

Example:

```
EFR:IDN?
//Result:
Rohde&Schwarz,FE44S,1234.5678K00/123456,0.8.0
```

Usage: Query only

[SENSe:]EFRontend:NETWork <IPAddress>, <Subnet>, <DHCP State>

Sets or queries the network information for the frontend.

This information is also indicated on the electronic paper display on the side panel of the device.

Beware that if you change the network setting to DHCP = ON, the connection is aborted and you must re-establish a connection to the frontend (see [\[SENSe:\]EFRontend:CONNection\[:STATe\]](#) on page 162).

Parameters:

<IPAddress> string in double quotes
 IP address of the frontend

<Subnet>	string in double quotes Subnet mask of the frontend
<DHCP State>	ON OFF 0 1 Indicates whether a DHCP server is used. OFF 0 DHCP off ON 1 DHCP on *RST: 0

Example: EFR:NETW?
//Result: "123.456.78.90", "255.255.255.0", ON

[SENSe:]EFRontend[:STATe] <State>

Enables or disables the general use of an external frontend for the application.

Parameters:

<State>	ON OFF 0 1 OFF 0 The frontend is disconnected. The application adapts the measurement settings to the common settings supported by the R&S FSV/A. ON 1 The R&S FSV/A allows you to configure and connect an external frontend for the application. The application adapts the available measurement settings to the connected frontend. The channel bar indicates "Inp: ExtFe". *RST: 0
---------	--

Example: EFR ON

Commands for test, alignment, and diagnosis

The following commands are required to test and optimize the connection after it has initially been set up.

[SENSe:]EFRontend:ALIGnment<ch>:FILE.....	170
[SENSe:]EFRontend:ALIGnment<ch>:STATe.....	171
[SENSe:]EFRontend:FWUPdate.....	171
[SENSe:]EFRontend<fe>:SELFtest?.....	171
[SENSe:]EFRontend<fe>:SELFtest:RESult?.....	172

[SENSe:]EFRontend:ALIGnment<ch>:FILE <File>

Selects or queries the touchstone file that contains correction data to compensate for signal losses in the cable occurring at different IF signal frequencies.

Suffix:

<ch> 1..n
Currently irrelevant

Parameters:

<File> string in double quotes
Path and file name of the correction data file. The file must be in s2p format.
If the specified file is not found or does not have the correct format, an error message is returned (-256, "File name not found", -150, "String data error").

Example:

EFR:ALIG:FILE "FE44S.s2p"

[SENSe:]EFRontend:ALIGNment<ch>:STATe <State>

Activates correction of the IF signal due to cable loss from the frontend to the analyzer. Specify the file with correction data using [SENSe:]EFRontend:ALIGNment<ch>:FILE on page 170.

Suffix:

<ch> 1..n
Currently irrelevant

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
Switches the function off
ON | 1
Switches the function on
*RST: 0

[SENSe:]EFRontend:FWUpdate

Updates the firmware on the external frontend. Note that this process can take some time.

Usage: Event

[SENSe:]EFRontend<fe>:SELFtest?

Performs a selftest on the frontend to compare the current performance and characteristic values with the specified values for the frontend.

As a result, the success is returned.

Suffix:
 <fe> 1
 Connected frontend

Return values:
 <Result> 0
 No error
 >0
 Error
 *RST: 0

Example: EFR:SELF?
 //Result: 0

Usage: Query only

[SENSe:]EFRontend<fe>:SELFtest:RESult?

Queries the results of the selftest on the frontend.

Suffix:
 <fe> 1
 Connected frontend

Return values:
 <Result> string containing xml data in double quotes
 For details, see ["Format of selftest result file"](#) on page 172.

Example: EFR:SELF:RES?
 For details, see ["Format of selftest result file"](#) on page 172.

Usage: Query only

Format of selftest result file

As a result of the selftest, an XML file with test details is created. It contains the following information in the specified order. Mandatory elements and attributes are indicated in bold font.

Element	Attributes	Description
<Sequence>		Main element
	Name	= "Selftest"
	Description	Optional description of the test process
	FirmwareVersion	Firmware version of the controlling instrument (R&S FSV/A)
	FrontendLibrary-Version	Version of the <code>control.dll</code> with the format <code>x.y.z</code>
	FrontendServerVersion	Version of the RRH server with the format <code>x.y.z</code> (FE44A only)
	Date	Date the selftest was performed, with the format <code>dd/mm/yyyy</code>

Element	Attributes	Description
	Time	Time the selftest was performed, with the format hh:mm:ss
	State	Test result state, combined result of all <SequenceCategory>s: "PASSED"/ "FAILED"
	Version	For internal use.
	Comment	Optional comment on the test process
<SequenceCategory>		Set of test steps
	Name	Name of the test sequence, e.g. "Frontend voltages"
	Description	Optional description of the test sequence
	State	Test sequence result state, combined result of all <SequenceStep>s: "PASSED"/ "FAILED"
	Version	For internal use.
	Comment	Optional comment on the test sequence
<SequenceStep>		Subelement of <SequenceCategory> for an individual test step
	Name	Name of the individual test step, e.g. FE1_3V3
	Description	Optional description of the test step
	LimitLow	Optional: lower limit to be checked
	LimitHigh	Optional: upper limit to be checked
	MeasValue	Optional: measured value
	Unit	Optional: unit of the measured value
	State	Test step result state: "PASSED"/ "FAILED"
	Version	For internal use.
	Comment	Optional comment on the test step

Example for selftest result xml file

```

<?xml version="1.0" encoding="UTF-8"?>
<Sequence xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:noNamespaceSchemaLocation="SelfTest_Schema.xsd">
  <Name>DeviceCheck</Name>
  <FirmwareVersion>1.50-21.1.1.0 Beta</FirmwareVersion>
  <FrontendLibraryVersion>0.8.0</FrontendLibraryVersion>
  <Date>15/01/2021</Date>
  <Time>08:51:35</Time>
  <State>FAILED</State>
  <Version>1.0.0</Version>
  <SequenceCategory>
    <Name>Frontend Voltages</Name>
    <Description>test description</Description>
    <State>FAILED</State>
  
```

```

<Version>1.0.0</Version>
<Type>Diagnose</Type>
<SequenceStep>
  <Name>FE1_3V3</Name>
  <LimitLow>3.000</LimitLow>
  <LimitHigh>3.600</LimitHigh>
  <MeasValue>3.311</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>FE1_5V</Name>
  <LimitLow>4.250</LimitLow>
  <LimitHigh>5.750</LimitHigh>
  <MeasValue>5.027</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>FE1_6V5</Name>
  <LimitLow>6.000</LimitLow>
  <LimitHigh>7.000</LimitHigh>
  <MeasValue>5.893</MeasValue>
  <State>FAILED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>FE1_M3V3</Name>
  <LimitLow>-3.600</LimitLow>
  <LimitHigh>-3.000</LimitHigh>
  <MeasValue>3.347</MeasValue>
  <State>FAILED</State>
  <Version>1.0.0</Version>
</SequenceStep>
</SequenceCategory>
<SequenceCategory>
  <Name>Frontend Temperature</Name>
  <Description>test description</Description>
  <State>PASSED</State>
  <Version>1.0.0</Version>
  <Type>Diagnose</Type>
  <SequenceStep>
    <Name>TEMP_FE1</Name>
    <LimitLow>0.000</LimitLow>
    <LimitHigh>60.000</LimitHigh>
    <MeasValue>39.300</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
</SequenceCategory>

```

```

<SequenceCategory>
  <Name>Synthesizer Voltage</Name>
  <Description>test description</Description>
  <State>PASSED</State>
  <Version>1.0.0</Version>
  <Type>Diagnose</Type>
  <SequenceStep>
    <Name>SYNTH_3V4</Name>
    <LimitLow>3.200</LimitLow>
    <LimitHigh>3.910</LimitHigh>
    <MeasValue>3.576</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
  <SequenceStep>
    <Name>SYNTH_5V4_SYN</Name>
    <LimitLow>4.860</LimitLow>
    <LimitHigh>5.940</LimitHigh>
    <MeasValue>5.405</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
  <SequenceStep>
    <Name>SYNTH_5V4_YIG</Name>
    <LimitLow>4.860</LimitLow>
    <LimitHigh>5.940</LimitHigh>
    <MeasValue>5.438</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
  <SequenceStep>
    <Name>SYNTH_M5V</Name>
    <LimitLow>-5.500</LimitLow>
    <LimitHigh>-4.500</LimitHigh>
    <MeasValue>-4.948</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
  <SequenceStep>
    <Name>SYNTH_REF5V</Name>
    <LimitLow>4.500</LimitLow>
    <LimitHigh>5.500</LimitHigh>
    <MeasValue>5.031</MeasValue>
    <State>PASSED</State>
    <Version>1.0.0</Version>
  </SequenceStep>
</SequenceCategory>
<SequenceCategory>
  <Name>Supply Voltage</Name>
  <Description>test description</Description>

```

```

<State>PASSED</State>
<Version>1.0.0</Version>
<Type>Diagnose</Type>
<SequenceStep>
  <Name>SUPPLY_12V</Name>
  <LimitLow>10.800</LimitLow>
  <LimitHigh>13.200</LimitHigh>
  <MeasValue>11.909</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>SUPPLY_3V3D</Name>
  <LimitLow>2.970</LimitLow>
  <LimitHigh>3.630</LimitHigh>
  <MeasValue>3.318</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>SUPPLY_4V</Name>
  <LimitLow>3.650</LimitLow>
  <LimitHigh>4.460</LimitHigh>
  <MeasValue>4.053</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>SUPPLY_6V</Name>
  <LimitLow>5.400</LimitLow>
  <LimitHigh>6.600</LimitHigh>
  <MeasValue>6.076</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
<SequenceStep>
  <Name>SUPPLY_M5V</Name>
  <LimitLow>-6.050</LimitLow>
  <LimitHigh>-4.950</LimitHigh>
  <MeasValue>-5.507</MeasValue>
  <State>PASSED</State>
  <Version>1.0.0</Version>
</SequenceStep>
</SequenceCategory>
</Sequence>

```

Programming example: Configuring an external frontend

The following example describes how to configure RF frontend settings in remote operation.


```
// Prepare the instrument
// Preset
*RST
// Create new IQ-Analyzer channel
:INST:SEL IQ
// Enable 640MHz Reference
:ROSC:O640 ON

//Enable general use of external frontend
SENSe:EFRontend:STATe ON
//Configure connection to ext. FE named "FE44S-1000826"
SENSe:EFRontend:CONNection:CONFig "FE44S","FE44S-1000826"
//Activate exclusive use of frontend by R&S FSV/A.
SENSe:EFRontend:CONNection:STATe ON

//For demonstration purposes only: assign a static IP address
SENSe:EFRontend:NETWork "123.456.7.8", "255.255.255.00", OFF

// Query information about the connected RF frontend.
SENSe:EFRontend:CONNection:CSTATe?
// Response: 1 (connected)
SENSe:EFRontend:IDN?
////Result: Rohde&Schwarz,FE44S,1234.5678K00/123456,0.8.0

// Specify frontend settings
//Query available intermediate frequency bands
SENSe:EFRontend:FREQuency:BCONfig:LIST?
//Result: "IF HIGH", "IF LOW"
//Use high IF
SENSe:EFRontend:FREQuency:BCONfig:SElect "IF HIGH"
//Query used intermediate frequency
SENSe:EFRontend:FREQuency:IFRequency?
//Result: 8.595000000

//Query available reference frequencies
SENSe:EFRontend:FREQuency:REFeRence:LIST?
//Result: 10000000,640000000,1000000000
//Use 640 MHz reference
SENSe:EFRontend:FREQuency:REFeRence 640000000

// Query ranges of the operating frequency band.
SENSe:EFRontend:FREQuency:BAND1:LOWer?
// Response in Hz: "24000000000" (= 24 GHz)
SENSe:EFRontend:FREQuency:BAND1:UPPer?
// Response in Hz: "44000000000" (= 44 GHz)

// Add cable correction data by loading an *.s2p file.
SENSe:EFRontend:ALIGnment:FILE "C:\R_S\Instr\user\external_frontends\FE44S\
touchstonefiles\if_default_cable_1347_7552_00.s2p"
SENSe:EFRontend:ALIGnment:STATe ON
```

```
//Update FW version on frontend (only available if external frontend firmware
//is incompatible to R&S FSV/A firmware)
SENSe:EFRontend:FWUpdate
//Perform a selftest on the frontend and query results
SENSe:EFRontend:SELFtest?
//Result: 0 (no errors)
SENSe:EFRontend:SELFtest:RESult?
//Result: "<?xml version='1.0' encoding='UTF-8'?>
//<Sequence xmlns:xsi='http://www.w3.org/2001/XMLSchema-instance'
//xsi:noNamespaceSchemaLocation='
//<Name>DeviceCheck</Name>
//<FirmwareVersion>1.50-21.1.1.0 Beta</FirmwareVersion>
//..."
```

10.4.1.4 Configuring input via the optional Analog Baseband interface

The following commands are required to control the optional "Analog Baseband" interface in a remote environment. They are only available if this option is installed.

For more information on the "Analog Baseband" interface, see [Section 4.3, "Processing data from the Analog Baseband interface"](#), on page 27.

For a programming example, see [Section 10.12.3, "Data acquisition via the optional Analog Baseband interface"](#), on page 351.

Useful commands for Analog Baseband data described elsewhere:

- INP:SEL AIQ (see [INPut<ip>:SElect](#) on page 158)
- [SENSe:]FREQuency:CENTer on page 235

Commands for the Analog Baseband calibration signal are described in the R&S FSV/A User Manual.

Remote commands exclusive to Analog Baseband data input and output

INPut:IQ:BALanced[:STATe]	179
INPut:IQ:FULLscale:AUTO	179
INPut:IQ:FULLscale[:LEVel]	179
INPut:IQ:IMPedance	180
INPut:IQ:IMPedance:PTYPE	180
INPut:IQ:OPTimize	180
INPut:IQ:SKEW	181
INPut:IQ:TYPE	181
TRACe:IQ:APCon[:STATe]	182
TRACe:IQ:APCon:A	182
TRACe:IQ:APCon:B	182
TRACe:IQ:APCon:RESult?	183

INPut:IQ:BALEnc[STATE] <State>

Defines whether the input is provided as a differential signal via all 4 Analog Baseband connectors or as a plain I/Q signal via 2 single-ended lines.

Parameters:

<State> ON | OFF | 1 | 0

ON | 1

Differential

OFF | 0

Single ended

*RST: 1

Example: INP:IQ:BALEnc OFF

Manual operation: See ["Input Configuration"](#) on page 54

INPut:IQ:FULLscale:AUTO <State>

Defines whether the full scale level (i.e. the maximum input power on the Baseband Input connector) is defined automatically according to the reference level, or manually.

Parameters:

<State>

ON | 1

Automatic definition

OFF | 0

Manual definition according to [INPut:IQ:FULLscale\[LEVEL\]](#) on page 179

*RST: 1

Example: INP:IQ:FULLscale:AUTO OFF

Manual operation: See ["Full Scale Level Mode / Value"](#) on page 65

INPut:IQ:FULLscale[LEVEL] <PeakVoltage>

Defines the peak voltage at the Baseband Input connector if the full scale level is set to manual mode (see [INPut:IQ:FULLscale:AUTO](#) on page 179).

Parameters:

<PeakVoltage>

0.25 V | 0.5 V | 1 V | 2 V

Peak voltage level at the connector.

*RST: 1V

Default unit: V

Example: INP:IQ:FULL 0.5V

Manual operation: See ["Full Scale Level Mode / Value"](#) on page 65

INPut:IQ:IMPedance <Impedance>

Selects the nominal input impedance of the analog baseband input.

For input from the RF input, use the **INPut:IMPedance** command.

Parameters:

<Impedance> 50 | 75

numeric value
 User-defined impedance from 50 Ohm to 100000000 Ohm
 (=100 MOhm)
 User-defined values are only available for:
 Spectrum application
 I/Q Analyzer
 MSRA mode, primary application only

*RST: 50
 Default unit: OHM

Example: INP:IQ:IMP 75

Manual operation: See "[Impedance](#)" on page 49

INPut:IQ:IMPedance:PTYPE <PadType>

Defines the type of matching pad used for impedance conversion for analog baseband input.

For RF input, use the **INPut:IMPedance:PTYPE** command.

Parameters:

<PadType> SRESistor | MLPad

SRESistor
 Series-R

MLPad
 Minimum Loss Pad

*RST: SRESistor

Example: INP:IQ:IMP 100
 INP:IQ:IMP:PTYP MLP

Manual operation: See "[Impedance](#)" on page 49

INPut:IQ:OPTimize <OptMode>

Selects the priority for signal processing.

This function is only available under the following conditions:

- One of the options R&S FSV3-B200, B400, which provide a separate wideband processing path in the R&S FSV/A, is installed.

- An I/Q bandwidth that requires the wideband path is used.

Parameters:

<OptMode> LDIStortion | LNOise

LNOise
Optimized for high sensitivity and low noise levels

LDIStortion
Optimized for low distortion by avoiding intermodulation

Example: INP:IQ:OPT LNO

Manual operation: See "[Optimization](#)" on page 54

INPut:IQ:SKEW <Skew>

Defines a delay of the I vector in respect to the Q vector. A skew can be used to compensate a physical impairment of the two analog signals I and Q.

Parameters:

<Skew> numeric value

Range: -1E-9 to +1E-9

Default unit: S

Example: INP:IQ:SKEW 0.0000002

Manual operation: See "[Skew \(Delay I vs. Q\)](#)" on page 54

INPut:IQ:TYPE <DataType>

Defines the format of the input signal.

Parameters:

<DataType> IQ | I | Q

IQ
The input signal is filtered and resampled to the sample rate of the application.
Two input channels are required for each input signal, one for the in-phase component, and one for the quadrature component.

I
The in-phase component of the input signal is filtered and resampled to the sample rate of the application. If the center frequency is not 0, the in-phase component of the input signal is down-converted first (Low IF I).

Q
The quadrature component of the input signal is filtered and resampled to the sample rate of the application. If the center frequency is not 0, the quadrature component of the input signal is down-converted first (Low IF Q).

*RST: IQ

Example: INP:IQ:TYPE Q

Manual operation: See ["I/Q Mode"](#) on page 53

TRACe:IQ:APCon:STATe] <State>

If enabled, the average power consumption is calculated at the end of the I/Q data measurement. This command must be set *before* the measurement is performed!

The conversion factors A and B for the calculation are defined using [TRACe:IQ:APCon:A](#) and [TRACe:IQ:APCon:B](#).

The results can be queried using [TRACe:IQ:APCon:RESult?](#) on page 183.

For details see [Section 4.3.5, "Average power consumption"](#), on page 32.

Parameters:

<State> ON | OFF | 0 | 1
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on

Example:

```
*RST
TRAC:IQ:STAT ON
TRAC:IQ:SRAT 1MHZ
TRAC:IQ:RLEN 1000000
TRAC:IQ:APC:STAT ON
TRAC:IQ:APC:A 3.0
TRAC:IQ:APC:B 0.6
INIT;*WAI
TRAC:IQ:APC:RES?
```

TRACe:IQ:APCon:A <ConvFact>

Defines the conversion factor A for the calculation of the average power consumption.

For details see [Section 4.3.5, "Average power consumption"](#), on page 32.

Parameters:

<ConvFact> numeric value
 *RST: 1.0

TRACe:IQ:APCon:B <ConvFact>

Defines the conversion factor B for the calculation of the average power consumption.

For details see [Section 4.3.5, "Average power consumption"](#), on page 32.

Parameters:

<ConvFact> numeric value
 *RST: 0.0

TRACe:IQ:APCon:RESult?

Queries the average power consumption for an analog baseband input. This value is only calculated at the end of the I/Q data measurement if the **TRACe:IQ:APCon[:STATe]** command is set to ON *before* the measurement is performed!

For details see [Section 4.3.5, "Average power consumption"](#), on page 32.

Return values:

<Average> numeric value
 Default unit: W

Usage: Query only

10.4.1.5 Using external mixers

The commands required to work with external mixers in a remote environment are described here. Note that these commands require the R&S FSV/A to have an external mixer option installed and an external mixer to be connected to the R&S FSV/A.

For details on working with external mixers see the R&S FSV/A User Manual.

- [Basic settings](#)..... 183
- [Mixer settings](#)..... 184
- [Conversion loss table settings](#)..... 190
- [Programming example: working with an external mixer](#)..... 194

Basic settings

The basic settings concern general usage of an external mixer.

- [\[SENSe:\]MIXer<x>\[:STATe\]](#)..... 183
- [\[SENSe:\]MIXer<x>:BIAS:HIGH](#)..... 184
- [\[SENSe:\]MIXer<x>:BIAS\[:LOW\]](#)..... 184
- [\[SENSe:\]MIXer<x>:LOPower](#)..... 184

[SENSe:]MIXer<x>[:STATe] <State>

Activates or deactivates the use of a connected external mixer as input for the measurement. This command is only available if the optional External Mixer is installed and an external mixer is connected.

Suffix:

<x> 1..n
 irrelevant

Parameters:

<State> ON | OFF | 1 | 0
 *RST: 0

Example: MIX ON

[SENSe:]MIXer<x>:BIAS:HIGH <BiasSetting>

Defines the bias current for the high (last) range.

Is only available if the external mixer is active (see [SENSe:]MIXer<x>[:STATe] on page 183).

Suffix:

<x> 1..n
irrelevant

Parameters:

<BiasSetting> *RST: 0.0 A
Default unit: A

[SENSe:]MIXer<x>:BIAS[:LOW] <BiasSetting>

Defines the bias current for the low (first) range.

Is only available if the external mixer is active (see [SENSe:]MIXer<x>[:STATe] on page 183).

Suffix:

<x> 1..n
irrelevant

Parameters:

<BiasSetting> *RST: 0.0 A
Default unit: A

[SENSe:]MIXer<x>:LOPower <Level>

Specifies the LO level of the external mixer's LO port.

Suffix:

<x> 1..n
irrelevant

Parameters:

<Level> Range: 13.0 dBm to 17.0 dBm
Increment: 0.1 dB
*RST: 15.5 dBm
Default unit: DBM

Example: MIX:LOP 16.0dBm

Mixer settings

The following commands are required to configure the band and specific mixer settings.

[SENSe:]MIXer<x>:FREQuency:HANDoVer.....	185
[SENSe:]MIXer<x>:FREQuency:STARt.....	185
[SENSe:]MIXer<x>:FREQuency:STOP.....	185

[SENSe:]MIXer<x>:HARMonic:BAND:PRESet.....	186
[SENSe:]MIXer<x>:HARMonic:BAND.....	186
[SENSe:]MIXer<x>:HARMonic:HIGH:STATe.....	187
[SENSe:]MIXer<x>:HARMonic:HIGH[:VALue].....	187
[SENSe:]MIXer<x>:HARMonic:TYPE.....	187
[SENSe:]MIXer<x>:HARMonic[:LOW].....	188
[SENSe:]MIXer<x>:IF?.....	188
[SENSe:]MIXer<x>:LOSS:HIGH.....	188
[SENSe:]MIXer<x>:LOSS:TABLE:HIGH.....	188
[SENSe:]MIXer<x>:LOSS:TABLE[:LOW].....	189
[SENSe:]MIXer<x>:LOSS[:LOW].....	189
[SENSe:]MIXer<x>:PORTs.....	189
[SENSe:]MIXer<x>:RFOVerrange[:STATe].....	190

[SENSe:]MIXer<x>:FREQuency:HANDOver <Frequency>

Defines the frequency at which the mixer switches from one range to the next (if two different ranges are selected). The handover frequency for each band can be selected freely within the overlapping frequency range.

Is only available if the external mixer is active (see [SENSe:]MIXer<x>[:STATe] on page 183).

Suffix:

<x> 1..n
 irrelevant

Parameters:

<Frequency> Default unit: HZ

Example:

MIX ON
Activates the external mixer.
MIX:FREQ:HAND 78.0299GHz
Sets the handover frequency to 78.0299 GHz.

[SENSe:]MIXer<x>:FREQuency:START

Sets or queries the frequency at which the external mixer band starts.

Suffix:

<x> 1..n
 irrelevant

Example:

MIX:FREQ:STAR?
Queries the start frequency of the band.

[SENSe:]MIXer<x>:FREQuency:STOP

Sets or queries the frequency at which the external mixer band stops.

Suffix:

<x> 1..n
irrelevant

Example:

MIX:FREQ:STOP?
Queries the stop frequency of the band.

[SENSe:]MIXer<x>:HARMonic:BAND:PRESet

Restores the preset frequency ranges for the selected standard waveguide band.

Note: Changes to the band and mixer settings are maintained even after using the [PRESET] function. Use this command to restore the predefined band ranges.

Suffix:

<x> 1..n
irrelevant

Example:

MIX:HARM:BAND:PRESet
Presets the selected waveguide band.

[SENSe:]MIXer<x>:HARMonic:BAND <Band>

Selects the external mixer band. The query returns the currently selected band.

Is only available if the external mixer is active (see [SENSe:]MIXer<x>[:STATe] on page 183).

Suffix:

<x> 1..n
irrelevant

Parameters:

<Band> KA|Q|U|V|E|W|F|D|G|Y|J|USER
Standard waveguide band or user-defined band.

Table 10-3: Frequency ranges for pre-defined bands

Band	Frequency start [GHz]	Frequency stop [GHz]
KA (A) *)	26.5	40.0
Q	33.0	50.0
U	40.0	60.0
V	50.0	75.0
E	60.0	90.0
W	75.0	110.0
F	90.0	140.0
D	110.0	170.0
G	140.0	220.0
*) The band formerly referred to as "A" is now named "KA".		

Band	Frequency start [GHz]	Frequency stop [GHz]
J	220.0	325.0
Y	325.0	500.0
USER	32.18 (default)	68.22 (default)
*) The band formerly referred to as "A" is now named "KA".		

[SENSe:]MIXer<x>:HARMonic:HIGH:STATe <State>

Specifies whether a second (high) harmonic is to be used to cover the band's frequency range.

Suffix:

<x> 1..n

Parameters:

<State> ON | OFF
*RST: ON

Example: MIX:HARM:HIGH:STAT ON

[SENSe:]MIXer<x>:HARMonic:HIGH[:VALue] <HarmOrder>

Specifies the harmonic order to be used for the high (second) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<HarmOrder> Range: 3 to 128 (USER band); for other bands: see band definition

Example: MIX:HARM:HIGH:STAT ON
MIX:HARM:HIGH 3

[SENSe:]MIXer<x>:HARMonic:TYPE <OddEven>

Specifies whether the harmonic order to be used should be odd, even, or both.

Which harmonics are supported depends on the mixer type.

Suffix:

<x> 1..n
irrelevant

Parameters:

<OddEven> ODD | EVEN | EODD
ODD | EVEN | EODD
*RST: EVEN

Example: MIX:HARM:TYPE ODD

[SENSe:]MIXer<x>:HARMonic[:LOW] <HarmOrder>

Specifies the harmonic order to be used for the low (first) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<HarmOrder> Range: 3 to 128 (USER band); for other bands: see band definition
 *RST: 4 (for band U)

Example: MIX:HARM 3

[SENSe:]MIXer<x>:IF?

Queries the intermediate frequency currently used by the external mixer.

Suffix:

<x> 1..n
irrelevant

Example: MIX:IF?

Example: See ["Programming example: working with an external mixer"](#) on page 194.

Usage: Query only

[SENSe:]MIXer<x>:LOSS:HIGH <Average>

Defines the average conversion loss to be used for the entire high (second) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<Average> Range: 0 to 100
 *RST: 24.0 dB
 Default unit: dB

Example: MIX:LOSS:HIGH 20dB

[SENSe:]MIXer<x>:LOSS:TABLE:HIGH <FileName>

Defines the conversion loss table to be used for the high (second) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<FileName> String containing the path and name of the file, or the serial number of the external mixer whose file is required. The R&S FSV/A automatically selects the correct cvl file for the current IF. As an alternative, you can also select a user-defined conversion loss table (.acl file).

[SENSe:]MIXer<x>:LOSS:TABLE[:LOW] <FileName>

Defines the file name of the conversion loss table to be used for the low (first) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<FileName> String containing the path and name of the file, or the serial number of the external mixer whose file is required. The R&S FSV/A automatically selects the correct cvl file for the current IF. As an alternative, you can also select a user-defined conversion loss table (.acl file).

Example:

```
MIX:LOSS:TABL '101567'
MIX:LOSS:TABL?
//Result:
'101567_MAG_6_B5000_3G5.B5G'
```

[SENSe:]MIXer<x>:LOSS[:LOW] <Average>

Defines the average conversion loss to be used for the entire low (first) range.

Suffix:

<x> 1..n
irrelevant

Parameters:

<Average> Range: 0 to 100
*RST: 24.0 dB
Default unit: dB

Example:

```
MIX:LOSS 20dB
```

[SENSe:]MIXer<x>:PORTs <PortType>

Selects the mixer type.

Suffix:

<x> 1..n
irrelevant

Parameters:

<PortType> 2 | 3

2
Two-port mixer.

3
Three-port mixer.

*RST: 2

Example: MIX:PORT 3

[SENSe:]MIXer<x>:RFOVerrange[:STATe] <State>

If enabled, the band limits are extended beyond "RF Start" and "RF Stop" due to the capabilities of the used harmonics.

Suffix:

<x> 1..n
irrelevant

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

Conversion loss table settings

The following settings are required to configure and manage conversion loss tables.

[SENSe:]CORRection:CVL:BAND.....	190
[SENSe:]CORRection:CVL:BIAS.....	191
[SENSe:]CORRection:CVL:CATalog?.....	191
[SENSe:]CORRection:CVL:CLEar.....	191
[SENSe:]CORRection:CVL:COMMeNT.....	192
[SENSe:]CORRection:CVL:DATA.....	192
[SENSe:]CORRection:CVL:HARMonic.....	192
[SENSe:]CORRection:CVL:MIXer.....	193
[SENSe:]CORRection:CVL:PORTs.....	193
[SENSe:]CORRection:CVL:SELeCt.....	193
[SENSe:]CORRection:CVL:SNUMber.....	194

[SENSe:]CORRection:CVL:BAND <Band>

Defines the waveguide band for which the conversion loss table is to be used. This setting is checked against the current mixer setting before the table can be assigned to the range.

Before this command can be performed, the conversion loss table must be selected (see [SENSe:]CORRection:CVL:SELeCt on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<Band>

K | KA | Q | U | V | E | W | F | D | G | Y | J | USER

Standard waveguide band or user-defined band.

For a definition of the frequency range for the pre-defined bands, see [Table 10-3](#).

*RST: F (90 GHz - 140 GHz)

Example:

CORR:CVL:SEL 'LOSS_TAB_4'

Selects the conversion loss table.

CORR:CVL:BAND KA

Sets the band to KA (26.5 GHz - 40 GHz).

[SENSe:]CORRection:CVL:BIAS <BiasSetting>

Defines the bias setting to be used with the conversion loss table.

Before this command can be performed, the conversion loss table must be selected (see [\[SENSe:\]CORRection:CVL:SElect](#) on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<BiasSetting>

*RST: 0.0 A

Default unit: A

Example:

CORR:CVL:SEL 'LOSS_TAB_4'

Selects the conversion loss table.

CORR:CVL:BIAS 3A

[SENSe:]CORRection:CVL:CATalog?

Queries all available conversion loss tables saved in the C:\R_S\INSTR\USER\cvl\ directory on the instrument.

Is only available with option B21 (External Mixer) installed.

Return values:

<Files>

'string'

Comma-separated list of strings containing the file names.

Example:

CORR:CVL:CAT?

Usage:

Query only

[SENSe:]CORRection:CVL:CLEarDeletes the selected conversion loss table. Before this command can be performed, the conversion loss table must be selected (see [\[SENSe:\]CORRection:CVL:SElect](#) on page 193).

Is only available with option B21 (External Mixer) installed.

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:CLE

[SENSe:]CORRection:CVL:COMMeNT <Text>

Defines a comment for the conversion loss table. Before this command can be performed, the conversion loss table must be selected (see [\[SENSe:\]CORRection:CVL:SELEct](#) on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<Text>

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:COMM 'Conversion loss table for
 FS_Z60'

[SENSe:]CORRection:CVL:DATA {<Freq>, <Level>}...

Defines the reference values of the selected conversion loss tables. The values are entered as a set of frequency/level pairs. You can define a maximum of 500 frequency/level pairs. Before this command can be performed, you must select the conversion loss table (see [\[SENSe:\]CORRection:CVL:SELEct](#) on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<Freq> The frequencies have to be sent in ascending order.

Default unit: HZ

<Level> Default unit: DB

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:DATA 1MHZ,-30DB,2MHZ,-40DB

[SENSe:]CORRection:CVL:HARMonic <HarmOrder>

Defines the harmonic order for which the conversion loss table is to be used. This setting is checked against the current mixer setting before the table can be assigned to the range.

Before this command can be performed, the conversion loss table must be selected (see [\[SENSe:\]CORRection:CVL:SELEct](#) on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<HarmOrder> Range: 2 to 65

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:HARM 3

[SENSe:]CORRection:CVL:MIXer <Type>

Defines the mixer name in the conversion loss table. This setting is checked against the current mixer setting before the table can be assigned to the range.

Before this command can be performed, the conversion loss table must be selected (see [SENSe:]CORRection:CVL:SElect on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<Type> string
 Name of mixer with a maximum of 16 characters

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:MIX 'FS_Z60'

[SENSe:]CORRection:CVL:PORTs <PortType>

Defines the mixer type in the conversion loss table. This setting is checked against the current mixer setting before the table can be assigned to the range.

Before this command can be performed, the conversion loss table must be selected (see [SENSe:]CORRection:CVL:SElect on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<PortType> 2 | 3
 *RST: 2

Example: CORR:CVL:SEL 'LOSS_TAB_4'
 Selects the conversion loss table.
 CORR:CVL:PORT 3

[SENSe:]CORRection:CVL:SElect <FileName>

Selects the conversion loss table with the specified file name. If <file_name> is not available, a new conversion loss table is created.

Is only available with option B21 (External Mixer) installed.

Parameters:

<FileName> String containing the path and name of the file.

Example: CORR:CVL:SEL 'LOSS_TAB_4'

[SENSe:]CORRection:CVL:SNUMber <SerialNo>

Defines the serial number of the mixer for which the conversion loss table is to be used. This setting is checked against the current mixer setting before the table can be assigned to the range.

Before this command can be performed, the conversion loss table must be selected (see [SENSe:]CORRection:CVL:SElect on page 193).

Is only available with option B21 (External Mixer) installed.

Parameters:

<SerialNo> Serial number with a maximum of 16 characters

Example:

CORR:CVL:SEL 'LOSS_TAB_4'

Selects the conversion loss table.

CORR:CVL:MIX '123.4567'

Programming example: working with an external mixer

This example demonstrates how to work with an external mixer in a remote environment. It is performed in the Spectrum application in the default layout configuration. Note that without a real input signal and connected mixer, this measurement will not return useful results.

```
//-----Preparing the instrument -----
//Reset the instrument
*RST
//Activate the use of the connected external mixer.
SENS:MIX ON
//----- Configuring basic mixer behavior -----
//Set the LO level of the mixer's LO port to 15 dBm.
SENS:MIX:LOP 15dBm
//Set the bias current to -1 mA .
SENS:MIX:BIAS:LOW -1mA
//----- Configuring the mixer and band settings -----
//Use band "V" to full possible range extent for assigned harmonic (6).
SENS:MIX:HARM:BAND V
SENS:MIX:RFOV ON
//Query the possible range
SENS:MIX:FREQ:STAR?
//Result: 47480000000 (47.48 GHz)
SENS:MIX:FREQ:STOP?
//Result: 138020000000 (138.02 GHz)
//Use a 3-port mixer type
SENS:MIX:PORT 3
//Split the frequency range into two ranges;
//range 1 covers 47.48 GHz GHz to 80 GHz; harmonic 6, average conv. loss of 20 dB
//range 2 covers 80 GHz to 138.02 GHz; harmonic 8, average conv.loss of 30 dB
SENS:MIX:HARM:TYPE EVEN
SENS:MIX:HARM:HIGH:STAT ON
SENS:MIX:FREQ:HAND 80GHz
```

```

SENS:MIX:HARM:LOW 6
SENS:MIX:LOSS:LOW 20dB
SENS:MIX:HARM:HIGH 8
SENS:MIX:LOSS:HIGH 30dB
//----- Activating automatic signal identification functions -----
//Activate both automatic signal identification functions.
SENS:MIX:SIGN ALL
//Use auto ID threshold of 8 dB.
SENS:MIX:THR 8dB

//-----Performing the Measurement-----
//Select single sweep mode.
INIT:CONT OFF
//Initiate a basic frequency sweep and wait until the sweep has finished.
INIT;*WAI
//-----Retrieving Results-----
//Return the trace data for the input signal without distortions
//(default screen configuration)
TRAC:DATA? TRACE3

```

Configuring a conversion loss table for a user-defined band

```

//-----Preparing the instrument -----
//Reset the instrument
*RST
//Activate the use of the connected external mixer.
SENS:MIX ON
//-----Configuring a new conversion loss table -----
//Define cvl table for range 1 of band as described in previous example
// (extended V band)
SENS:CORR:CVL:SEL 'UserTable'
SENS:CORR:CVL:COMM 'User-defined conversion loss table for USER band'
SENS:CORR:CVL:BAND USER
SENS:CORR:CVL:HARM 6
SENS:CORR:CVL:BIAS -1mA
SENS:CORR:CVL:MIX 'FS_Z60'
SENS:CORR:CVL:SNUM '123.4567'
SENS:CORR:CVL:PORT 3
//Conversion loss is linear from 55 GHz to 75 GHz
SENS:CORR:CVL:DATA 55GHZ,-20DB,75GHZ,-30DB
//----- Configuring the mixer and band settings -----
//Use user-defined band and assign new cvl table.
SENS:MIX:HARM:BAND USER
//Define band by two ranges;
//range 1 covers 47.48 GHz to 80 GHz; harmonic 6, cvl table 'UserTable'
//range 2 covers 80 GHz to 138.02 GHz; harmonic 8, average conv.loss of 30 dB
SENS:MIX:HARM:TYPE EVEN
SENS:MIX:HARM:HIGH:STAT ON
SENS:MIX:FREQ:HAND 80GHz

```

```

SENS:MIX:HARM:LOW 6
SENS:MIX:LOSS:TABL:LOW 'UserTable'
SENS:MIX:HARM:HIGH 8

SENS:MIX:LOSS:HIGH 30dB
//Query the possible range
SENS:MIX:FREQ:STAR?
//Result: 47480000000 (47.48 GHz)
SENS:MIX:FREQ:STOP?
//Result: 138020000000 (138.02 GHz)

//-----Performing the Measurement-----
//Select single sweep mode.
INIT:CONT OFF
//Initiate a basic frequency sweep and wait until the sweep has finished.
INIT;*WAI
//-----Retrieving Results-----
//Return the trace data (default screen configuration)
TRAC:DATA? TRACel

```

10.4.1.6 Setting up probes

Modular probes can be connected to the RF input connector of the R&S FSV/A.

For details see the R&S FSV/A User Manual.

[SENSe:]PROBe<pb>:ID:PARTnumber?	196
[SENSe:]PROBe<pb>:ID:SRNumber?	197
[SENSe:]PROBe<pb>:SETup:ATTRatio	197
[SENSe:]PROBe<pb>:SETup:CMOOffset	197
[SENSe:]PROBe<pb>:SETup:DMOOffset	198
[SENSe:]PROBe<pb>:SETup:MODE	198
[SENSe:]PROBe<pb>:SETup:NAME?	198
[SENSe:]PROBe<pb>:SETup:NMOOffset	199
[SENSe:]PROBe<pb>:SETup:PMODE	199
[SENSe:]PROBe<pb>:SETup:PMOOffset	200
[SENSe:]PROBe<pb>:SETup:STATe?	200
[SENSe:]PROBe<pb>:SETup:TYPE?	200

[SENSe:]PROBe<pb>:ID:PARTnumber?

Queries the R&S part number of the probe.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Return values:

<PartNumber>

Example: //Query part number
PROB3:ID:PART?

Usage: Query only

[SENSe:]PROBe<pb>:ID:SRNumber?

Queries the serial number of the probe.

Suffix:

<pb> 1..n
Selects the connector:
3 = RF

Return values:

<SerialNo>

Example: //Query serial number
PROB3:ID:SRN?

Usage: Query only

[SENSe:]PROBe<pb>:SETup:ATTRatio <AttenuationRatio>

Defines the attenuation applied to the input at the probe. This setting is only available for modular probes.

Suffix:

<pb> 1..n
Selects the connector:
3 = RF

Parameters:

<AttenuationRatio> **10**
Attenuation by 20 dB (ratio= 10:1)
2
Attenuation by 6 dB (ratio= 2:1)
*RST: 10
Default unit: DB

[SENSe:]PROBe<pb>:SETup:CMOffset <CMOffset>

Sets the common mode offset. The setting is only available if a differential probe in CM-mode is connected to the R&S FSV/A.

If the probe is disconnected, the common mode offset of the probe is reset to 0.0 V.

Note that if the offset for DM-mode or CM-mode is changed, the offsets for the P-mode and N-mode are adapted accordingly, and vice versa.

For details see the R&S FSV/A User Manual.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Parameters:

<CMOffset> Offset of the mean voltage between the positive and negative input terminal vs. ground
 Range: -16 V to +16 V
 Default unit: V

[SENSe:]PROBe<pb>:SETup:DMOffset <DMOffset>

Sets the DM-mode offset. The setting is only available if a modular probe in DM-mode is connected to the R&S FSV/A.

If the probe is disconnected, the DM-mode offset of the probe is reset to 0.0 V.

Note that if the offset for DM-mode or CM-mode is changed, the offsets for the P-mode and N-mode are adapted accordingly, and vice versa.

For details see the R&S FSV/A User Manual.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Parameters:

<DMOffset> Voltage offset between the positive and negative input terminal
 Default unit: V

[SENSe:]PROBe<pb>:SETup:MODE <Mode>**Suffix:**

<pb> 1..n
 Selects the connector:
 3 = RF

Parameters:

<Mode> RSINgle | NOACtion
RSINgle
 Run single: starts one data acquisition.
NOACtion
 Nothing is started on pressing the micro button.

[SENSe:]PROBe<pb>:SETup:NAME?

Queries the name of the probe.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Return values:

<Name> String containing the name of the probe.

Example:

//Query name of the probe
 PROB3:SET:NAME?

Usage:

Query only

[SENSe:]PROBe<pb>:SETup:NMOffset <NMOffset>

Sets the N-mode offset. The setting is only available if a modular probe in N-mode is connected to the R&S FSV/A. The maximum voltage difference between the positive and negative input terminals is 16 V.

If the probe is disconnected, the N-mode offset of the probe is reset to 0.0 V.

Note that if the offset for DM-mode or CM-mode is changed, the offsets for the P-mode and N-mode are adapted accordingly, and vice versa.

For details see the R&S FSV/A User Manual.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Parameters:

<NMOffset> The voltage offset between the negative input terminal and ground.
 Default unit: V

[SENSe:]PROBe<pb>:SETup:PMODE <Mode>

Determines the mode of a multi-mode modular probe.

For details see the R&S FSV/A User Manual.

Suffix:

<pb> 1..n
 Selects the connector:
 3 = RF

Parameters:

<Mode> CM | DM | PM | NM
DM
 Voltage between the positive and negative input terminal
CM
 Mean voltage between the positive and negative input terminal vs. ground

PM

Voltage between the positive input terminal and ground

NM

Voltage between the negative input terminal and ground

Example:

```
SENS:PROB:SETU:PMOD PM
Sets the probe to P-mode.
```

[SENSe:]PROBe<pb>:SETup:PMOffset <PMOffset>

Sets the P-mode offset. The setting is only available if a modular probe in P-mode is connected to the R&S FSV/A. The maximum voltage difference between the positive and negative input terminals is 16 V.

If the probe is disconnected, the P-mode offset of the probe is reset to 0.0 V.

Note that if the offset for DM-mode or CM-mode is changed, the offsets for the P-mode and N-mode are adapted accordingly, and vice versa.

For details see the R&S FSV/A User Manual.

Suffix:

<pb> 1..n
Selects the connector:
3 = RF

Parameters:

<PMOffset> The voltage offset between the positive input terminal and ground.
Default unit: V

[SENSe:]PROBe<pb>:SETup:STATe?

Queries if the probe at the specified connector is active (detected) or not active (not detected).

Suffix:

<pb> 1..n
Selects the connector:
3 = RF

Return values:

<State> DETected | NDETECTED

Example:

```
//Query connector state
PROB3:SET:STAT?
```

Usage:

Query only

[SENSe:]PROBe<pb>:SETup:TYPE?

Queries the type of the probe.

Suffix:	
<pb>	1..n Selects the connector: 3 = RF
Return values:	
<Type>	String containing one of the following values: –"None" (no probe detected) –"active differential" –"active single-ended" –"active modular"
Example:	//Query probe type PROB3:SET:TYPE?
Usage:	Query only

10.4.1.7 External generator control

External generator control commands are available if the R&S FSV/A External Generator Control option (R&S FSV3-B10) is installed.

For each measurement channel, you can configure one external generator. To switch between different configurations, define multiple measurement channels.

For more information on external generator control, see the R&S FSV/A User Manual.

- [Measurement configuration](#).....201
- [Interface configuration](#).....205
- [Source calibration](#).....206
- [Programming example for external generator control](#).....209

Measurement configuration

The following commands are required to activate external generator control and to configure a calibration measurement with an external tracking generator.

SOURCE<si>:EXTERNAL<gen>:FREQUENCY	201
SOURCE<si>:EXTERNAL<gen>:FREQUENCY:COUPLING[:STATE]	202
SOURCE<si>:EXTERNAL<gen>:FREQUENCY[:FACTOR]:DENOMINATOR	202
SOURCE<si>:EXTERNAL<gen>:FREQUENCY[:FACTOR]:NUMERATOR	203
SOURCE<si>:EXTERNAL<gen>:FREQUENCY:OFFSET	203
SOURCE<si>:EXTERNAL<gen>:POWER[:LEVEL]	204
SOURCE<si>:EXTERNAL<gen>[:STATE]	204
SOURCE<si>:POWER[:LEVEL][:IMMEDIATE]:OFFSET	204

SOURCE<si>:EXTERNAL<gen>:FREQUENCY <Frequency>

Defines a fixed source frequency for the external generator.

Suffix:	
<si>	irrelevant
<gen>	

Parameters:

<Frequency> Source frequency of the external generator.

*RST: 1100050000

Default unit: HZ

Example:

//Define frequency of the generator

SOUR:EXT:FREQ 10MHz

SOURce<si>:EXTernal<gen>:FREQuency:COUPling[:STATe] <State>

Couples the frequency of the external generator output to the R&S FSV/A.

Suffix:

<si> irrelevant

<gen>

Parameters:

<State> ON | OFF | 0 | 1

ON | 1

Default setting: a series of frequencies is defined (one for each sweep point), based on the current frequency at the RF input of the R&S FSV/A. The RF frequency range covers the currently defined span of the R&S FSV/A (unless limited by the range of the signal generator).

OFF | 0

The generator uses a single fixed frequency, defined by

[SOURce<si>:EXTernal<gen>:FREQuency](#).

*RST: 1

Example:

SOUR:EXT:FREQ:COUP ON

SOURce<si>:EXTernal<gen>:FREQuency[:FACTor]:DENominator <Value>

Defines the denominator of the factor with which the analyzer frequency is multiplied to obtain the transmit frequency of the selected generator.

Select the multiplication factor such that the frequency range of the generator is not exceeded if the following formula is applied to the start and stop frequency of the analyzer:

$$F_{Generator} = \left| F_{Analyzer} * \frac{Numerator}{Denominator} + F_{Offset} \right|$$

Suffix:

<si> irrelevant

<gen>

Parameters:

<Value> <numeric value>
 *RST: 1

Example:

//Define multiplication factor of 4/3; the transmit frequency of the generator is 4/3 times the analyzer frequency
 SOUR:EXT:FREQ:NUM 4
 SOUR:EXT:FREQ:DEN 3

SOURce<si>:EXTernal<gen>:FREQuency[:FACTor]:NUMerator <Value>

Defines the numerator of the factor with which the analyzer frequency is multiplied to obtain the transmit frequency of the selected generator.

Select the multiplication factor such that the frequency range of the generator is not exceeded if the following formula is applied to the start and stop frequency of the analyzer:

$$F_{\text{Generator}} = \left| F_{\text{Analyzer}} * \frac{\text{Numerator}}{\text{Denominator}} + F_{\text{Offset}} \right|$$

Suffix:

<si> irrelevant

<gen>

Parameters:

<Value> <numeric value>
 *RST: 1

Example:

//Define multiplication factor of 4/3; the transmit frequency of the generator is 4/3 times the analyzer frequency
 SOUR:EXT:FREQ:NUM 4
 SOUR:EXT:FREQ:DEN 3

SOURce<si>:EXTernal<gen>:FREQuency:OFFSet <Offset>

Defines the frequency offset of the generator with reference to the analyzer frequency.

Select the offset such that the frequency range of the generator is not exceeded if the following formula is applied to the start and stop frequency of the analyzer:

$$F_{\text{Generator}} = \left| F_{\text{Analyzer}} * \frac{\text{Numerator}}{\text{Denominator}} + F_{\text{Offset}} \right|$$

Suffix:

<si> irrelevant

<gen>

Parameters:

<Offset> <numeric value>, specified in Hz, kHz, MHz or GHz, rounded to the nearest Hz

*RST: 0 Hz

Default unit: HZ

Example:

//Define an offset between generator output frequency and analyzer frequency

SOUR:EXT:FREQ:OFFS 10HZ

SOURce<si>:EXTeRnal<gen>:POWeR[:LEVel] <Level>

Sets the output power of the selected generator.

Suffix:

<si> irrelevant

<gen>

Parameters:

<Level> <numeric value>

*RST: -20 dBm

Default unit: DBM

Example:

//Define generator output level

SOUR:EXT:POW -30dBm

SOURce<si>:EXTeRnal<gen>[:STATe] <State>

Activates or deactivates the connected external generator.

Suffix:

<si> irrelevant

<gen>

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

SOURce<si>:POWeR[:LEVel][:IMMediate]:OFFSet <Offset>**Suffix:**

<si> irrelevant

Parameters:

<Offset> Range: -200 dB to +200 dB

*RST: 0dB

Default unit: DB

Example: `SOUR:POW:OFFS -10dB`

Interface configuration

The following commands are required to configure the interface for the connection to the external generator.

<code>SOURce<si>:EXTErnal<gen>:ROSCillator[:SOURce]</code>	205
<code>SYSTem:COMMunicate:RDEVice:GENerator<gen>:INTerface</code>	205
<code>SYSTem:COMMunicate:RDEVice:GENerator<gen>:TYPE</code>	205
<code>SYSTem:COMMunicate:TCPIp:RDEVice:GENerator<gen>:ADDRess</code>	206

SOURce<si>:EXTErnal<gen>:ROSCillator[:SOURce] <Source>

Controls selection of the reference oscillator for the external generator.

If the external reference oscillator is selected, the reference signal must be connected to the rear panel of the instrument.

Suffix:

<code><si></code>	irrelevant
<code><gen></code>	irrelevant

Parameters:

<code><Source></code>	INTernal Uses the internal reference.
	EXTernal Uses the external reference; if none is available, an error flag is displayed in the status bar.
<code>*RST:</code>	INT

Example: `//Select an external reference oscillator`
`SOUR:EXT:ROSC EXT`

SYSTem:COMMunicate:RDEVice:GENerator<gen>:INTerface <Type>

Defines the interface used for the connection to the external generator.

Is only available if external generator control is active (see `SOURce<si>:EXTErnal<gen>[:STATe]` on page 204).

Suffix:

`<gen>`

Parameters:

`<Type>` **TCPIp**

Example: `SYST:COMM:RDEV:GEN:INT TCP`

SYSTem:COMMunicate:RDEVice:GENerator<gen>:TYPE <Type>

Selects the type of external generator.

For a list of the available generator types, see the specifications document.

Suffix:

<gen>

Parameters:

<Name> <Generator name as string value>

*RST: SMU02

Example:

//Select an external generator

SYST:COMM:RDEV:GEN:TYPE 'SMW06'

SYSTem:COMMunicate:TCPIp:RDEvice:GENerator<gen>:ADDRess <Address>

Configures the TCP/IP address for the external generator.

Suffix:

<gen>

Parameters:

<Address> TCP/IP address between 0.0.0.0 and 0.255.255.255

*RST: 0.0.0.0

Example:

SYST:COMM:TCP:RDEV:GEN:ADDR 130.094.122.195

Source calibration

The following commands are required to activate the calibration functions of the external tracking generator. However, they are only available if external generator control is active (see [SOURCE<si>:EXTERNAL<gen>\[:STATe\]](#) on page 204).

Useful commands for source calibration described elsewhere:

- [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:Y\[:SCALe\]:RVALue](#).....
[RPOSITION](#) on page 234

Remote commands exclusive to source calibration:

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RVALue	206
[SENSe:]CORRection:COLLect[:ACQuire]	207
[SENSe:]CORRection:METHod	207
[SENSe:]CORRection:RECall	208
[SENSe:]CORRection[:STATe]	208
[SENSe:]CORRection:TRANsducer:GENerate	208

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RVALue <Value>

Defines the reference value assigned to the reference position in the specified window. Separate reference values are maintained for the various displays.

Suffix:

<n> [Window](#)

<w> subwindow

<t> irrelevant

Parameters:

<Value> Default unit: DB

Example:

DISP:TRAC:Y:RVAL 0

Sets the value assigned to the reference position to 0 Hz

[SENSe:]CORRection:COLLect[:ACQuire] <MeasType>

Initiates a reference measurement (calibration). The reference measurement is the basis for the measurement normalization. The result depends on whether a reflection measurement or transmission measurement is performed (see [SENSe:]CORRection:METHod on page 207).

To obtain a correct reference measurement, a complete sweep with synchronization to the end of the sweep must have been carried out. This is only possible in the single sweep mode.

Is only available if external generator control is active (see SOURce<si>:EXTernal<gen>[:STATe] on page 204).

Setting parameters:

<MeasType> THROugh | OPEN

THROugh

"TRANsmission" mode: calibration with direct connection between generator and device input

"REFLection" mode: calibration with short circuit at the input

OPEN

only allowed in "REFLection" mode: calibration with open input

Example:

INIT:CONT OFF

Selects single sweep operation

CORR:METH TRAN

Selects a transmission measurement.

CORR:COLL THR;*WAI

Starts the measurement of reference data using direct connection between generator and device input and waits for the sweep end.

Usage: Setting only

[SENSe:]CORRection:METHod <Type>

Selects the type of measurement to be performed with the generator.

Is only available if external generator control is active (see SOURce<si>:EXTernal<gen>[:STATe] on page 204).

Parameters:

<Type> REFLection

Selects reflection measurements.

TRANsmission

Selects transmission measurements.

*RST: TRANsmission

Example:

CORR:METH TRAN

Sets the type of measurement to "transmission".

[SENSe:]CORRection:RECall

Restores the measurement configuration used for calibration.

Is only available if external generator control is active (see [SOURce<si>: EXTeRnal<gen>\[:STATe\]](#) on page 204).

Example:

CORR:REC

[SENSe:]CORRection[:STATe] <State>

Turns correction of measurement results (normalization) on and off.

The command is available after you have created a reference trace for the selected measurement type with [\[SENSe:\]CORRection:COLLect\[:ACQuire\]](#) on page 207.

Is only available if external generator control is active (see [SOURce<si>: EXTeRnal<gen>\[:STATe\]](#) on page 204).

Parameters:

<State>

ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

*RST: 1

Example:

CORR ON

Activates normalization.

[SENSe:]CORRection:TRANsducer:GENerate <Name>

Uses the normalized measurement data to generate a transducer factor with up to 1001 points. The trace data is converted to a transducer with unit dB and stored in a file with the specified name and the suffix .trd under C:\Program Files\Rohde-Schwarz\FSV3000\<version>\trd. The frequency points are allocated in equidistant steps between start and stop frequency.

The generated transducer factor can be further adapted using the commands described in the "Remote Commands > Configuring the R&S FSV/A > Working with Transducers" section in the R&S FSV/A User Manual.

Parameters:

<Name>

'<name>'

Example: CORR:TRAN:GEN 'MyGenerator'
 Creates the transducer file
 C:\r_s\instr\trd\MyGenerator.trd.

Programming example for external generator control

The following example demonstrates how to work with an external generator in a remote environment.

```
//-----Preparing the instrument -----

//Reset the instrument
*RST

//Set the frequency span.
SENS:FREQ:STAR 10HZ
SENS:FREQ:STOP 1MHZ

//-----Configuring the interface -----

//Set the generator type to SMW06 with a frequency range of 100 kHz to 3GHz
SYST:COMM:RDEV:GEN:TYPE 'SMA01A'

//Set the interface used to the TCP/IP address 130.094.122.195
SYST:COMM:RDEV:GEN:INT TCP
SYST:COMM:TCP:RDEV:GEN:ADDR 130.094.122.195

//Activate the use of the external reference frequency at 10 MHz on the generator
SOUR:EXT:ROSC EXT

//-----Configuring the calibration measurement -----

//Activate external generator control.
SOUR:EXT:STAT ON
//Set the generator output level to -10 dBm.
SOUR:EXT:POW -10DBM
//Set the frequency coupling to automatic
SOUR:EXT:FREQ:COUP:STAT ON

//-----Configuring the generator frequency range -----

//Define a series of frequencies (one for each sweep point) based on the current
//frequency at the RF input of the analyzer; the generator frequency is half the
//frequency of the analyzer, with an offset of 100 kHz;
// analyzer start:          10 Hz
// analyzer stop:           1 MHz
// analyzer span:           999.99 KHz
// generator frequency start: 100.005 KHz
// generator frequency stop:  600 KHz
// generator span:          499.995 KHz
```

```

SOUR:EXT:FREQ:FACT:NUM 1
SOUR:EXT:FREQ:FACT:DEN 2
SOUR:EXT:FREQ:OFFS 100KHZ

//-----Performing the calibration measurement -----

//Perform a transmission measurement with direct connection between the generator
//and the analyzer and wait till the end
SENS:CORR:METH TRAN
SENS:CORR:COLL:ACQ THR; *WAI

//-----Retrieving the calibration trace results -----

//Retrieve the measured frequencies (10 Hz - 600 kHz)
TRAC:DATA:X? TRACE1

//Retrieve the measured power levels; = 0 between 10 Hz and 100 kHz (below
//generator minimum frequency); nominal -5dBm as of 100 kHz;
TRAC:DATA? TRACE1

//-----Normalizing the calibration trace results -----

//Retrieve the normalized power levels (= power offsets from calibration results)
//Should be 0 for all sweep points directly after calibration
SENS:CORR:STAT ON
TRAC:DATA? TRACE1

//-----Changing the display of the calibration results -----
//Shift the reference line so the -5 dB level is displayed in the center
DISP:TRAC:Y:SCAL:RVAL -5DB
DISP:TRAC:Y:SCAL:RPOS 50PCT

```

10.4.1.8 Working with power sensors

The following commands describe how to work with power sensors.

These commands require the use of a Rohde & Schwarz power sensor. For a list of supported sensors, see the specifications document.

Using power sensors requires the option R&S FSV3-K9.

- [Configuring power sensors](#).....211
- [Configuring power sensor measurements](#).....212
- [Configuring power sensor math](#).....218
- [Triggering with power sensors](#).....219

Configuring power sensors

SYSTem:COMMunicate:RDEvice:PMETer<p>:CONFigure:AUTO[:STATe].....	211
SYSTem:COMMunicate:RDEvice:PMETer<p>:COUNt?.....	211
SYSTem:COMMunicate:RDEvice:PMETer<p>:DEFine.....	211

SYSTem:COMMunicate:RDEvice:PMETer<p>:CONFigure:AUTO[:STATe] <State>

Turns automatic assignment of a power sensor to the power sensor index on and off.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

*RST: 1

Example:

SYST:COMM:RDEV:PMET:CONF:AUTO OFF

SYSTem:COMMunicate:RDEvice:PMETer<p>:COUNt?

Queries the number of power sensors currently connected to the R&S FSV/A.

Suffix:

<p> Power sensor index

Return values:

<NumberSensors> Number of connected power sensors.

Example:

SYST:COMM:RDEV:PMET:COUN?

Usage:

Query only

SYSTem:COMMunicate:RDEvice:PMETer<p>:DEFine <Placeholder>, <Type>, <Interface>, <SerialNo>

Assigns the power sensor with the specified serial number to the selected power sensor index (configuration).

The query returns the power sensor type and serial number of the sensor assigned to the specified index.

Suffix:

<p> Power sensor index

Parameters:

<Placeholder> Currently not used

<Type> Detected power sensor type, e.g. "NRP-Z81".

<Interface> Interface the power sensor is connected to; always "USB"

<SerialNo> Serial number of the power sensor assigned to the specified index

Example:

```
SYST:COMM:RDEV:PMET2:DEF
'', 'NRP-Z81', '', '123456'
```

Assigns the power sensor with the serial number '123456' to the configuration "Power Sensor 2".

```
SYST:COMM:RDEV:PMET2:DEF?
```

Queries the sensor assigned to "Power Sensor 2".

Result:

```
'', 'NRP-Z81', 'USB', '123456'
```

The NRP-Z81 power sensor with the serial number '123456' is assigned to the "Power Sensor 2".

Configuring power sensor measurements

CALibration:PMETer<p>:ZERO:AUTO ONCE	212
CALCulate<n>:PMETer<p>:RELative[:MAGNitude]	213
CALCulate<n>:PMETer<p>:RELative[:MAGNitude]:AUTO ONCE	213
CALCulate<n>:PMETer<p>:RELative:STATe	213
FETCh:PMETer<p>?	214
READ:PMETer<p>?	214
[SENSe:]PMETer<p>:DCYCLe[:STATe]	214
[SENSe:]PMETer<p>:DCYCLe:VALue	214
[SENSe:]PMETer<p>:FREQuency	215
[SENSe:]PMETer<p>:FREQuency:LINK	215
[SENSe:]PMETer<p>:MTIme	215
[SENSe:]PMETer<p>:MTIme:AVERAge:COUNT	216
[SENSe:]PMETer<p>:MTIme:AVERAge[:STATe]	216
[SENSe:]PMETer<p>:ROFFset[:STATe]	216
[SENSe:]PMETer<p>:SOFFset	217
[SENSe:]PMETer<p>[:STATe]	217
[SENSe:]PMETer<p>:UPDate[:STATe]	217
UNIT<n>:PMETer<p>:POWer	218
UNIT<n>:PMETer<p>:POWer:RATio	218

CALibration:PMETer<p>:ZERO:AUTO ONCE

Zeroes the power sensor.

Note that you have to disconnect the signals from the power sensor input before you start to zero the power sensor. Otherwise, results are invalid.

Suffix:

<p> Power sensor index

Example:

```
CAL:PMET2:ZERO:AUTO ONCE;*WAI
```

Starts zeroing the power sensor 2 and delays the execution of further commands until zeroing is concluded.

Usage:

Event

CALCulate<n>:PMETer<p>:RELative[:MAGNitude] <RefValue>

Defines the reference value for relative measurements.

Suffix:

<n> [Window](#)

<p> Power sensor index

Parameters:

<RefValue> Range: -200 dBm to 200 dBm
 *RST: 0
 Default unit: DBM

Example: `CALC:PMET2:REL -30`
 Sets the reference value for relative measurements to -30 dBm for power sensor 2.

CALCulate<n>:PMETer<p>:RELative[:MAGNitude]:AUTO ONCE

Sets the current measurement result as the reference level for relative measurements.

Suffix:

<n> [Window](#)

<p> Power sensor index

Example: `CALC:PMET2:REL:AUTO ONCE`
 Takes the current measurement value as reference value for relative measurements for power sensor 2.

Usage: Event

CALCulate<n>:PMETer<p>:RELative:STATE <State>

Turns relative power sensor measurements on and off.

Suffix:

<n> [Window](#)

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
 Switches the function off
ON | 1
 Switches the function on

Example: `CALC:PMET2:REL:STAT ON`
 Activates the relative display of the measured value for power sensor 2.

FETCH:PMETer<p>?

Queries the results of power sensor measurements.

Suffix:

<p> Power sensor index

Usage: Query only

READ:PMETer<p>?

Initiates a power sensor measurement and queries the results.

Suffix:

<p> Power sensor index

Usage: Query only

[SENSe:]PMETer<p>:DCYClE[:STATe] <State>

Turns the duty cycle correction on and off.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example: PMET2:DCYC:STAT ON

[SENSe:]PMETer<p>:DCYClE:VALue <Percentage>

Defines the duty cycle for the correction of pulse signals.

The power sensor uses the duty cycle in combination with the mean power to calculate the power of the pulse.

Suffix:

<p> Power sensor

Parameters:

<Percentage> Range: 0.001 to 99.999

*RST: 99.999

Default unit: %

Example: PMET2:DCYC:STAT ON
Activates the duty cycle correction.
PMET2:DCYC:VAL 0.5
Sets the correction value to 0.5%.

[SENSe:]PMETer<p>:FREQuency <Frequency>

Defines the frequency of the power sensor.

Suffix:

<p> Power sensor index

Parameters:

<Frequency>

The available value range is specified in the specifications document of the power sensor in use.

*RST: 50 MHz

Default unit: HZ

Example:

PMET2:FREQ 1GHZ

Sets the frequency of the power sensor to 1 GHz.

[SENSe:]PMETer<p>:FREQuency:LINK <Coupling>

Selects the frequency coupling for power sensor measurements.

Suffix:

<p> Power sensor index

Parameters:

<Coupling>

CENTer

Couples the frequency to the center frequency of the analyzer

MARKer1

Couples the frequency to the position of marker 1

OFF

Switches the frequency coupling off

*RST: CENTer

Example:

PMET2:FREQ:LINK CENT

Couples the frequency to the center frequency of the analyzer

[SENSe:]PMETer<p>:MTIMe <Duration>

Selects the duration of power sensor measurements.

Suffix:

<p> Power sensor index

Parameters:

<Duration>

SHORt | NORMaL | LONG

*RST: NORMaL

Example:

PMET2:MTIM SHOR

Sets a short measurement duration for measurements of stationary high power signals for the selected power sensor.

[SENSe:]PMETer<p>:MTIMe:AVERage:COUNT <NumberReadings>

Sets the number of power readings included in the averaging process of power sensor measurements.

Extended averaging yields more stable results for power sensor measurements, especially for measurements on signals with a low power, because it minimizes the effects of noise.

Suffix:

<p> Power sensor index

Parameters:

<NumberReadings> An average count of 0 or 1 performs one power reading.

Range: 0 to 256

Increment: binary steps (1, 2, 4, 8, ...)

Example:

PMET2:MTIM:AVER ON

Activates manual averaging.

PMET2:MTIM:AVER:COUN 8

Sets the number of readings to 8.

[SENSe:]PMETer<p>:MTIMe:AVERage[:STATe] <State>

Turns averaging for power sensor measurements on and off.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

PMET2:MTIM:AVER ON

Activates manual averaging.

[SENSe:]PMETer<p>:ROFFset[:STATe] <State>

Includes or excludes the reference level offset of the analyzer for power sensor measurements.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

PMET2:ROFF OFF

Takes no offset into account for the measured power.

[SENSe:]PMETer<p>:SOFFset <SensorOffset>

Takes the specified offset into account for the measured power. Only available if [SENSe:]PMETer<p>:ROFFset[:STATe] is disabled.

Suffix:

<p> Power sensor index

Parameters:

<SensorOffset> Default unit: DB

Example:

PMET2:SOFF 0.001

[SENSe:]PMETer<p>[:STATe] <State>

Turns a power sensor on and off.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

PMET1 ON

Switches the power sensor measurements on.

[SENSe:]PMETer<p>:UPDate[:STATe] <State>

Turns continuous update of power sensor measurements on and off.

If on, the results are updated even if a single sweep is complete.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example: `PMET1:UPD ON`
The data from power sensor 1 is updated continuously.

UNIT<n>:PMETer<p>:POWer <Unit>

Selects the unit for absolute power sensor measurements.

Suffix:

<n> irrelevant
<p> Power sensor index

Parameters:

<Unit> DBM | WATT | W | DB | PCT
*RST: DBM

Example: `UNIT:PMET:POW DBM`

UNIT<n>:PMETer<p>:POWer:RATio <Unit>

Selects the unit for relative power sensor measurements.

Suffix:

<n> irrelevant
<p> Power sensor index

Parameters:

<Unit> DB | PCT
*RST: DB

Example: `UNIT:PMET:POW:RAT DB`

Configuring power sensor math

You can determine the difference between results from multiple power sensors in one measurement using the power sensor math function.

[SENSe:]PMETer:MATH[:STATe]	218
[SENSe:]PMETer:MATH:MINuend	219
[SENSe:]PMETer:MATH:SUBTrahend	219
[SENSe:]PMETer:MATH:RESult?	219

[SENSe:]PMETer:MATH[:STATe] <State>

Enables or disables the math function for the power sensor results.

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
Switches the function off
ON | 1
Switches the function on

```

Example:      *RST:      0
              PMET:MATH:STAT ON
              //Subtract the value for power sensor 2 from power sensor 1.
              PMET:MATH:MIN 1
              PMET:MATH:SUBT 2
              PMET:MATH:RES?

```

[SENSe:]PMETer:MATH:MINuend <Minuend>

Defines the minuend for power sensor math (<Sensor A> in <Sensor A> - <Sensor B>).

Parameters:

<Minuend> 1 | 2 | 3 | 4

[SENSe:]PMETer:MATH:SUBTrahend <Subtrahend>

Defines the subtrahend for power sensor math (<Sensor B> in <Sensor A> - <Sensor B>).

Parameters:

<Subtrahend> 1 | 2 | 3 | 4

[SENSe:]PMETer:MATH:RESult?

Returns the difference between results from multiple power sensors (<Sensor A> - <Sensor B>).

Return values:

<MathResult>

Usage: Query only

Triggering with power sensors

[SENSe:]PMETer<p>:TRIGger:DTIME.....	219
[SENSe:]PMETer<p>:TRIGger:HOLDoff.....	220
[SENSe:]PMETer<p>:TRIGger:HYSTeresis.....	220
[SENSe:]PMETer<p>:TRIGger:LEVel.....	220
[SENSe:]PMETer<p>:TRIGger:SLOPe.....	221
[SENSe:]PMETer<p>:TRIGger[:STATe].....	221

[SENSe:]PMETer<p>:TRIGger:DTIME <Time>

Defines the time period that the input signal has to stay below the IF power trigger level before the measurement starts.

Suffix:

<p> Power sensor index

Parameters:

<Time> Range: 0 s to 1 s
 Increment: 100 ns
 *RST: 100 µs
 Default unit: S

Example: PMET2:TRIG:DTIME 0.001

[SENSe:]PMETer<p>:TRIGger:HOLDoff <Holdoff>

Defines the trigger holdoff for external power triggers.

Suffix:

<p> Power sensor index

Parameters:

<Holdoff> Time period that has to pass between the trigger event and the start of the measurement, in case another trigger event occurs.
 Range: 0 s to 1 s
 Increment: 100 ns
 *RST: 0 s
 Default unit: S

Example: PMET2:TRIG:HOLD 0.1
 Sets the holdoff time of the trigger to 100 ms

[SENSe:]PMETer<p>:TRIGger:HYSTeresis <Hysteresis>

Defines the trigger hysteresis for external power triggers.

The hysteresis in dB is the value the input signal must stay below the IF power trigger level to allow a trigger to start the measurement.

Suffix:

<p> Power sensor index

Parameters:

<Hysteresis> Range: 3 dB to 50 dB
 Increment: 1 dB
 *RST: 0 dB
 Default unit: DB

Example: PMET2:TRIG:HYST 10
 Sets the hysteresis of the trigger to 10 dB.

[SENSe:]PMETer<p>:TRIGger:LEVel <Level>

Defines the trigger level for external power triggers.

Suffix:

<p> Power sensor index

Parameters:

<Level> -20 to +20 dBm
 Range: -20 dBm to 20 dBm
 *RST: -10 dBm
 Default unit: DBM

Example:

PMET2:TRIG:LEV -10 dBm
 Sets the level of the trigger

[SENSe:]PMETer<p>:TRIGger:SLOPe <Edge>

Selects the trigger condition for external power triggers.

Suffix:

<p> Power sensor index

Parameters:

<Edge> **POSitive**
 The measurement starts in case the trigger signal shows a positive edge.
NEGative
 The measurement starts in case the trigger signal shows a negative edge.
 *RST: POSitive

Example:

PMET2:TRIG:SLOP NEG

[SENSe:]PMETer<p>:TRIGger[:STATe] <State>

Turns the external power trigger on and off.

Suffix:

<p> Power sensor index

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
 Switches the function off
ON | 1
 Switches the function on

Example:

PMET2:TRIG ON
 Switches the external power trigger on

10.4.1.9 Configuring the outputs

The following commands are required to provide output from the R&S FSV/A.



Configuring trigger input/output is described in [Section 10.4.4.2, "Configuring the trigger output"](#), on page 243.

DIAGnostic:SERvice:NSource.....	222
OUTPut:IF:IFFrequency.....	222
OUTPut:IF:STATe.....	222
OUTPut:UPORt:STATe.....	223
OUTPut:UPORt[:VALue].....	223
OUTPut:VIDeo:STATe.....	223
SYSTem:SPEaker[:STATe].....	224
SYSTem:SPEaker:MAXVolume.....	224
SYSTem:SPEaker:MUTE.....	224
SYSTem:SPEaker:VOLume.....	224

DIAGnostic:SERvice:NSource <State>

Turns the 28 V supply of the BNC connector labeled [noise source control] on the R&S FSV/A on and off.

Parameters:

<State> ON | OFF | 0 | 1
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on

Example: DIAG:SERV:NSO ON

Manual operation: See ["Noise Source Control"](#) on page 56

OUTPut:IF:IFFrequency <Frequency>

Defines the frequency for the IF output of the R&S FSV/A. The IF frequency of the signal is converted accordingly.

This command is only available in the time domain.

Parameters:

<Frequency> *RST: 32.0 MHz
 Default unit: HZ

OUTPut:IF:STATe <State>

Enables or disables output of the measured IF value at the "IF" output connector.

Parameters:

<State> ON | OFF | 0 | 1
 OFF | 0
 Switches the function off

ON | 1

Switches the function on

***RST:** 0**Example:** `OUTP:IF:STAT ON`**OUTPut:UPORt:STATe <State>**

Toggles the control lines of the user ports for the **AUX PORT** connector. This 9-pole SUB-D male connector is located on the rear panel of the R&S FSV/A.

For the R&S FSV/A, this command is maintained for compatibility reasons only. The **AUX PORT** connector only supports output.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

User port is switched to INPut

ON | 1

User port is switched to OUTPut

Example: `OUTP:UPOR:STAT ON`**OUTPut:UPORt[:VALue] <Value>**

Sets the control lines of the user ports.

The assignment of the pin numbers to the bits is as follows:

Bit	7	6	5	4	3	2	1	0
Pin	N/A	N/A	19	18	17	16	15	14

Bits 7 and 6 are not assigned to pins and must always be 0.

The user port is written to with the given binary pattern.

Parameters:

<Value> bit values in hexadecimal format

TTL type voltage levels (max. 5V)

Range: #B00000000 to #B00111111

Example: `OUTP:UPOR #B00100100`

Sets pins 5 and 7 to 5 V.

OUTPut:VIDeo:STATe <State>

Enables or disables output of the displayed video signal (i.e. the filtered and detected IF signal) at the "Video" output connector.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

***RST:** 0**Example:** `OUTP:VID:STAT ON`**SYSTem:SPEaker[:STATe] <State>**

Switches the built-in loudspeaker on or off for demodulated signals. This setting applies only to the current application.

The command is available in the time domain in Spectrum mode and in Analog Modulation Analysis mode.

To set the volume, use the `SYSTem:SPEaker:VOLume` command.

Parameters:**<State>** ON | OFF | 0 | 1**OFF | 0**

Switches the function off

ON | 1

Switches the function on

Example: `SYST:SPE ON`
`SYST:SPE:VOL 0.5`
 Sets the loudspeaker to half the full volume.

SYSTem:SPEaker:MAXVolume <Volume>

Defines the maximum volume to be output as a percentage of the maximum possible volume.

Parameters:**<Volume>** percentage**Example:** `SYST:SPE:MAXV 50`**SYSTem:SPEaker:MUTE**

Temporarily disables the audio output via the built-in loudspeakers.

Example: `SYST:SPE:MUTE`**SYSTem:SPEaker:VOLume <Volume>**

Defines the volume of the built-in loudspeaker for demodulated signals. This setting is maintained for all applications.

The command is available in the time domain in Spectrum mode and in Analog Modulation Analysis mode.

Parameters:

<Volume> Percentage of the maximum possible volume.
 Range: 0 to 1
 *RST: 0.5

Example:

SYST:SPE:VOL 0
 Switches the loudspeaker to mute.

10.4.2 Configuring the vertical axis (amplitude, scaling)

The following commands are required to configure the amplitude and vertical axis settings in a remote environment.

- [Amplitude settings](#).....225
- [Configuring the attenuation](#).....227
- [Configuring a preamplifier](#).....230
- [Scaling the Y-axis](#).....232

10.4.2.1 Amplitude settings

Useful commands for amplitude configuration described elsewhere:

- [\[SENSe:\]ADJust:LEVel](#) on page 257

Remote commands exclusive to amplitude configuration:

CALCulate<n>:MARKer<m>:FUNCTION:REFERENCE	225
UNIT<n>:POWER	226
CALCulate<n>:UNIT:POWER	226
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALE]:RLEVel	226
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALE]:RLEVel:OFFSet	226

CALCulate<n>:MARKer<m>:FUNCTION:REFERENCE

Matches the reference level to the power level of a marker.

If you use the command in combination with a delta marker, that delta marker is turned into a normal marker.

Suffix:

<n> [Window](#)
 <m> [Marker](#)

Example:

CALC:MARK2:FUNC:REF
 Sets the reference level to the level of marker 2.

Manual operation: See "[Reference Level = Marker Level](#)" on page 115

UNIT<n>:POWer <Unit>
CALCulate<n>:UNIT:POWer <Unit>

Selects the power unit.

The unit applies to all power-based measurement windows with absolute values.

In addition, the unit of the reference level is adapted to the same unit.

Suffix:

<n> irrelevant

Parameters:

<Unit> DBM | V | A | W | DBPW | WATT | DBUV | DBMV | VOLT |
DBUA | AMPere

*RST: dBm

Example:

CALC:UNIT:POW DBM

Sets the power unit to dBm.

Manual operation: See "Unit" on page 59

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RLEVel
 <ReferenceLevel>

Defines the reference level (for all traces in all windows).

With a reference level offset $\neq 0$, the value range of the reference level is modified by the offset.

Suffix:

<n> irrelevant

<w> subwindow
Not supported by all applications

<t> irrelevant

Parameters:

<ReferenceLevel> The unit is variable.

Range: see specifications document

*RST: 0 dBm

Default unit: DBM

Example:

DISP:TRAC:Y:RLEV -60dBm

Manual operation: See "Reference Level" on page 58

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RLEVel:OFFSet
 <Offset>

Defines a reference level offset (for all traces in all windows).

Suffix:

<n> irrelevant

<w> subwindow
Not supported by all applications

<t> irrelevant

Parameters:

<Offset> Range: -200 dB to 200 dB
*RST: 0dB
Default unit: DB

Example: DISP:TRAC:Y:RLEV:OFFS -10dB

Manual operation: See "Shifting the Display (Offset)" on page 59

10.4.2.2 Configuring the attenuation

INPut:ATTenuation.....	227
INPut:ATTenuation:AUTO.....	228
INPut:ATTenuation:AUTO:MODE.....	228
INPut:EATT.....	228
INPut:EATT:AUTO.....	229
INPut:EATT:STATe.....	229
INPut:SANalyzer:ATTenuation.....	229
INPut:SANalyzer:ATTenuation:AUTO.....	230

INPut:ATTenuation <Attenuation>

Defines the total attenuation for RF input.

If an electronic attenuator is available and active, the command defines a mechanical attenuation (see INPut:EATT:STATe on page 229).

If you set the attenuation manually, it is no longer coupled to the reference level, but the reference level is coupled to the attenuation. Thus, if the current reference level is not compatible with an attenuation that has been set manually, the command also adjusts the reference level.

If an external frontend is active (see [SENSe:]EFRontend[:STATe] on page 170), you can configure the attenuation of the external frontend and the analyzer separately. See also INPut:SANalyzer:ATTenuation:AUTO on page 230 and INPut:SANalyzer:ATTenuation on page 229.

Parameters:

<Attenuation> Range: see specifications document
Increment: 5 dB (with optional electr. attenuator: 1 dB)
*RST: 10 dB (AUTO is set to ON)
Default unit: DB

Example: INP:ATT 30dB
Defines a 30 dB attenuation and decouples the attenuation from the reference level.

Manual operation: See "External Frontend Attenuation" on page 60
See "Attenuation Mode / Value" on page 60

INPut:ATTenuation:AUTO <State>

Couples or decouples the attenuation to the reference level. Thus, when the reference level is changed, the R&S FSV/A determines the signal level for optimal internal data processing and sets the required attenuation accordingly.

If an external frontend is active (see [\[SENSe:\]EFrontend\[:STATe\]](#) on page 170), you can configure the attenuation of the external frontend and the analyzer separately. See also [INPut:SANalyzer:ATTenuation:AUTO](#) on page 230 and [INPut:SANalyzer:ATTenuation](#) on page 229.

Parameters:

<State> ON | OFF | 0 | 1
 *RST: 1

Example: INP:ATT:AUTO ON
 Couples the attenuation to the reference level.

Manual operation: See ["External Frontend Attenuation"](#) on page 60
 See ["Attenuation Mode / Value"](#) on page 60

INPut:ATTenuation:AUTO:MODE <OptMode>

Selects the priority for signal processing *after* the RF attenuation has been applied.

Parameters:

<OptMode> LNOise | LDISTortion
LNOise
 Optimized for high sensitivity and low noise levels
LDISTortion
 Optimized for low distortion by avoiding intermodulation
 *RST: LDISTortion (WLAN application: LNOise)

Example: INP:ATT:AUTO:MODE LNO

Manual operation: See ["Optimization"](#) on page 61

INPut:EATT <Attenuation>

Defines an electronic attenuation manually. Automatic mode must be switched off (INP:EATT:AUTO OFF, see [INPut:EATT:AUTO](#) on page 229).

If the current reference level is not compatible with an attenuation that has been set manually, the command also adjusts the reference level.

Requires the electronic attenuation hardware option.

Parameters:

<Attenuation> attenuation in dB
 Range: see specifications document
 Increment: 1 dB
 *RST: 0 dB (OFF)
 Default unit: DB

Example: `INP:EATT:AUTO OFF`
 `INP:EATT 10 dB`

Manual operation: See ["Using Electronic Attenuation"](#) on page 61

INPut:EATT:AUTO <State>

Turns automatic selection of the electronic attenuation on and off.

If on, electronic attenuation reduces the mechanical attenuation whenever possible.

Requires the electronic attenuation hardware option.

Parameters:

<State> `ON | OFF | 0 | 1`
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on
 ***RST:** `1`

Example: `INP:EATT:AUTO OFF`

Manual operation: See ["Using Electronic Attenuation"](#) on page 61

INPut:EATT:STATE <State>

Turns the electronic attenuator on and off.

Requires the electronic attenuation hardware option.

Parameters:

<State> `ON | OFF | 0 | 1`
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on
 ***RST:** `0`

Example: `INP:EATT:STAT ON`
 Switches the electronic attenuator into the signal path.

Manual operation: See ["Using Electronic Attenuation"](#) on page 61

INPut:SANalyzer:ATTenuation <Attenuation>

Configures attenuation at the analyzer input for an active external frontend manually.

Parameters:

<Attenuation> **Range:** see specifications document
 Increment: `1 dB`
 Default unit: `DB`

Manual operation: See ["Attenuation Mode / Value"](#) on page 60

INPut:SANalyzer:ATTenuation:AUTO <State>

Enables or disables automatic configuration of attenuation at the analyzer input for an active external frontend.

By default, the attenuation settings are applied at the input of the external frontend.

See [INPut:ATTenuation:AUTO](#) on page 228 and [INPut:ATTenuation](#) on page 227.

Parameters:

<State>

ON | OFF | 0 | 1

OFF | 0

Auto mode for analyzer attenuation is disabled. Allows you to configure attenuation at the analyzer using [INPut:SANalyzer:ATTenuation](#) on page 229.

ON | 1

Auto mode for analyzer attenuation is enabled. No attenuation is configured at the analyzer.

*RST: 0

Example:

```
//Enable external frontend
EFR ON
//Query the currently configured RF attenuation
INP:ATT?
//Result: 10 dB
//Disable auto mode for analyzer attenuation
INP:SAN:ATT:AUTO OFF
//Configure 10 dB attenuation at the analyzer
INP:SAN:ATT 10
//Query the currently configured RF attenuation at the ext. FE
INP:ATT?
//Result: 0 dB
```

Manual operation: See ["Attenuation Mode / Value"](#) on page 60

10.4.2.3 Configuring a preamplifier

INPut:EGAIN[:STATe]	230
INPut:GAIN:STATe	231
INPut:GAIN[:VALue]	231

INPut:EGAIN[:STATe] <State>

Before this command can be used, the external preamplifier must be connected to the R&S FSV/A. See the preamplifier's documentation for details.

When activated, the R&S FSV/A automatically compensates the magnitude and phase characteristics of the external preamplifier in the measurement results.

Note that when an optional external preamplifier is activated, the internal preamplifier is automatically disabled, and vice versa.

When deactivated, no compensation is performed even if an external preamplifier remains connected.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0
No data correction is performed based on the external preamplifier

ON | 1
Performs data corrections based on the external preamplifier

*RST: 0

Example: INP:EGA ON

INPut:GAIN:STATe <State>

Turns the internal preamplifier on and off. It requires the optional preamplifier hardware.

For R&S FSV/A44 models, note the restrictions described in ["Preamplifier"](#) on page 62.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0
Switches the function off

ON | 1
Switches the function on

*RST: 0

Example: INP:GAIN:STAT ON
INP:GAIN:VAL 15
Switches on 15 dB preamplification.

Manual operation: See ["Preamplifier"](#) on page 62

INPut:GAIN[:VALue] <Gain>

Selects the "gain" if the preamplifier is activated (INP:GAIN:STAT ON, see [INPut:GAIN:STATe](#) on page 231).

The command requires the additional preamplifier hardware option.

For R&S FSV/A44 or higher models, note the restrictions described in ["Preamplifier"](#) on page 62.

Parameters:

<Gain> The following settings are available:
 15 dB and 30 dB
 All other values are rounded to the nearest of these two.
 Default unit: DB

Example:

INP:GAIN:STAT ON
 INP:GAIN:VAL 30
 Switches on 30 dB preamplification.

Manual operation: See ["Preamplifier"](#) on page 62

10.4.2.4 Scaling the Y-axis

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe].....	232
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:AUTO ONCE.....	232
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:MODE.....	233
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:PDIVision.....	233
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RPOSition.....	234
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y:SPACing.....	234

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe] <Range>

Defines the display range of the y-axis (for all traces).

Note that the command works only for a logarithmic scaling. You can select the scaling with [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:Y:SPACing](#).

Suffix:

<n> [Window](#)
 <w> subwindow
 Not supported by all applications
 <t> irrelevant

Parameters:

<Range> Range: 1 dB to 200 dB
 *RST: 100 dB
 Default unit: HZ

Example:

DISP:TRAC:Y 110dB

Manual operation: See ["Range"](#) on page 66
 See ["Y-Axis Max"](#) on page 66

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:AUTO ONCE

Automatic scaling of the y-axis is performed once, then switched off again (for all traces).

Suffix:

<n> [Window](#)

<t> irrelevant

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:MODE <Mode>

Selects the type of scaling of the y-axis (for all traces).

When the display update during remote control is off, this command has no immediate effect.

Suffix:

<n> [Window](#)

<w> subwindow

<t> irrelevant

Parameters:

<Mode> **ABSolute**
absolute scaling of the y-axis

RELative
relative scaling of the y-axis

*RST: ABSolute

Example: DISP:TRAC:Y:MODE REL

Manual operation: See ["Scaling"](#) on page 66

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:PDIVision <Value>

This remote command determines the grid spacing on the Y-axis for all diagrams, where possible.

In spectrum displays, for example, this command is not available.

Suffix:

<n> [Window](#)

<w> subwindow
Not supported by all applications

<t> irrelevant

Parameters:

<Value> numeric value WITHOUT UNIT (unit according to the result display)

Defines the range per division (total range = 10*<Value>)

*RST: depends on the result display

Default unit: DBM

Example: DISP:TRAC:Y:PDIV 10
Sets the grid spacing to 10 units (e.g. dB) per division

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y[:SCALe]:RPOSition
 <Position>

Defines the vertical position of the reference level on the display grid (for all traces).

The R&S FSV/A adjusts the scaling of the y-axis accordingly.

For measurements with the optional external generator control, the command defines the position of the reference value.

Suffix:

<n>	Window
<w>	subwindow Not supported by all applications
<t>	irrelevant

Parameters:

<Position>	0 PCT corresponds to the lower display border, 100% corresponds to the upper display border. *RST: 100 PCT = frequency display; 50 PCT = time display Default unit: PCT
------------	---

Example: `DISP:TRAC:Y:RPOS 50PCT`

Manual operation: See ["Ref Level Position"](#) on page 66

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:Y:SPACing <ScalingType>

Selects the scaling of the y-axis (for all traces, <t> is irrelevant).

Suffix:

<n>	Window
<w>	subwindow
<t>	Trace

Parameters:

<ScalingType>	LOGarithmic Logarithmic scaling. LINear Linear scaling in %. LDB Linear scaling in the specified unit. PERCent Linear scaling in %. *RST: LOGarithmic
---------------	---

Example: `DISP:TRAC:Y:SPAC LIN`
 Selects linear scaling in %.

Manual operation: See ["Scaling"](#) on page 66

10.4.3 Frequency

CALCulate<n>:MARKer<m>:FUNCTION:CENTer.....	235
[SENSe:]FREQuency:CENTer.....	235
[SENSe:]FREQuency:CENTer:STEP.....	235
[SENSe:]FREQuency:CENTer:STEP:AUTO.....	236
[SENSe:]FREQuency:OFFSet.....	236

CALCulate<n>:MARKer<m>:FUNCTION:CENTer

Matches the center frequency to the frequency of a marker.

If you use the command in combination with a delta marker, that delta marker is turned into a normal marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Example:

CALC:MARK2:FUNC:CENT

Sets the center frequency to the frequency of marker 2.

Manual operation: See ["Center Frequency = Marker Frequency"](#) on page 115

[SENSe:]FREQuency:CENTer <Frequency>

Defines the center frequency.

CW, pulsed and VCO measurements:

This command defines or queries (in case of automatic frequency search) the current signal frequency.

Transient measurement:

This command defines the center frequency of the transient measurement.

Parameters:

<Frequency> For the allowed range and f_{max} , refer to the specifications document.

*RST: $f_{max}/2$

Default unit: Hz

Example:

FREQ:CENT 100 MHz

FREQ:CENT:STEP 10 MHz

FREQ:CENT UP

Sets the center frequency to 110 MHz.

Manual operation: See ["Center Frequency"](#) on page 55

See ["Center Frequency"](#) on page 67

[SENSe:]FREQuency:CENTer:STEP <StepSize>

Defines the center frequency step size.

You can increase or decrease the center frequency quickly in fixed steps using the `SENS:FREQ UP` AND `SENS:FREQ DOWN` commands, see [\[SENSe:\]FREQuency:CENTer](#) on page 235.

Parameters:

`<StepSize>` For $f_{\max}$, refer to the specifications document.
 Range: 1 to $f_{\max}$
 *RST: 0.1 x span
 Default unit: Hz

Example:

```
//Set the center frequency to 110 MHz.
FREQ:CENT 100 MHz
FREQ:CENT:STEP 10 MHz
FREQ:CENT UP
```

Manual operation: See ["Center Frequency Stepsize"](#) on page 67

[SENSe:]FREQuency:CENTer:STEP:AUTO <State>

Couples or decouples the center frequency step size to the span.

In time domain (zero span) measurements, the center frequency is coupled to the RBW.

Parameters:

`<State>` ON | OFF | 0 | 1
 *RST: 1

Example:

```
FREQ:CENT:STEP:AUTO ON
Activates the coupling of the step size to the span.
```

[SENSe:]FREQuency:OFFSet <Offset>

Defines a frequency offset.

If this value is not 0 Hz, the application assumes that the input signal was frequency shifted outside the application. All results of type "frequency" will be corrected for this shift numerically by the application.

See also ["Frequency Offset"](#) on page 68.

Parameters:

`<Offset>` Range: -1 THz to 1 THz
 *RST: 0 Hz
 Default unit: HZ

Example:

```
FREQ:OFFS 1GHZ
```

Manual operation: See ["Frequency Offset"](#) on page 68

10.4.4 Triggering

The following remote commands are required to configure a triggered measurement in a remote environment. More details are described for manual operation in [Section 5.6, "Trigger settings"](#), on page 68.



*OPC should be used after requesting data. This will hold off any subsequent changes to the selected trigger source, until after the sweep is completed and the data is returned.

- [Configuring the triggering conditions](#).....237
- [Configuring the trigger output](#).....243
- [Configuring I/Q gating](#).....245

10.4.4.1 Configuring the triggering conditions

The following commands are required to configure a triggered measurement.

TRIGger[:SEquence]:CTPis	237
TRIGger[:SEquence]:DTIME	238
TRIGger[:SEquence]:HOLDoff[:TIME]	238
TRIGger[:SEquence]:IFPower:HOLDoff	239
TRIGger[:SEquence]:IFPower:HYSteresis	239
TRIGger[:SEquence]:LEVel:BBPower	239
TRIGger[:SEquence]:LEVel[:EXternal<port>]	240
TRIGger[:SEquence]:LEVel:IFPower	240
TRIGger<undef>[:SEquence]:LEVel:IQPower	240
TRIGger[:SEquence]:SLOPe	241
TRIGger[:SEquence]:SOURce	241
TRIGger[:SEquence]:TIME:RINTerval	242

TRIGger[:SEquence]:CTPis <State>

The trigger event is set to the actual trigger point within the sample (TPIS). See also [TRACe:IQ:TPISample?](#) on page 253.

This setting is available for all trigger sources except "Free Run".

This setting only becomes effective for the next new measurement.

TPIS correction is not supported if user frequency response correction (R&S FSV3-K544) is active. If you activate frequency response correction, TPIS correction is automatically deactivated in all channels.

For details see frequency response correction in the R&S FSV/A user manual.

Parameters:

<State> ON | OFF | 0 | 1
 OFF | 0
 Switches the function off

ON | 1

Switches the function on

***RST:** 0**Example:**

```
// Query the current trigger offset within the sample
TRAC:IQ:TPIS?
//Result: 0.000002
//Correct the trigger event
TRIG:CTP ON
//Start a new measurement
INIT:IMM; *WAI
//Query the trigger offset within the sample again
TRAC:IQ:TPIS?
//Result: 0.000000
```

Manual operation: See ["Correct TPIS"](#) on page 74**TRIGger[:SEquence]:DTIME <DropoutTime>**

Defines the time the input signal must stay below the trigger level before a trigger is detected again.

For input from the "Analog Baseband" interface using the baseband power trigger (BBP), the default drop out time is set to 100 ns to avoid unintentional trigger events (as no hysteresis can be configured in this case).

Parameters:

<DropoutTime> Dropout time of the trigger.
 Range: 0 s to 10.0 s
***RST:** 0 s
 Default unit: S

Manual operation: See ["Drop-Out Time"](#) on page 73**TRIGger[:SEquence]:HOLDoff[:TIME] <Offset>**

Defines the time offset between the trigger event and the start of the sweep.

Parameters:

<Offset> For measurements in the frequency domain, the range is 0 s to 30 s.
 For measurements in the time domain, the range is the negative sweep time to 30 s.
***RST:** 0 s
 Default unit: S

Example: TRIG:HOLD 500us**Manual operation:** See ["Trigger Offset"](#) on page 73

TRIGger[:SEQuence]:IFPower:HOLDoff <Period>

Defines the holding time before the next trigger event.

Note that this command can be used for **any trigger source**, not just IF Power (despite the legacy keyword).

Note: If you perform gated measurements in combination with the IF Power trigger, the R&S FSV/A ignores the holding time for frequency sweep, FFT sweep, zero span and I/Q data measurements.

Parameters:

<Period> Range: 0 s to 10 s
 *RST: 0 s
 Default unit: S

Example:

TRIG:SOUR EXT
Sets an external trigger source.
TRIG:IFP:HOLD 200 ns
Sets the holding time to 200 ns.

Manual operation: See ["Trigger Holdoff"](#) on page 74

TRIGger[:SEQuence]:IFPower:HYSTeresis <Hysteresis>

Defines the trigger hysteresis, which is only available for "IF Power" trigger sources.

Parameters:

<Hysteresis> Range: 3 dB to 50 dB
 *RST: 3 dB
 Default unit: DB

Example:

TRIG:SOUR IFP
Sets the IF power trigger source.
TRIG:IFP:HYST 10DB
Sets the hysteresis limit value.

Manual operation: See ["Hysteresis"](#) on page 73

TRIGger[:SEQuence]:LEVel:BBPower <Level>

Sets the level of the baseband power trigger.

Is available for the optional "Analog Baseband" interface.

Parameters:

<Level> Range: -50 dBm to +20 dBm
 *RST: -20 dBm
 Default unit: DBM

Example:

TRIG:LEV:BBP -30DBM

Manual operation: See ["Trigger Level"](#) on page 72

TRIGger[:SEquence]:LEVel[:EXternal<port>] <TriggerLevel>

Defines the level the external signal must exceed to cause a trigger event.

In the I/Q Analyzer application, only EXTERNAL1 is supported.

Suffix:

<port>	Selects the trigger port. 1 = trigger port 1 (TRIGGER INPUT connector on front panel) 2 = trigger port 2 (TRIGGER INPUT/OUTPUT connector on front panel) 3 = trigger port 3 (TRIGGER3 INPUT/OUTPUT connector on rear panel)
<port>	Selects the trigger port. 1 = trigger port 1 (TRIG IN connector on rear panel) 2 = trigger port 2 (TRIG AUX connector on rear panel)

Parameters:

<TriggerLevel>	Range: 0.5 V to 3.5 V *RST: 1.4 V Default unit: V
----------------	--

Example: TRIG:LEV 2V

Manual operation: See ["Trigger Level"](#) on page 72

TRIGger[:SEquence]:LEVel:IFPower <TriggerLevel>

Defines the power level at the third intermediate frequency that must be exceeded to cause a trigger event.

Note that any RF attenuation or preamplification is considered when the trigger level is analyzed. If defined, a reference level offset is also considered.

For compatibility reasons, this command is also available for the "Baseband Power" trigger source when using the "Analog Baseband" interface.

Parameters:

<TriggerLevel>	For details on available trigger levels and trigger bandwidths, see the specifications document. *RST: -20 dBm Default unit: DBM
----------------	---

Example: TRIG:LEV:IFP -30DBM

Manual operation: See ["Trigger Level"](#) on page 72

TRIGger<undef>[:SEquence]:LEVel:IQPower <TriggerLevel>

Defines the magnitude the I/Q data must exceed to cause a trigger event.

Note that any RF attenuation or preamplification is considered when the trigger level is analyzed. If defined, a reference level offset is also considered.

Suffix:

<undef> 1..n

Parameters:

<TriggerLevel> Range: -130 dBm to 30 dBm
 *RST: -20 dBm
 Default unit: DBM

Example: TRIG:LEV:IQP -30DBM**Manual operation:** See ["Trigger Level"](#) on page 72**TRIGger[:SEquence]:SLOPe <Type>**

For all trigger sources except time, you can define whether triggering occurs when the signal rises to the trigger level or falls down to it.

Parameters:

<Type> POSitive | NEGative

POSitive

Triggers when the signal rises to the trigger level (rising edge).

NEGative

Triggers when the signal drops to the trigger level (falling edge).

*RST: POSitive

Example: TRIG:SLOP NEG**Manual operation:** See ["Slope"](#) on page 73**TRIGger[:SEquence]:SOURce <Source>**

Selects the trigger source.

For details on trigger sources, see ["Trigger Source"](#) on page 69.**Note on external triggers:**

If a measurement is configured to wait for an external trigger signal in a remote control program, remote control is blocked until the trigger is received and the program can continue. Make sure that this situation is avoided in your remote control programs.

Parameters:

<Source>

IMMediate

Free Run

EXternal

Trigger signal from the "Trigger Input" connector.

Trigger signal from the "Trigger In" connector.

EXT2

Trigger signal from the "Trigger Input/Output" connector.

Note: Connector must be configured for "Input".

Trigger signal from the "Trigger AUX" connector.

RFPower

First intermediate frequency

(Frequency and time domain measurements only.)

Not available for input from the optional "Analog Baseband" interface.

IFPower

Second intermediate frequency

For input from the optional "Analog Baseband" interface, this parameter is interpreted as **BBPower** for compatibility reasons.

IQPower

Magnitude of sampled I/Q data

For applications that process I/Q data, such as the I/Q Analyzer or optional applications.

TIME

Time interval

VIDeo

Video mode is available in the time domain and only in the Spectrum application.

BBPower

Baseband power

For input from the optional "Analog Baseband" interface.

PSEN

External power sensor

*RST: IMMEDIATE

Example:

TRIG:SOUR EXT

Selects the external trigger input as source of the trigger signal

Manual operation:

See ["Trigger Source"](#) on page 69

See ["Free Run"](#) on page 70

See ["External Trigger 1/2"](#) on page 70

See ["IF Power"](#) on page 70

See ["Baseband Power"](#) on page 71

See ["I/Q Power"](#) on page 71

See ["RF Power"](#) on page 71

See ["Power Sensor"](#) on page 72

See ["Time"](#) on page 72

TRIGger[:SEquence]:TIME:RINTerval <Interval>

Defines the repetition interval for the time trigger.

Parameters:

<Interval>

numeric value

Range: 1 us to 15 s

*RST: 10 ms

Default unit: S

Example: `TRIG:SOUR TIME`
 Selects the time trigger input for triggering.
`TRIG:TIME:RINT 5`
 The sweep starts every 5 s.

Manual operation: See ["Repetition Interval"](#) on page 72

10.4.4.2 Configuring the trigger output

The following commands are required to send the trigger signal to one of the variable "TRIGGER INPUT/OUTPUT" connectors on the R&S FSV/A.

<code>OUTPut:TRIGger<tp>:DIRection</code>	243
<code>OUTPut:TRIGger<tp>:LEVel</code>	243
<code>OUTPut:TRIGger<tp>:OTYPe</code>	244
<code>OUTPut:TRIGger<tp>:PULSe:IMMediate</code>	244
<code>OUTPut:TRIGger<tp>:PULSe:LENGth</code>	244

`OUTPut:TRIGger<tp>:DIRection <Direction>`

Selects the trigger direction for trigger ports that serve as an input as well as an output.

Suffix:

`<tp>`

Parameters:

`<Direction>` INPut | OUTPut

INPut

Port works as an input.

OUTPut

Port works as an output.

*RST: INPut

Manual operation: See ["Trigger 1/2"](#) on page 74

`OUTPut:TRIGger<tp>:LEVel <Level>`

Defines the level of the (TTL compatible) signal generated at the trigger output.

Works only if you have selected a user-defined output with `OUTPut:TRIGger<tp>:OTYPe`.

Suffix:

`<tp>`

1..n

Selects the trigger port to which the output is sent.

2 = trigger port 2 (front)

3 = trigger port 3 (rear)

2 = Trigger 2 Input / Output

Parameters:

`<Level>`

HIGH

5 V

LOW

0 V

*RST: LOW

Example: OUTPut:TRIG2:LEV HIGH**Manual operation:** See ["Level"](#) on page 75**OUTPut:TRIGger<tp>:OTYPe <OutputType>**

Selects the type of signal generated at the trigger output.

Suffix:

<tp>

1..n

Selects the trigger port to which the output is sent.

2 = trigger port 2 (front)

3 = trigger port 3 (rear)

2 = Trigger 2 Input / Output

Parameters:

<OutputType>

DEVIce

Sends a trigger signal when the R&S FSV/A has triggered internally.

TARMed

Sends a trigger signal when the trigger is armed and ready for an external trigger event.

UDEFI nedSends a user-defined trigger signal. For more information, see [OUTPut:TRIGger<tp>:LEVel](#).

*RST: DEVIce

Manual operation: See ["Output Type"](#) on page 75**OUTPut:TRIGger<tp>:PULSe:IMMediate**

Generates a pulse at the trigger output.

Suffix:

<tp>

1..n

Selects the trigger port to which the output is sent.

2 = trigger port 2 (front)

3 = trigger port 3 (rear)

2 = Trigger 2 Input / Output

Manual operation: See ["Send Trigger"](#) on page 76**OUTPut:TRIGger<tp>:PULSe:LENGth <Length>**

Defines the length of the pulse generated at the trigger output.

Suffix:

<tp>

Selects the trigger port to which the output is sent.

2 = trigger port 2 (front)

3 = trigger port 3 (rear)

2 = Trigger 2 Input / Output

Parameters:

<Length>

Pulse length in seconds.

Default unit: S

Example:

OUTP:TRIG2:PULS:LENG 0.02

Manual operation: See ["Pulse Length"](#) on page 76**10.4.4.3 Configuring I/Q gating**

Usually in spectrum analysis, measurements are based on a certain length of time called the gate area. With I/Q gating, you can define the gate area using the gate length, the distance between the capture periods and the number of periods. The gate length and the distance between the capture periods are specified in samples.

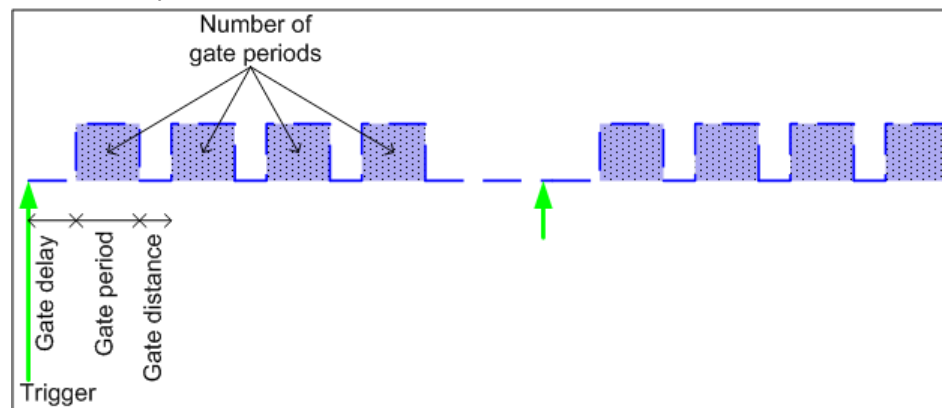


I/Q gating is only available using remote commands; manual configuration is not possible.

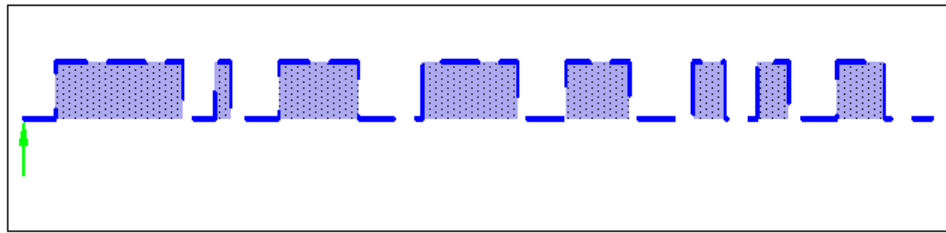
It is only possible up to a bandwidth of 10 MHz.

Using I/Q gating, the gate area can be defined using the following methods:

- **Edge triggered capturing**
After a trigger signal, the gate period is defined by a gate length and a gate distance. All data in the gate period is captured until the required number of samples has been captured.



- **Level triggered capturing**
After a trigger signal, all data is captured in which the gate signal is set to 1, which means it has exceeded a level. In this case, the gate signal can be generated by the IFP trigger, for example: each time the IFP level is exceeded, the IFP trigger signal is set to 1 and the samples in this area are captured as gate samples.



The number of complex samples to be captured prior to the trigger event can be selected (see [TRACe:IQ:SET](#) on page 251) for all available trigger sources, except for "Free Run".

TRACe:IQ:EGATe[:STATe] <State>

Turns gated measurements with the I/Q analyzer on and off.

Before you can use the command you have to turn on the I/Q analyzer and select an external or IF power trigger source.

Parameters:

<State> ON | OFF

Example: TRAC:IQ:EGAT ON

TRACe:IQ:EGATe:GAP <Samples>

Defines the interval between several gate periods for gated measurements with the I/Q analyzer.

Parameters:

<Samples> <numeric value>

Max = (440 MS * sample rate/200MHz) -1
pretrigger samples defined by [TRACe:IQ:SET](#);
sample rate defined by [TRACe:IQ:SRATe](#))

Range: 1...Max (samples)

*RST: 1

Example: TRAC:IQ:EGAT:GAP 2

TRACe:IQ:EGATe:LENGth <GateLength>

Defines the gate length for gated measurements with the I/Q analyzer.

Parameters:

<GateLength> <numeric value>

Max = (440 MS * sample rate/200MHz) -1
pretrigger samples defined by [TRACe:IQ:SET](#);
sample rate defined by [TRACe:IQ:SRATe](#))

Range: 1...Max (samples)

*RST: 100

Example: TRAC:IQ:EGAT:LENG 2000

TRACe:IQ:EGATe:NOF <Number>

Defines the number of gate periods after the trigger signal for gated measurements with the I/Q analyzer.

Parameters:

<Number> Range: 1 to 1023
 *RST: 1

Example: TRAC:IQ:EGAT:NOF 2

TRACe:IQ:EGATe:TYPE <Type>

Selects the gate mode for gated measurements with the I/Q analyzer.

Note: The IF power trigger holdoff time is ignored if you are using the "Level" gate mode in combination with an IF Power trigger.

Parameters:

<Type> **LEVel**
 EDGE
 *RST: EDGE

Example: TRAC:IQ:EGAT:TYPE LEV

10.4.5 Configuring data acquisition

The following commands are required to capture data in the I/Q Analyzer.

Useful commands for I/Q data acquisition described elsewhere

- [SENSe:]SWEep:COUNT on page 270
- [SENSe:]SWEep[:WINDow<n>]:POINTs on page 270
- [SENSe:]SWEep:TIME on page 271

Remote commands exclusive to I/Q data acquisition

[SENSe:]IQ:BANDwidth:MODE.....	248
[SENSe:]IQ:BWIDth:MODE.....	248
[SENSe:]IQ:BANDwidth:RESolution.....	248
[SENSe:]IQ:BWIDth:RESolution.....	248
[SENSe:]IQ:FFT:ALGorithm.....	249
[SENSe:]IQ:FFT:LENGth.....	249
[SENSe:]IQ:FFT:WINDow:LENGth.....	249
[SENSe:]IQ:FFT:WINDow:OVERlap.....	250
[SENSe:]IQ:FFT:WINDow:TYPE.....	250
[SENSe:]SWAPiq.....	250
TRACe:IQ:BWIDth.....	251
TRACe:IQ:RLENGth.....	251
TRACe:IQ:SET.....	251
TRACe:IQ:SRATe.....	252

TRACe:IQ:TPISample?.....	253
TRACe:IQ:WBAND[:STATE].....	253
TRACe:IQ:WBAND:MBWidth.....	254

[SENSe:]IQ:BANDwidth:MODE <Mode>

[SENSe:]IQ:BWIDth:MODE <Mode>

Defines how the resolution bandwidth is determined.

Parameters:

<Mode> AUTO | MANual | FFT

AUTO

(Default) The RBW is determined automatically depending on the sample rate and record length.

MANual

The user-defined RBW is used and the (FFT) window length (and possibly the sample rate) are adapted accordingly. The RBW is defined using the [SENSe:]IQ:BWIDth:RESolution command.

FFT

The RBW is determined by the FFT parameters.

*RST: AUTO

Example:

IQ:BAND:MODE MAN

Switches to manual RBW mode.

IQ:BAND:RES 120000

Sets the RBW to 120 kHz.

Manual operation: See "RBW" on page 79

[SENSe:]IQ:BANDwidth:RESolution <Bandwidth>

[SENSe:]IQ:BWIDth:RESolution <Bandwidth>

Defines the resolution bandwidth manually if [SENSe:]IQ:BWIDth:MODE is set to MAN.

Defines the resolution bandwidth. The available RBW values depend on the sample rate and record length.

Parameters:

<Bandwidth> refer to specifications document

*RST: RBW: AUTO mode is used

Default unit: HZ

Example:

IQ:BAND:MODE MAN

Switches to manual RBW mode.

IQ:BAND:RES 120000

Sets the RBW to 120 kHz.

Manual operation: See "RBW" on page 79

[SENSe:]IQ:FFT:ALGorithm <Method>

Defines the FFT calculation method.

Parameters:

<Method>

SINGle

One FFT is calculated for the entire record length; if the FFT length is larger than the record length (see [\[SENSe:\]IQ:FFT:LENGth](#) and [TRACe:IQ:RLENGth](#)), zeros are appended to the captured data.

AVERAge

Several overlapping FFTs are calculated for each record; the results are averaged to determine the final FFT result for the record.

The user-defined window length and window overlap are used. See [\[SENSe:\]IQ:FFT:WINDow:LENGth](#) and [\[SENSe:\]IQ:FFT:WINDow:OVERlap](#).

*RST: AVER

Example: IQ:FFT:ALG SING

Manual operation: See ["Transformation Algorithm"](#) on page 80

[SENSe:]IQ:FFT:LENGth <NoOfBins>

Defines the number of frequency points determined by each FFT calculation. The more points are used, the higher the resolution in the spectrum becomes, but the longer the calculation takes.

Parameters:

<NoOfBins>

integer value

Range: 3 to 524288

*RST: 4096

Example: IQ:FFT:LENG 2048

Manual operation: See ["FFT Length"](#) on page 80

[SENSe:]IQ:FFT:WINDow:LENGth <NoOfFFT>

Defines the number of samples to be included in a single FFT window when multiple FFT windows are used.

Parameters:

<NoOfFFT>

integer value

Range: 3 to 4096

*RST: record length

Example: IQ:FFT:WIND:LENG 500

Manual operation: See ["Window Length"](#) on page 81

[SENSe:]IQ:FFT:WINDow:OVERlap <Rate>

Defines the part of a single FFT window that is re-calculated by the next FFT calculation.

Parameters:

<Rate> double value
 Percentage rate
 Range: 0 to 1
 *RST: 0.75

Example:

IQ:FFT:WIND:OVER 0.5
 Half of each window overlaps the previous window in FFT calculation.

Manual operation: See ["Window Overlap"](#) on page 81

[SENSe:]IQ:FFT:WINDow:TYPE <Function>

In the I/Q Analyzer you can select one of several FFT window types.

Parameters:

<Function> **BLACKharris**
 Blackman-Harris
 FLATtop
 Flattop
 GAUSSian
 Gauss
 RECTangular
 Rectangular
 P5
 5-Term
 *RST: FLAT

Example:

IQ:FFT:WIND:TYPE GAUS

Manual operation: See ["Window Function"](#) on page 80

[SENSe:]SWAPiq <State>

Defines whether or not the recorded I/Q pairs should be swapped ($I \leftrightarrow Q$) before being processed. Swapping I and Q inverts the sideband.

This is useful if the DUT interchanged the I and Q parts of the signal; then the R&S FSV/A can do the same to compensate for it.

Parameters:

<State> **ON | 1**
 I and Q signals are interchanged
 Inverted sideband, $Q+jI$

OFF | 0

I and Q signals are not interchanged

Normal sideband, $I+j*Q$

*RST: 0

Manual operation: See ["Swap I/Q"](#) on page 79**TRACe:IQ:BWIDth**

Defines or queries the bandwidth of the resampling filter.

The bandwidth of the resampling filter depends on the sample rate.

Parameters:

<Bandwidth> Default unit: HZ

Manual operation: See ["Analysis Bandwidth"](#) on page 78**TRACe:IQ:RLEngh <NoOfSamples>**

Sets the record length for the acquired I/Q data.

Increasing the record length also increases the measurement time.

Note: Alternatively, you can define the measurement time using the `SENS:SWE:TIME` command.**Parameters:**

<NoOfSamples> Number of samples to record.

*RST: 1001

Example: `TRAC:IQ:RLEN 256`**Manual operation:** See ["Record Length"](#) on page 79**TRACe:IQ:SET <NORM>, <0>, <SampleRate>, <TriggerMode>, <TriggerSlope>, <PretriggerSamp>, <NumberSamples>**

Sets up the R&S FSV/A for I/Q measurements.

If you do not use this command to set up I/Q measurements, the R&S FSV/A will use its current settings for I/Q measurements.

If the I/Q Analyzer has not been turned on previously, the command also switches to the I/Q Analyzer.

Note: If you use the default settings with `TRACe:IQ:DATA??`, the following minimum buffer sizes for the response data are recommended:

ASCII format: 10 kBytes

Binary format: 2 kBytes

Parameters:<NORM> This value is always `NORM`.

<0>	Default unit: HZ This value is always 0.
<SampleRate>	Sample rate for the data acquisition. Range: 100 Hz to 10 GHz, continuously adjustable *RST: 32000000 Default unit: HZ
<TriggerMode>	Selection of the trigger source used for the measurement. IMMediate EXTernal EXT2 EXT3 IFPower For IMM mode, gating is automatically deactivated. *RST: IMM
<TriggerSlope>	Used trigger slope. POSitive NEGative *RST: POS
<PretriggerSamp>	Defines the trigger offset in terms of pretrigger samples. Negative values correspond to a trigger delay. This value also defines the interval between the trigger signal and the gate edge in samples. Range: -1399999999 to 1399999999 *RST: 0
<NumberSamples>	Number of measurement values to record (including the pretrigger samples). *RST: 1001
Example:	TRAC:IQ:SET NORM,0,32MHz,EXT,POS,0,2048 Reads 2048 I/Q-values starting at the trigger point. sample rate = 32 MHz trigger = External slope = Positive TRAC:IQ:SET NORM,0,4 MHz,EXT,POS,1024,512 Reads 512 I/Q-values from 1024 measurement points before the trigger point. filter type = NORMAL sample rate = 4 MHz trigger = External slope = Positive

Manual operation: See "[Record Length](#)" on page 79

TRACe:IQ:SRATe <SampleRate>

Sets the final user sample rate for the acquired I/Q data. Thus, the user sample rate can be modified without affecting the actual data capturing settings on the R&S FSV/A.

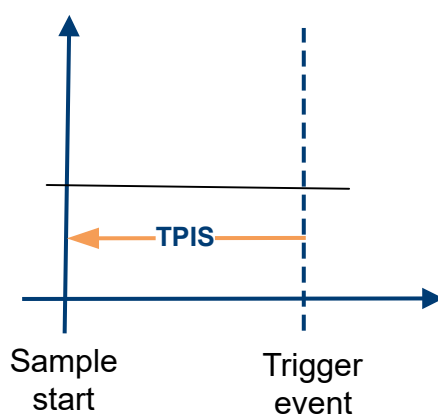
Parameters:

<SampleRate>	*RST: 32 MHz Default unit: HZ
--------------	----------------------------------

Manual operation: See ["Sample Rate"](#) on page 77

TRACe:IQ:TPISample?

Queries the time offset from the sample start to the trigger event (trigger point in sample = TPIS). Since the R&S FSV/A usually samples with a much higher sample rate than the specific application actually requires, the trigger point determined internally is much more precise than the one determined from the (downsampled) data in the application. Thus, the TPIS indicates the offset from the sample start to the actual trigger event.



This value can only be determined in triggered measurements using external or IFPower triggers, otherwise the value is 0.

Return values:

<TPIS> numeric value
 Default unit: s

Example:

TRAC:IQ:TPIS?
 Result for a sample rate of 1 MHz: between 0 and 1/1 MHz, i.e. between 0 and 1 μ s (the duration of 1 sample).

Usage:

Query only

TRACe:IQ:WBAND[:STATe] <State>

Determines whether the wideband provided by bandwidth extension options is used or not (if installed).

Parameters:

<State> ON | OFF | 0 | 1
 OFF | 0
 Switches the function off
 ON | 1
 Switches the function on

Manual operation: See ["Maximum Bandwidth"](#) on page 78

TRACe:IQ:WBANd:MBWidth <Limit>

Defines the maximum analysis bandwidth. Any value can be specified; the next higher fixed bandwidth is used.

Defining a value other than "MAX" is useful if you want to specify the sample rate directly and at the same time, ensure a minimum bandwidth is available.

(See [Section 4.1.1](#), "Sample rate and maximum usable I/Q bandwidth for RF input", on page 20).

Parameters:

<Limit>

40 MHz

Restricts the analysis bandwidth to a maximum of 40 MHz. Larger bandwidth extension options are deactivated.

200 MHz

Restricts the analysis bandwidth to a maximum of 200 MHz. Larger bandwidth extension options are deactivated.

400 MHz

Restricts the analysis bandwidth to a maximum of 400 MHz. Larger bandwidth extension options are deactivated.

600 MHz

Restricts the analysis bandwidth to a maximum of 600 MHz. The bandwidth extension option R&S FSV3-B1000/B1001 is deactivated.

*RST: maximum available

Default unit: Hz

Example:

```
TRAC:IQ:WBAN:MBW 82 MHz
```

```
TRAC:IQ:WBAN:MBW?
```

Result if R&S FSV3-B200 is active:

```
200000000
```

Manual operation: See ["Maximum Bandwidth"](#) on page 78

10.4.6 Adjusting settings automatically

The commands required to adjust settings automatically in a remote environment are described here.

[SENSe:]ADJust:ALL.....	255
[SENSe:]ADJust:CONFigure:LEVel:DURation.....	255
[SENSe:]ADJust:CONFigure:LEVel:DURation:MODE.....	255
[SENSe:]ADJust:CONFigure:HYSTeresis:LOWer.....	256
[SENSe:]ADJust:CONFigure:HYSTeresis:UPPer.....	256
[SENSe:]ADJust:CONFigure:SMODE.....	256
[SENSe:]ADJust:CONFigure:TRIGger.....	256
[SENSe:]ADJust:FREQuency.....	257
[SENSe:]ADJust:LEVel.....	257
CALCulate<n>:MARKer<m>:FUNCTion:POWEr<sb>:AUTO:LIST.....	257

[SENSe:]ADJust:ALL

Initiates a measurement to determine and set the ideal settings for the current task automatically (only once for the current measurement).

This includes:

- Reference level

Example: ADJ:ALL

Manual operation: See ["Adjusting all Determinable Settings Automatically \(Auto All\)"](#) on page 86

[SENSe:]ADJust:CONFigure:LEVel:DURation <Duration>

To determine the ideal reference level, the R&S FSV/A performs a measurement on the current input data. This command defines the length of the measurement if [\[SENSe:\]ADJust:CONFigure:LEVel:DURation:MODE](#) is set to MANual.

Parameters:

<Duration> Numeric value in seconds
 Range: 0.001 to 16000.0
 *RST: 0.001
 Default unit: s

Example: ADJ:CONF:DUR:MODE MAN
 Selects manual definition of the measurement length.
 ADJ:CONF:LEV:DUR 5ms
 Length of the measurement is 5 ms.

Manual operation: See ["\(Meas Time Manual\)"](#) on page 87

[SENSe:]ADJust:CONFigure:LEVel:DURation:MODE <Mode>

To determine the ideal reference level, the R&S FSV/A performs a measurement on the current input data. This command selects the way the R&S FSV/A determines the length of the measurement .

Parameters:

<Mode> **AUTO**
 The R&S FSV/A determines the measurement length automatically according to the current input data.
 MANual
 The R&S FSV/A uses the measurement length defined by [\[SENSe:\]ADJust:CONFigure:LEVel:DURation](#) on page 255.
 *RST: AUTO

Manual operation: See ["Resetting the Automatic Measurement Time \(Meas Time Auto\)"](#) on page 87
 See ["\(Meas Time Manual\)"](#) on page 87

[SENSe:]ADJJust:CONFigure:HYSTeresis:LOWer <Threshold>**Parameters:**

<Threshold> Range: 0 dB to 200 dB
 *RST: +1 dB
 Default unit: dB

Example:

SENS:ADJ:CONF:HYST:LOW 2

For an input signal level of currently 20 dBm, the reference level is only adjusted when the signal level falls below 18 dBm.

Manual operation: See ["Lower Level Hysteresis"](#) on page 87

[SENSe:]ADJJust:CONFigure:HYSTeresis:UPPer <Threshold>**Parameters:**

<Threshold> Range: 0 dB to 200 dB
 *RST: +1 dB
 Default unit: dB

Example:

SENS:ADJ:CONF:HYST:UPP 2

Example:

For an input signal level of currently 20 dBm, the reference level is only adjusted when the signal level rises above 22 dBm.

Manual operation: See ["Upper Level Hysteresis"](#) on page 87

[SENSe:]ADJJust:CONFigure:SMODE <Search Mode>

Determines the search mode for the automatic measurement performed to determine the optimal measurement configuration.

Parameters:

<Search Mode> FAST | POPTimized

FAST

The measurement is optimized for speed.

POPTimized

The measurement is optimized to analyze pulse signals adequately.

Example:

ADJ:CONF:SMOD POPT

Manual operation: See ["Search Mode"](#) on page 88

[SENSe:]ADJJust:CONFigure:TRIGger <State>

Defines the behavior of a triggered measurement when adjusting a setting automatically (using SENS:ADJ:LEV ON, for example).

See ["Adjusting settings automatically during triggered measurements"](#) on page 85.

Parameters:

<State> ON | OFF | 0 | 1

ON | 1

(default:) The measurement for adjustment waits for the next trigger.

OFF | 0

The measurement for adjustment is performed without waiting for a trigger (corresponds to "Continue" in manual operation).

*RST: 0

Example:

```
//Use default ref level at 0.00 dBm.
//Define an RF power trigger at -20 dBm
:TRIG:SEQ:SOUR RFP
:TRIG:SEQ:LEV:RFP -20
//Perform adjustment measurement without waiting for trigger
SENS:ADJ:CONF:TRIG OFF
//Perform auto level adjustment
:SENS:ADJ:LEV;*WAI
```

[SENSe:]ADJust:FREQuency

Sets the center frequency to the frequency with the highest signal level in the current frequency range.

Example: ADJ:FREQ

Manual operation: See ["Adjusting the Center Frequency Automatically \(Auto Frequency\)"](#) on page 86

[SENSe:]ADJust:LEVel

Initiates a single (internal) measurement that evaluates and sets the ideal reference level for the current input data and measurement settings. Thus, the settings of the RF attenuation and the reference level are optimized for the signal level. The R&S FSV/A is not overloaded and the dynamic range is not limited by an S/N ratio that is too small.

Example: ADJ:LEV

Manual operation: See ["Setting the Reference Level Automatically \(Auto Level\)"](#) on page 59

CALCulate<n>:MARKer<m>:FUNCTION:POWer<sb>:AUTO:LIST <List>

Defines the standard to be loaded for an automatic measurement, that is: after [\[SENSe:\]ADJust:ALL](#) on page 255.

The query returns the list of available standards for the measurement.

Is only available for ACLR and SEM measurements.

Suffix:

<n> [Window](#)

<m> [Marker](#)

<sb> irrelevant

Parameters:

<List> String containing the name of the standard to be loaded.
The query returns a comma-separated list of available standards.

Example: :CALCulate1:MARKer1:FUNCTION:POWER1:AUTO:LIST?
//Result:
//L20S, USER, L05R, L05S, RW3G, FW3G, L03S,
L03R, WiMax, RTCD, FTCD, FIS95C0, S2CD,
RIS95C1, FIS95C1, R19C, F19C, F8CD, R8CD,
RIS95C0, L14R, L14S, RFID14443, GSM, PHS,
WiBRo, CDPD, TETRA, PDC, PAPCo25, NONE, L10R,
L10S, L15R, L15S, L20R, AWLAN, BWLAN
CALCulate1:MARKer1:FUNCTION:POWER1:AUTO:LIST
'L10R'

10.5 Configuring the result display

The commands required to configure the screen display in a remote environment are described here.

- [General window commands](#).....258
- [Working with windows in the display](#).....259

10.5.1 General window commands

The following commands are required to configure general window layout, independent of the application.

Note that the suffix <n> always refers to the window *in the currently selected channel* (see [INSTrument\[:SElect\]](#) on page 154).

[DISPlay:FORMat](#)..... 258
[DISPlay\[:WINDow<n>\]:SIZE](#)..... 259

DISPlay:FORMat <Format>

Determines which tab is displayed.

Parameters:

<Format>

SPLit

Displays the MultiView tab with an overview of all active channels

SINGle

Displays the measurement channel that was previously focused.

*RST: SING

Example: DISP:FORM SPL

DISPlay[:WINDow<n>]:SIZE <Size>

Maximizes the size of the selected result display window *temporarily*. To change the size of several windows on the screen permanently, use the `LAY:SPL` command (see `LAYout:SPLitter` on page 263).

Suffix:

<n> `Window`

Parameters:

<Size>

LARGE

Maximizes the selected window to full screen.
Other windows are still active in the background.

SMALL

Reduces the size of the selected window to its original size.
If more than one measurement window was displayed originally, these are visible again.

*RST: `SMALL`

Example:

`DISP:WIND2:SIZE LARGE`

10.5.2 Working with windows in the display

The following commands are required to change the evaluation type and rearrange the screen layout for a channel as you do using the SmartGrid in manual operation. Since the available evaluation types depend on the selected application, some parameters for the following commands also depend on the selected channel.

Note that the suffix <n> always refers to the window *in the currently selected channel*.

(See `INSTrument[:SElect]` on page 154).

<code>LAYout:ADD[:WINDow]?</code>	259
<code>LAYout:CATalog[:WINDow]?</code>	261
<code>LAYout:IDENtify[:WINDow]?</code>	261
<code>LAYout:MOVE[:WINDow]</code>	262
<code>LAYout:REMove[:WINDow]</code>	262
<code>LAYout:REPLace[:WINDow]</code>	262
<code>LAYout:SPLitter</code>	263
<code>LAYout:WINDow<n>:ADD?</code>	264
<code>LAYout:WINDow<n>:REMove</code>	265
<code>LAYout:WINDow<n>:REPLace</code>	265
<code>LAYout:WINDow<n>:TYPE</code>	265

LAYout:ADD[:WINDow]? <WindowName>, <Direction>, <WindowType>

Adds a window to the display in the active channel.

Is always used as a query so that you immediately obtain the name of the new window as a result.

To replace an existing window, use the `LAYout:REPLace[:WINDow]` command.

Query parameters:

<WindowName>	String containing the name of the existing window the new window is inserted next to. By default, the name of a window is the same as its index. To determine the name and index of all active windows, use the LAYout:CATalog[:WINDow]? query .
<Direction>	LEFT RIGHT ABOVE BELOW Direction the new window is added relative to the existing window.
<WindowType>	text value Type of result display (evaluation method) you want to add. See the table below for available parameter values.

Return values:

<NewWindowName>	When adding a new window, the command returns its name (by default the same as its number) as a result.
-----------------	---

Example:

LAY:ADD? '1', LEFT, MTAB

Result:

'2'

Adds a new window named '2' with a marker table to the left of window 1.

Usage:

Query only

Manual operation:

See ["Magnitude"](#) on page 14
 See ["Spectrum"](#) on page 15
 See ["I/Q-Vector"](#) on page 15
 See ["Real/Imag \(I/Q\)"](#) on page 16
 See ["Phase vs. Time"](#) on page 16
 See ["Marker Table"](#) on page 17
 See ["Marker Peak List"](#) on page 17

Table 10-4: <WindowType> parameter values for IQ Analyzer application

Parameter value	Window type
Basic I/Q measurement:	
FREQ	"Spectrum"
MAGN	"Magnitude"
MTABle	"Marker table"
PEAKlist	"Marker peak list"
PHASe	"Phase vs. time"
RIMAG	"Real/Imag (I/Q)"
VECT	"I/Q Vector"
Frequency and time domain measurements:	
DIAGram	"Diagram"

Parameter value	Window type
MTABle	"Marker table"
PEAKlist	"Marker peak list"
RSUMmary	"Result summary"

LAYout:CATalog[:WINDow]?

Queries the name and index of all active windows in the active channel from top left to bottom right. The result is a comma-separated list of values for each window, with the syntax:

<WindowName_1>,<WindowIndex_1>..

Return values:

<WindowName> string
Name of the window.
In the default state, the name of the window is its index.

<WindowIndex> **numeric value**
Index of the window.

Example:

LAY:CAT?

Result:

'2',2,'1',1

Two windows are displayed, named '2' (at the top or left), and '1' (at the bottom or right).

Usage: Query only

LAYout:IDENtify[:WINDow]? <WindowName>

Queries the **index** of a particular display window in the active channel.

Query parameters:

<WindowName> String containing the name of a window.

Return values:

<WindowIndex> Index number of the window.

Example:

LAY:IDEN:WIND? '2'

Queries the index of the result display named '2'.

Response:

2

Usage: Query only

LAYout:MOVE[:WINDow] <WindowName>, <WindowName>, <Direction>**Setting parameters:**

<WindowName>	String containing the name of an existing window that is to be moved. By default, the name of a window is the same as its index. To determine the name and index of all active windows in the active channel, use the LAYout:CATalog[:WINDow]? query.
<WindowName>	String containing the name of an existing window the selected window is placed next to or replaces. By default, the name of a window is the same as its index. To determine the name and index of all active windows in the active channel, use the LAYout:CATalog[:WINDow]? query.
<Direction>	LEFT RIGHT ABOVE BELOW REPLACE Destination the selected window is moved to, relative to the reference window.

Example: `LAY:MOVE '4','1',LEFT`
Moves the window named '4' to the left of window 1.

Example: `LAY:MOVE '1','3',REPL`
Replaces the window named '3' by window 1. Window 3 is deleted.

Usage: Setting only

LAYout:REMove[:WINDow] <WindowName>

Removes a window from the display in the active channel.

Setting parameters:

<WindowName>	String containing the name of the window. In the default state, the name of the window is its index.
--------------	--

Example: `LAY:REM '2'`
Removes the result display in the window named '2'.

Usage: Setting only

LAYout:REPLace[:WINDow] <WindowName>, <WindowType>

Replaces the window type (for example from "Diagram" to "Result Summary") of an already existing window in the active channel while keeping its position, index and window name.

To add a new window, use the [LAYout:ADD\[:WINDow\]?](#) command.

Setting parameters:

<WindowName>	String containing the name of the existing window. By default, the name of a window is the same as its index. To determine the name and index of all active windows in the active channel, use the LAYout:CATalog[:WINDow]? query.
--------------	---

<WindowType> Type of result display you want to use in the existing window. See [LAYout:ADD\[:WINDow\]?](#) on page 259 for a list of available window types.

Example: `LAY:REPL:WIND '1',MTAB`
Replaces the result display in window 1 with a marker table.

Usage: Setting only

LAYout:SPLitter <Index1>, <Index2>, <Position>

Changes the position of a splitter and thus controls the size of the windows on each side of the splitter.

Note that windows must have a certain minimum size. If the position you define conflicts with the minimum size of any of the affected windows, the command does not work, but does not return an error.

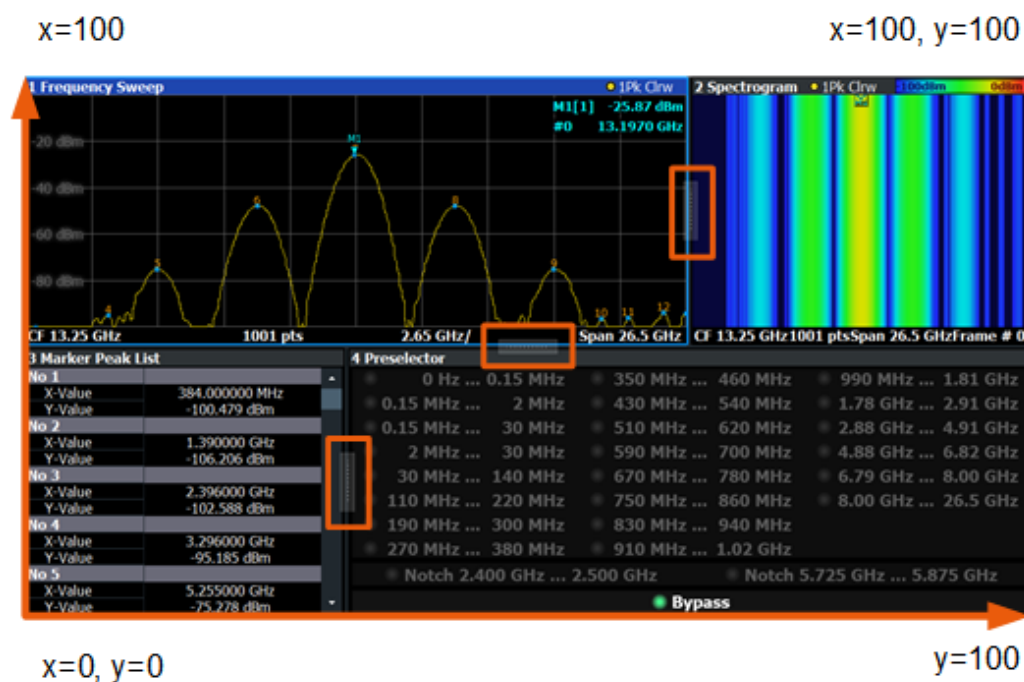


Figure 10-1: SmartGrid coordinates for remote control of the splitters

Setting parameters:

<Index1> The index of one window the splitter controls.
<Index2> The index of a window on the other side of the splitter.

<Position>	New vertical or horizontal position of the splitter as a fraction of the screen area (without channel and status bar and softkey menu). The point of origin (x = 0, y = 0) is in the lower left corner of the screen. The end point (x = 100, y = 100) is in the upper right corner of the screen. (See Figure 10-1.) The direction in which the splitter is moved depends on the screen layout. If the windows are positioned horizontally, the splitter also moves horizontally. If the windows are positioned vertically, the splitter also moves vertically. Range: 0 to 100
Example:	LAY:SPL 1,3,50 Moves the splitter between window 1 ('Frequency Sweep') and 3 ('Marker Table') to the center (50%) of the screen, i.e. in the figure above, to the left.
Example:	LAY:SPL 1,4,70 Moves the splitter between window 1 ('Frequency Sweep') and 3 ('Marker Peak List') towards the top (70%) of the screen. The following commands have the exact same effect, as any combination of windows above and below the splitter moves the splitter vertically. LAY:SPL 3,2,70 LAY:SPL 4,1,70 LAY:SPL 2,1,70
Usage:	Setting only

LAYout:WINDow<n>:ADD? <Direction>,<WindowType>

Adds a measurement window to the display. Note that with this command, the suffix <n> determines the existing window next to which the new window is added. Unlike [LAYout:ADD\[:WINDow\]?](#), for which the existing window is defined by a parameter.

To replace an existing window, use the [LAYout:WINDow<n>:REPLace](#) command.

Is always used as a query so that you immediately obtain the name of the new window as a result.

Suffix:

<n> [Window](#)

Query parameters:

<Direction> LEFT | RIGHT | ABOVE | BELOW

<WindowType> Type of measurement window you want to add.
See [LAYout:ADD\[:WINDow\]?](#) on page 259 for a list of available window types.

Return values:

<NewWindowName> When adding a new window, the command returns its name (by default the same as its number) as a result.

Example: `LAY:WIND1:ADD? LEFT,MTAB`
Result:
 '2'
 Adds a new window named '2' with a marker table to the left of window 1.

Usage: Query only

LAYout:WINDow<n>:REMove

Removes the window specified by the suffix <n> from the display in the active channel.
 The result of this command is identical to the `LAYout:REMove[:WINDow]` command.

Suffix:
 <n> [Window](#)

Example: `LAY:WIND2:REM`
 Removes the result display in window 2.

Usage: Event

LAYout:WINDow<n>:REPLace <WindowType>

Changes the window type of an existing window (specified by the suffix <n>) in the active channel.

The effect of this command is identical to the `LAYout:REPLace[:WINDow]` command.

To add a new window, use the `LAYout:WINDow<n>:ADD?` command.

Suffix:
 <n> [Window](#)

Setting parameters:

<WindowType> Type of measurement window you want to replace another one with.
 See `LAYout:ADD[:WINDow]?` on page 259 for a list of available window types.

Example: `LAY:WIND2:REPL MTAB`
 Replaces the result display in window 2 with a marker table.

Usage: Setting only

LAYout:WINDow<n>:TYPE <WindowType>

Queries or defines the window type of the window specified by the index <n>. The window type determines which results are displayed. For a list of possible window types, see `LAYout:ADD[:WINDow]?` on page 259.

Note that this command is not available in all applications and measurements.

Suffix:

<n> 1..n
Window

Parameters:

<WindowType>

Example:

LAY:WIND2:TYPE?

10.6 Capturing data and performing sweeps



Different measurement procedures

Two different procedures to capture I/Q data remotely are available:

- Measurement and result query with one command (see [TRACe:IQ:DATA?](#) on page 334)
This method causes the least delay between measurement and output of the result data, but it requires the control computer to wait actively for the response data.
- Setting up the instrument, starting the measurement via `INIT` and querying the result list at the end of the measurement (see [TRACe:IQ:DATA:MEMory?](#) on page 335)
With this method, the control computer can be used for other activities during the measurement. However, the additional time needed for synchronization via service request must be taken into account.

ABORT	266
INITiate<n>:CONMeas	267
INITiate<n>:CONTinuous	268
INITiate<n>[:IMMediate]	268
INITiate:SEQuencer:ABORT	269
INITiate:SEQuencer:IMMediate	269
INITiate:SEQuencer:MODE	269
[SENSe:]SWEep:COUNt	270
[SENSe:]SWEep:COUNt:CURREnt?	270
[SENSe:]SWEep[:WINDow<n>]:POINTs	270
[SENSe:]SWEep:TIME	271
SYSTem:SEQuencer	271

ABORT

Aborts the measurement in the current channel and resets the trigger system.

To prevent overlapping execution of the subsequent command before the measurement has been aborted successfully, use the `*OPC?` or `*WAI` command after `ABOR` and before the next command.

For details on overlapping execution see [Remote control via SCPI](#).

To abort a sequence of measurements by the Sequencer, use the `INITiate:SEQuencer:ABORt` command.

Note on blocked remote control programs:

If a sequential command cannot be completed, for example because a triggered sweep never receives a trigger, the remote control program will never finish and the remote channel to the R&S FSV/A is blocked for further commands. In this case, you must interrupt processing on the remote channel first in order to abort the measurement.

To do so, send a "Device Clear" command from the control instrument to the R&S FSV/A on a parallel channel to clear all currently active remote channels. Depending on the used interface and protocol, send the following commands:

- **Visa:** `viClear()`

Now you can send the `ABORt` command on the remote channel performing the measurement.

Example: `ABOR; :INIT:IMM`
Aborts the current measurement and immediately starts a new one.

Example: `ABOR; *WAI`
`INIT:IMM`
Aborts the current measurement and starts a new one once abortion has been completed.

Usage: Event

INITiate<n>:CONMeas

Restarts a (single) measurement that has been stopped (using `ABORt`) or finished in single sweep mode.

The measurement is restarted at the beginning, not where the previous measurement was stopped.

Suffix:
<n> irrelevant

Example: `INIT:CONT OFF`
Switches to single sweep mode.
`DISP:WIND:TRAC:MODE AVER`
Switches on trace averaging.
`SWE:COUN 20`
Setting the sweep counter to 20 sweeps.
`INIT; *WAI`
Starts the measurement and waits for the end of the 20 sweeps.
`INIT:CONM; *WAI`
Continues the measurement (next 20 sweeps) and waits for the end.
Result: Averaging is performed over 40 sweeps.

Usage: Asynchronous command

Manual operation: See ["Continue Single Sweep"](#) on page 83

INITiate<n>:CONTinuous <State>

Controls the sweep mode for an individual channel.

Note that in single sweep mode, you can synchronize to the end of the measurement with *OPC, *OPC? or *WAI. In continuous sweep mode, synchronization to the end of the measurement is not possible. Thus, it is not recommended that you use continuous sweep mode in remote control, as results like trace data or markers are only valid after a single sweep end synchronization.

For details on synchronization see [Remote control via SCPI](#).

If the sweep mode is changed for a channel while the Sequencer is active (see [INITiate:SEQuencer:IMMediate](#) on page 269), the mode is only considered the next time the measurement in that channel is activated by the Sequencer.

Suffix:

<n> irrelevant

Parameters:

<State> ON | OFF | 0 | 1

ON | 1

Continuous sweep

OFF | 0

Single sweep

*RST: 1 (some applications can differ)

Example:

INIT:CONT OFF

Switches the sweep mode to single sweep.

INIT:CONT ON

Switches the sweep mode to continuous sweep.

Manual operation: See ["Continuous Sweep / Run Cont"](#) on page 82

INITiate<n>[:IMMediate]

Starts a (single) new measurement.

With sweep count or average count > 0, this means a restart of the corresponding number of measurements. With trace mode MAXHold, MINHold and AVERage, the previous results are reset on restarting the measurement.

You can synchronize to the end of the measurement with *OPC, *OPC? or *WAI.

For details on synchronization see [Remote control via SCPI](#).

Suffix:

<n> irrelevant

Example: `INIT:CONT OFF`
 Switches to single sweep mode.
 `DISP:WIND:TRAC:MODE AVER`
 Switches on trace averaging.
 `SWE:COUN 20`
 Sets the sweep counter to 20 sweeps.
 `INIT;*WAI`
 Starts the measurement and waits for the end of the 20 sweeps.

Usage: Asynchronous command

Manual operation: See ["Single Sweep / Run Single"](#) on page 83

INITiate:SEQuencer:ABORt

Stops the currently active sequence of measurements.

You can start a new sequence any time using [INITiate:SEQuencer:IMMediate](#) on page 269.

Usage: Event

INITiate:SEQuencer:IMMediate

Starts a new sequence of measurements by the Sequencer.

Before this command can be executed, the Sequencer must be activated (see [SYSTem:SEQuencer](#) on page 271).

Example: `SYST:SEQ ON`
 Activates the Sequencer.
 `INIT:SEQ:MODE SING`
 Sets single sequence mode so each active measurement is performed once.
 `INIT:SEQ:IMM`
 Starts the sequential measurements.

INITiate:SEQuencer:MODE <Mode>

Defines the capture mode for the entire measurement sequence and all measurement groups and channels it contains.

Note: To synchronize to the end of a measurement sequence using `*OPEC`, `*OPEC?` or `*WAI`, use `SINGLE` Sequencer mode.

Parameters:

<Mode>

SINGLE

Each measurement group is started one after the other in the order of definition. All measurement channels in a group are started simultaneously and performed once. After *all* measurements are completed, the next group is started. After the last group, the measurement sequence is finished.

CONTInuous

Each measurement group is started one after the other in the order of definition. All measurement channels in a group are started simultaneously and performed once. After *all* measurements are completed, the next group is started. After the last group, the measurement sequence restarts with the first one and continues until it is stopped explicitly.

*RST: CONTInuous

[SENSe:]SWEep:COUNT <SweepCount>

Defines the number of sweeps that the application uses to average traces.

In continuous sweep mode, the application calculates the moving average over the average count.

In single sweep mode, the application stops the measurement and calculates the average after the average count has been reached.

Example: SWE:COUN 64
Sets the number of sweeps to 64.
INIT:CONT OFF
Switches to single sweep mode.
INIT;*WAI
Starts a sweep and waits for its end.

Manual operation: See "[Sweep/Average Count](#)" on page 82

[SENSe:]SWEep:COUNT:CURRent?

This query returns the current number of started sweeps or measurements. This command is only available if a sweep count value is defined and the instrument is in single sweep mode.

Return values:
<CurrentCount>

Example: SWE:COUNT 64
Sets sweep count to 64
INIT:CONT OFF
Switches to single sweep mode
INIT
Starts a sweep (without waiting for the sweep end!)
SWE:COUN:CURR?
Queries the number of started sweeps

Usage: Query only

[SENSe:]SWEep[:WINDow<n>]:POINTs

This command defines the number of sweep points to analyze after a sweep.

Suffix:

<n>

Example: SWE:POIN 251**Manual operation:** See ["Sweep Points"](#) on page 82**[SENSe:]SWEep:TIME <Time>**

Defines the sweep time. It automatically decouples the time from any other settings.

Parameters:

<Time>

refer to specifications document

*RST: depends on current settings (determined automatically)

Default unit: S

Manual operation: See ["Meas Time"](#) on page 79**SYSTem:SEQuencer <State>**

Turns the Sequencer on and off. The Sequencer must be active before any other Sequencer commands (INIT:SEQ...) are executed, otherwise an error occurs.

A detailed programming example is provided in the "Operating Modes" section in the R&S FSV/A User Manual.

Parameters:

<State>

ON | OFF | 0 | 1

ON | 1

The Sequencer is activated and a sequential measurement is started immediately.

OFF | 0

The Sequencer is deactivated. Any running sequential measurements are stopped. Further Sequencer commands (INIT:SEQ...) are not available.

*RST: 0

Example:

SYST:SEQ ON

Activates the Sequencer.

INIT:SEQ:MODE SING

Sets single Sequencer mode so each active measurement is performed once.

INIT:SEQ:IMM

Starts the sequential measurements.

SYST:SEQ OFF

10.7 I/Q Analysis

General result analysis settings concerning the trace, markers, etc. can be configured using the following commands. They are identical to the analysis functions in the Spectrum application except for the special marker functions.

- [Configuring standard traces](#).....272
- [Configuring spectrograms](#).....279
- [Using markers](#).....287

10.7.1 Configuring standard traces

Useful commands for trace configuration described elsewhere

- [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:Y:SPACing](#)
on page 234
- [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:Y\[:SCALE\]](#)
on page 232

Remote commands exclusive to trace configuration

DISPlay[:WINDow<n>]:TRACe<t>:LABel[:STATe]	272
DISPlay[:WINDow<n>]:TRACe<t>:LABel:TEXT	273
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:MODE	273
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:MODE:HCONTinuous	274
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:PRESet	275
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>[:STATe]	275
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing:APERture	276
DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing[:STATe]	276
[SENSe:]AVERAge<n>:TYPE	277
[SENSe:][:WINDow<n>]:DETector<t>[:FUNCTION]	277
[SENSe:][:WINDow<n>]:DETector<t>[:FUNCTION]:AUTO	278
TRACe<n>:COPY	278
[SENSe:]AVERAge<n>:COUNT	278
TRACe:IQ:AVERAge:COUNT	278
[SENSe:]AVERAge<n>[:STATe<t>]	279
TRACe:IQ:AVERAge[:STATe]	279

DISPlay[:WINDow<n>]:TRACe<t>:LABel[:STATe] <State>

Turns on the display of a descriptive label for the specified trace instead of the default "Trace <x>" label.

Define the label using the [DISPlay\[:WINDow<n>\]:TRACe<t>:LABel:TEXT](#) command.

You can only configure labels for active traces and for traces whose "State" is enabled.

Suffix:

<n> 1..n
Window

<t> 1..n
Trace

Parameters:

<State> **OFF**
Switch the function off
ON
Switch the function on
*RST: OFF

Example:

```
DISP:WIND2:TRAC1:LAB ON
DISP:WIND2:TRAC1:LAB:TEXT 'MaxTrace'
```

Manual operation: See "Trace Labels" on page 93

DISPlay[:WINDow<n>]:TRACe<t>:LABel:TEXT <Text>

Defines a descriptive label for the specified trace instead of the default "Trace <x>" label.

Enable the label using the `DISPlay[:WINDow<n>]:TRACe<t>:LABel[:STATe]` command.

You can only configure labels for active traces and for traces whose "State" is enabled.

Suffix:

<n> 1..n
Window

<t> 1..n
Trace

Parameters:

<Text> String containing the trace label.

Example:

```
DISP:WIND2:TRAC1:LAB ON
DISP:WIND2:TRAC1:LAB:TEXT 'MaxTrace'
```

Manual operation: See "Trace Labels" on page 93

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:MODE <Mode>

Selects the trace mode. If necessary, the selected trace is also activated.

For max hold, min hold or average trace mode, you can set the number of single measurements with `[SENSe:]SWEep:COUNT`. Note that synchronization to the end of the measurement is possible only in single sweep mode.

Suffix:

<n> Window

<w>	subwindow Not supported by all applications
<t>	Trace
Parameters:	
<Mode>	WRITe (default:) Overwrite mode: the trace is overwritten by each sweep. AVERage The average is formed over several sweeps. The "Sweep/Average Count" determines the number of averaging procedures. MAXHold The maximum value is determined over several sweeps and displayed. The R&S FSV/A saves the sweep result in the trace memory only if the new value is greater than the previous one. MINHold The minimum value is determined from several measurements and displayed. The R&S FSV/A saves the sweep result in the trace memory only if the new value is lower than the previous one. VIEW The current contents of the trace memory are frozen and displayed. BLANK Hides the selected trace. *RST: Trace 1: WRITe, Trace 2-6: BLANK
Example:	INIT:CONT OFF Switching to single sweep mode. SWE:COUN 16 Sets the number of measurements to 16. DISP:TRAC3:MODE WRIT Selects clear/write mode for trace 3. INIT;*WAI Starts the measurement and waits for the end of the measurement.

Manual operation: See "Trace Mode" on page 90

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:MODE:HCONTinuous
<State>

Turns an automatic reset of a trace on and off after a parameter has changed.

The reset works for trace modes min hold, max hold and average.

Note that the command has no effect if critical parameters like the span have been changed to avoid invalid measurement results

Suffix:<n> [Window](#)

<w> subwindow

<t> [Trace](#)**Parameters:**

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

DISP:WIND:TRAC3:MODE:HCON ON
 Switches off the reset function.

Manual operation: See ["Hold"](#) on page 92**DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:PRESet <ResultType>**

Applies predefined, commonly required trace settings to the selected window.

Suffix:<n> 1..n
[Window](#)<w> 1..n
subwindow<t> 1..n
[Trace](#)**Parameters:**<ResultType> **ALL**
Preset All Traces**MAM**

Max | Avg | Min

MCM

Max | ClrWrite | Min

Example:

DISP:WIND3:TRAC:PRESet MCM
 In window 3, the traces are set to the following modes:
 Trace 1: Max Hold
 Trace 2: Clear Write
 Trace 3: Min Hold

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>[:STATe] <State>

Turns a trace on and off.

The measurement continues in the background.

Suffix:

<n>	Window
<w>	subwindow Not supported by all applications
<t>	Trace

Parameters:

<State>	ON OFF 0 1 OFF 0 Switches the function off ON 1 Switches the function on
---------	--

Example: DISP:TRAC3 ON

Manual operation: See ["Trace 1/Trace 2/Trace 3/Trace 4/Trace 5/Trace 6"](#) on page 90
See ["Trace 1/ Trace 2/ Trace 3/ Trace 4 \(Softkeys\)"](#) on page 93

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing:APERture
<Aperture>

Defines the degree (aperture) of the trace smoothing, if [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:SMOothing\[:STATE\]](#) TRUE.

Suffix:

<n>	Window
<w>	subwindow
<t>	Trace

Parameters:

<Aperture>	Range: 1 to 50 *RST: 2 Default unit: PCT
------------	--

Example: DISP3:TRAC2:SMO:APER 5
Defines an aperture of 5% for trace 2 in window 3

Manual operation: See ["Smoothing"](#) on page 92

DISPlay[:WINDow<n>][:SUBWindow<w>]:TRACe<t>:SMOothing[:STATE] <State>

Turns trace smoothing for a particular trace on and off.

If enabled, the trace is smoothed by the value specified using [DISPlay\[:WINDow<n>\]\[:SUBWindow<w>\]:TRACe<t>:SMOothing:APERture](#) on page 276.

For more information see the R&S FSV/A User Manual.

Suffix:

<n>	Window
-----	------------------------

<w> subwindow

<t> [Trace](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0
Switches the function off

ON | 1
Switches the function on

Example: DISP3:TRAC2:SMO ON
Turns on trace smoothing for trace 2 in window 3

Manual operation: See "[Smoothing](#)" on page 92

[SENSe:]AVERage<n>:TYPE <Mode>

Selects the trace averaging mode.

Suffix:

<n> 1..n
[Window](#)

Parameters:

<Mode> **LOGarithmic**
The logarithmic power values are averaged.

LINear
The power values are averaged before they are converted to logarithmic values.

POWER
The power level values are converted into unit Watt prior to averaging. After the averaging, the data is converted back into its original unit.

Example: AVER:TYPE LIN
Switches to linear average calculation.

Manual operation: See "[Average Mode](#)" on page 92

[SENSe:][WINDow<n>:]DETector<t>[:FUNCTion] <Detector>

Defines the trace detector to be used for trace analysis.

Suffix:

<n> [Window](#)

<t> [Trace](#)

Example: DET POS
Sets the detector to "positive peak".

Manual operation: See "[Detector](#)" on page 91

[SENSe:][WINDow<n>:]DETector<t>[:FUNCTION]:AUTO <State>

Couples and decouples the detector to the trace mode.

Suffix:

<n> [Window](#)

<t> [Trace](#)

Parameters:

<State> ON | OFF | 0 | 1
*RST: 1

Example: DET:AUTO OFF
The selection of the detector is not coupled to the trace mode.

Manual operation: See ["Detector"](#) on page 91

TRACe<n>:COPY <TraceNumber>, <TraceNumber>

Copies data from one trace to another.

Suffix:

<n> [Window](#)

Parameters:

<TraceNumber> **TRACE1 | TRACE2 | TRACE3 | TRACE4 | TRACE5 | TRACE6**
The first parameter is the destination trace, the second parameter is the source.
(Note the 'e' in the parameter is required!)

Example: TRAC:COPY TRACE1, TRACE2
Copies the data from trace 2 to trace 1.

Manual operation: See ["Copy Trace"](#) on page 93

[SENSe:]AVERage<n>:COUNT <AverageCount>

TRACe:IQ:AVERage:COUNT <NumberSets>

This command defines the number of I/Q data sets that the averaging is based on.

Parameters:

<NumberSets> Range: 0 to 32767
*RST: 0

Example: TRAC:IQ ON
Switches on acquisition of I/Q data.
TRAC:IQ:AVER ON
Enables averaging of the I/Q measurement data
TRAC:IQ:AVER:COUN 10
Selects averaging over 10 data sets
TRAC:IQ:DATA?
Starts the measurement and reads out the averaged data.

[SENSe:]AVERage<n>[:STATe<t>] <State>

TRACe:IQ:AVERage[:STATe] <State>

This command turns averaging of the I/Q data on and off.

If averaging is on, the maximum amount of I/Q data that can be recorded is 512kS (524288 samples).

Parameters:

<State>

ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

TRAC:IQ ON

Switches on acquisition of I/Q data.

TRAC:IQ:AVER ON

Enables averaging of the I/Q measurement data.

TRAC:IQ:AVER:COUN 10

Selects averaging over 10 data sets.

TRAC:IQ:DATA?

Starts the measurement and reads out the averaged data.

10.7.2 Configuring spectrograms

In addition to the standard "level versus frequency" or "level versus time" spectrum traces, the R&S FSV/A also provides a spectrogram display of the measured data. A spectrogram shows how the spectral density of a signal varies over time. The x-axis shows the frequency, the y-axis shows the time. The commands required to configure spectrograms in a remote environment are described here. For details and manual operation see [Section 6.2, "Spectrogram settings"](#), on page 94.



When configuring spectrograms, the window suffix is irrelevant. The settings are always applied to the spectrogram window, or to all spectrogram windows, if several are active for the same channel.

For commands to set markers in spectrograms, see [Section 10.7.3.4, "Marker search \(spectrograms\)"](#), on page 299.

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10.7.2.1 Configuring a spectrogram measurement

CALCulate<n>:SGRam:CLEar[:IMMediate].....	280
CALCulate<n>:SPECtrogram:CLEar[:IMMediate].....	280
CALCulate<n>:SGRam:CONTinuous.....	280
CALCulate<n>:SPECtrogram:CONTinuous.....	280

CALCulate<n>:SGRam:FRAMe:COUNT.....	281
CALCulate<n>:SPECtrogram:FRAMe:COUNT.....	281
CALCulate<n>:SGRam:FRAMe:SElect.....	281
CALCulate<n>:SPECtrogram:FRAMe:SElect.....	281
CALCulate<n>:SGRam:HDEPth.....	282
CALCulate<n>:SPECtrogram:HDEPth.....	282
CALCulate<n>:SGRam:LAYout.....	282
CALCulate<n>:SPECtrogram:LAYout.....	282
CALCulate<n>:SGRam[:STATe].....	282
CALCulate<n>:SPECtrogram[:STATe].....	282
CALCulate<n>:SGRam:THReedim[:STATe].....	283
CALCulate<n>:SPECtrogram:THReedim[:STATe].....	283
CALCulate<n>:SGRam:TRACe.....	283
CALCulate<n>:SPECtrogram:TRACe.....	283
CALCulate<n>:SGRam:TSTamp:DATA?.....	283
CALCulate<n>:SPECtrogram:TSTamp:DATA?.....	283
CALCulate<n>:SGRam:TSTamp[:STATe].....	284
CALCulate<n>:SPECtrogram:TSTamp[:STATe].....	284

CALCulate<n>:SGRam:CLEAr[:IMMediate]**CALCulate<n>:SPECtrogram:CLEAr[:IMMediate]**

Resets the spectrogram and clears the history buffer.

Suffix:

<n> [Window](#)

Example:

```
//Reset the result display and clear the memory
CALC:SGR:CLE
```

Manual operation: See ["Clear Spectrogram"](#) on page 84

CALCulate<n>:SGRam:CONTInuous <State>**CALCulate<n>:SPECtrogram:CONTInuous <State>**

Determines whether the results of the last measurement are deleted before starting a new measurement in single sweep mode.

This setting applies to all spectrograms in the channel.

Suffix:

<n> [Window](#)

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
 Switches the function off
ON | 1
 Switches the function on

Example:

```
INIT:CONT OFF
Selects single sweep mode.
INIT;*WAI
Starts the sweep and waits for the end of the sweep.
CALC:SGR:CONT ON
Repeats the single sweep measurement without deleting the
results of the last measurement.
```

Manual operation: See ["Continue Frame"](#) on page 84

CALCulate<n>:SGRam:FRAME:COUNT <Frames>
CALCulate<n>:SPECtrogram:FRAME:COUNT <Frames>

Defines the number of frames to be recorded in a single sweep.

This value applies to all spectrograms in the channel.

Suffix:

<n> [Window](#)

Parameters:

<Frames> The maximum number of frames depends on the history depth.
 Range: 1 to history depth
 Increment: 1
 *RST: 1

Example:

```
//Select single sweep mode
INIT:CONT OFF
//Set the number of frames to 200
CALC:SGR:FRAM:COUN 200
```

Manual operation: See ["Frame Count"](#) on page 84

CALCulate<n>:SGRam:FRAME:SElect <Frame> | <Time>
CALCulate<n>:SPECtrogram:FRAME:SElect <Frame> | <Time>

Selects a specific frame for further analysis.

The command is available if no measurement is running or after a single sweep has ended.

Suffix:

<n> [Window](#)

Parameters:

<Frame> Selects a frame directly by the frame number. Valid if the time stamp is off.
 The range depends on the history depth.
 Default unit: S

<Time> Selects a frame via its time stamp. Valid if the time stamp is on.
 The number is the distance to frame 0 in seconds. The range depends on the history depth.

Example: `INIT:CONT OFF`
 Stop the continuous sweep.
 `CALC:SGR:FRAM:SEL -25`
 Selects frame at -25 s.

Manual operation: See ["Select Frame"](#) on page 83

CALCulate<n>:SGRam:HDEPth <History>
CALCulate<n>:SPECtrogram:HDEPth <History>

Defines the number of frames to be stored in the R&S FSV/A memory.

Suffix:

<n> [Window](#)

Parameters:

<History> The maximum number of frames depends on the number of sweep points.

Range: 781 to 20000

Increment: 1

*RST: 3000

Example: `//Set the history depth to 1500`
 `CALC:SGR:SPEC 1500`

Manual operation: See ["History Depth"](#) on page 96

CALCulate<n>:SGRam:LAYout <State>
CALCulate<n>:SPECtrogram:LAYout <State>

This command selects the state and size of spectrograms.

The command is available for result displays that support spectrograms.

Suffix:

<n> [Window](#)

Example: `CALC4:SPEC:LAY FULL`
 Shows the spectrogram in window 4. The corresponding trace diagram is hidden.

Manual operation: See ["State"](#) on page 95

CALCulate<n>:SGRam[:STATe] <State>
CALCulate<n>:SPECtrogram[:STATe] <State>

Turns the spectrogram on and off.

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:SGR ON

Activates the Spectrogram result display.

CALCulate<n>:SGRam:THReedim[:STATe] <State>**CALCulate<n>:SPECtrogram:THReedim[:STATe] <State>**

Activates or deactivates a 3-dimensional spectrogram for the selected result display.

Suffix:

<n>

[Window](#)**Parameters:**

<State>

ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

*RST: 0

Example:

CALC:SPEC:THR:STAT ON

Manual operation: See ["3D Spectrogram State"](#) on page 96**CALCulate<n>:SGRam:TRACe <Trace>****CALCulate<n>:SPECtrogram:TRACe <Trace>**

This command determines the trace in the result display the Spectrogram is based on.

Suffix:

<n>

[Window](#)**Parameters:**

<Trace>

TRACE1 | TRACE2 | TRACE3 | TRACE4 | TRACE5 | TRACE6

How many traces are available depends on the selected result display.

Example:

CALC2:SPEC:TRAC TRACE3

CALCulate<n>:SGRam:TSTamp:DATA? <Frames>**CALCulate<n>:SPECtrogram:TSTamp:DATA? <Frames>**

Queries the starting time of the frames.

The return values consist of four values for each frame. If the "Spectrogram" is empty, the command returns '0,0,0,0'. The times are given as delta values, which simplifies evaluating relative results; however, you can also calculate the absolute date and time as displayed on the screen.

The frame results themselves are returned with TRAC:DATA? SGR

Suffix:<n> [Window](#)**Query parameters:**

<Frames>

CURRent

Returns the starting time of the current frame.

ALL

Returns the starting time for all frames. The results are sorted in descending order, beginning with the current frame.

Return values:

<Seconds>

Number of seconds that have passed since 01.01.1970 until the frame start

<Nanoseconds>

Number of nanoseconds that have passed *in addition to the* <Seconds> since 01.01.1970 until the frame start.

<Reserved>

The third value is reserved for future uses.

<Reserved>

The fourth value is reserved for future uses.

Example:

CALC:SGR:TST:DATA? ALL

Returns the starting times of all frames sorted in a descending order.

Usage:

Query only

Manual operation: See ["Time Stamp"](#) on page 96**CALCulate<n>:SGRam:TSTamp[:STATe] <State>****CALCulate<n>:SPECTrogram:TSTamp[:STATe] <State>**

Activates and deactivates the time stamp.

If the time stamp is active, some commands do not address frames as numbers, but as (relative) time values:

- [CALCulate<n>:DELTaMarker<m>:SPECTrogram:FRAMe](#) on page 305
- [CALCulate<n>:MARKer<m>:SPECTrogram:FRAMe](#) on page 300
- [CALCulate<n>:SPECTrogram:FRAMe:SELection](#) on page 281

Suffix:<n> 1..n
[Window](#)**Parameters:**

<State>

ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

//Activates the time stamp

CALC:SGR:TST ON

Manual operation: See ["Time Stamp"](#) on page 96

10.7.2.2 Configuring the color map

DISPlay[:WINDow<n>]:SGRam:COLor:DEFault	285
DISPlay[:WINDow<n>]:SPEctrogram:COLor:DEFault	285
DISPlay[:WINDow<n>]:SGRam:COLor:LOWer	285
DISPlay[:WINDow<n>]:SPEctrogram:COLor:LOWer	285
DISPlay[:WINDow<n>]:SGRam:COLor:SHAPE	285
DISPlay[:WINDow<n>]:SPEctrogram:COLor:SHAPE	285
DISPlay[:WINDow<n>]:SGRam:COLor:UPPer	286
DISPlay[:WINDow<n>]:SPEctrogram:COLor:UPPer	286
DISPlay[:WINDow<n>]:SGRam:COLor:STYLe	286
DISPlay[:WINDow<n>]:SPEctrogram:COLor:STYLe	286

DISPlay[:WINDow<n>]:SGRam:COLor:DEFault

DISPlay[:WINDow<n>]:SPEctrogram:COLor:DEFault

Restores the original color map.

Suffix:

<n> [Window](#)

Manual operation: See ["Set to Default"](#) on page 99

DISPlay[:WINDow<n>]:SGRam:COLor:LOWer <Percentage>

DISPlay[:WINDow<n>]:SPEctrogram:COLor:LOWer <Percentage>

Defines the starting point of the color map.

Suffix:

<n> [Window](#)

Parameters:

<Percentage> Statistical frequency percentage.

Range: 0 to 66

*RST: 0

Default unit: %

Example:

DISP:WIND:SGR:COL:LOW 10

Sets the start of the color map to 10%.

Manual operation: See ["Start / Stop"](#) on page 99

DISPlay[:WINDow<n>]:SGRam:COLor:SHAPE <Shape>

DISPlay[:WINDow<n>]:SPEctrogram:COLor:SHAPE <Shape>

Defines the shape and focus of the color curve for the spectrogram result display.

Suffix:

<n> [Window](#)

Parameters:

<Shape> Shape of the color curve.
 Range: -1 to 1
 *RST: 0

Manual operation: See ["Shape"](#) on page 99

DISPlay[:WINDow<n>]:SGRam:COLor:UPPer <Percentage>

DISPlay[:WINDow<n>]:SPECtrogram:COLor:UPPer <Percentage>

Defines the end point of the color map.

Suffix:

<n> [Window](#)

Parameters:

<Percentage> Statistical frequency percentage.
 Range: 0 to 66
 *RST: 0
 Default unit: %

Example:

DISP:WIND:SGR:COL:UPP 95

Sets the start of the color map to 95%.

Manual operation: See ["Start / Stop"](#) on page 99

DISPlay[:WINDow<n>]:SGRam:COLor[:STYLE] <ColorScheme>

DISPlay[:WINDow<n>]:SPECtrogram:COLor[:STYLE] <ColorScheme>

Selects the color scheme.

Parameters:

<ColorScheme>

HOT

Uses a color range from blue to red. Blue colors indicate low levels, red colors indicate high ones.

COLD

Uses a color range from red to blue. Red colors indicate low levels, blue colors indicate high ones.

RADar

Uses a color range from black over green to light turquoise with shades of green in between.

GRAYscale

Shows the results in shades of gray.

*RST: HOT

Example:

DISP:WIND:SPEC:COL GRAY

Changes the color scheme of the spectrogram to black and white.

Manual operation: See ["Hot/Cold/Radar/Grayscale"](#) on page 99

10.7.3 Using markers

The following commands are available for marker settings and functions in the I/Q Analyzer application.



For "I/Q Vector" displays markers are not available.

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• Configuring and performing a marker search.....	295
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• Measuring the time domain power.....	322

10.7.3.1 Setting up individual markers

The following commands define the position of markers in the diagram.

CALCulate<n>:DELTamarker<m>:AOFF.....	287
CALCulate<n>:DELTamarker<m>:LINK.....	288
CALCulate<n>:DELTamarker<ms>:LINK:TO:DELTa<md>.....	288
CALCulate<n>:DELTamarker<ms>:LINK:TO:MARKer<md>.....	289
CALCulate<n>:DELTamarker<m>:MODE.....	289
CALCulate<n>:DELTamarker<m>:MREference.....	290
CALCulate<n>:DELTamarker<m>[:STATe].....	290
CALCulate<n>:DELTamarker<m>:TRACe.....	290
CALCulate<n>:DELTamarker<m>:X.....	291
CALCulate<n>:MARKer<m>:AOFF.....	291
CALCulate<n>:MARKer<ms>:LINK:TO:DELTa<md>.....	291
CALCulate<n>:MARKer<ms>:LINK:TO:MARKer<md>.....	292
CALCulate<n>:MARKer<m>[:STATe].....	292
CALCulate<n>:MARKer<m>:TRACe.....	293
CALCulate<n>:MARKer<m>:X.....	293

CALCulate<n>:DELTamarker<m>:AOFF

Turns off *all* delta markers.

Suffix:

<n> [Window](#)

<m> irrelevant

Example:

CALC:DELT:AOFF

Turns off all delta markers.

CALCulate<n>:DELTamarker<m>:LINK <State>

Links delta marker <m> to marker 1.

If you change the horizontal position (x-value) of marker 1, delta marker <m> changes its horizontal position to the same value.

Tip: to link any marker to a different marker than marker 1, use the

[CALCulate<n>:DELTamarker<ms>:LINK:TO:MARKer<md>](#) or [CALCulate<n>:MARKer<ms>:LINK:TO:MARKer<md>](#) commands.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example: `CALC:DELT2:LINK ON`

Manual operation: See ["Linking to Another Marker"](#) on page 105

CALCulate<n>:DELTamarker<ms>:LINK:TO:DELTa<md> <State>

Links the delta source marker <ms> to any active destination delta marker <md>.

If you change the horizontal position of marker <md>, marker <ms> changes its horizontal position to the same value.

Suffix:

<n> [Window](#)

<ms> source marker, see [Marker](#)

<md> destination marker, see [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example: `CALC:DELT2:LINK:TO:DELT3 ON`
Links D2 and D3.

CALCulate<n>:DELTamarker<ms>:LINK:TO:MARKer<md> <State>

Links the delta source marker <ms> to any active destination marker <md> (normal or delta marker). If you move <md> along the x-axis, <ms> follows to the same horizontal position.

- If <md> is a delta marker, it is turned into a normal marker. To maintain the delta marker, use [CALCulate<n>:DELTamarker<ms>:LINK:TO:DELTa<md>](#) on page 288.
- If <md> is the reference marker for the delta marker <ms>, the delta marker is initially set to the same horizontal position as the normal marker. To change the relative distance (delta) value again, use [CALCulate<n>:DELTamarker<m>:X](#) on page 291. Then the delta between the two markers is maintained when you move the normal marker.

Suffix:

<n>	Window
<ms>	source marker, see Marker
<md>	destination marker, see Marker

Parameters:

<State>	ON OFF 0 1
	OFF 0
	Switches the function off
	ON 1
	Switches the function on

Example:

`CALC:DELT4:LINK:TO:MARK2 ON`
Links the delta marker 4 to the marker 2.

Manual operation: See "[Linking to Another Marker](#)" on page 105

CALCulate<n>:DELTamarker<m>:MODE <Mode>

Defines whether the position of a delta marker is provided as an absolute value or relative to a reference marker. Note that this setting applies to *all* windows.

Note that when the position of a delta marker is *queried*, the result is always an absolute value (see [CALCulate<n>:DELTamarker<m>:X](#) on page 291)!

Suffix:

<n>	irrelevant
<m>	irrelevant

Parameters:

<Mode>	ABSolute
	Delta marker position in absolute terms.
	RELative
	Delta marker position in relation to a reference marker.
*RST:	RELative

Example: `CALC:DELT:MODE ABS`
Absolute delta marker position.

CALCulate<n>:DELTamarker<m>:MREference <Reference>

Selects a reference marker for a delta marker other than marker 1.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<Reference> **1 to 16**
Selects markers 1 to 16 as the reference.

Example: `CALC:DELT3:MREF 2`
Specifies that the values of delta marker 3 are relative to marker 2.

Manual operation: See ["Reference Marker"](#) on page 105

CALCulate<n>:DELTamarker<m>[:STATe] <State>

Turns delta markers on and off.

If necessary, the command activates the delta marker first.

No suffix at DELTmarker turns on delta marker 1.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> **ON | OFF | 0 | 1**
OFF | 0
Switches the function off
ON | 1
Switches the function on

Example: `CALC:DELT2 ON`
Turns on delta marker 2.

Manual operation: See ["Marker State"](#) on page 104
See ["Marker Type"](#) on page 105
See ["Select Marker"](#) on page 106

CALCulate<n>:DELTamarker<m>:TRACe <Trace>

Selects the trace a delta marker is positioned on.

Note that the corresponding trace must have a trace mode other than "Blank".

If necessary, the command activates the marker first.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<Trace> Trace number the marker is assigned to.

Example:

CALC:DELT2:TRAC 2
Positions delta marker 2 on trace 2.

CALCulate<n>:DELTamarker<m>:X <Position>

Moves a delta marker to a particular coordinate on the x-axis.

If necessary, the command activates the delta marker and positions a reference marker to the peak power.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<Position> Numeric value that defines the marker position on the x-axis.

Range: The value range and unit depend on the measurement and scale of the x-axis.

Example:

CALC:DELT:X?
Outputs the absolute x-value of delta marker 1.

Manual operation: See ["Marker Position X-value"](#) on page 105

CALCulate<n>:MARKer<m>:AOFF

Turns off all markers.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Example:

CALC:MARK:AOFF
Switches off all markers.

Manual operation: See ["All Markers Off"](#) on page 106

CALCulate<n>:MARKer<ms>:LINK:TO:DELTa<md> <State>

Links the normal source marker <ms> to any active delta destination marker <md>.

If you change the horizontal position of marker <md>, marker <ms> changes its horizontal position to the same value.

Suffix:<n> [Window](#)<ms> source marker, see [Marker](#)<md> destination marker, see [Marker](#)**Parameters:**

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK4:LINK:TO:DELT2 ON
 Links marker 4 to delta marker 2.

CALCulate<n>:MARKer<ms>:LINK:TO:MARKer<md> <State>

Links the normal source marker <ms> to any active destination marker <md> (normal or delta marker).

If you change the horizontal position of marker <md>, marker <ms> changes its horizontal position to the same value.

Suffix:<n> [Window](#)<ms> source marker, see [Marker](#)<md> destination marker, see [Marker](#)**Parameters:**

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK4:LINK:TO:MARK2 ON
 Links marker 4 to marker 2.

Manual operation: See "[Linking to Another Marker](#)" on page 105

CALCulate<n>:MARKer<m>[:STATe] <State>

Turns markers on and off. If the corresponding marker number is currently active as a delta marker, it is turned into a normal marker.

Suffix:<n> [Window](#)

<m>	Marker
Parameters:	
<State>	ON OFF 0 1 OFF 0 Switches the function off ON 1 Switches the function on
Example:	CALC:MARK3 ON Switches on marker 3.
Manual operation:	See "Marker State" on page 104 See "Marker Type" on page 105 See "Select Marker" on page 106

CALCulate<n>:MARKer<m>:TRACe <Trace>

Selects the trace the marker is positioned on.

Note that the corresponding trace must have a trace mode other than "Blank".

If necessary, the command activates the marker first.

Suffix:

<n> Window

<m> Marker

Parameters:

<Trace>

Example: //Assign marker to trace 1
CALC:MARK3:TRAC 1

Manual operation: See "Assigning the Marker to a Trace" on page 106

CALCulate<n>:MARKer<m>:X <Position>

Moves a marker to a specific coordinate on the x-axis.

If necessary, the command activates the marker.

If the marker has been used as a delta marker, the command turns it into a normal marker.

Suffix:

<n> Window

<m> Marker

Parameters:

<Position> Numeric value that defines the marker position on the x-axis.
The unit depends on the result display.
Range: The range depends on the current x-axis range.
Default unit: Hz

Example: `CALC:MARK2:X 1.7MHz`
Positions marker 2 to frequency 1.7 MHz.

Manual operation: See ["Marker Table"](#) on page 17
See ["Marker Peak List"](#) on page 17
See ["Marker Position X-value"](#) on page 105

10.7.3.2 General marker settings

The following commands control general marker functionality.

Remote commands exclusive to general marker functionality

<code>DISPlay[:WINDow<n>]:MTABle</code>	294
<code>DISPlay[:WINDow<n>]:MINFo[:STATe]</code>	294
<code>CALCulate<n>:MARKer<m>:X:SSIZe</code>	295

`DISPlay[:WINDow<n>]:MTABle <DisplayMode>`

Turns the marker table on and off.

Suffix:

`<n>` irrelevant

Parameters:

`<DisplayMode>` **ON | 1**
Turns on the marker table.

OFF | 0
Turns off the marker table.

AUTO
Turns on the marker table if 3 or more markers are active.

***RST:** AUTO

Example: `DISP:MTAB ON`
Activates the marker table.

Manual operation: See ["Marker Table Display"](#) on page 107

`DISPlay[:WINDow<n>]:MINFo[:STATe] <State>`

Turns the marker information in all diagrams on and off.

Suffix:

`<n>` irrelevant

Parameters:

`<State>` **ON | 1**
Displays the marker information in the diagrams.

OFF | 0
Hides the marker information in the diagrams.

***RST:** 1

Example: `DISP:MINF OFF`
Hides the marker information.

Manual operation: See ["Marker Info"](#) on page 107

CALCulate<n>:MARKer<m>:X:SSIZe <StepSize>

Selects the marker step size mode for *all* markers in *all* windows.

It therefore takes effect in manual operation only.

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<StepSize>

STANdard

the marker moves from one pixel to the next

POINTs

the marker moves from one sweep point to the next

*RST: POINTs

Example: `CALC:MARK:X:SSIZ STAN`
Sets the marker step size to one pixel.

Manual operation: See ["Marker Stepsize"](#) on page 108

10.7.3.3 Configuring and performing a marker search

The following commands control the marker search.

CALCulate<n>:MARKer<m>:LOEXclude	295
CALCulate<n>:MARKer<m>:PEXCursion	296
CALCulate<n>:MARKer<m>:SEARch	296
CALCulate<n>:MARKer<m>:X:SLIMits[:STATe]	297
CALCulate<n>:MARKer<m>:X:SLIMits:LEFT	297
CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT	298
CALCulate<n>:MARKer<m>:X:SLIMits:ZOOM[:STATe]	298
CALCulate<n>:THReshold	299
CALCulate<n>:THReshold:STATe	299

CALCulate<n>:MARKer<m>:LOEXclude <State>

Turns the suppression of the local oscillator during automatic marker positioning on and off (for *all* markers in *all* windows).

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<State> ON | OFF | 0 | 1
 *RST: 1

Example: CALC:MARK:LOEX ON

Manual operation: See ["Exclude LO"](#) on page 110

CALCulate<n>:MARKer<m>:PEXCursion <Excursion>

Defines the peak excursion (for *all* markers in *all* windows).

The peak excursion sets the requirements for a peak to be detected during a peak search.

The unit depends on the measurement.

Application/Result display	Unit
Spectrum	dB

Suffix:

<n> irrelevant
 <m> irrelevant

Example: CALC:MARK:PEXC 10dB
 Defines peak excursion as 10 dB.

Manual operation: See ["Peak Excursion"](#) on page 110

CALCulate<n>:MARKer<m>:SEARch <MarkReallmag>

Selects the trace type a marker search is performed on.

Suffix:

<n> irrelevant
 <m> irrelevant

Parameters:

<MarkReallmag>

REAL

Marker search functions are performed on the real trace of the "I/Q" measurement.

IMAG

Marker search functions are performed on the imaginary trace of the "I/Q" measurement.

MAGN

Marker search functions are performed on the magnitude of the I and Q data.

*RST: REAL

Example: CALC4:MARK:SEAR IMAG

Manual operation: See ["Branch for Peaksearch"](#) on page 111

CALCulate<n>:MARKer<m>:X:SLIMits[:STATe] <State>

Turns marker search limits on and off for *all* markers in *all* windows.

If you perform a measurement in the time domain, this command limits the range of the trace to be analyzed.

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK:X:SLIM ON

Switches on search limitation.

Manual operation: See ["Search Limits \(Left / Right\)"](#) on page 110
 See ["Search Limits Off"](#) on page 111
 See ["Limit State"](#) on page 126

CALCulate<n>:MARKer<m>:X:SLIMits:LEFT <SearchLimit>

Defines the left limit of the marker search range for *all* markers in *all* windows.

If you perform a measurement in the time domain, this command limits the range of the trace to be analyzed.

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<SearchLimit> The value range depends on the frequency range or sweep time.
 The unit is Hz for frequency domain measurements and s for time domain measurements.

*RST: left diagram border

Default unit: HZ

Example:

CALC:MARK:X:SLIM ON

Switches the search limit function on.

CALC:MARK:X:SLIM:LEFT 10MHz

Sets the left limit of the search range to 10 MHz.

Manual operation: See ["Search Limits \(Left / Right\)"](#) on page 110
 See ["Left Limit / Right Limit"](#) on page 126

CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT <SearchLimit>

Defines the right limit of the marker search range for *all* markers in *all* windows.

If you perform a measurement in the time domain, this command limits the range of the trace to be analyzed.

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<Limit> The value range depends on the frequency range or sweep time.
 The unit is Hz for frequency domain measurements and s for time domain measurements.
 *RST: right diagram border
 Default unit: HZ

Example:

CALC:MARK:X:SLIM ON
 Switches the search limit function on.
 CALC:MARK:X:SLIM:RIGH 20MHz
 Sets the right limit of the search range to 20 MHz.

Manual operation: See ["Search Limits \(Left / Right\)"](#) on page 110
 See ["Left Limit / Right Limit"](#) on page 126

CALCulate<n>:MARKer<m>:X:SLIMits:ZOOM[:STATe] <State>

Adjusts the marker search range to the zoom area for *all* markers in *all* windows.

Suffix:

<n> irrelevant

<m> irrelevant

Parameters:

<State> ON | OFF | 0 | 1
OFF | 0
 Switches the function off
ON | 1
 Switches the function on

Example:

CALC:MARK:X:SLIM:ZOOM ON
 Switches the search limit function on.
 CALC:MARK:X:SLIM:RIGH 20MHz
 Sets the right limit of the search range to 20 MHz.

Manual operation: See ["Use Zoom Limits"](#) on page 111

CALCulate< n>:THReshold <Level>

Defines a threshold level for the marker peak search (for *all* markers in *all* windows).

Note that you must enable the use of the threshold using [CALCulate< n>:THReshold:STATe](#) on page 299.

Suffix:

<n> irrelevant

Parameters:

<Level> Numeric value. The value range and unit are variable.

*RST: -120 dBm

Default unit: DBM

Example: CALC:THR:STAT ON

Example: CALC:THR -82DBM

Enables the search threshold and sets the threshold value to -82 dBm.

Manual operation: See ["Search Threshold"](#) on page 110

CALCulate< n>:THReshold:STATe <State>

Turns a threshold for the marker peak search on and off (for *all* markers in *all* windows).

Suffix:

<n> irrelevant

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example: CALC:THR:STAT ON

Switches on the threshold line.

Manual operation: See ["Search Threshold"](#) on page 110

See ["Search Limits Off"](#) on page 111

10.7.3.4 Marker search (spectrograms)

The following commands automatically define the marker and delta marker position in the spectrogram.

Using markers

The following commands control spectrogram markers.

Useful commands for spectrogram markers described elsewhere

The following commands define the horizontal position of the markers.

- `CALCulate<n>:MARKer<m>:MAXimum:LEFT` on page 309
- `CALCulate<n>:MARKer<m>:MAXimum:NEXT` on page 309
- `CALCulate<n>:MARKer<m>:MAXimum[:PEAK]` on page 310
- `CALCulate<n>:MARKer<m>:MAXimum:RIGHT` on page 310
- `CALCulate<n>:MARKer<m>:MINimum:LEFT` on page 311
- `CALCulate<n>:MARKer<m>:MINimum:NEXT` on page 311
- `CALCulate<n>:MARKer<m>:MINimum[:PEAK]` on page 311
- `CALCulate<n>:MARKer<m>:MINimum:RIGHT` on page 311

Remote commands exclusive to spectrogram markers

<code>CALCulate<n>:MARKer<m>:SGRam:FRAME</code>	300
<code>CALCulate<n>:MARKer<m>:SPECTrogram:FRAME</code>	300
<code>CALCulate<n>:MARKer<m>:SGRam:SARea</code>	301
<code>CALCulate<n>:MARKer<m>:SPECTrogram:SARea</code>	301
<code>CALCulate<n>:MARKer<m>:SGRam:XY:MAXimum[:PEAK]</code>	301
<code>CALCulate<n>:MARKer<m>:SPECTrogram:XY:MAXimum[:PEAK]</code>	301
<code>CALCulate<n>:MARKer<m>:SGRam:XY:MINimum[:PEAK]</code>	301
<code>CALCulate<n>:MARKer<m>:SPECTrogram:XY:MINimum[:PEAK]</code>	301
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:ABOVE</code>	302
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:ABOVE</code>	302
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:BELOW</code>	302
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:BELOW</code>	302
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:NEXT</code>	302
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:NEXT</code>	302
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum[:PEAK]</code>	302
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum[:PEAK]</code>	302
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:ABOVE</code>	303
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:ABOVE</code>	303
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:BELOW</code>	303
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:BELOW</code>	303
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:NEXT</code>	303
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:NEXT</code>	303
<code>CALCulate<n>:MARKer<m>:SGRam:Y:MINimum[:PEAK]</code>	303
<code>CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum[:PEAK]</code>	303

`CALCulate<n>:MARKer<m>:SGRam:FRAME` <Frame>

`CALCulate<n>:MARKer<m>:SPECTrogram:FRAME` <Frame> | <Time>

Positions a marker on a particular frame.

Suffix:

<n> Window

<m> Marker

Parameters:

- <Frame> Selects a frame directly by the frame number. Valid if the time stamp is off.
 The range depends on the history depth.
 Default unit: S
- <Time> Selects a frame via its time stamp. Valid if the time stamp is on.
 The number is the (negative) distance to frame 0 in seconds.
 The range depends on the history depth.

Example:

CALC:MARK:SGR:FRAM -20
 Sets the marker on the 20th frame before the present.
 CALC:MARK2:SGR:FRAM -2s
 Sets second marker on the frame 2 seconds ago.

Manual operation: See ["Frame \(Spectrogram only\)"](#) on page 105

CALCulate<n>:MARKer<m>:SGRam:SARea <SearchArea>

CALCulate<n>:MARKer<m>:SPECTrogram:SARea <SearchArea>

Defines the marker search area for all spectrogram markers in the channel.

Parameters:

- <SearchArea> **VISible**
 Performs a search within the visible frames.
 Note that the command does not work if the spectrogram is not visible for any reason (e.g. if the display update is off).
- MEMory**
 Performs a search within all frames in the memory.
 *RST: VISible

Manual operation: See ["Marker Search Area"](#) on page 113

CALCulate<n>:MARKer<m>:SGRam:XY:MAXimum[:PEAK]

CALCulate<n>:MARKer<m>:SPECTrogram:XY:MAXimum[:PEAK]

Moves a marker to the highest level of the spectrogram.

Suffix:

- <n> [Window](#)
- <m> [Marker](#)

CALCulate<n>:MARKer<m>:SGRam:XY:MINimum[:PEAK]

CALCulate<n>:MARKer<m>:SPECTrogram:XY:MINimum[:PEAK]

Moves a marker to the minimum level of the spectrogram.

Suffix:

- <n> [Window](#)
- <m> [Marker](#)

CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:ABOVE**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:ABOVE**

Moves a marker vertically to the next lower peak level for the current frequency.

The search includes only frames above the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:BELOW**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:BELOW**

Moves a marker vertically to the next lower peak level for the current frequency.

The search includes only frames below the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum:NEXT**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum:NEXT**

Moves a marker vertically to the next lower peak level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MAXimum[:PEAK]**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MAXimum[:PEAK]**

Moves a marker vertically to the highest level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

If the marker hasn't been active yet, the command looks for the peak level in the whole spectrogram.

Suffix:<n> [Window](#)<m> [Marker](#)**CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:ABOVE****CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:ABOVE**

Moves a marker vertically to the next higher minimum level for the current frequency.

The search includes only frames above the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:BELOW**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:BELOW**

Moves a marker vertically to the next higher minimum level for the current frequency.

The search includes only frames below the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MINimum:NEXT**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum:NEXT**

Moves a marker vertically to the next higher minimum level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:MARKer<m>:SGRam:Y:MINimum[:PEAK]**CALCulate<n>:MARKer<m>:SPECTrogram:Y:MINimum[:PEAK]**

Moves a marker vertically to the minimum level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

If the marker hasn't been active yet, the command first looks for the peak level for all frequencies and moves the marker vertically to the minimum level.

Suffix:

<n> Window

<m> Marker

Using delta markers

The following commands control spectrogram delta markers.

Useful commands for spectrogram markers described elsewhere

The following commands define the horizontal position of the delta markers.

- `CALCulate<n>:DELTamarker<m>:MAXimum:LEFT` on page 312
- `CALCulate<n>:DELTamarker<m>:MAXimum:NEXT` on page 312
- `CALCulate<n>:DELTamarker<m>:MAXimum[:PEAK]` on page 313
- `CALCulate<n>:DELTamarker<m>:MAXimum:RIGHT` on page 313
- `CALCulate<n>:DELTamarker<m>:MINimum:LEFT` on page 313
- `CALCulate<n>:DELTamarker<m>:MINimum:NEXT` on page 313
- `CALCulate<n>:DELTamarker<m>:MINimum[:PEAK]` on page 314
- `CALCulate<n>:DELTamarker<m>:MINimum:RIGHT` on page 314

Remote commands exclusive to spectrogram markers

<code>CALCulate<n>:DELTamarker<m>:SGRam:FRAME</code>	305
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:FRAME</code>	305
<code>CALCulate<n>:DELTamarker<m>:SGRam:SARea</code>	305
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:SARea</code>	305
<code>CALCulate<n>:DELTamarker<m>:SGRam:XY:MAXimum[:PEAK]</code>	306
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:XY:MAXimum[:PEAK]</code>	306
<code>CALCulate<n>:DELTamarker<m>:SGRam:XY:MINimum[:PEAK]</code>	306
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:XY:MINimum[:PEAK]</code>	306
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:ABOVE</code>	306
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:ABOVE</code>	306
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:BELOW</code>	306
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:BELOW</code>	306
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:NEXT</code>	306
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:NEXT</code>	306
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum[:PEAK]</code>	307
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum[:PEAK]</code>	307
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum:ABOVE</code>	307
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:ABOVE</code>	307
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum:BELOW</code>	307
<code>CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:BELOW</code>	307
<code>CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum:NEXT</code>	308

CALCulate<n>:DELTamarker<m>:SPECtrogram:Y:MINimum:NEXT.....	308
CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum[:PEAK].....	308
CALCulate<n>:DELTamarker<m>:SPECtrogram:Y:MINimum[:PEAK].....	308

CALCulate<n>:DELTamarker<m>:SGRam:FRAME <Frame>

CALCulate<n>:DELTamarker<m>:SPECtrogram:FRAME <Frame>

Positions a delta marker on a particular frame. The frame is relative to the position of marker 1.

The command is available for the spectrogram.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<Frame> Selects a frame either by its frame number or time stamp.
 The frame number is available if the time stamp is off. The range depends on the history depth.
 The time stamp is available if the time stamp is on. The number is the distance to frame 0 in seconds. The range depends on the history depth.
 Default unit: S

Example:

CALC:DELT4:SGR:FRAM -20

Sets fourth deltamarker 20 frames below marker 1.

CALC:DELT4:SGR:FRAM 2 s

Sets fourth deltamarker 2 seconds above the position of marker 1.

Manual operation: See ["Frame \(Spectrogram only\)"](#) on page 105

CALCulate<n>:DELTamarker<m>:SGRam:SARea <SearchArea>

CALCulate<n>:DELTamarker<m>:SPECtrogram:SARea <SearchArea>

Defines the marker search area for *all* spectrogram markers in the channel.

Parameters:

<SearchArea>

VISible

Performs a search within the visible frames.

Note that the command does not work if the spectrogram is not visible for any reason (e.g. if the display update is off).

MEMory

Performs a search within all frames in the memory.

*RST: VISible

Manual operation: See ["Marker Search Area"](#) on page 113

CALCulate<n>:DELTamarker<m>:SGRam:XY:MAXimum[:PEAK]**CALCulate<n>:DELTamarker<m>:SPECTrogram:XY:MAXimum[:PEAK]**

Moves a marker to the highest level of the spectrogram over all frequencies.

Suffix:<n> [Window](#)<m> [Marker](#)

CALCulate<n>:DELTamarker<m>:SGRam:XY:MINimum[:PEAK]**CALCulate<n>:DELTamarker<m>:SPECTrogram:XY:MINimum[:PEAK]**

Moves a delta marker to the minimum level of the spectrogram over all frequencies.

Suffix:<n> [Window](#)<m> [Marker](#)

CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:ABOVE**CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:ABOVE**

Moves a marker vertically to the next higher level for the current frequency.

The search includes only frames above the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:BELOW**CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:BELOW**

Moves a marker vertically to the next higher level for the current frequency.

The search includes only frames below the current marker position. It does not change the horizontal position of the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum:NEXT**CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum:NEXT**

Moves a delta marker vertically to the next higher level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTamarker<m>:SGRam:Y:MAXimum[:PEAK]

CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MAXimum[:PEAK]

Moves a delta marker vertically to the highest level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

If the marker hasn't been active yet, the command looks for the peak level in the whole spectrogram.

Suffix:

<n> [Window](#)

<m> [Marker](#)

CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum:ABOVE

CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:ABOVE

Moves a delta marker vertically to the next minimum level for the current frequency.

The search includes only frames above the current marker position. It does not change the horizontal position of the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTamarker<m>:SGRam:Y:MINimum:BELOW

CALCulate<n>:DELTamarker<m>:SPECTrogram:Y:MINimum:BELOW

Moves a delta marker vertically to the next minimum level for the current frequency.

The search includes only frames below the current marker position. It does not change the horizontal position of the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTaMarker<m>:SGRam:Y:MINimum:NEXT
CALCulate<n>:DELTaMarker<m>:SPECTrogram:Y:MINimum:NEXT

Moves a delta marker vertically to the next minimum level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Mode for Next Peak in Y-Direction"](#) on page 112

CALCulate<n>:DELTaMarker<m>:SGRam:Y:MINimum[:PEAK]
CALCulate<n>:DELTaMarker<m>:SPECTrogram:Y:MINimum[:PEAK]

Moves a delta marker vertically to the minimum level for the current frequency.

The search includes all frames. It does not change the horizontal position of the marker.

If the marker hasn't been active yet, the command first looks for the peak level in the whole spectrogram and moves the marker vertically to the minimum level.

Suffix:

<n> [Window](#)

<m> [Marker](#)

10.7.3.5 Positioning the marker

This section contains remote commands necessary to position the marker on a trace.

- [Positioning normal markers](#).....308
- [Positioning delta markers](#).....312

Positioning normal markers

The following commands position markers on the trace.

CALCulate<n>:MARKer<m>:MAXimum:AUTO	309
CALCulate<n>:MARKer<m>:MAXimum:LEFT	309
CALCulate<n>:MARKer<m>:MAXimum:NEXT	309
CALCulate<n>:MARKer<m>:MAXimum[:PEAK]	310
CALCulate<n>:MARKer<m>:MAXimum:RIGHT	310
CALCulate<n>:MARKer<m>:MINimum:AUTO	310
CALCulate<n>:MARKer<m>:MINimum:LEFT	311
CALCulate<n>:MARKer<m>:MINimum:NEXT	311
CALCulate<n>:MARKer<m>:MINimum[:PEAK]	311
CALCulate<n>:MARKer<m>:MINimum:RIGHT	311

CALCulate<n>:MARKer<m>:MAXimum:AUTO <State>

Turns an automatic marker peak search for a trace maximum on and off (using marker 1). The R&S FSV/A performs the peak search after each sweep.

Suffix:

<n> [Window](#)

<m> irrelevant

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK:MAX:AUTO ON

Activates the automatic peak search function for marker 1 at the end of each particular sweep.

Manual operation: See ["Auto Max Peak Search / Auto Min Peak Search"](#) on page 110

CALCulate<n>:MARKer<m>:MAXimum:LEFT

Moves a marker to the next positive peak.

The search includes only measurement values to the left of the current marker position.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Peak"](#) on page 115

CALCulate<n>:MARKer<m>:MAXimum:NEXT

Moves a marker to the next positive peak.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Peak"](#) on page 115

CALCulate<n>:MARKer<m>:MAXimum[:PEAK]

Moves a marker to the highest level.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

If the marker is not yet active, the command first activates the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Peak Search"](#) on page 115

CALCulate<n>:MARKer<m>:MAXimum:RIGHT

Moves a marker to the next positive peak.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Peak"](#) on page 115

CALCulate<n>:MARKer<m>:MINimum:AUTO <State>

Turns an automatic marker peak search for a trace minimum on and off (using marker 1). The R&S FSV/A performs the peak search after each sweep.

Suffix:

<n> [Window](#)

<m> irrelevant

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK:MIN:AUTO ON

Activates the automatic minimum value search function for marker 1 at the end of each particular sweep.

Manual operation: See ["Auto Max Peak Search / Auto Min Peak Search"](#) on page 110

CALCulate<n>:MARKer<m>:MINimum:LEFT

Moves a marker to the next minimum peak value.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Minimum"](#) on page 115

CALCulate<n>:MARKer<m>:MINimum:NEXT

Moves a marker to the next minimum peak value.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Minimum"](#) on page 115

CALCulate<n>:MARKer<m>:MINimum[:PEAK]

Moves a marker to the minimum level.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

If the marker is not yet active, the command first activates the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Minimum"](#) on page 115

CALCulate<n>:MARKer<m>:MINimum:RIGHT

Moves a marker to the next minimum peak value.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Next Minimum"](#) on page 115**Positioning delta markers**

The following commands position delta markers on the trace.

CALCulate<n>:DELTamarker<m>:MAXimum:LEFT	312
CALCulate<n>:DELTamarker<m>:MAXimum:NEXT	312
CALCulate<n>:DELTamarker<m>:MAXimum[:PEAK]	313
CALCulate<n>:DELTamarker<m>:MAXimum:RIGHT	313
CALCulate<n>:DELTamarker<m>:MINimum:LEFT	313
CALCulate<n>:DELTamarker<m>:MINimum:NEXT	313
CALCulate<n>:DELTamarker<m>:MINimum[:PEAK]	314
CALCulate<n>:DELTamarker<m>:MINimum:RIGHT	314

CALCulate<n>:DELTamarker<m>:MAXimum:LEFT

Moves a delta marker to the next positive peak value.

The search includes only measurement values to the left of the current marker position.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Next Peak"](#) on page 115**CALCulate<n>:DELTamarker<m>:MAXimum:NEXT**

Moves a marker to the next positive peak value.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:<n> 1..n
[Window](#)<m> 1..n
[Marker](#)**Manual operation:** See ["Search Next Peak"](#) on page 115

CALCulate<n>:DELTaMarker<m>:MAXimum[:PEAK]

Moves a delta marker to the highest level.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

If the marker is not yet active, the command first activates the marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Peak Search"](#) on page 115

CALCulate<n>:DELTaMarker<m>:MAXimum:RIGHT

Moves a delta marker to the next positive peak value on the trace.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the maximum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Peak"](#) on page 115

CALCulate<n>:DELTaMarker<m>:MINimum:LEFT

Moves a delta marker to the next minimum peak value.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Manual operation: See ["Search Next Minimum"](#) on page 115

CALCulate<n>:DELTaMarker<m>:MINimum:NEXT

Moves a marker to the next minimum peak value.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Next Minimum"](#) on page 115**CALCulate<n>:DELTaMarker<m>:MINimum[:PEAK]**

Moves a delta marker to the minimum level.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

If the marker is not yet active, the command first activates the marker.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Minimum"](#) on page 115**CALCulate<n>:DELTaMarker<m>:MINimum:RIGHT**

Moves a delta marker to the next minimum peak value.

The search includes only measurement values to the right of the current marker position.

In the spectrogram, the command moves a marker horizontally to the minimum level in the currently selected frame. The vertical marker position remains the same.

Suffix:<n> [Window](#)<m> [Marker](#)**Manual operation:** See ["Search Next Minimum"](#) on page 115**10.7.3.6 Band power marker**

The following commands control the marker for band power measurements.

Using markers

CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:AOff	314
CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:MODE	315
CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:RESult?	315
CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:SPAN	316
CALCulate<n>:MARKer<m>:FUNCTION:BPOWER[:STATe]	316

CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:AOff

Removes all band power markers in the specified window.

Suffix:<n> [Window](#)

<m> irrelevant

Example: `CALC:MARK:FUNC:BPOW:AOff`

CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:MODE <Mode>

Selects the way the results for a band power marker are displayed.

(Note: relative power results are only available for delta markers, see [CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:MODE](#) on page 316)**Suffix:**<n> [Window](#)<m> [Marker](#)**Parameters:**

<Mode>

POWERResult is displayed as an absolute power. The power unit depends on the [CALCulate<n>:UNIT:POWER](#) setting.**DENSITY**

Result is displayed as a density in dBm/Hz.

*RST: POWER

Example: `CALC:MARK4:FUNC:BPOW:MODE DENS`
Configures marker 4 to show the measurement results in dBm/Hz.**Manual operation:** See ["Power Mode"](#) on page 122

CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:RESult?

Queries the results of the band power measurement.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<Power> Signal power over the marker bandwidth.

Example:

Activate the band power marker:

`CALC:MARK:FUNC:BPOW:STAT ON`

Select the density mode for the result:

`CALC:MARK:FUNC:BPOW:MODE DENS`

Query the result:

`CALC:MARK:FUNC:BPOW:RES?`

Response:

20dBm/Hz

Usage: Query only

**CALCulate<n>:MARKer<m>:FUNCTION:BPOWER:SPAN **

Defines the bandwidth around the marker position.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

 Frequency. The maximum span depends on the marker position and R&S FSV/A model.

*RST: 5% of current span

Default unit: Hz

Example:

CALC:MARK:FUNC:BPOW:SPAN 2MHz

Measures the band power over 2 MHz around the marker.

Manual operation: See ["Span"](#) on page 121

CALCulate<n>:MARKer<m>:FUNCTION:BPOWER[:STATe] <State>

Turns markers for band power measurements on and off.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK4:FUNC:BPOW:STAT ON

Activates or turns marker 4 into a band power marker.

Manual operation: See ["Band Power Measurement State"](#) on page 121
See ["Switching All Band Power Measurements Off"](#) on page 122

Using delta markers

CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:MODE	316
CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:RESult?	317
CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:SPAN	317
CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER[:STATe]	318

CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWER:MODE <Mode>

Selects the way the results for a band power delta marker are displayed.

Suffix:<n> [Window](#)<m> [Marker](#)**Parameters:**

<Mode>

POWer

Result is displayed as an absolute power. The power unit depends on the [CALCulate<n>:UNIT:POWer](#) setting.

DENSity

Result is displayed as a density in dBm/Hz.

RPOWer

This setting is only available for a delta band power marker. The result is the difference between the absolute power in the band around the delta marker and the absolute power for the reference marker. The powers are subtracted logarithmically, so the result is a dB value.

[Relative band power (Delta2) in dB] = [absolute band power (Delta2) in dBm] - [absolute (band) power of reference marker in dBm]

For details see "[Relative band power markers](#)" on page 119.

*RST: POWer

Manual operation: See "[Power Mode](#)" on page 122

CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWer:RESult?

Queries the results of the band power measurement.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<Power> Signal power over the delta marker bandwidth.

Usage: Query only

**CALCulate<n>:DELTamarker<m>:FUNCTION:BPOWer:SPAN **

Defines the bandwidth around the delta marker position.

Suffix:<n> [Window](#)<m> [Marker](#)**Parameters:**

Frequency. The maximum span depends on the marker position and R&S FSV/A model.

*RST: 5% of current span

Default unit: Hz

Manual operation: See ["Span"](#) on page 121

CALCulate<n>:DELTaMarker<m>:FUNction:BPOWer[:STATe] <State>

Turns delta markers for band power measurements on and off.

If necessary, the command also turns on a reference marker.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Manual operation: See ["Band Power Measurement State"](#) on page 121

See ["Switching All Band Power Measurements Off"](#) on page 122

10.7.3.7 Marker peak lists

Useful commands for peak lists described elsewhere

- [CALCulate<n>:MARKer<m>:PEXCursion](#) on page 296
- [MMEMory:STORe<n>:PEAK](#) on page 321

Remote commands exclusive to peak lists

CALCulate<n>:MARKer<m>:FUNction:FPEaks:ANNotation:LABel[:STATe]	318
CALCulate<n>:MARKer<m>:FUNction:FPEaks:COUNt?	319
CALCulate<n>:MARKer<m>:FUNction:FPEaks[:IMMediate]	319
CALCulate<n>:MARKer<m>:FUNction:FPEaks:LIST:SIZE	320
CALCulate<n>:MARKer<m>:FUNction:FPEaks:SORT	320
CALCulate<n>:MARKer<m>:FUNction:FPEaks:STATe	320
CALCulate<n>:MARKer<m>:FUNction:FPEaks:X?	321
CALCulate<n>:MARKer<m>:FUNction:FPEaks:Y?	321
MMEMory:STORe<n>:PEAK	321

CALCulate<n>:MARKer<m>:FUNction:FPEaks:ANNotation:LABel[:STATe] <State>

Turns labels for peaks found during a peak search on and off.

The labels correspond to the marker number in the marker peak list.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1
*RST: 1

Example: CALC:MARK:FUNC:FPE:ANN:LAB:STAT OFF
Removes the peak labels from the diagram

Manual operation: See ["Display Marker Numbers"](#) on page 130

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:COUNT?

Queries the number of peaks that have been found during a peak search.

The actual number of peaks that have been found may differ from the number of peaks you have set to be found because of the peak excursion.

Suffix:

<n> irrelevant

<m> irrelevant

Return values:

<NumberOfPeaks>

Example: CALC:MARK:FUNC:FPE:COUN?
Queries the number of peaks.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks[:IMMEDIATE] <Peaks>

Initiates a peak search.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<Peaks> This parameter defines the number of peaks to find during the search.
Note that the actual number of peaks found during the search also depends on the peak excursion you have set with [CALCulate<n>:MARKer<m>:PEXCursion](#).
Range: 1 to 200

Example: CALC:MARK:PEXC 5
Defines a peak excursion of 5 dB, i.e. peaks must be at least 5 dB apart to be detected as a peak.
CALC:MARK:FUNC:FPE 10
Initiates a search for 10 peaks on the current trace.

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:LIST:SIZE <MaxNoPeaks>

Defines the maximum number of peaks that the R&S FSV/A looks for during a peak search.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<MaxNoPeaks> Maximum number of peaks to be determined.

Range: 1 to 500

*RST: 50

Example:

CALC:MARK:FUNC:FPE:LIST:SIZE 10

The marker peak list will contain a maximum of 10 peaks.

Manual operation: See ["Maximum Number of Peaks"](#) on page 130

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:SORT <SortMode>

Selects the order in which the results of a peak search are returned.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<SortMode> **X**

Sorts the peaks according to increasing position on the x-axis.

Y

Sorts the peaks according to decreasing position on the y-axis.

*RST: X

Example:

CALC:MARK:FUNC:FPE:SORT Y

Sets the sort mode to decreasing y values

Manual operation: See ["Sort Mode"](#) on page 130

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:STATE <State>

Turns a peak search on and off.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 0 | 1

OFF | 0

Switches the function off

ON | 1

Switches the function on

Example:

CALC:MARK:FUNC:FPE:STAT ON

Activates marker peak search

Manual operation:See ["Peak List State"](#) on page 129**CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:X?**

Queries the position of the peaks on the x-axis.

The order depends on the sort order that has been set with [CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:SORT](#).

Suffix:

<n> irrelevant

<m> irrelevant

Return values:

<PeakPosition> Position of the peaks on the x-axis. The unit depends on the measurement.

Usage:

Query only

CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:Y?

Queries the position of the peaks on the y-axis.

The order depends on the sort order that has been set with [CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:SORT](#).

Suffix:

<n> irrelevant

<m> irrelevant

Return values:

<PeakPosition> Position of the peaks on the y-axis. The unit depends on the measurement.

Usage:

Query only

MMEMory:STORe<n>:PEAK <FileName>

Exports the marker peak list to a file.

Secure User Mode

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Suffix:

<n> [Window](#)

Parameters:

<FileName> String containing the path, name and extension of the target file.

Example:

MMEM:STOR:PEAK 'test.dat'

Saves the current marker peak list in the file test.dat.

Manual operation: See ["Export Peak List"](#) on page 130

10.7.3.8 Measuring the time domain power

All remote control commands specific to time domain power measurements are described here.

- [Configuring the measurement](#)..... 322
- [Performing a time domain power measurement](#)..... 326
- [Retrieving measurement results](#)..... 326

Configuring the measurement

The following remote commands measure the time domain power.

Useful commands for time domain power measurements described elsewhere

- [CALCulate<n>:MARKer<m>:X:SLIMits:LEFT](#)
- [CALCulate<n>:MARKer<m>:X:SLIMits:RIGHT](#)
- [CALCulate<n>:MARKer<m>:X:SLIMits\[:STATe\]](#)

Remote commands exclusive to time domain power measurements

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AOff	322
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERage	323
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MODE	323
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:REFerence:AUTO	324
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLd	324
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY[:STATe]	324
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN[:STATe]	325
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEak[:STATe]	325
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS[:STATe]	325
CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:SDEViation[:STATe]	326

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AOff

Turns all time domain power evaluation modes off.

Suffix:

<n> [Window](#)

<m> [Marker](#)

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERage <State>

Switches on or off averaging for the active power measurement in zero span in the window specified by the suffix <n>. If activated, a time domain value is calculated from the trace after each sweep; in the end, all values are averaged to calculate the final result.

The number of results required for the calculation of average is defined with [\[SENSe:\] AVERage<n>:COUNT](#).

Averaging is reset by switching it off and on again.

Synchronization to the end of averaging is only possible in single sweep mode.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

Example:

```
INIT:CONT OFF
Switches to single sweep mode.
CALC:MARK:FUNC:SUMM:AVER ON
Switches on the calculation of average.
AVER:COUN 200
Sets the measurement counter to 200.
INIT;*WAI
Starts a sweep and waits for the end.
```

Manual operation: See ["Power"](#) on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MODE <Mode>

Selects whether time domain power summary results are reported in absolute units or relative to the stored reference.

Suffix:

<n> 1..n
[Window](#)

<m> 1..n
[Marker](#)

Parameters:

<Mode> ABSolute | RELative
ABSolute
Shows values as absolute levels.
RELative
Shows values that are relative to the stored reference.

Example: //Set mode to relative
 CALC:MARK:FUNC:SUMM:MODE REL

Manual operation: See "Power" on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:REFERENCE:AUTO <Event>

Stores the current measurement as the reference for relative display.

Suffix:

<n> 1..n
[Window](#)

<m> 1..n
[Marker](#)

Parameters:

<Event> ONCE

Example: //Set reference
 CALC:MARK:FUNC:SUMM:REF:AUTO ONCE

Manual operation: See "Power" on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLD <State>

Switches on or off the peak-hold function for the active power measurement in zero span in the window specified by the suffix <n>. If activated, the peak for each sweep is compared to the previously stored peak; the maximum of the two is stored as the current peak.

The peak-hold function is reset by switching it off and on again.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
 *RST: 0

Manual operation: See "Power" on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY[:STATE] <State>

Turns time domain power measurements on and off. This measurement is only available in zero span.

When you turn the measurement on, the R&S FSV/A activates a marker and positions it on the peak power level in the marker search range.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN[:STATe] <State>

Turns the evaluation to determine the mean time domain power on and off.

The R&S FSV/A performs the measurement on the trace marker 1 is positioned on.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

Manual operation: See ["Results"](#) on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEAK[:STATe] <State>

Turns the evaluation to determine the positive peak time domain power on and off.

The R&S FSV/A performs the measurement on the trace marker 1 is positioned on.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

Manual operation: See ["Results"](#) on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS[:STATe] <State>

Turns the evaluation to determine the RMS time domain power on and off.

The R&S FSV/A performs the measurement on the trace marker 1 is positioned on.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0
*RST: 0

Manual operation: See ["Results"](#) on page 125

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:SDEVIation[:STATe] <State>

Turns the evaluation to determine the standard deviation of the time domain power on and off.

The R&S FSV/A performs the measurement on the trace marker 1 is positioned on.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Parameters:

<State> ON | OFF | 1 | 0

*RST: 0

Performing a time domain power measurement

The following commands are required to perform a Time Domain Power measurement:

[INITiate<n>\[:IMMediate\]](#) on page 268

Retrieving measurement results

The following commands query the results for time domain measurements.

Measuring the mean power

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:AVERage:RESult?](#)..... 326

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:PHOLd:RESult?](#)..... 327

[CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:RESult?](#)..... 327

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:AVERage:RESult?

Queries the average mean time domain power. The query is only possible if averaging has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERage](#) on page 323.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<MeanPower> Mean power of the signal during the measurement time.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:PHOLD:RESULT?

Queries the maximum mean time domain power. The query is only possible if the peak hold function has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLD](#).

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<MeanPower> Mean power of the signal during the measurement time.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:MEAN:RESULT?

Queries the mean time domain power.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<MeanPower> Mean power of the signal during the measurement time.

Usage: Query only

Manual operation: See ["Results"](#) on page 125

Measuring the peak power**CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEAK:AVERAGE:RESULT?**

Queries the average positive peak time domain power. The query is only possible if averaging has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERAGE](#) on page 323.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<PeakPower> Peak power of the signal during the measurement time.

Usage: Query only**CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEak:PHOLd:RESult?**

Queries the maximum positive peak time domain power. The query is only possible if the peak hold function has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLd](#).

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<PeakPower> Peak power of the signal during the measurement time.

Usage: Query only**CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PPEak:RESult?**

Queries the positive peak time domain power.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<PeakPower> Peak power of the signal during the measurement time.

Usage: Query only**Manual operation:** See ["Results"](#) on page 125

Measuring the RMS power

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS:AVERage:RESult?

Queries the average RMS of the time domain power. The query is only possible if averaging has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERage](#) on page 323.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<RMSPower> RMS power of the signal during the measurement time.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS:PHOLd:RESult?

Queries the maximum RMS of the time domain power. The query is only possible if the peak hold function has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLd](#).

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<RMSPower> RMS power of the signal during the measurement time.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:RMS:RESult?

Queries the RMS of the time domain power.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<RMSPower> RMS power of the signal during the measurement time.

Usage: Query only**Manual operation:** See ["Results"](#) on page 125**Measuring the standard deviation****CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:SDEVIation:AVERage:RESult?**

Queries the average standard deviation of the time domain power. The query is only possible if averaging has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:AVERage](#) on page 323.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTInuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<StandardDeviation> Standard deviation of the signal during the measurement time.

Usage: Query only**CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:SDEVIation:PHOLd:RESult?**

Queries the maximum standard deviation of the time domain power. The query is only possible if the peak hold function has been activated previously using [CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:PHOLd](#).

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTInuous](#) on page 268.

Suffix:<n> [Window](#)<m> [Marker](#)**Return values:**

<StandardDeviation> Standard deviation of the signal during the measurement time.

Usage: Query only

CALCulate<n>:MARKer<m>:FUNCTION:SUMMARY:SDEVIation:RESult?

Queries the standard deviation of the time domain power.

To get a valid result, you have to perform a complete measurement with synchronization to the end of the measurement before reading out the result. This is only possible for single sweep mode.

See also [INITiate<n>:CONTinuous](#) on page 268.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<StandardDeviation> Standard deviation of the signal during the measurement time.

Usage: Query only

10.8 Configuring I/Q noise cancellation

If you have an input file (.wv or .iq.tar) with the reference signal, you can remove the internal noise from the measured raw I/Q data in the I/Q Analyzer application remotely. The resulting data is stored in a specified iq.tar file, which you can load in any other I/Q application for analysis.



Set the same sample rate and level settings in the I/Q Analyzer as in the application you require for analysis.

For details, see "[I/Q noise cancellation via remote control](#)" on page 40.

[SENSe:]ADJust:NCANcel:AVERage[:COUNT]	331
[SENSe:]ADJust:NCANcel:ERRor?	331
[SENSe:]ADJust:NCANcel:FILE:REFerence	332
[SENSe:]ADJust:NCANcel:FILE:RESult	332
[SENSe:]ADJust:NCANcel:STARt	333

[SENSe:]ADJust:NCANcel:AVERage[:COUNT] <Count>

Defines the number of measurements that are performed on the captured I/Q data to determine the average noise density due to the spectrum analyzer.

Parameters:

<Count> integer
Number of measurements

[SENSe:]ADJust:NCANcel:ERRor?

Returns any errors that occur during the I/Q noise cancellation process.

See also [Section 9.1, "Troubleshooting I/Q noise cancellation"](#), on page 142.

Return values:

<ErrorNumber> numeric value
 Error number

<ErrorText> Error description.

Usage: Query only

Table 10-5: Possible errors

Error number	Error text
1	I/Q noise cancellation successful
-1	Option K575 not available
-2	Wrong application. Function only available in I/Q Analyzer
-3	Number of averages needs to be at least 2
-7	Unable to open reference file
-9	Invalid format of reference file. Expected file type .wv or .iq.tar
-11	Invalid format of result file. Expected file type .iq.tar
-12	Writing results to file failed. Invalid file path
-13	K575 error

[SENSe:]ADJust:NCANcel:FILE:REFeRence <RefFile>

Reference signal to be used by the I/Q noise cancellation process.

Parameters:

<RefFile> string
 Path and file name of the reference signal in .wv or .iq.tar
 format.

[SENSe:]ADJust:NCANcel:FILE:RESult <ResultFile>

Exports the I/Q data with the analyzer noise removed to a file.

Prerequisites for this command:

- Turn on I/Q noise cancellation.
 (See [\[SENSe:\]ADJust:NCANcel:START](#) on page 333).

Parameters:

<ResultFile> String containing the path and name of the file. The specified
 path must exist already.
 The file extension must be .iq.tar.

[SENSe:]ADJ:NCANcel:STARt

Starts an I/Q noise cancellation process for the input signal using the specified reference signal. The result displays are not changed.

Example:

```
//Perform 8 averaging operations
ADJ:NCAN:AVER:COUN 8
//Use reference file "ideal_iqdata.iq.tar"
ADJ:NCAN:FILE:REF'c:\ideal_iqdata.iq.tar'
//Store corrected I/Q data to "corrected_iqdata.iq.tar"
ADJ:NCAN:FILE:RESult'c:\corrected_iqdata.iq.tar'
//Start correction process
ADJ:NCAN:STAR
//Query possible errors
ADJ:NCAN:ERR?
```

Usage: Event

10.9 Retrieving results

The following commands can be used to retrieve the results of the I/Q Analyzer measurement.



Storing large amounts of I/Q data

When storing large amounts of I/Q data to a file, consider the following tips to improve performance:

- If capturing and storing the I/Q data is the main goal of the measurement and evaluation functions are not required, use the basic I/Q data acquisition mode (see [TRACe: IQ\[:STATe\]](#) on page 155).
- Use a HiSlip or raw socket connection to export the data from the R&S FSV/A to a PC.
- Export the data in binary format rather than ASCII format (see [Section A, "Formats for returned values: ASCII format and binary format"](#), on page 355).
- Use the "Compatible" or "IQPair" data mode (see [Section B, "Reference: format description for I/Q data files"](#), on page 356).
- If only an extract of the available data is relevant, use the [TRACe<n>\[:DATA\]: MEMoRy?](#) command to store only the required section of data.

• Retrieving captured I/Q data	334
• Retrieving I/Q trace data	337
• Exporting traces and data	339
• Retrieving marker results	343

10.9.1 Retrieving captured I/Q data

The raw captured I/Q data is output in the form of a list.

TRACe:IQ:DATA?	334
TRACe:IQ:DATA:FORMat	334
TRACe:IQ:DATA:MEMory?	335

TRACe:IQ:DATA?

Initiates a measurement with the current settings and returns the captured data from I/Q measurements.

Corresponds to:

```
INIT:IMM;*WAI;:TRACe:IQ:DATA:MEMory?
```

However, the TRACe:IQ:DATA? command is quicker in comparison.

Note: Using the command with the *RST values for the TRACe:IQ:SET command, the following minimum buffer sizes for the response data are recommended: ASCII format 10 kBytes, binary format: 2 kBytes

Return values:

<Results> Measured voltage for I and Q component for each sample that has been captured during the measurement.
 The number of samples depends on TRACe:IQ:SET. In ASCII format, the number of results is 2* the number of samples.
 The data format depends on TRACe:IQ:DATA:FORMat on page 334.
 Default unit: V

Example:

```
TRAC:IQ:STAT ON
Enables acquisition of I/Q data
TRAC:IQ:SET NORM,10MHz,32MHz,EXT,POS,0,4096
Measurement configuration:
Sample Rate = 32 MHz
Trigger Source = External
Trigger Slope = Positive
Pretrigger Samples = 0
Number of Samples = 4096
FORMat REAL,32
Selects format of response data
TRAC:IQ:DATA?
Starts measurement and reads results
```

Usage: Query only

TRACe:IQ:DATA:FORMat <Format>

Selects the order of the I/Q data.

For details see [Section B, "Reference: format description for I/Q data files"](#), on page 356.

Parameters:

<Format>	COMPAtible IQBLoCk IQPaIr
	COMPAtible
	I and Q values are separated and collected in blocks: A block (512k) of I values is followed by a block (512k) of Q values, followed by a block of I values, followed by a block of Q values etc. (I,I,I,I,Q,Q,Q,Q,I,I,I,I,Q,Q,Q,Q...)
	IQBLoCk
	First all I-values are listed, then the Q-values (I,I,I,I,I,...Q,Q,Q,Q,Q,Q)
	IQPaIr
	One pair of I/Q values after the other is listed (I,Q,I,Q,I,Q...).
*RST:	IQP

TRACe:IQ:DATA:MEMory? [<OffsetSamples>,<NoOfSamples>]

Queries the I/Q data currently stored in the capture buffer of the R&S FSV/A.

By default, the command returns all I/Q data in the memory. You can, however, narrow down the amount of data that the command returns using the optional parameters.

If no parameters are specified with the command, the entire trace data is retrieved.

In this case, the command returns the same results as [TRACe:IQ:DATA?](#). (Note, however, that the [TRAC:IQ:DATA?](#) command initiates a new measurement before returning the captured values, rather than returning the existing data in the memory.)

The command returns a comma-separated list of the measured values in floating point format (comma-separated values = CSV). The number of values returned is 2 * the number of complex samples.

The total number of complex samples is displayed in the channel bar in manual operation and can be calculated as:

$$<\text{SampleRate}> * <\text{CaptureTime}>$$

(See [TRACe:IQ:SET](#), [TRACe:IQ:SRATe](#) on page 252 and [\[SENSe:\]SWEep:TIME](#) on page 271)

Query parameters:

<OffsetSamples>	Selects an offset at which the output of data should start in relation to the first data. If omitted, all captured samples are output, starting with the first sample.
Range:	0 to <# of samples> – 1, with <# of samples> being the maximum number of captured values
*RST:	0

<NoOfSamples> Number of samples you want to query, beginning at the offset you have defined. If omitted, all captured samples (starting at offset) are output.

Range: 1 to <# of samples> - <offset samples> with <# of samples> maximum number of captured values

*RST: <# of samples>

Return values:

<IQData> Measured value pair (I,Q) for each sample that has been recorded.

By default, the first half of the list contains the I values, the second half the Q values. The order can be configured using [TRACe:IQ:DATA:FORMat](#).

The data format of the individual values depends on [FORMat\[:DATA\]](#) on page 337.

Default unit: V

Example:

```
TRAC:IQ:STAT ON
Enables acquisition of I/Q data
TRAC:IQ:SET NORM,10MHz,32MHz,EXT,POS,100,4096
Measurement configuration:
Sample Rate = 32 MHz
Trigger Source = External
Trigger Slope = Positive
Pretrigger Samples = 100
Number of Samples = 4096
INIT;*WAI
Starts measurement and wait for sync
FORMat REAL,32
Determines output format
To read the results:
TRAC:IQ:DATA:MEM?
Reads all 4096 I/Q data
TRAC:IQ:DATA:MEM? 0,2048
Reads 2048 I/Q data starting at the beginning of data acquisition
TRAC:IQ:DATA:MEM? 2048,1024
Reads 1024 I/Q data from half of the recorded data
TRAC:IQ:DATA:MEM? 100,512
Reads 512 I/Q data starting at the trigger point (<Pretrigger Samples> was 100)
```

Example:

```
// Perform a single I/Q capture.
INIT;*WAI
// Determine output format (binary float32)
FORMat REAL,32
// Read 1024 I/Q samples starting at sample 2048.
TRAC:IQ:DATA:MEM? 2048,1024
```

Usage:

Query only

10.9.2 Retrieving I/Q trace data

In addition to the raw captured I/Q data, the results from I/Q analysis as shown in the result displays can also be retrieved.

FORMat[:DATA].....	337
TRACe<n>[:DATA]?.....	337
TRACe<n>[:DATA]:MEMory?.....	339
TRACe<n>[:DATA]:X?.....	339

FORMat[:DATA] <Format>[, <BitLength>]

Selects the data format that is used to transmit trace data from the R&S FSV/A to the controlling computer.

Note that the command has no effect for data that you send to the R&S FSV/A. The R&S FSV/A automatically recognizes the data it receives, regardless of the format.

Parameters:

<Format>

AScii

AScii format, separated by commas.

This format is almost always suitable, regardless of the actual data format. However, the data is not as compact as other formats can be.

REAL

Floating-point numbers (according to IEEE 754) in the "definite length block format".

The data is stored as binary data (definite length block data according to IEEE 488.2), each measurement value being formatted in 16-bit/32-bit/64-bit IEEE 754 floating-point-format.

<BitLength>

Length in bits for floating-point results

16

16-bit floating-point numbers.

Compared to REAL, 32 format, half as many numbers are returned.

32

32-bit floating-point numbers

For I/Q data, 8 bytes per sample are returned for this format setting.

64

64-bit floating-point numbers

Compared to REAL, 32 format, twice as many numbers are returned.

Example:

FORM REAL, 32

TRACe<n>[:DATA]? <ResultType>

This command queries current trace data and measurement results.

The data format depends on [FORMat \[:DATA\]](#) on page 337.

Suffix:

<n> [Window](#)

Query parameters:

<ResultType> Selects the type of result to be returned.
See [Table 10-6](#).

Example:

TRAC? TRACE3
Queries the data of trace 3.

Manual operation:

See ["Magnitude"](#) on page 14
See ["Spectrum"](#) on page 15
See ["I/Q-Vector"](#) on page 15
See ["Real/Imag \(I/Q\)"](#) on page 16

Table 10-6: Return values for result type parameters

Parameter	Result display / measurement	Results
TRACE1 ... TRACE6		Returns the sweep point values as shown in the result display. For the auto peak detector, the command returns positive peak values only. (To retrieve negative peak values, define a second trace with a negative peak detector.)
	"Magnitude"	Magnitude of the I and Q values (I+jQ) for each sweep point (=1001 values)
	"Spectrum"	
	"Real/Imag (I/Q)"	First the real parts for each trace point, then the imaginary parts (I ₁ ,...,I ₁₀₀₁ , Q ₁ ,...,Q ₁₀₀₁).
	"I/Q Vector"	The I and Q values for each trace point are returned (1001 pairs of I and Q values).
LIST	SEM measurements	Peak list evaluation, one peak per range is returned.
	Spurious emission measurements	Peak list evaluation For each peak, the command returns 11 values in the following order: • <No>: range number • <StartFreq>,<StopFreq>: start and stop frequency of the range • <RBW>: resolution bandwidth • <PeakFreq>: frequency of the peak in a range • <PowerAbs>: absolute power of the peak in dBm • <PowerRel>: power of the peak in relation to the channel power in dBc • <PowerDelta>: distance from the peak to the limit line in dB, positive values indicate a failed limit check • <LimitCheck>: state of the limit check (0 = PASS, 1 = FAIL) • <Unused1>,<Unused2>: reserved (0.0)
SPURious		Peak list evaluation of Spurious Emission measurements.
SPECTrogram SGRam		For every frame in the spectrogram, the command returns the power levels that have been measured, one for each sweep point. The number of frames depends on the size of the history depth. The power level depends on the configured unit. Only REAL, 32 format is supported.

TRACe<n>[:DATA]:MEMory? <Trace>,<OffsSwPoint>,<NoOfSwPoints>

Queries the previously captured trace data for the specified trace from the memory. As an offset and number of sweep points to be retrieved can be specified, the trace data can be retrieved in smaller portions, making the command faster than the `TRAC:DATA?` command. This is useful if only specific parts of the trace data are of interest.

If no parameters are specified with the command, the entire trace data is retrieved; in this case, the command returns the same results as `TRAC:DATA? TRACE1`.

Suffix:

<n> [Window](#)

Query parameters:

<Trace> TRACE1 | TRACE2 | TRACE3 | TRACE4 | TRACE5 | TRACE6

<OffsSwPoint> The offset in sweep points related to the start of the measurement at which data retrieval is to start.

<NoOfSwPoints> Number of sweep points to be retrieved from the trace.

Return values:

<SweepPointValues>

Example:

`TRAC:DATA:MEM? TRACE1,25,100`

Retrieves 100 sweep points from trace 1, starting at sweep point 25.

Usage:

Query only

TRACe<n>[:DATA]:X? <TraceNumber>

This command queries the horizontal trace data for each sweep point in the specified window, for example the frequency in frequency domain or the time in time domain measurements.

Suffix:

<n> [Window](#)

Query parameters:

<TraceNumber> Trace number.

TRACE1 | ... | TRACE6

Example:

`TRAC3:X? TRACE1`

Returns the x-values for trace 1 in window 3.

Usage:

Query only

10.9.3 Exporting traces and data

The following commands are required to export traces and spectrograms.

FORMat:DEXPort:CSEPARATOR.....	340
FORMat:DEXPort:DSEPARATOR.....	340
FORMat:DEXPort:FORMat.....	341
FORMat:DEXPort:HEADer.....	341
FORMat:DEXPort:TRACes.....	341
MMEMory:STORe<n>:SPECTrogram.....	341
MMEMory:STORe<n>:TRACe.....	342

FORMat:DEXPort:CSEPARATOR <Separator>

Selects the column separator for exported trace data.

The selected value is not affected by a preset. The command therefore has no reset value.

Parameters:

<Separator>

COMMa

Selects a comma as a separator.

SEMIColon

Selects a semicolon as a separator.

TAB

Selects a tabulator as a separator.

*RST: n/a

Example:

//Select column separator

FORM:DEXP:CSEP TAB

Manual operation: See ["Column Separator"](#) on page 46

FORMat:DEXPort:DSEPARATOR <Separator>

Selects the decimal separator for data exported in ASCII format.

Parameters:

<Separator>

POINT | COMMa

COMMa

Uses a comma as decimal separator, e.g. 4,05.

POINT

Uses a point as decimal separator, e.g. 4.05.

*RST: *RST has no effect on the decimal separator.
Default is POINT.

Example:

FORM:DEXP:DSEP POIN

Sets the decimal point as separator.

Manual operation: See ["Decimal Separator"](#) on page 46
See ["Export Peak List"](#) on page 130

FORMat:DEXPort:FORMat <FileFormat>

Determines the format of the ASCII file to be imported or exported. Depending on the external program that creates the data file or evaluates it, a comma-separated list (CSV) or a plain data format (DAT) file is required.

Parameters:

<FileFormat> CSV | DAT
 *RST: DAT

Example: FORM:DEXP:FORM CSV

Manual operation: See ["File Type"](#) on page 45

FORMat:DEXPort:HEADer <State>

If enabled, additional instrument and measurement settings are included in the header of the export file for result data. If disabled, only the pure result data from the selected traces and tables is exported.

Parameters:

<State> ON | OFF | 0 | 1
 *RST: 1

Manual operation: See ["Include Instrument & Measurement Settings"](#) on page 100

FORMat:DEXPort:TRACes <Selection>

Selects the data to be included in a data export file (see [MMEMory:STORe<n>:TRACe](#) on page 342).

Parameters:

<Selection> SINGle | ALL

SINGle

Only a single trace is selected for export, namely the one specified by the [MMEMory:STORe<n>:TRACe](#) command.

ALL

Selects all active traces and result tables (e.g. "Result Summary", marker peak list etc.) in the current application for export to an ASCII file.

The <trace> parameter for the [MMEMory:STORe<n>:TRACe](#) command is ignored.

*RST: SINGle

Manual operation: See ["Export all Traces and all Table Results"](#) on page 100

MMEMory:STORe<n>:SPECtrogram <FileName>

Exports spectrogram data to an ASCII file.

The file contains the data for every frame in the history buffer. The data corresponding to a particular frame begins with information about the frame number and the time that frame was recorded.

Note that, depending on the size of the history buffer, the process of exporting the data can take a while.

Secure User Mode

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Suffix:

<n> [Window](#)

Parameters:

<FileName> String containing the path and name of the target file.

Example:

MMEM:STOR:SGR 'Spectrogram'
Copies the spectrogram data to a file.

Manual operation: See ["Export Spectrogram to ASCII File"](#) on page 102

MMEMory:STORe<n>:TRACe <Trace>, <FileName>

Exports trace data from the specified window to an ASCII file.

Secure User Mode

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Suffix:

<n> [Window](#)

Parameters:

<Trace> Number of the trace to be stored

<FileName> String containing the path and name of the target file.

Example:

MMEM:STOR1:TRAC 1, 'C:\TEST.ASC'
Stores trace 1 from window 1 in the file TEST.ASC.

Manual operation: See ["Export Trace to ASCII File"](#) on page 44

10.9.4 Retrieving marker results

The following commands are required to retrieve the results of markers.

Useful commands for retrieving marker results described elsewhere:

- [CALCulate<n>:DELTamarker<m>:X](#) on page 291
- [CALCulate<n>:MARKer<m>:X](#) on page 293
- [CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:X?](#) on page 321
- [CALCulate<n>:MARKer<m>:FUNCTION:FPEaks:Y?](#) on page 321

Remote commands exclusive to retrieving marker results:

CALCulate<n>:DELTamarker<m>:X:RELative?	343
CALCulate<n>:DELTamarker<m>:Y?	343
CALCulate<n>:MARKer<m>:Y?	344
MMEMory:STORe<n>:LIST	344

CALCulate<n>:DELTamarker<m>:X:RELative?

Queries the relative position of a delta marker on the x-axis.

If necessary, the command activates the delta marker first.

Suffix:

<n> [Window](#)

<m> [Marker](#)

Return values:

<Position> Position of the delta marker in relation to the reference marker.

Example:

CALC:DELT3:X:REL?

Outputs the frequency of delta marker 3 relative to marker 1 or relative to the reference position.

Usage: Query only

CALCulate<n>:DELTamarker<m>:Y?

Queries the result at the position of the specified delta marker.

Suffix:

<n> 1..n

<m> 1..n

Return values:

<Result> Result at the position of the delta marker.
The unit is variable and depends on the one you have currently set.

Default unit: DBM

Usage: Query only

CALCulate<n>:MARKer<m>:Y?

Queries the result at the position of the specified marker.

Suffix:

<n> 1..n

<m> 1..n

Return values:

<Result> Default unit: DBM

Usage: Query only

Manual operation: See "[Marker Table](#)" on page 17
See "[Marker Peak List](#)" on page 17

MMEMory:STORe<n>:LIST <FileName>

Exports the SEM and spurious emission list evaluation to a file.

The file format is *.dat.

Secure User Mode

In secure user mode, settings that are stored on the instrument are stored to volatile memory, which is restricted to 256 MB. Thus, a "memory limit reached" error can occur although the hard disk indicates that storage space is still available.

To store data permanently, select an external storage location such as a USB memory device.

For details, see "Protecting Data Using the Secure User Mode" in the "Data Management" section of the R&S FSV3000/ FSVA3000 base unit user manual.

Suffix:

<n> [Window](#)

Parameters:

<FileName> String containing the path and name of the target file.

Example:

MMEM:STOR:LIST 'test'

Stores the current list evaluation results in the test.dat file.

10.10 Importing and exporting I/Q data and results

Alternatively to capturing I/Q data by the I/Q Analyzer itself, stored I/Q data from previous measurements or other applications can be imported to the I/Q Analyzer. Furthermore, I/Q data processed in the I/Q Analyzer can be stored to a file for further evaluation in other applications.



I/Q data can only be exported in applications that process I/Q data, such as the I/Q Analyzer or optional applications.

For details see [Section 7, "I/Q data import and export"](#), on page 132.

MMEMory:STORe<n>:IQ:COMMeNt	345
MMEMory:STORe<n>:IQ:FORMat	345
MMEMory:STORe<n>:IQ:STATe	345

MMEMory:STORe<n>:IQ:COMMeNt <Comment>

Adds a comment to a file that contains I/Q data.

Suffix:

<n> irrelevant

Parameters:

<Comment> String containing the comment.

Example:

```
MMEM:STOR:IQ:COMM 'Device test 1b'
Creates a description for the export file.
MMEM:STOR:IQ:STAT 1, 'C:
\R_S\Instr\user\data.iq.tar'
Stores I/Q data and the comment to the specified file.
```

Manual operation: See ["I/Q Export"](#) on page 46

MMEMory:STORe<n>:IQ:FORMat <Format>,<DataFormat>

Sets or queries the format of the I/Q data to be stored.

Suffix:

<n> irrelevant

Parameters:

<Format> **FLOat32**
32-bit floating point format.

INT32
32-bit integer format.
*RST: FLOat32

<DataFormat> **COMPLex**
Exports complex data.

REAL
Exports real data.
*RST: COMPLex

Example: `MMEM:STOR:IQ:FORM INT32,REAL`

MMEMory:STORe<n>:IQ:STATe <1>,<FileName>

Writes the captured I/Q data to a file.

By default, the contents of the file are in 32-bit floating point format.

Suffix:

<n> 1..n

Parameters:

<1>

<FileName>

String containing the path and name of the target file.
 The file type is determined by the file extension. If no file extension is provided, the file type is assumed to be .iq.tar.
 For .mat files, Matlab® v4 is assumed.

Example:

```
MMEM:STOR:IQ:STAT 1, 'C:\R_S\Instr\user\data.iq.tar'
```

Stores the captured I/Q data to the specified file.

Usage:

Asynchronous command

Manual operation: See ["I/Q Export"](#) on page 46

10.11 Querying the status registers

The R&S FSV3-I/Q Analyzer uses the standard status registers of the R&S FSV/A.

In addition, the `STATus:QUEStionable:SYNC` register indicates errors concerning I/Q data processing.

For details on the common R&S FSV/A status registers refer to the description of remote control basics in the R&S FSV/A User Manual.



*RST does not influence the status registers.

- [STATus:QUEStionable:SYNC register](#)..... 346

10.11.1 STATus:QUEStionable:SYNC register

This register contains information about the state of the I/Q data acquisition.

The status of the `STATus:QUEStionable:SYNC` register is indicated in bit 11 of the `STATus:QUEStionable` register.

You can read out the state of the register with [STATus:QUEStionable:SYNC:CONDition?](#) on page 347 and [STATus:QUEStionable:SYNC\[:EVENT\]?](#) on page 348.

Bit No.	Meaning
0-6	not used
7	I/Q noise cancellation error This bit is set if an error occurs during I/Q noise cancellation, e.g. due to failed synchronization. See Section 9.1, "Troubleshooting I/Q noise cancellation" , on page 142.
8	I/Q data acquisition error This bit is set if an error occurs during I/Q data acquisition because the input sample rates or number of samples between the signal source and the R&S FSV/A do not match.
9-14	not used
15	This bit is always set to 0.

STATus:QUESTionable:SYNC:CONDition?	347
STATus:QUESTionable:SYNC:ENABle	347
STATus:QUESTionable:SYNC:NTRansition	347
STATus:QUESTionable:SYNC:PTRansition	348
STATus:QUESTionable:SYNC[:EVENT]	348

STATus:QUESTionable:SYNC:CONDition? <ChannelName>

Reads out the CONDition section of the status register.

The command does not delete the contents of the EVENT section.

Query parameters:

<ChannelName> String containing the name of the channel.
The parameter is optional. If you omit it, the command works for the currently active channel.

Usage: Query only

STATus:QUESTionable:SYNC:ENABle <BitDefinition>, <ChannelName>

Controls the ENABle part of a register.

The ENABle part allows true conditions in the EVENT part of the status register to be reported in the summary bit. If a bit is 1 in the enable register and its associated event bit transitions to true, a positive transition will occur in the summary bit reported to the next higher level.

Parameters:

<BitDefinition> Range: 0 to 65535
<ChannelName> String containing the name of the channel.
The parameter is optional. If you omit it, the command works for the currently active channel.

STATus:QUESTionable:SYNC:NTRansition <BitDefinition>[, <ChannelName>]

Controls the Negative TRansition part of a register.

Setting a bit causes a 1 to 0 transition in the corresponding bit of the associated register. The transition also writes a 1 into the associated bit of the corresponding EVENT register.

Parameters:

<BitDefinition> Range: 0 to 65535

<ChannelName> String containing the name of the channel.
The parameter is optional. If you omit it, the command works for the currently active channel.

STATus:QUEStionable:SYNC:PTRansition <BitDefinition>[,<ChannelName>]

These commands control the Positive TRansition part of a register.

Setting a bit causes a 0 to 1 transition in the corresponding bit of the associated register. The transition also writes a 1 into the associated bit of the corresponding EVENT register.

Parameters:

<BitDefinition> Range: 0 to 65535

<ChannelName> String containing the name of the channel.
The parameter is optional. If you omit it, the command works for the currently active channel.

STATus:QUEStionable:SYNC[:EVENT]? <ChannelName>

Reads out the EVENT section of the status register.

The command also deletes the contents of the EVENT section.

Query parameters:

<ChannelName> String containing the name of the channel.
The parameter is optional. If you omit it, the command works for the currently active channel.

Usage: Query only

10.12 Programming examples

The following programming examples demonstrate how to capture I/Q data and perform I/Q data analysis using the I/Q Analyzer in a remote environment.

Optional interfaces for I/Q data input are also demonstrated in the I/Q Analyzer.

- [I/Q analysis with graphical evaluation](#)..... 349
- [Basic I/Q analysis with improved performance](#)..... 350
- [Data acquisition via the optional Analog Baseband interface](#)..... 351

10.12.1 I/Q analysis with graphical evaluation

This example demonstrates how to configure and perform a basic I/Q data acquisition and analyze the data using the I/Q Analyzer in a remote environment.

```
//-----Activating the I/Q Analyzer application -----
*RST
//Reset the instrument
INST:CRE IQ,'IQANALYZER'
//Creates a new measurement channel named 'IQANALYZER'.
INIT:CONT OFF
//Switches to single sweep mode

//-----Configuring Data Acquisition-----
TRAC:IQ:SRAT 32MHZ
//Defines the sample rate.
TRAC:IQ:RLEN 1000
//Sets the record length (number of samples to capture) to 1000 samples.
TRAC:IQ:BWID?
//Queries the bandwidth of the resampling filter, determined by the sample rate
FORM:DATA REAL,32
//Formats the data as 32-byte real values.
TRAC:IQ:DATA:FORM IQBL
//Lists all I values first, then all Q values in the trace results.

//-----Configuring the Trace-----
TRAC:IQ:AVER ON
//Defines averaging for the I/Q trace.
TRAC:IQ:AVER:COUN 10
//Defines an average over 10 sweeps.

DISP:TRAC1:MODE WRIT
DISP:TRAC2:MODE MAXH
DISP:TRAC3:MODE MINH
//Changes the trace modes.

//-----Performing the Measurement-----
INIT;*WAI
//Initiates a new measurement and waits until the sweep has finished.

//-----Retrieving Results-----

TRAC:DATA? TRACE1
TRAC:DATA? TRACE2
TRAC:DATA? TRACE3
//Returns the magnitude for each sweep point

LAY:REPL:WIND '1',RIMAG
//Changes the result display to Real/Imag (I/Q)
```

```
CALC:MARK:SEAR MAGN
//Configures searches to search both I and Q branches.
CALC:MARK:Y?
//Queries the result of the peak search on both branches.

TRAC:IQ:DATA:MEM? 0,500
//Returns the first 500 samples of the stored I/Q data for the measurement.
//For each sample, first the I-value, then the Q-value is listed.

TRAC:IQ:DATA:MEM? 500,500
//Returns the second half of the 1000 captured sample values.
```

10.12.2 Basic I/Q analysis with improved performance

This example demonstrates how to configure and perform a basic I/Q data acquisition and analyze the data using the I/Q Analyzer functionality in a remote environment.

```
//-----Activating the I/Q Analyzer application -----
*RST
//Reset the instrument

INIT:CONT OFF
//Switches to single sweep mode
TRACE:IQ ON
//Switches the (internal) operating mode of the current measurement channel to
//simple I/Q data acquisition mode while retaining the relevant parameters
//from the Spectrum mode.

//-----Configuring Data Acquisition-----
TRACE:IQ:SET NORM,0,32000000,IQP,POS,0,1000
//Configures the sample rate as 32 MHz, IQP trigger, positive trigger slope,
//no pretrigger samples, 1000 samples to capture
FORM REAL,32
//The data is formatted as real values.

//-----Configuring I/Q Gating-----
TRAC:IQ:EGAT ON
//Turns on gated measurement.
TRAC:IQ:EGAT:TYPE LEV
//Select the level gate type.
TRAC:IQ:EGAT:LENG 20
//Sets the gate length to 20 samples.
TRAC:IQ:EGAT:GAP 20
//Sets the interval between gate periods to 20 samples.
TRAC:IQ:EGAT:NOF 2
//Sets the number of gate periods after the trigger signal to 2.
TRIG:SOUR IQP
//Defines the magnitude of the sampled I/Q data to be used as a trigger.
```

```

TRIG:LEV:IQP -30dbm
//Sets the trigger level.

//-----Performing the Measurement and Retrieving Results-----
TRAC:IQ:DATA?; *WAI;
//Performs a measurement and returns the RF input voltage at each sample point
//(first 1000 I-values, then 1000 Q-values).

TRAC:IQ:DATA:MEM? 0,500
//Returns the first 500 samples of the stored trace data for the measurement.
//For each sample, first the I-value, then the Q-value is listed.

TRAC:IQ:DATA:MEM? 500,500
//Returns the second half of the 1000 captured sample values.

```

10.12.3 Data acquisition via the optional Analog Baseband interface

This example demonstrates how to capture I/Q data via the optional "Analog Baseband" interface using the I/Q Analyzer in a remote environment. As an input signal, a differential probe is assumed to be connected to the R&S FSV/A.

```

//-----Activating the I/Q Analyzer application -----
*RST
//Reset the instrument
INST:CRE IQ,'IQANALYZER'
//Creates a new measurement channel named 'IQANALYZER'.
INIT:CONT OFF
//Switches to single sweep mode

//-----Activating the Analog Baseband Interface-----
INP:SEL AIQ
//Selects the analog baseband interface as the input source
INP:IQ:TYPE I
//Only the signal on I input is analyzed (I only mode)
INP:IQ:BAL ON
//Differential input signal
INP:IQ:FULL:AUTO OFF
INP:IQ:FULL:LEV 2V
//Peak voltage at connector is set manually to the maximum of 2V
FREQ:CEN 1MHz
//Shift center frequency to 1 MHz (Low IF I)

//-----Configuring Data Acquisition-----
TRIG:SOUR BBP
TRIG:SEQ:LEV:BBP -20
//Trigger on baseband power of -20 dBm.
TRAC:IQ:SRAT 32MHZ

```

```

//Defines the sample rate.
TRAC:IQ:RLEN 1000
//Sets the record length (number of samples to capture) to 1000 samples.
TRAC:IQ:BWID?
//Queries the bandwidth of the resampling filter, determined by the sample rate.

//-----Adding result displays-----
LAY:ADD? '1',RIGH,FREQ
//Spectrum display in window 2, to the right of Magnitude results
LAY:ADD? '1',BEL,RIMAG
//Real I display in window 3, below Magnitude results

//-----Configuring the Trace-----
TRAC:IQ:AVER ON
//Defines averaging for the magnitude trace of I component.
TRAC:IQ:AVER:COUN 10
//Defines an average over 10 sweeps.

DISP:TRAC1:MODE WRIT
DISP:TRAC2:MODE MAXH
DISP:TRAC3:MODE MINH
//Changes the trace modes.

//-----Performing the Measurement-----
INIT;*WAI
//Initiates a new measurement and waits until the sweep has finished.

//-----Retrieving Results-----
TRAC:IQ:DATA:FORM IQBL
TRAC:IQ:DATA?
//Retrieves the captured I samples (1000 values), followed by the captured
//Q samples (1000 values); Q samples are all 0 because of I/Q mode: Low IF
TRAC2:DATA? TRACE1
//Returns the power levels for each sample (y-values from Spectrum display)
TRAC2:DATA:X? TRACE1
//Returns the frequency for each sample (x-values from Spectrum display)

```

10.13 Deprecated commands

The following commands are provided for compatibility to other signal analyzers only. For new remote control programs use the specified alternative commands.

[CALCulate<n>:FORMat](#).....352

CALCulate<n>:FORMat <Evaluation>

This command selects the evaluation method of the measured data that is to be displayed in the specified window.

Note that for the I/Q analyzer application, this command is maintained for compatibility reasons only. Use the `LAYout` commands for new remote control programs (see [Section 10.5.2, "Working with windows in the display"](#), on page 259).

Suffix:

<n> 1..n

Parameters:

<Evaluation> Type of evaluation you want to display.
See the table below for available parameter values.

Example:

```
INST:SEL IQ
Activates I/Q Analyzer.
CALC:FORM FREQ
Selects the display of the I/Q data spectrum.
```

Table 10-7: <Evaluation> parameter values for the I/Q Analyzer application

Parameter value	Window type
FREQ	Spectrum
MAGN	Magnitude
MTABLE	Marker table
PEAKlist	Marker peak list
RIMAG	Real/Imag (I/Q)
VECT	I/Q Vector

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A Formats for returned values: ASCII format and binary format

When trace data is retrieved using the `TRAC:DATA` or `TRAC:IQ:DATA` command, the data is returned in the format defined using the `FORMat[:DATA]` on page 337. The possible formats are described here.

- **ASCII Format (FORMat ASCII):**
The data is stored as a list of comma-separated values (CSV) of the measured values in floating point format.
- **Binary Format (FORMat REAL,16/32/64):**
The data is stored as binary data (definite length block data according to IEEE 488.2), each measurement value being formatted in 16-bit/32-bit/64-bit IEEE 754 floating-point-format.
The schema of the result string is as follows:
#<Length of length><Length of data><value1><value2>...<value n>
with:

<Length of length>	Number of digits of the following number of data bytes
<Length of data>	Number of following data bytes
<Value>	2-byte/4-byte/8-byte floating point value

Example: #41024<Data>... contains 1024 data bytes

Data blocks larger than 999,999,999 bytes

According to SCPI, the header of the block data format allows for a maximum of 9 characters to describe the data length. Thus, the maximum REAL 32 data that can be represented is 999,999,999 bytes. However, the R&S FSV/A is able to send larger data blocks. In this case, the length of the data block is placed in brackets, e.g.
#(1234567890)<value1><value2>...



Reading out data in binary format is quicker than in ASCII format. Thus, binary format is recommended for large amounts of data.

B Reference: format description for I/Q data files

This section describes how I/Q data is transferred to the memory during remote control (see [TRACe: IQ: DATA: FORMat](#) command).

For details on the format of the individual values, see [Section A, "Formats for returned values: ASCII format and binary format"](#), on page 355.

For details on the format of I/Q export files (using the "I/Q Export" function), see the R&S FSV/A User Manual.

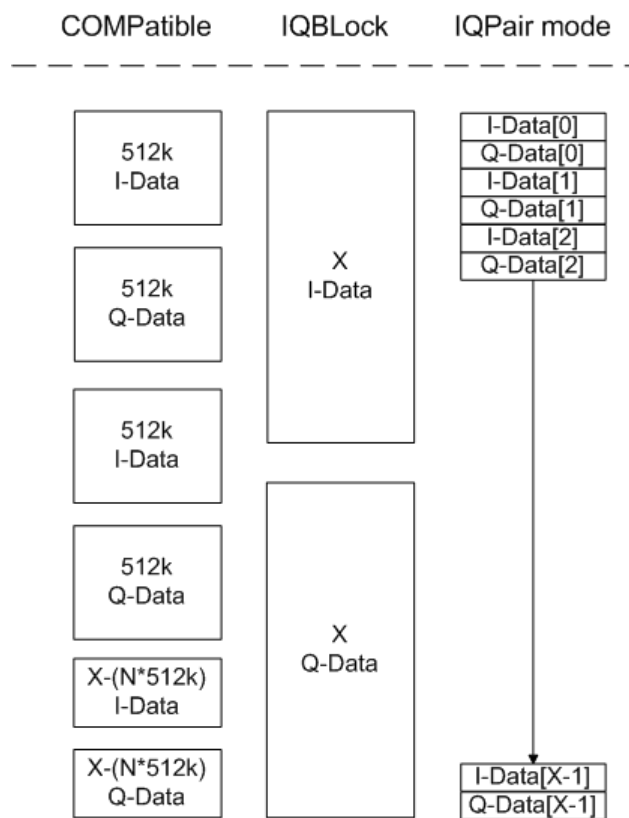


Figure B-1: I/Q data formats

Note: 512k corresponds to 524288 samples

For maximum performance, the formats "Compatible" or "IQPair" should be used. Furthermore, for large amounts of data, the data should be in binary format to improve performance.

In binary format, the number of I- and Q-data can be calculated as follows:

$$\# \text{ of I-Data} = \# \text{ of Q-Data} = \frac{\# \text{ of DataBytes}}{8}$$

For the format "QBLock", the offset of Q-data in the output buffer can be calculated as follows:

$$Q - Data - Offset = \frac{(\# \text{ of } DataBytes)}{2} + LengthIndicatorDigits$$

with "LengthIndicatorDigits" being the number of digits of the length indicator including the #. In the example above (#41024...), this results in a value of 6 for "LengthIndicatorDigits" and the offset for the Q-data results in $512 + 6 = 518$.

C Reference: supported I/Q file formats

Various file types are supported for I/Q data import and export. The most important characteristics for each format are described here.

See ["Select I/Q data file"](#) on page 51.



For best performance and to ensure comprehensive meta data is available, use the `iq.tar` format. This is a widely used file format for Rohde & Schwarz products.

Table C-1: Characteristics of data file formats

File format	File extension	Comment
IQ.tar	<code>.iq.tar</code>	An <code>IQ.tar</code> file contains I/Q data in binary format together with meta information that describes the nature and the source of data, e.g. the sample rate. The objective of the <code>IQ.tar</code> file format is to separate I/Q data from the meta information while still having both inside one file. In addition, the file format allows a preview of the I/Q data in a web browser, and inclusion of user-specific data. Several streams of data can be provided in one file.
IQW	<code>.iqw</code>	A binary file format containing one channel of complex IQ data. The file contains float32 data in a binary format (interleaved IQIQ or in blocks, IIIQQQ). The file does not contain any additional information as a header. This format requires setting the sample rate and measurement time or record length manually.
CSV	<code>.csv</code>	A file containing I/Q data as comma-separated values (CSV). Additional metadata can be included.
Simple CSV	<code>.csv</code>	(Import only) Simple CSV contains I/Q data only, without any header or meta data. That is, the file contains only (I,Q) data pairs, separated by commas. Several streams of data can be provided in one file. This format requires setting the sample rate and measurement time or record length manually.
Matlab® v4	<code>.mat</code>	A file containing I/Q data in Matlab® file format v4. Channel-related information is stored in matlab variables with names starting with 'ChX_'. 'x' represents the number of the channel with a lower bound of 1, e.g. the variable <code>Ch1_ChannelName</code> contains the name of the first channel. The corresponding data is contained in <code>ChX_Data</code> . Optional user data can be saved to variables named <code>UserDataX</code> , where 'x' starts at 0. The variable <code>UserData_Count</code> contains the number of <code>UserData</code> variables. For compatibility reasons user data needs to be saved as a 2xN char array, where the first row contains the key of the user data and the second row the actual value. Both rows must have the same column count and are therefore right-padded with white spaces. Variables can be written to the <code>*.mat</code> files in arbitrary order. Limitations: In general, the file format is limited to a maximum of 2 GB. A maximum of 100000000 values can be stored in a single variable, e.g. 50000000 complex data samples.
Matlab® v7.3	<code>.mat</code>	A file containing I/Q data in Matlab® file format v7.3.
Simple Matlab®	<code>.mat</code>	(Import only) Simple Matlab® format contains I/Q data only, without any meta data. That is, the file contains only variables (double, double) for the corresponding channel data. This format requires setting the sample rate and measurement time or record length manually.

File format	File extension	Comment
AMMOS intermediate frequency data format	.aid	Format used to transmit real or complex baseband signals. The IF signal is sent along with information that characterizes the datastream and datastream source. All datastreams have a frame-based structure, consisting of a global frame header coupled with a data-type specific frame body (i.e. the frame payload).
wv	.wv	(Import only) Proprietary file format used by Rohde & Schwarz signal generators to store waveform data. A waveform file contains a header and raw I/Q samples.

- [I/Q data file format \(iq-tar\)](#)..... 359
- [CSV file format](#)..... 367
- [IQW file format](#)..... 369
- [Matlab® v. 4 / v. 7.3 file format](#)..... 370
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C.1 I/Q data file format (iq-tar)

I/Q data is packed in a file with the extension `.iq.tar`. An `iq-tar` file contains I/Q data in binary format together with meta information that describes the nature and the source of data, e.g. the sample rate. The objective of the `iq-tar` file format is to separate I/Q data from the meta information while still having both inside one file. In addition, the file format allows you to include user-specific data and to preview the I/Q data in a web browser (not supported by all web browsers).

The `iq-tar` container packs several files into a single `.tar` archive file. Files in `.tar` format can be unpacked using standard archive tools (see http://en.wikipedia.org/wiki/Comparison_of_file_archivers) available for most operating systems. The advantage of `.tar` files is that the archived files inside the `.tar` file are not changed (not compressed) and thus it is possible to read the I/Q data directly within the archive without the need to unpack (untar) the `.tar` file first.



Sample iq-tar files

Some sample `iq-tar` files are provided in the `C:\R_S\INSTR\USER\Demo\` directory on the R&S FSV/A.



An application note on converting Rohde & Schwarz I/Q data files is available from the Rohde & Schwarz website:

[1EF85: Converting R&S I/Q data files](#)

Contained files

An `iq-tar` file must contain the following files:

- **I/Q parameter XML file**, e.g. `xyz.xml`

Contains meta information about the I/Q data (e.g. sample rate). The filename can be defined freely, but there must be only one single I/Q parameter XML file inside an iq-tar file.

- **I/Q data binary file**, e.g. xyz.complex.float32

Contains the binary I/Q data of all channels. There must be only one single I/Q data binary file inside an iq-tar file.

Optionally, an iq-tar file can contain the following file:

- **I/Q preview XSLT file**, e.g. open_IqTar_xml_file_in_web_browser.xslt
Contains a stylesheet to display the I/Q parameter XML file and a preview of the I/Q data in a web browser (not supported by all web browsers).

A sample stylesheet is available at http://www.rohde-schwarz.com/file/open_IqTar_xml_file_in_web_browser.xslt.

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- [I/Q data binary file](#)..... 364

C.1.1 I/Q parameter XML file specification



The content of the I/Q parameter XML file must comply with the XML schema RsIqTar.xsd available at: <http://www.rohde-schwarz.com/file/RsIqTar.xsd>.

In particular, the order of the XML elements must be respected, i.e. iq-tar uses an "ordered XML schema". For your own implementation of the iq-tar file format make sure to validate your XML file against the given schema.

The following example shows an I/Q parameter XML file. The XML elements and attributes are explained in the following sections.

Sample I/Q parameter XML file: xyz.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl"
href="open_IqTar_xml_file_in_web_browser.xslt"?>
<RS_IQ_TAR_FileFormat fileFormatVersion="1"
xsi:noNamespaceSchemaLocation="RsIqTar.xsd"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <Name>R&S FSV/A</Name>
  <Comment>Here is a comment</Comment>
  <DateTime>2011-01-24T14:02:49</DateTime>
  <Samples>68751</Samples>
  <Clock unit="Hz">6.5e+006</Clock>
  <Format>complex</Format>
  <DataType>float32</DataType>
  <ScalingFactor unit="V">1</ScalingFactor>
  <NumberOfChannels>1</NumberOfChannels>
  <DataFilename>xyz.complex.float32</DataFilename>
  <UserData>
    <UserDefinedElement>Example</UserDefinedElement>
  </UserData>
</RS_IQ_TAR_FileFormat>
```



```

</UserData>
  <PreviewData>...</PreviewData>
</RS_IQ_TAR_FileFormat>

```

C.1.1.1 Minimum data elements

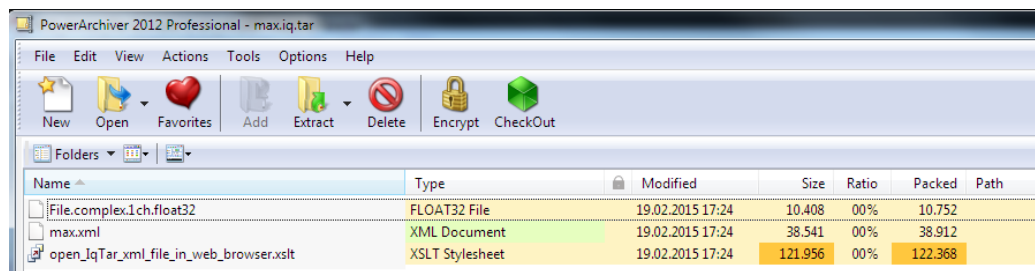
The following data elements are the minimum required for a valid `iq-tar` file. They are always provided by an `iq-tar` file export from a Rohde & Schwarz product. If not specified otherwise, it must be available in all `iq-tar` files used to import data to a Rohde & Schwarz product.

Element	Possible Values	Description
<RS_IQ_TAR_FileFormat>	-	The root element of the XML file. It must contain the attribute <code>fileFormatVersion</code> that contains the number of the file format definition.
<Name>	string	Optional: describes the device or application that created the file.
<Comment>	string	Optional: contains text that further describes the contents of the file.
<DateTime>	yyyy-mm-ddThh:mm:ss	Contains the date and time of the creation of the file. Its type is <code>xs:dateTime</code> (see <code>RsIqTar.xsd</code>).
<Samples>	integer	Contains the number of samples of the I/Q data. For multi-channel signals all channels have the same number of samples. One sample can be: • A complex number represented as a pair of I and Q values • A complex number represented as a pair of magnitude and phase values • A real number represented as a single real value See also <Format> element.
<Clock>	double	Contains the clock frequency in Hz, i.e. the sample rate of the I/Q data. A signal generator typically outputs the I/Q data at a rate that equals the clock frequency. If the I/Q data was captured with a signal analyzer, the signal analyzer used the clock frequency as the sample rate. The attribute <code>unit</code> must be set to "Hz".
<Format>	complex real polar	Specifies how the binary data is saved in the I/Q data binary file (see <DataFilename> element). Every sample must be in the same format. The format can be one of the following: • <code>complex</code>: Complex number in cartesian format, i.e. I and Q values interleaved. I and Q are unitless • <code>real</code>: Real number (unitless) • <code>polar</code>: Complex number in polar format, i.e. magnitude (unitless) and phase (rad) values interleaved. Requires <code>DataType = float32 or float64</code>
<DataType>	int8 int16 int32 float32 float64	Specifies the binary format used for samples in the I/Q data binary file (see <DataFilename> element and Section C.1.2, "I/Q data binary file" , on page 364). The following data types are allowed: • <code>int8</code>: 8 bit signed integer data • <code>int16</code>: 16 bit signed integer data • <code>int32</code>: 32 bit signed integer data • <code>float32</code>: 32 bit floating point data (IEEE 754) • <code>float64</code>: 64 bit floating point data (IEEE 754)

Element	Possible Values	Description
<ScalingFactor>	double	Optional: describes how the binary data can be transformed into values in the unit Volt. The binary I/Q data itself has no unit. To get an I/Q sample in the unit Volt the saved samples have to be multiplied by the value of the <ScalingFactor>. For polar data only the magnitude value has to be multiplied. For multi-channel signals the <ScalingFactor> must be applied to all channels. The attribute <code>unit</code> must be set to "v". The <ScalingFactor> must be > 0. If the <ScalingFactor> element is not defined, a value of 1 V is assumed.
<NumberOfChannels>	integer	Optional: specifies the number of channels, e.g. of a MIMO signal, contained in the I/Q data binary file. For multi-channels, the I/Q samples of the channels are expected to be interleaved within the I/Q data file (see Section C.1.2, "I/Q data binary file" , on page 364). If the <NumberOfChannels> element is not defined, one channel is assumed.
<DataFilename>		Contains the filename of the I/Q data binary file that is part of the iq-tar file. It is recommended that the filename uses the following convention: <xyz>.<Format>.<Channels>ch.<Type> • <xyz> = a valid Windows file name • <Format> = complex, polar or real (see <code>Format</code> element) • <Channels> = Number of channels (see <code>NumberOfChannels</code> element) • <Type> = float32, float64, int8, int16, int32 or int64 (see <code>DataType</code> element) Examples: • xyz.complex.1ch.float32 • xyz.polar.1ch.float64 • xyz.real.1ch.int16 • xyz.complex.16ch.int8
<UserData>	xml	Optional: contains user, application or device-specific XML data which is not part of the iq-tar specification. This element can be used to store additional information, e.g. the hardware configuration. User data must be valid XML content.
<PreviewData>	xml	Optional: contains further XML elements that provide a preview of the I/Q data. The preview data is determined by the routine that saves an iq-tar file (e.g. R&S FSV/A). For the definition of this element refer to the <code>RsIqTar.xsd</code> schema. Note that the preview can be only displayed by current web browsers that have JavaScript enabled and if the XSLT stylesheet <code>open_IqTar_xml_file_in_web_browser.xslt</code> is available.

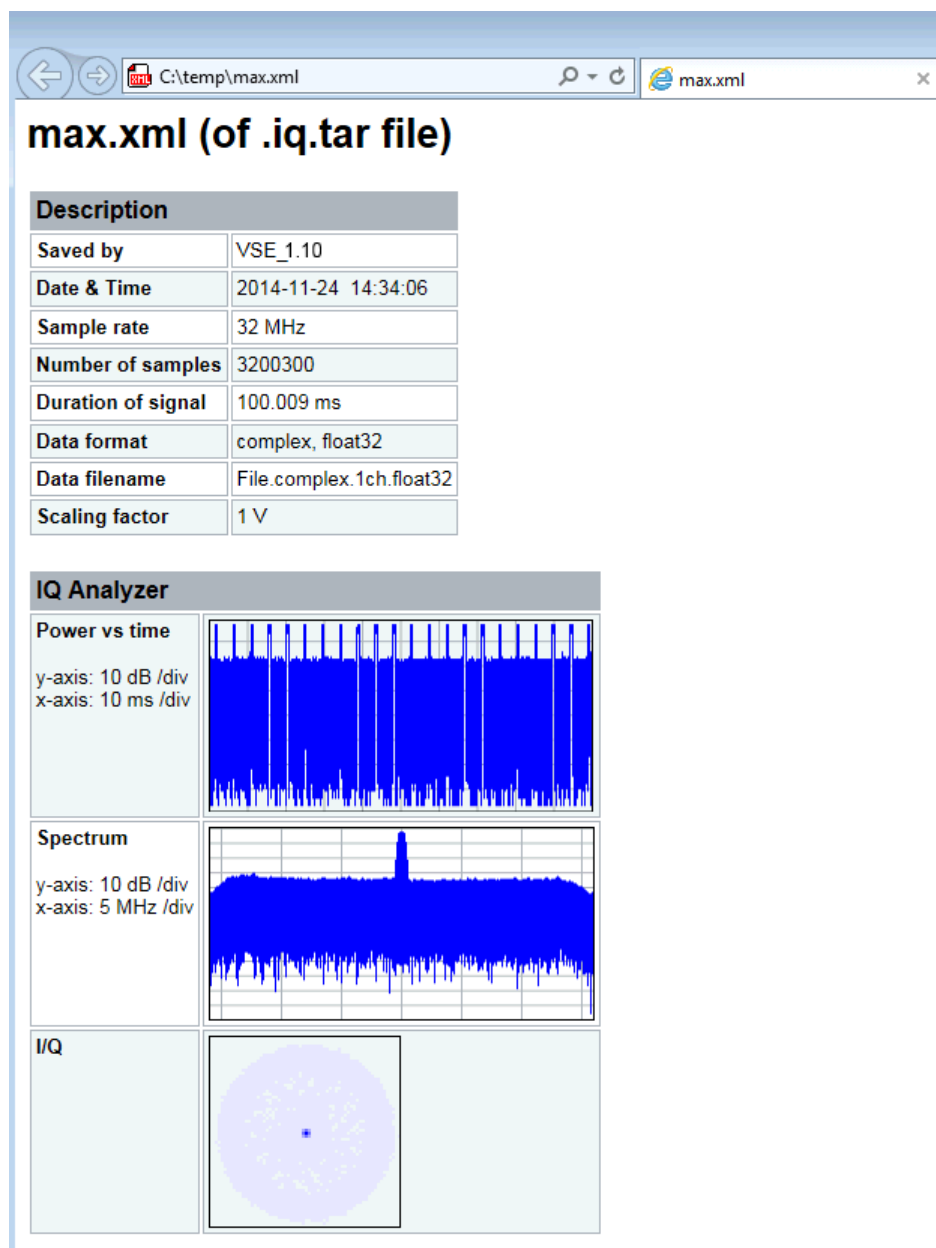
C.1.1.2 Example

The following example demonstrates the XML description inside the iq-tar file. Note that this preview is not supported by all web browsers.



Name	Type	Modified	Size	Ratio	Packed	Path
File.complex.1ch.float32	FLOAT32 File	19.02.2015 17:24	10.408	00%	10.752	
max.xml	XML Document	19.02.2015 17:24	38.541	00%	38.912	
open_IqTar_xml_file_in_web_browser.xslt	XSLT Stylesheet	19.02.2015 17:24	121.956	00%	122.368	

Open the xml file in a web browser. If the stylesheet `open_IqTar_xml_file_in_web_browser.xslt` is in the same directory, the web browser displays the xml file in a readable format.

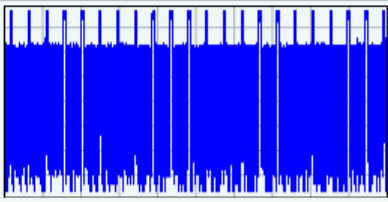


max.xml (of .iq.tar file)

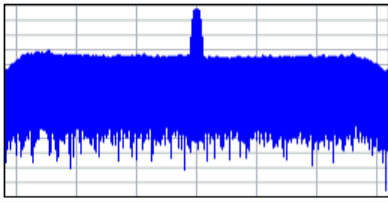
Description	
Saved by	VSE_1.10
Date & Time	2014-11-24 14:34:06
Sample rate	32 MHz
Number of samples	3200300
Duration of signal	100.009 ms
Data format	complex, float32
Data filename	File.complex.1ch.float32
Scaling factor	1 V

IQ Analyzer

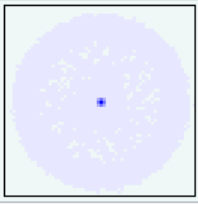
Power vs time
y-axis: 10 dB /div
x-axis: 10 ms /div



Spectrum
y-axis: 10 dB /div
x-axis: 5 MHz /div



I/Q



```

<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="open_IqTar_xml_file_in_web_browser.xslt"?>
<RS_IQ_TAR_FileFormat fileFormatVersion="1" xsi:noNamespaceSchemaLocation=
"http://www.rohde-schwarz.com/file/RsIqTar.xsd" xmlns:xsi=
"http://www.w3.org/2001/XMLSchema-instance">
  <Name>VSE_1.10a 29 Beta</Name>
  <Comment></Comment>
  <DateTime>2015-02-19T15:24:58</DateTime>
  <Samples>1301</Samples>
  <Clock unit="Hz">32000000</Clock>
  <Format>complex</Format>
  <DataType>float32</DataType>
  <ScalingFactor unit="V">1</ScalingFactor>
  <NumberOfChannels>1</NumberOfChannels>
  <DataFilename>File.complex.1ch.float32</DataFilename>

  <UserData>
    <RohdeSchwarz>
      <DataImportExport_MandatoryData>
        <ChannelNames>
          <ChannelName>IQ Analyzer</ChannelName>
        </ChannelNames>
        <CenterFrequency unit="Hz">0</CenterFrequency>
      </DataImportExport_MandatoryData>
      <DataImportExport_OptionalData>
        <Key name="Ch1_NumberOfPostSamples">150</Key>
        <Key name="Ch1_NumberOfPreSamples">150</Key>
      </DataImportExport_OptionalData>
    </RohdeSchwarz>
  </UserData>

</RS_IQ_TAR_FileFormat>

```

Example: ScalingFactor

Data stored as int16 and a desired full scale voltage of 1 V

ScalingFactor = 1 V / maximum int16 value = 1 V / 2^{15} = 3.0517578125e-5 V

Scaling Factor	Numerical value	Numerical value x ScalingFactor
Minimum (negative) int16 value	- 2^{15} = - 32768	-1 V
Maximum (positive) int16 value	$2^{15}-1$ = 32767	0.999969482421875 V

C.1.2 I/Q data binary file

The I/Q data is saved in binary format according to the format and data type specified in the XML file (see <Format> element and <DataType> element). To allow reading and writing of streamed I/Q data, all data is interleaved, i.e. complex values are inter-

leaved pairs of I and Q values and multi-channel signals contain interleaved (complex) samples for channel 0, channel 1, channel 2 etc. If the <NumberOfChannels> element is not defined, one channel is presumed.

Example: Element order for real data (1 channel)

```
I[0],           // Real sample 0
I[1],           // Real sample 1
I[2],           // Real sample 2
...
```

Example: Element order for complex cartesian data (1 channel)

```
I[0], Q[0],     // Real and imaginary part of complex sample 0
I[1], Q[1],     // Real and imaginary part of complex sample 1
I[2], Q[2],     // Real and imaginary part of complex sample 2
...
```

Example: Element order for complex polar data (1 channel)

```
Mag[0], Phi[0], // Magnitude and phase part of complex sample 0
Mag[1], Phi[1], // Magnitude and phase part of complex sample 1
Mag[2], Phi[2], // Magnitude and phase part of complex sample 2
...
```

Example: Element order for complex cartesian data (3 channels)

Complex data: I[channel no][time index], Q[channel no][time index]

```
I[0][0], Q[0][0], // Channel 0, Complex sample 0
I[1][0], Q[1][0], // Channel 1, Complex sample 0
I[2][0], Q[2][0], // Channel 2, Complex sample 0

I[0][1], Q[0][1], // Channel 0, Complex sample 1
I[1][1], Q[1][1], // Channel 1, Complex sample 1
I[2][1], Q[2][1], // Channel 2, Complex sample 1

I[0][2], Q[0][2], // Channel 0, Complex sample 2
I[1][2], Q[1][2], // Channel 1, Complex sample 2
I[2][2], Q[2][2], // Channel 2, Complex sample 2
...
```

Example: Element order for complex cartesian data (1 channel)

This example demonstrates how to store complex cartesian data in float32 format using MATLAB®.

```
% Save vector of complex cartesian I/Q data, i.e. iqiqliq...
N = 100
iq = randn(1,N)+1j*randn(1,N)
fid = fopen('xyz.complex.float32','w');
for k=1:length(iq)
```

```

        fwrite(fid, single(real(iq(k))), 'float32');
        fwrite(fid, single(imag(iq(k))), 'float32');
    end
    fclose(fid)

```

Example: PreviewData in XML

```

<PreviewData>
  <ArrayOfChannel length="1">
    <Channel>
      <PowerVsTime>
        <Min>
          <ArrayOfFloat length="256">
            <float>-134</float>
            <float>-142</float>
            ...
            <float>-140</float>
          </ArrayOfFloat>
        </Min>
        <Max>
          <ArrayOfFloat length="256">
            <float>-70</float>
            <float>-71</float>
            ...
            <float>-69</float>
          </ArrayOfFloat>
        </Max>
      </PowerVsTime>
      <Spectrum>
        <Min>
          <ArrayOfFloat length="256">
            <float>-133</float>
            <float>-111</float>
            ...
            <float>-111</float>
          </ArrayOfFloat>
        </Min>
        <Max>
          <ArrayOfFloat length="256">
            <float>-67</float>
            <float>-69</float>
            ...
            <float>-70</float>
            <float>-69</float>
          </ArrayOfFloat>
        </Max>
      </Spectrum>
      <IQ>
        <Histogram width="64" height="64">0123456789...0</Histogram>
      </IQ>
    </Channel>
  </ArrayOfChannel>
</PreviewData>

```

```

        </Channel>
    </ArrayOfChannel>
</PreviewData>

```

C.2 CSV file format

CSV files contain I/Q data as comma-separated values. Additional metadata can be saved.

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C.2.1 Mandatory data elements

Parameter Name	Possible Values
Name	String
Comment	String
DateTime	Year-Month-DayTHour:Min:Sec
Format	complex
DataType	float32
NumberOfChannels	Integer
Ch<n>_ChannelName	String
Ch<n>_Samples	Integer
Ch<n>_Clock[Hz]	double
Ch<n>_CenterFrequency[Hz]	Double
IQ Data Header	<Channel Name>_I; <Channel Name>_Q (IQ data value)
-----	Double ; Double (IQ data I/Q pairs)

C.2.2 Optional data elements

Parameter name	Possible Values
Ch<n>_AttenuElecState	ON OFF
Ch<n>_AttenuElecValue[dB]	Integer
Ch<n>_AttenuMech[dB]	Integer

Parameter name	Possible Values
Ch<n>_CalibrationState	ON OFF
Ch<n>_DeviceHwInfo	String
Ch<n>_DeviceId	String
Ch<n>_DeviceOptions	String
Ch<n>_DeviceVersions	String
Ch<n>_FilterSettings	FLAT GAUSS OFF
Ch<n>_HighPassFilterState	ON OFF
Ch<n>_Impedance[Ohm]	50 75
Ch<n>_InputCoupling	AC DC
Ch<n>_InputPath	RF
Ch<n>_MeasBandwidth[Hz]	double
Ch<n>_NumberOfPostSamples	Integer
Ch<n>_NumberOfPreSamples	Integer
Ch<n>_PreampGain[dB]	Integer
Ch<n>_PreampState	ON OFF
Ch<n>_RefLevelOffset[dB]	Double
Ch<n>_RefLevel[dBm]	Double
Ch<n>_RefOscillatorInput	OFF ON
Ch<n>_RefOscillatorFreq[Hz]	Double
Ch<n>_TrgSource	Extern <1..4> I/Q Power IF Power RF Power Power Sensor Time
Ch<n>_TrgLevel[dB]	Double
Ch<n>_TrgHysteresis[dB]	Double
Ch<n>_TrgTpis[s]	Double
Ch<n>_TrgOffset[s]	Double
Ch<n>_TrgSlope	Rising Falling Rising/Falling
Ch<n>_TrgHoldoff[s]	Double
Ch<n>_TrgDropOut[s]	Double
Ch<n>_YigPreSelectorState	ON OFF

C.2.3 Example

```

DataImportExport_MandatoryData;
Name;ExampleFile
Comment;Example Comment

```



```

DateTime;2015-02-19T15:26:33
Format;complex
DataType;float32
NumberOfChannels;1
Ch1_ChannelName;Example_Channel
Ch1_Samples;10
Ch1_Clock[Hz];3,2000000E+007
Ch1_CenterFrequency[Hz];100,0000000E+007
DataImportExport_EndHeaderSection;
Example_Channel_I;Example_Channel_Q
-5,9390777E-006;-3,4644620E-006
9,8984629E-007;-8,4631858E-005
-5,9885701E-005;4,1078620E-005
2,0786772E-005;7,8692778E-005
-4,9492314E-006;-1,5095156E-004
1,6332464E-005;1,8312156E-005
-5,4936470E-005;4,5532928E-005
-4,8997390E-005;9,7004937E-005
-1,1383232E-005;4,5532928E-005
-8,2157239E-005;3,2170003E-005

```

C.2.4 Simple CSV format

The simple .CSV format contains I/Q data only, without any header or meta data. That is, the file contains only (I,Q) data pairs, separated by commas. Several streams of data can be provided in one file, one after the other.

Example:

```

7.0663854e-003,1.7059683e-005,
7.0817876e-003,7.5836733e-006,
7.0711789e-003,-1.2189972e-005,

```



This format requires setting the sample rate and measurement time or record length manually.

C.3 IQW file format

IQW is a binary file format containing one channel of complex IQ data.

Format description details:

- IQDataFormat: Complex
- IQDataType: Float32
- Byte order: Intel
- Data order: IQIQIQ (I/Q paired or interleaved) or IIIQQQ (I/Q blocks, default)



This format requires setting the sample rate and measurement time or record length manually.

Mandatory Data Elements

Only the binary I/Q data.

Optional Data Elements

None.

C.4 Matlab® v. 4 / v. 7.3 file format

In Matlab® files, channel-related information is stored in Matlab® variables with names starting with 'ChX_'. 'X' represents the number of the channel with a lower bound of 1, e.g. the variable `Ch1_ChannelName` contains the name of the first channel. The corresponding data is contained in `ChX_Data`.

Optional user data can be saved to variables named `UserDataX`, where 'X' starts at 0. The variable `UserData_Count` contains the number of `UserData` variables. For compatibility reasons user data needs to be saved as a 2xN char array, where the first row contains the key of the user data and the second row the actual value. Both rows must have the same column count and are therefore right-padded with white spaces.

Variables can be written to the *.mat files in arbitrary order.



The Matlab® v7.3. file format requires the Matlab® Compiler Runtime (MCR) to be installed on the system and registered in the `PATH` environment variable.



This format requires setting the sample rate and measurement time or record length manually.

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C.4.1 Mandatory data elements

Variable name	Class	Format / possible values
Name	char	
Comment	char	
DateTime	char	Year-Month-DayTHour:Min:Sec

Variable name	Class	Format / possible values
Format	char	complex
DataType	char	float32
NumberOfChannels	Double	
Ch<n>_ChannelName	char	
Ch<n>_Samples	double	
Ch<n>_Clock_Hz	double	
Ch<n>_CFrequency_Hz	Double	
Ch<n>_Data	Double, Double	I,Q
UserData_Count	Double	(Number of optional user data variables)

C.4.2 Optional data elements

Optional user data can be saved to variables named `UserDataX`, where 'x' starts at 0. The variable `UserData_Count` contains the number of `UserData` variables. For compatibility reasons user data needs to be saved as a 2xN char array, where the first row contains the key of the user data and the second row the actual value. Both rows must have the same column count and are therefore right-padded with white spaces.

Variable name	Class	Format
UserData<n>	char	Optional Data Parameter name, Value

Table C-2: Optional parameter names to be defined in `UserData<n>` variables

Parameter name	Possible Values
Ch<n>_AttenuElecState	ON OFF
Ch<n>_AttenuElecValue_dB	Integer
Ch<n>_AttenuMech_dB	Integer
Ch<n>_CalibrationState	ON OFF
Ch<n>_DeviceHwInfo	String
Ch<n>_DeviceId	String
Ch<n>_DeviceOptions	String
Ch<n>_DeviceVersions	String
Ch<n>_FilterSettings	FLAT GAUSS OFF
Ch<n>_HighPassFilterState	ON OFF
Ch<n>_Impedance_Ohm	50 75
Ch<n>_InputCoupling	AC DC
Ch<n>_InputPath	RF

Parameter name	Possible Values
Ch<n>_MeasBandwidth_Hz	double
Ch<n>_NumberOfPostSamples	Integer
Ch<n>_NumberOfPreSamples	Integer
Ch<n>_PreampGain_dB	Integer
Ch<n>_PreampState	ON OFF
Ch<n>_RefLevelOffset_dB	Double
Ch<n>_RefLevel_dBm	Double
Ch<n>_RefOscillatorInput	OFF ON
Ch<n>_RefOscillatorFreq_Hz	Double
Ch<n>_TrgSource	Extern <1 ..4> I/Q Power IF Power RF Power Power Sensor Time
Ch<n>_TrgLevel_dB	Double
Ch<n>_TrgHysteresis_dB	Double
Ch<n>_TrgTpis_s	Double
Ch<n>_TrgOffset_s	Double
Ch<n>_TrgSlope	Rising Falling Rising/Falling
Ch<n>_TrgHoldoff_s	Double
Ch<n>_TrgDropOut_s	Double
Ch<n>_YigPreSelectorState	ON OFF

C.4.3 Example

Workspace						
Stack: Base						
Select data to plot						
Name	Value	Size	Min	Max	Class	Range
Ch1_CFrequency_Hz	4.0000e+09	1x1	4.0000...	4.0000...	double	0
Ch1_ChannelName	'IQ Analyzer'	1x11			char	
Ch1_Clock_Hz	32000000	1x1	32000...	32000...	double	0
Ch1_Data	<1301x2 double>	1301x2	-2.128...	2.6082...	double	4.7364e-04
Ch1_Samples	1301	1x1	1301	1301	double	0
Comment	"	1x0			char	
DataType	'float32'	1x7			char	
DateTime	'2015-02-19T15:25:58'	1x19			char	
Format	'complex'	1x7			char	
Name	'VSE_1.10a 29 Beta'	1x17			char	
NumberOfChannels	1	1x1	1	1	double	0
UserData0	<2x19 char>	2x19			char	
UserData1	<2x23 char>	2x23			char	
UserData10	<2x17 char>	2x17			char	
UserData11	<2x13 char>	2x13			char	
UserData12	<2x20 char>	2x20			char	
UserData13	<2x23 char>	2x23			char	
UserData14	<2x22 char>	2x22			char	
UserData15	<2x18 char>	2x18			char	
UserData16	<2x15 char>	2x15			char	
UserData17	<2x22 char>	2x22			char	
UserData18	<2x17 char>	2x17			char	
UserData19	<2x22 char>	2x22			char	
UserData2	<2x18 char>	2x18			char	
UserData20	<2x13 char>	2x13			char	
UserData21	<2x23 char>	2x23			char	
UserData3	<2x20 char>	2x20			char	
UserData4	<2x744 char>	2x744			char	
UserData5	<2x44 char>	2x44			char	
UserData6	<2x192 char>	2x192			char	
UserData7	<2x18 char>	2x18			char	
UserData8	<2x23 char>	2x23			char	
UserData9	<2x18 char>	2x18			char	
UserData_Count	22	1x1	22	22	double	0

C.4.4 Simple matlab® format

As of R&S FSV/A software version 1.50, a simple `.mat` format is supported. This format contains I/Q data only, without any meta data. That is, the file contains only variables (double, double) for the corresponding channel data.



When you load a simple Matlab® file, you must define the used sample rate (and optionally analysis bandwidth) manually.

C.5 AID format

AID is a format used to transmit real or complex baseband signals. The IF signal is sent along with information that characterizes the datastream and datastream source.

All datastreams have a frame based structure using the same format, consisting of a global *Frame header* coupled with a data-type specific *Frame body* (i.e. the frame payload).

The header and the body of the frame consist of a number of 32-bit words. The *Frame header* has a predefined structure and size. The size and structure of the *Frame body* depends on the payload type. This is an important factor in the choice of the frame size.

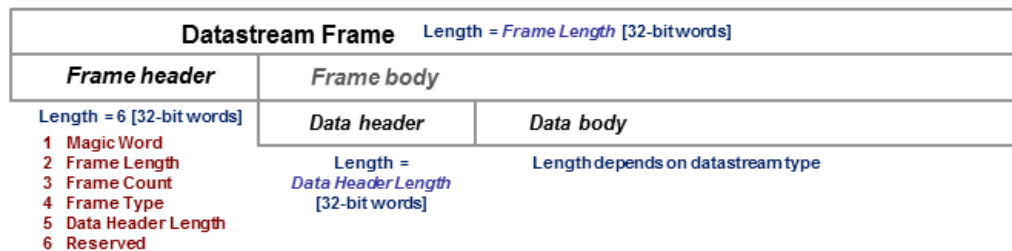


Figure C-1: Generic Datastream Frame structure

Global Frame header

The *Frame header* contains information used for frame synchronization, frame sequencing, payload identification and frame sizing. It consists of six 32-bit words as depicted in the following figure and is defined in `rs_gx40x_global_frame_header_if_defs.h`

Table C-3: Global Frame header (structure name: `typFRH_FRAMEHEADER`)

Word position in frame	Member name Member type	Description
1	uintMagicWord ptypUINT	Magic Word - 32-bit word, always identical (<code>0xFB746572</code>), defines the start of the <i>Frame header</i> and is used for frame synchronization. The <i>Magic Word</i> and the <i>Frame Length</i> are used to identify the beginning of each frame.
2	uintFrameLength ptypUINT	Frame Length - gives the length of the frame including both <i>Frame header</i> and <i>Frame body</i> . The length is expressed in 32-bit words. The minimum length is six in case the <i>Frame body</i> is empty and the maximum length is limited to the value: <code>kFRH_FRAME_LENGTH_MAX = 0x100000</code> ($1048576 = 2^{20}$) in case of normal frames <code>kFRH_FRAME_LENGTH_MAX_EX = 0x400000</code> ($64 \cdot 1048576 = 2^{26}$) in case of extended frames (an extended frame is marked by Bit#0 of the Reserved word of the frame header). Only some datastream types allow the extended frame size, see the definitions in the <code>rs_gx40x_global_frame_types_if_defs.h</code>. The next <i>Magic Word</i> which denotes the next frame in this data stream will occur <code>uintFrameLength</code> [32-bit words] after the <i>Magic Word</i> in this frame.
3	uintFrameCount ptypUINT	Frame Count - sequence counter modulo 2^{32} . Determines the position of this frame in the datastream and is used for sequencing and lost frame detection.

Word position in frame	Member name Member type	Description
4	uintFrameType ptypUINT	Frame Type - identifies the data type contained in this frame and gives the specific structure of the frame payload. The complete list of frame types (i.e. datastream types) can be found in the following header file: <code>rs_gx40x_global_frame_types_if_defs.h</code>
5	uintDataHeaderLength ptypUINT	Data Header Length - gives the length of the <i>data header</i> positioned at the beginning of the <i>Frame body</i> . The length is expressed in 32-bit words (0 means no data header). This information can be used by the software to recognize the version of the datastream format and thus its compatibility to read and correctly interpret the datastream. It enables forward-compatibility with future datastream versions. This value will not vary for a continuous data stream.
6	uintReserved ptypUINT	- Bits #31 to #1 - Reserved (not yet used, must be 0) - Bit #0 - Marks the frame with extended size (up to <code>KFRH_FRAME_LENGTH_MAX_EX</code> 32-bit words).



The *Data Header Length* information is very important for the correct addressing of the data samples. This information gives the exact position in the frame where the *Data body* begins independent of the version of the *data header* (different versions consist of different number of parameters). From the frame beginning (indicated by the *Magic Word*), the first six 32-bit words represent the *Frame header* and the next *Data Header Length* 32-bit words represent the *data header*. After $6 + \text{uintDataHeaderLength}$ 32-bit words starts the *Data body*, i.e. the first data sample.

Frame body

The *Frame body* contains the payload of the frame and its structure depends on the datastream type, as defined by the *Frame Type* element in the *Frame header*.

The *Frame body* is structured into a *data header* followed by the *Data body*. The *data header* contains datastream specific information of the payload.



Bit numbering

Throughout this format description it is assumed that bit #0 is the bit of least numeric significance.

C.5.1 Data body

The IF Data format is valid for the following datastream types:

Table C-4: IF Datastream types

Datastream type ID	Description	Sample data type
ekFRH_DATASTREAM__IFDATA_32RE_32IM_FIX ekFRH_DATASTREAM__IFDATA_32RE_32IM_FIX_RESCALED	Complex IF Data samples, 32-bit real-part and 32-bit imaginary-part, fixed point	typIFD_SAMPLE_32RE_32IM_FIX
ekFRH_DATASTREAM__IFDATA_16RE_16IM_FIX	Complex IF Data samples, 16-bit real-part and 16-bit imaginary-part, fixed point	typIFD_SAMPLE_16RE_16IM_FIX
ekFRH_DATASTREAM__IFDATA_16RE_16RE_FIX	Real IF Data samples, 16-bit real-part, two samples in each 32-bit word, fixed point	typIFD_SAMPLE_16RE_16RE_FIX
ekFRH_DATASTREAM__IFDATA_32RE_32IM_FLOAT_RESCALED	Complex IF Data samples, 32-bit real-part und 32-bit imaginary-part, floating point	typIFD_SAMPLE_32RE_32IM_FLOAT

For the datastream types defined in Table C-4, the same frame body structure is used, the only difference is the carried sample data type.

IF Data Frame Structure

The structure of the IF Datastream is defined in the `rs_gx40x_global_ifdata_header_if_defs.h` header file.

The Data Frame consists of the global *frame header* of type `typFRH_FRAMEHEADER`, as described in "Global Frame header" on page 374, followed by the datastream-specific *Frame body*.

The corresponding "Frame Type" value from the *frame header* for this datastream type can be found in the global frame types header file: `rs_gx40x_global_frame_types_if_defs.h`.

The *frame body* consists of: the *data header* which describes the datastream payload and the *data body* which contains the actual datastream payload.

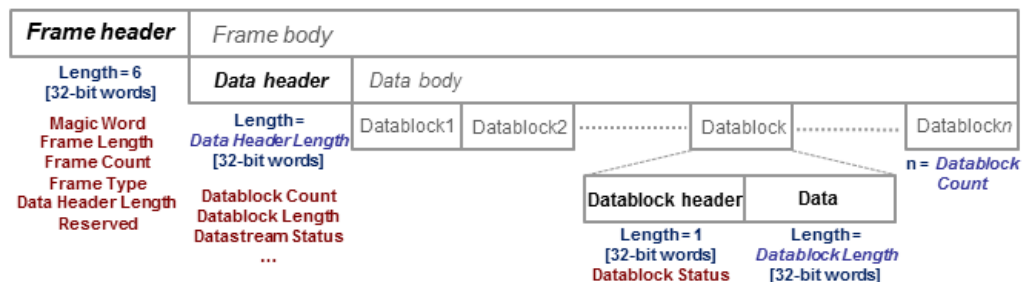


Figure C-2: IF Data frame format

IF data header

The data header describes the datastream payload (such as number of data samples contained in this frame), and contains common properties of the data samples.

The **basic** data header contains several fields that are always sent.

The **extended** data header contains extra information fields sent after the fields of the basic structure.

The length of the data header, as specified by the `uintDataHeaderLength` parameter from the frame header. This parameter provides information about which data header type is used - i.e. the basic header or the extended header.

The IF data header structure, of type `typIFD_IFDATAHEADER`, is described in the following table (data header length = 14 [32-bit words]).

Table C-5: IF DATA header (`typIFD_IFDATAHEADER`)

Word position in frame	Member name Member type	Description
7	uintDatablockCount ptypUINT	Datablock Count - represents the number of IF signal data blocks in the IF data frame.
8	uintDatablockLength ptypUINT	Datablock Length - The number 32-bit words in each IF signal data block excluding the data block header (has to be of the form 2^N with $N \geq 2$). This may not be the same as the number of IF signal data samples, as the size of a sample may be 16, 32 or 64 bits.
9 10	bigtimeTimeStamp ptypBIGTIME	64-bit Timestamp [μs] - Absolute time of the first IF signal data sample, in the first data block of IF signal data in this frame.
11	uintStatusword ptypUINT	Status Word - extra information that help the receiver react by parameter changes. • Bit #31 - Reserved • Bit #30 - dBFS flag – 1 indicates that all samples in this frame are considered to be dBFS (dB full scale). – 0 indicates that the values <i>Antenna Voltage Reference</i> and <i>Reciprocal gain correction</i> (see the <i>Status Word</i> description of the datablock header) can be used to calculate the corresponding level for each sample. • Bits #29 to #8 - Reserved (not yet used, must be 0) • Bits #7 to #0 - User flags for special signaling between IF Data processing components.
12	uintSignalSourceID ptypUINT	Signal Source Identifier or antenna identifier (value 0x0 if not used)

Word position in frame	Member name Member type	Description
13	uintSignalSourceState ptypUINT	Current Signal Source State (value 0×0 if not used) - gives the <i>Configuration Set Identifier</i> of the Task Data Set (in GX400) currently being applied by the IF signal source OR - the current <i>Scan Step Number</i> in the case of scan operation In the case of memory scanning, the scan step number starts at 0 for the scan channel (memory location) configured with the lowest frequency, and increments (+1) for every channel configured for scanning in the memory scan list. In the case of frequency scanning, the scan step number starts at 0 for the scan step at the lowest frequency, and increments (+1) for every step taken within the configured frequency scan range.
14 15	uintTunerFrequency_Low uintTunerFrequency_High ptypUINT	64-bit Tuner Center Frequency [Hz] - least significant 32 bits (uintTunerFrequency_Low) followed by most significant 32 bits (uintTunerFrequency_High)
16	uintBandwidth ptypUINT	IF signal 3dB Bandwidth [Hz]
17	uintSamplerate ptypUINT	Sample Rate of the AD Converter [samples / second] - due to digital filtering within the source, the resulting sample rate of the samples within this frame is: Sample Rate × Interpolation / Decimation
18	uintInterpolation ptypUINT	Interpolation Factor referred to the ADC signal sample rate. The value 0×1 indicates no interpolation

Word position in frame	Member name Member type	Description
19	uintDecimation ptypUINT	Decimation Factor referred to the ADC signal sample rate. The value 0x1 indicates no decimation
20	intAntennaVoltageRef ptypINT	Antenna Voltage Reference (Ant-VoltRef) is the device specific correction value for the tuner front-end Rx attenuation (expresses anything from antenna input connector to ADC) and is expressed in [0.1 dBμV]. This is the level which, while the AGC amplification is at maximum attenuation, is required at the antenna input to produce the full scale value at the ADC. Using this value together with the Recip-Gain (Reciprocal Gain) value, one can calculate the true signal level at the antenna input connector (see "Data samples" on page 381). The RecipGain value is given in the Status Word of the IF Datablock header Table C-4

The extended IF data header structure, of type `typIFD_IFDATAHEADER_EX`, is described in the following table (total data header length = 19 [32-bit words]).

Table C-6: Extended IF data header (typIFD_IFDATAHEADER_EX) - extra fields only

Word position in frame	Member name Member type	Description
21 22	bigtimeStartTimeStamp ptypBIGTIME_NS	64-bit Timestamp [ns] - Absolute time of the first sample of the datastream since starting the datastream ("Sample Counter" == 0). This value remains constant until the datastream is stopped and started again or until the tuner performs an internal synchronization.
23 24	uintSampleCounter_Low uintSampleCounter_High ptypUINT	Sample Count - 64-bit counter from the first sample of the first datablock in this frame. Note that this value can be reset when the datastream is stopped and started again or when the tuner performs an internal synchronization. The Sample Count of the next IF frame can be deduced from Datablock Count, Datablock Length and the number of 32-bit words per sample. In this way the number of sample Dropouts can be estimated (that can be replaced with Null values). The exact time is given by $t = \text{Start Time} + \text{Sample Count} * \text{Decimation} / (\text{Sample rate} * \text{Interpolation})$.
25	intKFactor ptypINT	kFactor - Correction factor of the current antenna, given in 0.1dB/m. Used to determine the field strength (in dBμV/m) at the antenna from the voltage level at the antenna input of the receiver. Contains antenna gain, cable attenuation, antenna switch matrix attenuation and anything else from air to antenna input. (the value 0x80000000 is used if no kFactor is defined).



The values contained in the data header fields represent the status at the beginning of the frame. A modification happening during the transmission of a frame will only be noted in the data header of the next frame.

IF Data Body

The IF data body contains zero or more IF Data samples arranged as an array of `typIFD_DATABLOCK` data blocks (the actual IF signal datastream payload). The number of datablocks is specified by the `Datablock Count` parameter from the data header.

Each datablock (`typIFD_DATABLOCK`) has its own datablock header: `datablockheaderDatablockHeader` (of type `typIFD_DATABLOCKHEADER`) and a datablock body that contains the actual data sample.

Table C-7: IF Datablock header (`typIFD_DATABLOCKHEADER`)

Member name Member type	Description
<code>uintStatusword</code> <code>ptypUINT</code>	Status of the Datablock - Bits #31 to #16 - RecipGain - Automatic Gain Control (AGC) Reciprocal Gain Correction value that was applied when generating the following IF Data samples. The RecipGain is represented as 16-bit unsigned decimal value (the 16-bit unsigned decimal has to be divided by $2^{16} = 65535$ to obtain the unsigned fractional between 0 and 1). For example a correction value of -17.5dB gives a value for RecipGain of 0.1333 which will be represented as <code>0x2220</code>. Using this value together with the value for the antenna voltage reference, one can calculate the true signal level at the antenna input connector (see "Data samples" on page 381). - Bits #15 to #8 - Reserved (must be 0). - Bits #7 to #2 - User flags for special signaling between IF Data processing components. Set to 0 if not used. - Bit #1 - Blanking flag - this flag is set (1) to indicate that the data in this block may have been falsified by some external event. - Bit #0 - Invalidity flag - this flag is set (1) to indicate that the data within this block may be corrupt (e.g. the input signal exceeded the range of the AD converter, or the analog signal input from which the data was converted was overloaded), OR any one of the fields in the IF datastream header does not represent the data in this block correctly.

The datablock body is defined as an array of size `uintDatablockLength` with `uintData` elements interpreted using the corresponding sample type format ("`typIFD_SAMPLE....`" as described in the following table). The actual IF data samples have to be extracted from the array. Their structure and size is given by the IF datastream format (Table C-4). The possible IF data sample formats are described in the table below:

Table C-8: IF data sample format

Sample type	Sample format	Most significant bits	Least significant bits	Data type
<code>typIFD_SAMPLE_32RE_32IM_FIX</code>	64-bit I/Q format	First 32-bit Real component		<code>ptypINT</code> or <code>ptyp-FLOAT_SP</code>
<code>typIFD_SAMPLE_32RE_32IM_FLOAT</code>		Second 32-bit Imaginary component		<code>ptypINT</code> or <code>ptyp-FLOAT_SP</code>

Sample type	Sample format	Most significant bits	Least significant bits	Data type
typIFD_SAMPLE_16RE_16IM_FIX	32-bit I/Q format	16-bit Imaginary component	16-bit Real component	ptypINT
typIFD_SAMPLE_16RE_16RE_FIX	16-bit Real format	16-bit sample number I+1	16-bit sample number I	ptypINT

The term 'fix' ('fixed' point) indicates signed (2s-complement) fixed point fractional numbers.

Data samples

The absolute signal level in [dBμV] may be calculated as follows:

$$\text{Level [dB}\mu\text{V]} = 10 \cdot \log(I_{\text{rel}}^2 + Q_{\text{rel}}^2) [\text{dB}] + 20 \cdot \log(\text{RecipGain} / 2^{16}) [\text{dB}] + 0.1 \cdot \text{AntVoltRef} [\text{dB}\mu\text{V}]$$

where I and Q are the real and imaginary parts of each signal sample.

The absolute signal level in [μV] may be calculated as follows:

$$I [\mu\text{V}] = I_{\text{rel}} \cdot (\text{RecipGain} / 2^{16}) \cdot \text{AntVoltLin}$$

$$Q [\mu\text{V}] = Q_{\text{rel}} \cdot (\text{RecipGain} / 2^{16}) \cdot \text{AntVoltLin}$$

$$\text{where AntVoltLin} [\mu\text{V}] = 10^{(0.1 \cdot \text{AntVoltRef}) / 20}$$

Depending on the sample format, as presented in [Table C-8](#), I and Q values can be represented as signed integers on 32-bits (I_{int32}) or 16-bits (I_{int16}) or as 32-bit float values (I_{float}). The relative values of I and Q can be calculated with the following formulas (same applies for Q_{rel}):

- $I_{\text{rel}} = I_{\text{int32}} / (2^{31} - 1)$ where I_{int32} is a signed integer, the most significant bit gives the sign (0 is positive, 1 is negative)
- $I_{\text{rel}} = I_{\text{int16}} / (2^{15} - 1)$ where I_{int16} is a signed integer, the most significant bit gives the sign (0 is positive, 1 is negative)
- $I_{\text{rel}} = I_{\text{float}}$

In the first two cases I_{rel} and Q_{rel} represent relative signal level values between -1 and 1. The absolute signal levels are retrieved through the parameter AntVoltRef as presented above. In the third case, I_{rel} and Q_{rel} can represent directly the absolute signal levels - in this case the RecipGain and AntVoltRef are not used (and are set to RecipGain=1, AntVoltRef=0).

Example

Word position in frame	Frame component name	Hex value	Description
1	uintMagicWord	FB746572	Frame synchronisation
2	uintFrameLength	0000001E	Entire frame length = 30 (in 32-bit units)

Word position in frame	Frame component name	Hex value	Description
3	uintFrameCount	000000FE	Running frame number = 254
4	uintFrameType	00000004	The type of data contained in this frame
5	uintDataHeaderLength	0000000E	data header length = 14 (in 32-bit units)
6	uintReserved	00000000	Reserved field
7	uintDatablockCount	00000002	Number of data blocks in this frame = 2
8	uintDatablockLength	00000004	The data block length (in 32-bit units) excluding the data block header = 4. Every data block in this frame will have the same length.
9 10	bigtimeTimeStamp	00035CED 1D63F4D0	Absolute time [µs] of the first IF signal data sample in this frame
11	uintStatusword	00000000	No status change indications.
12	uintSignalSourceID	00000003	Antenna ID = 3
13	uintSignalSourceState	00000A73	Tuner scan status = position 2675
14	uintTunerFrequency_Low	42E9EC0	Tuner center frequency = 1,123 GHz
15	uintTunerFrequency_High	00000000	
16	uintBandwidth	01312d00	The IF Data bandwidth = 20 MHz
17	uintSamplerate	0493E000	ADC sample rate = 76,8 Msample/s
18	uintInterpolation	00000001	Interpolation factor = none
19	uintDecimation	00000003	Decimation factor referred to the ADC sample rate = 3
20	intAntennaVoltageRef	0000001E	Antenna reference voltage = 3dBµV
21	uintStatusword	22200000	Beginning of the first Datablock. Statusword contains AGC correction factor = 0.1333 and no flags indications.
22	uintData	23873454	Real part of first sample
23	uintData	34234523	Imaginary part of first sample
24	uintData	56567543	Real part of second sample
25	uintData	34563456	Imaginary part of second sample
26	uintStatusword	41000004	Beginning of the second Datablock. Statusword contains AGC correction factor = 0.2539 and one user flag indication.
27	uintData	45345222	Real part of third sample
28	uintData	546672ab	Imaginary part of third sample
29	uintData	5BB25346	Real part of fourth sample
30	uintData	BBF7673e	Imaginary part of fourth sample

C.6 WV format

WV is a format used by Rohde & Schwarz signal generators to store waveforms. A waveform file contains a header and raw I/Q samples.

C.6.1 Mandatory elements

Each waveform file must begin with the TYPE tag. The sequence of the remaining tags is arbitrary.

Element	Description
TYPE	Designates the file type and source of creation (instrument type). Also includes an ASCII-coded checksum of the data part of the WAVEFORM tag, used to detect transmission errors.
CLOCK	The clock frequency (sample rate), in Hz
EMPTYTAG-Length	Length is an ASCII integer value that specifies the number of bytes in the EMPTYTAG, i.e. defines the number of bytes from the colon : to the end bracket }
WAVEFORM-Length	The actual waveform data (I/Q stream) Length specifies the number of bytes in a WAVEFORM tag and is calculated as follows: Length = Number of I/Q pairs * 4 (2 bytes per I and 2 bytes per Q value) + 1 byte (the length of the #) The binary data is represented by 16-bit signed integer in 2's complement notation. It contains the I and Q component alternately, starting with the I component. Each component consists of 2 bytes in Little endian format representation, i.e. least significant byte (LSB) first. The values of the 2 bytes in an I component and a Q component are in the range 0x00 to 0xFFFF (-32767 to +32767).

C.6.2 Optional elements

The following elements are optional in a .wv file.

Element	Description
DATE	Date and time at which the file was created Syntax: yyyy-mm-dd;hh:mm:ss
LEVEL OFFS	Offset of RMS and peak level relative to the 16-bit full scale modulation (-32767 to + 32767) = 0 dB.
SAMPLES	Number of I/Q samples in the waveform in ASCII format

List of commands (I/Q Analyzer)

[SENSe:] [WINDow<n>:] DETector<t>[:FUNCTION]	277
[SENSe:] [WINDow<n>:] DETector<t>[:FUNCTION]:AUTO	278
[SENSe:]ADJust:ALL	255
[SENSe:]ADJust:CONFigure:HYSTeresis:LOWer	256
[SENSe:]ADJust:CONFigure:HYSTeresis:UPPer	256
[SENSe:]ADJust:CONFigure:LEVel:DURation	255
[SENSe:]ADJust:CONFigure:LEVel:DURation:MODE	255
[SENSe:]ADJust:CONFigure:SMODE	256
[SENSe:]ADJust:CONFigure:TRIGger	256
[SENSe:]ADJust:FREQuency	257
[SENSe:]ADJust:LEVel	257
[SENSe:]ADJust:NCANcel:AVERage[:COUNT]	331
[SENSe:]ADJust:NCANcel:ERRor?	331
[SENSe:]ADJust:NCANcel:FILE:REference	332
[SENSe:]ADJust:NCANcel:FILE:RESult	332
[SENSe:]ADJust:NCANcel:STARt	333
[SENSe:]AVERage<n>:COUNT	278
[SENSe:]AVERage<n>:TYPE	277
[SENSe:]AVERage<n>[:STATe<t>]	279
[SENSe:]CORRection:COLLect[:ACQuire]	207
[SENSe:]CORRection:CVL:BAND	190
[SENSe:]CORRection:CVL:BIAS	191
[SENSe:]CORRection:CVL:CATalog?	191
[SENSe:]CORRection:CVL:CLEar	191
[SENSe:]CORRection:CVL:COMMeNt	192
[SENSe:]CORRection:CVL:DATA	192
[SENSe:]CORRection:CVL:HARMonic	192
[SENSe:]CORRection:CVL:MIXer	193
[SENSe:]CORRection:CVL:PORTs	193
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